HARDWARE RELEASE REPORT

Traction Inverter

Chassis: Chassis2
BOM lines: 106 (98 priced)
Generated: 2026-07-17

Cost at quantity

Units	Estimated total
1	$2,403.52
2	$4,739.58
3	$7,070.09
5	$11,744.59
10	$23,433.45

University of California, Santa Cruz — Corzine Lab

Contents

1. Order Summary
2. Fabrication Package
3. Control Board — schematic & PCB layers
4. DC Bus Capacitor Board — schematic & PCB layers
5. Filter Board — schematic & PCB layers
6. Gate Driver Board — schematic & PCB layers
7. IO Board — schematic & PCB layers
8. Capacitor Temp Sense Wiring Harness — wiring harness
9. Current Sensor Wiring Harness — wiring harness
10. Gate Drive Wiring Harness — wiring harness
11. IGBT Temperature Wiring Harness — wiring harness
12. Thermal Analysis — DC Link Module Standoff Heat Path — design document
13. Hazard Analysis & Risk Assessment — design document
14. User Manual — design document
15. Software Architecture Document — design document
16. Threat Analysis and Risk Assessment — design document

Order Summary

Digi-Key

Qty	Internal P/N	Description	Part Number	Unit	Total
1	HW-C2-HW-3M156058-ND-A	3M156058-ND	3M156058-ND	$46.9800	$46.98
				Subtotal	$46.98

McMaster-Carr

Qty	Internal P/N	Description	Part Number	Unit	Total
8 (1pk)	HW-C2-MECH-90128A232-A	90128A232	90128A232	$10.5500/pk	$10.55
5 (1pk)	HW-C2-MECH-90225A102-A	90225A102	90225A102	$12.8000/pk	$12.80
18 (1pk)	HW-C2-MECH-91292A134-A	91292A134	91292A134	$11.7200/pk	$11.72
4 (1pk)	HW-C2-MECH-91294A220-A	91294A220	91294A220	$12.7400/pk	$12.74
6 (1pk)	HW-C2-MECH-91502A180-A	91502A180	91502A180	$19.6500/pk	$19.65
6 (1pk)	HW-C2-MECH-91502A183-A	91502A183	91502A183	$17.2700/pk	$17.27
6	HW-C2-MECH-94669A190-A	94669A190	94669A190	$1.3900	$8.34
6	HW-C2-MECH-94669A196-A	94669A196	94669A196	$2.9600	$17.76
6	HW-C2-MECH-94669A199-A	94669A199	94669A199	$3.1000	$18.60
6	HW-C2-MECH-94669A202-A	94669A202	94669A202	$3.2200	$19.32
25 (1pk)	HW-C2-MECH-98044A224-A	98044A224	98044A224	$14.8800/pk	$14.88
				Subtotal	$163.63

Mouser

Qty	Internal P/N	Description	Part Number	Unit	Total
8	HW-C2-HW-0022011042-A	0022011042	-	N/A	N/A
32	HW-C2-HW-08-50-0113-A	08-50-0113	-	N/A	N/A
70	HW-C2-CAP-100N-A	100nF 100V	80-C1210C104K1RAC	$0.1610	$11.27
27	HW-C2-RES-10K-A	10k 1210	603-AC1210JR-0710KL	$0.0500	$1.35
21	HW-C2-CAP-10N-A	10nF 500V	187-CL32B103KGFNNNE	$0.1430	$3.00
37	HW-C2-CAP-10U-A	10uF 25V	187-CL32A106KAULNNE	$0.1340	$4.96

Qty	Internal P/N	Description	Part Number	Unit	Total
4	HW-C2-RES-120-A	120 1210	603-AC1210FR-07120RL	$0.1000	$0.40
17	HW-C2-RES-1K-A	1k 1210	603-RC1210FR-071KL	$0.0490	$0.83
8	HW-C2-CAP-1U-A	1uf 50V	187-CL32B105KBHNNNE	$0.2400	$1.92
2	HW-C2-CAP-2U2-A	2.2uf 1210 100v	80-C1210C225J1RAUTO	$0.5200	$1.04
12	HW-C2-RES-2R7-A	2.7 ohm 1210	667-ERJ-14BQF2R7U	$0.2340	$2.81
16	HW-C2-RES-20K-A	20k 1210 resistor	667-ERJ-P14F2002U	$0.2450	$3.92
4	HW-C2-HW-22-04-1041-A	22-04-1041	538-22-04-1041	$0.8900	$3.56
1	HW-C2-RES-220-A	220 1210	603-RC1210FR-07220RL	$0.1000	$0.10
6	HW-C2-CAP-220P-A	220pF 100V C0G 1210	80-C1210C221J2G	$0.8800	$5.28
4	HW-C2-CAP-22P-A	22pF 50V	80-C1210C220J5G	$0.7400	$2.96
3	HW-C2-RES-3K-A	3k 1210 1%	603-RC1210FR-073KL	$0.1000	$0.30
3	HW-C2-RES-4K7-A	4.7k 1210	71-CRCW12104K70FKEAH	$0.5900	$1.77
3	HW-C2-HW-455-2266-ND-A	455-2266-ND	306-XHP-2	$0.1000	$0.30
25	HW-C2-HW-464762-E-A	464762-E	305-464762	$0.3970	$9.93
1	HW-C2-RES-499K-A	499kohm 1210 resistor	667-ERJ-P14F4993U	$0.4400	$0.44
5	HW-C2-HW-524265-E-A	524265-E	305-524265-E	$7.8700	$39.35
5	HW-C2-HW-544959-E-A	544959-E	-	N/A	N/A
2	HW-C2-RES-560-A	560ohm 1210 resistor	667-ERJ-P14F5600U	$0.4400	$0.88
1	HW-C2-HW-776231-1-A	776231-1	571-776231-1	$16.6900	$16.69
2	HW-C2-IC-8MHZ-A	8mhz hc49-4h crystal	710-830003156B	$0.5000	$1.00
8	HW-C2-HW-971150581-A	971150581	710-971150581	$0.6000	$4.80
8	HW-C2-HW-971250581-A	971250581	710-971250581	$1.1000	$8.80
8	HW-C2-HW-971350581-A	971350581	710-971350581	$1.4800	$11.84
8	HW-C2-HW-9774030592R-A	9774030592R	710-9774030592R	$1.5700	$12.56
2	HW-C2-HW-ACT45B-101-2P-TL003-A	ACT45B-101-2P-TL003	810-ACT45B1012PTL003	$1.5300	$3.06
2	HW-C2-HW-AWHSH105D00G-A	AWHSH105D00G	523-AWHSH105D00G	$1.1100	$2.22
3	HW-C2-HW-B2B-XH-AM_LF__SN_-A	B2B-XH-AM_LF__SN_	306-B2B-XH-AMLFSNP	$0.1000	$0.30
12	HW-C2-HW-B58031U9254M062-A	B58031U9254M062	871-B58031U9254M062	$4.6500	$55.80
12	HW-C2-HW-BAT54WS-7-F-A	BAT54WS-7-F	621-BAT54WS-F	$0.1240	$1.49
5	HW-C2-HW-BTS462TATMA1-A	BTS462TATMA1	726-BTS462TATMA1	$2.6600	$13.30
3	HW-C2-HW-CM600DY-24T-A	CM600DY-24T	CM600DY-24T	$323.5000	$970.50
1	HW-C2-IC-CY15B102Q-SXET-A	CY15B102Q-SXET	727-CY15B102Q-SXET	$18.1400	$18.14

Qty	Internal P/N	Description	Part Number	Unit	Total
1	HW-C2-HW-DRDNB21D-7-A	DRDNB21D-7	621-DRDNB21D-7	$0.5600	$0.56
1	HW-C2-HW-EC7BW-110S12-A	EC7BW-110S12	418-EC7BW-110S12	$48.8600	$48.86
12	HW-C2-HW-FDFNYD1-110-5-A	FDFNYD1-110(5)	161-FDFNYD1-1105	$0.4800	$5.76
4	HW-C2-IC-ISO1042BDWVR-A	ISO1042BDWVR	595-ISO1042BDWVR	$4.4400	$17.76
1	HW-C2-IC-ISO1212DBQ-A	ISO1212DBQ	595-ISO1212DBQ	$1.7300	$1.73
4	HW-C2-HW-LA37S600S05KM-A	LA37S600S05KM	838-LA37S600S05KM	$19.9600	$79.84
1	HW-C2-HW-LT3080IQ-PBF-A	LT3080IQ#PBF	584-LT3080IQ#PBF	$8.5400	$8.54
2	HW-C2-HW-LTST-C230CKT-A	LTST-C230CKT	859-LTST-C230CKT	$0.4100	$0.82
2	HW-C2-HW-LTST-C230GKT-A	LTST-C230GKT	859-LTST-C230GKT	$0.3600	$0.72
1	HW-C2-IC-MAX22530AWE-A	MAX22530AWE+	700-MAX22530AWE+	$15.6000	$15.60
1	HW-C2-HW-MF-RHT200_32-2-A	MF-RHT200_32-2	652-MF-RHT200/32-2	$0.6100	$0.61
6	HW-C2-HW-MGJ2D121509MPC-R7-A	MGJ2D121509MPC-R7	580-MGJ2D121509MPCR7	$6.7700	$40.62
6	HW-C2-HW-MKP1848S61010JY5B-A	MKP1848S61010JY5B	75-MKP1848S61010JY5B	$7.5400	$45.24
5	HW-C2-HW-MM3Z3V0ST1G-A	MM3Z3V0ST1G	863-MM3Z3V0ST1G	$0.1700	$0.85
6	HW-C2-IC-NCV57100DWR2G-A	NCV57100DWR2G	863-NCV57100DWR2G	$2.9800	$17.88
1	HW-C2-HW-NXFT15XH103FEAB021-A	NXFT15XH103FEAB021	81-NXFT15XH103FEAB21	$0.3400	$0.34
1	HW-C2-HW-RD7-12S033R-A	RD7-12S033R	737-RD7-12S033R	$6.7700	$6.77
3	HW-C2-HW-RKE-1205S_H-A	RKE-1205S_H	919-RKE-1205S/H	$5.1700	$15.51
2	HW-C2-HW-RTS103C1R2M6L201-A	RTS103C1R2M6L201	527-RTS103C1R2M6L201	$2.2700	$4.54
2	HW-C2-HW-S3N-A	S3N	512-S3N	$0.6700	$1.34
2	HW-C2-HW-SSW-103-01-L-D-A	SSW-103-01-L-D	200-SSW10301LD	$1.3400	$2.68
1	HW-C2-HW-SSW-104-01-L-D-A	SSW-104-01-L-D	200-SSW10401LD	$1.6900	$1.69
2	HW-C2-HW-SSW-108-01-L-D-A	SSW-108-01-L-D	200-SSW10801LD	$2.3500	$4.70
1	HW-C2-CONN-SSW-110-01-F-D-A	SSW-110-01-F-D	200-SSW11001FD	$2.4100	$2.41
1	HW-C2-HW-SSW-110-01-F-D-A	SSW-110-01-F-D	-	N/A	N/A
1	HW-C2-IC-STM32G474RCT3-A	STM32G474RCT3	511-STM32G474RCT3	$8.3100	$8.31
1	HW-C2-IC-STM32H723ZGT6-A	STM32H723ZGT6	511-STM32H723ZGT6	$13.2500	$13.25

Qty	Internal P/N	Description	Part Number	Unit	Total
1	HW-C2-HW-STS41A-AWLB-R3-A	STS41A-AWLB-R3	403-STS41A-AWLB-R3	$1.1300	$1.13
6	HW-C2-HW-STTH112A-A	STTH112A	511-STTH112A	$0.5900	$3.54
6	HW-C2-HW-SXH-001T-P0-6-A	SXH-001T-P0.6	306-SXH-001T-P0.6	$0.1000	$0.60
4	HW-C2-HW-SZMMSZ4678T1G-A	SZMMSZ4678T1G	863-SZMMSZ4678T1G	$0.4300	$1.72
21	HW-C2-HW-TESTPOINT-TH-RED-A	Testpoint TH, RED	534-5000	$0.2930	$6.15
16	HW-C2-RES-250K-A	Thick Film Resistors - SMD 1/2watt 249Kohms 1%	71-CRCW1210-249K-E3	$0.0700	$1.12
1	HW-C2-HW-TPS389006ADJRTER-A	TPS389006ADJRTER	595-TPS389006ADJRTER	$5.1000	$5.10
2	HW-C2-HW-TS40-66-50-R-260-SMT-TR-A	TS40-66-50-R-260-SMT-TR	179-TS406650R260S	$0.4800	$0.96
2	HW-C2-HW-TSW-103-18-G-D-A	TSW-103-18-G-D	200-TSW10318GD	$1.0100	$2.02
2	HW-C2-HW-TSW-108-18-F-D-A	TSW-108-18-F-D	200-TSW10818FD	$1.4500	$2.90
1	HW-C2-HW-TSW-110-18-T-D-A	TSW-110-18-T-D	200-TSW11018TD	$1.5800	$1.58
1	HW-C2-CONN-TSW-110-21-L-D-LL-A	TSW-110-21-L-D-LL	200-TSW11021LDLL	$2.2700	$2.27
13	HW-C2-HW-UCM1H101MCL1GS-A	UCM1H101MCL1GS	647-UCM1H101MCL1GS	$0.3250	$4.23
60	HW-C2-HW-UCS2D331MHD-A	UCS2D331MHD	647-UCS2D331MHD	$2.1600	$129.60
1	HW-C2-HW-UJ2-B-HR-G-V-A	UJ2-B-HR-G-V	179-UJ2-B-HR-G-V	$1.7000	$1.70
18	HW-C2-HW-VY1222M47Y5UQ6TV0-A	VY1222M47Y5UQ6TV0	72-VY1222M47Y5UQ6TV0	$0.2710	$4.88
				Subtotal	$1,733.29

PCB Fabrication

Qty	Internal P/N	Description	Part Number	Unit	Total
1	HW-C2-PCB-CTRL-A	ControlBoard	ControlBoard	$25.2400	$25.24
1	HW-C2-PCB-DCBUSCAP-A	DCBusCapacitorBoard	DCBusCapacitorBoard	$7.5400	$7.54
1	HW-C2-PCB-DCFLT-A	DCBusFilter	DCBusFilter	$8.4400	$8.44
1	HW-C2-PCB-GD-A	GateDriver	GateDriver	$11.6600	$11.66
1	HW-C2-PCB-IO-A	IOBoard	IOBoard	$11.6600	$11.66
				Subtotal	$64.54

SendCutSend

Qty	Internal P/N	Description	Part Number	Unit	Total
1	HW-C2-PLT-BSP-A	HW-C2-BSP-A	HW-C2-BSP-A	$201.6100	$201.61
1	HW-C2-PLT-CHSP-A	HW-C2-CHSP-A	HW-C2-CHSP-A	$33.3100	$33.31

Qty	Internal P/N	Description	Part Number	Unit	Total
2	HW-C2-BB-DCLBB-A	HW-C2-DCLBB-A	HW-C2-DCLBB-A	$50.7100	$101.42
3	HW-C2-BB-PBB-A	HW-C2-PBB-A	HW-C2-PBB-A	$19.5800	$58.74
				Subtotal	$395.08

Cost by Assembly

Assembly	Kind	Qty/chassis	Parts	Cost (1 unit)
PCB assembly: Control Board	PCB assembly	1	43	$210.39
PCB assembly: DC Bus Capacitor Board	PCB assembly	1	2	$149.70
PCB assembly: Filter Board	PCB assembly	1	4	$122.23
PCB assembly: Gate Driver Board	PCB assembly	1	22	$138.47
PCB assembly: IO Board	PCB assembly	1	15	$105.24
chassis hardware: Mechanical	chassis hardware	1	13	$1,118.82
Capacitor Temp Sense Wiring Harness	wiring harness	1	3	$0.64
Current Sensor Wiring Harness	wiring harness	4	3	$79.84
Gate Drive Wiring Harness	wiring harness	1	3	$15.69
IGBT Temperature Wiring Harness	wiring harness	2	3	$5.14
fabricated part: Fab	fabricated part	1	1	$201.61
fabricated part: Fab	fabricated part	1	1	$33.31
fabricated part: Fab	fabricated part	2	1	$101.42
fabricated part: Fab	fabricated part	3	1	$58.74
			Total	$2,341.23

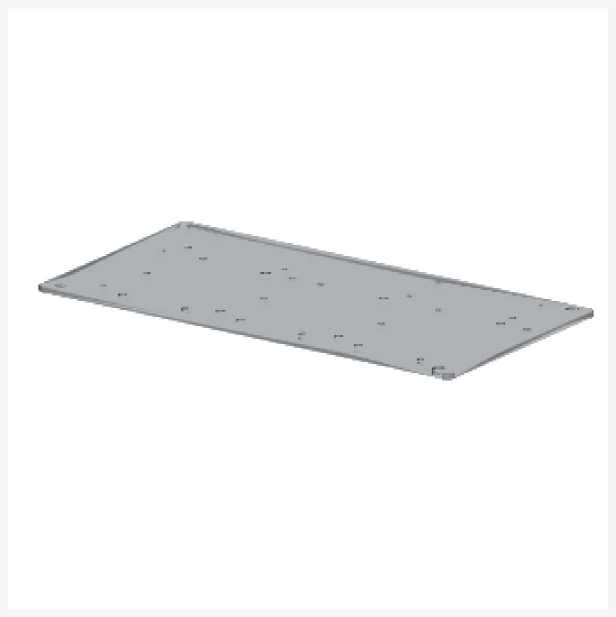
Costs are per-piece at current prices (pack prices divided out); assembly labor and PCB fab quantities at other build counts follow the pack math in the vendor sections.

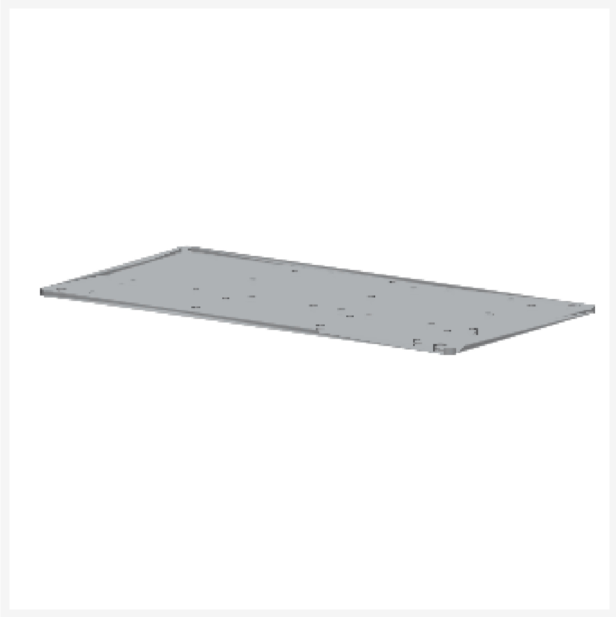
Fabrication Package

Item	Kind	Qty	Rev	Price	Ready
ControlBoard	PCB	1	A	$25.24	yes
DCBusCapacitorBoard	PCB	1	A	$7.54	yes
DCBusFilter	PCB	1	A	$8.44	yes
GateDriver	PCB	1	A	$11.66	yes
IOBoard	PCB	1	A	$11.66	yes
HW-C2-BSP-A	fab part	1	A	$201.61	yes
HW-C2-CHSP-A	fab part	1	A	$33.31	yes
HW-C2-DCLBB-A	fab part	2	A	$50.71	yes
HW-C2-PBB-A	fab part	3	A	$19.58	yes
CapacitorTempSenseHarness	harness	1	A	-	yes
CurrentSenseHarness	harness	4	A	-	yes
GDVSHarness	harness	1	A	-	yes
TempSenseHarness	harness	2	A	-	yes

HW-C2-BSP-A

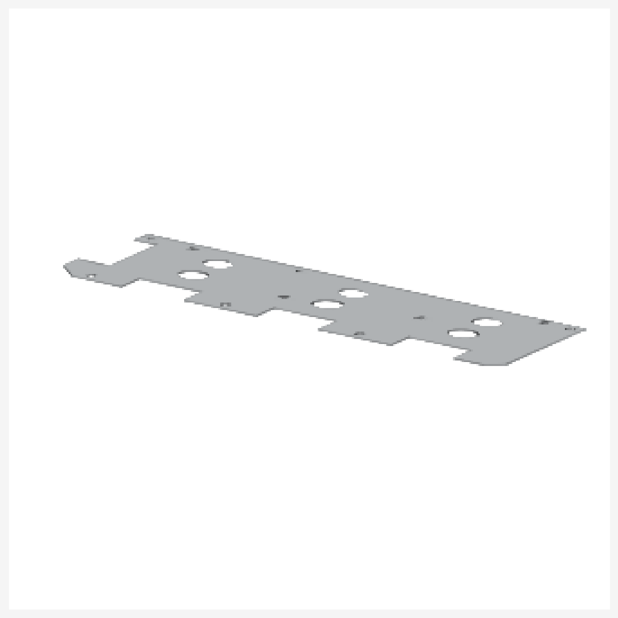
6061 T6 Aluminum / 9.52 mm / Sheet Cutting / Deburring, Tapping

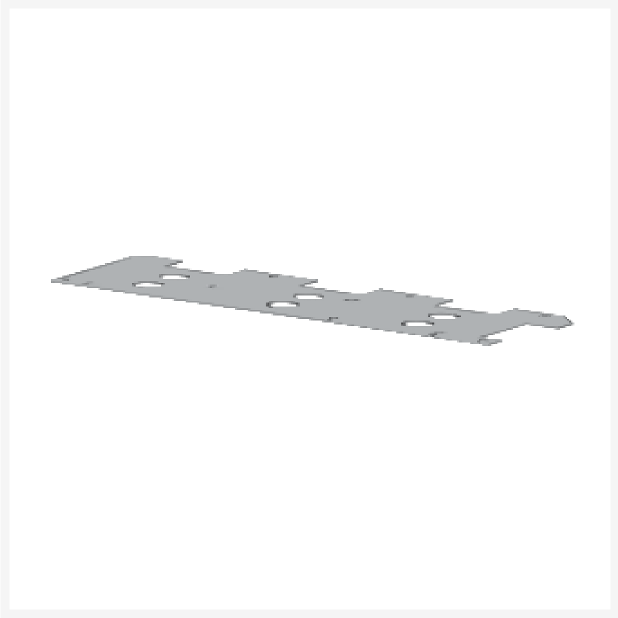




HW-C2-CHSP-A

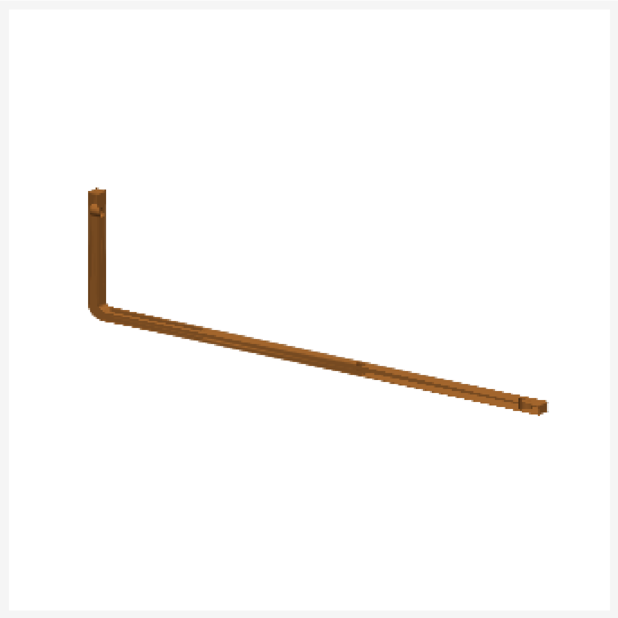
6061 T6 Aluminum / 3.17 mm / Sheet Cutting / Deburring

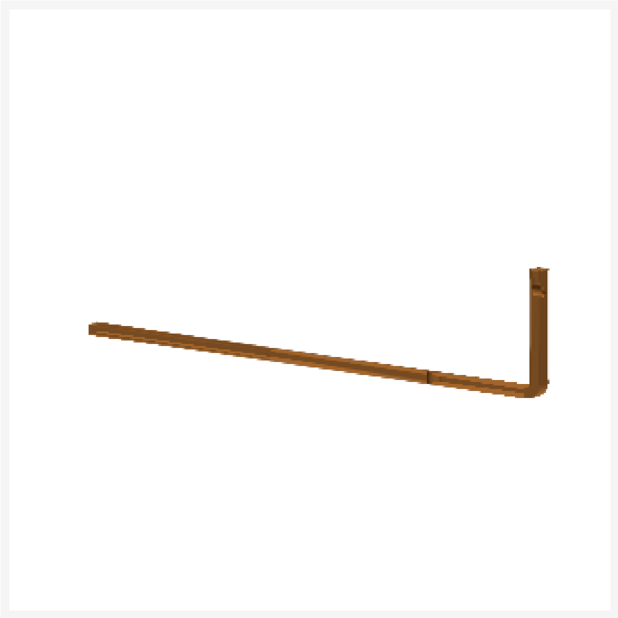




HW-C2-DCLBB-A

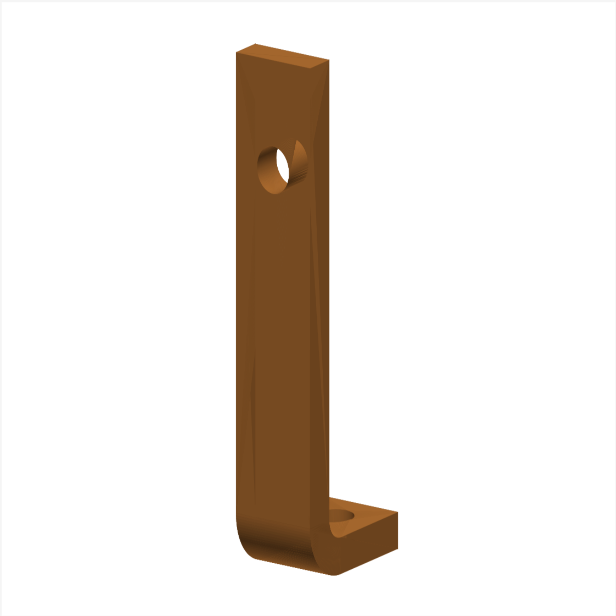
Copper / 4.75 mm / Sheet Cutting / Bending

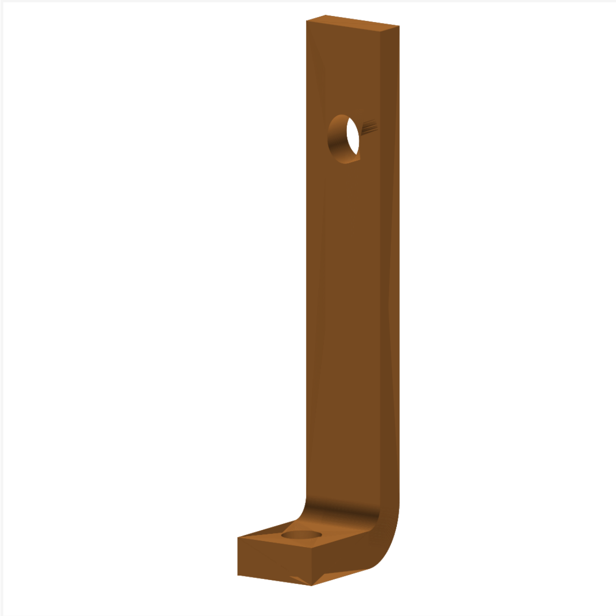




HW-C2-PBB-A

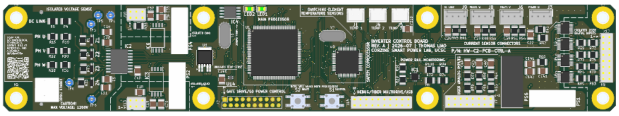
Copper / 4.75 mm / Sheet Cutting / Bending

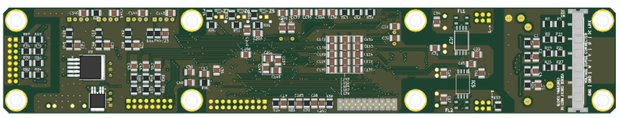


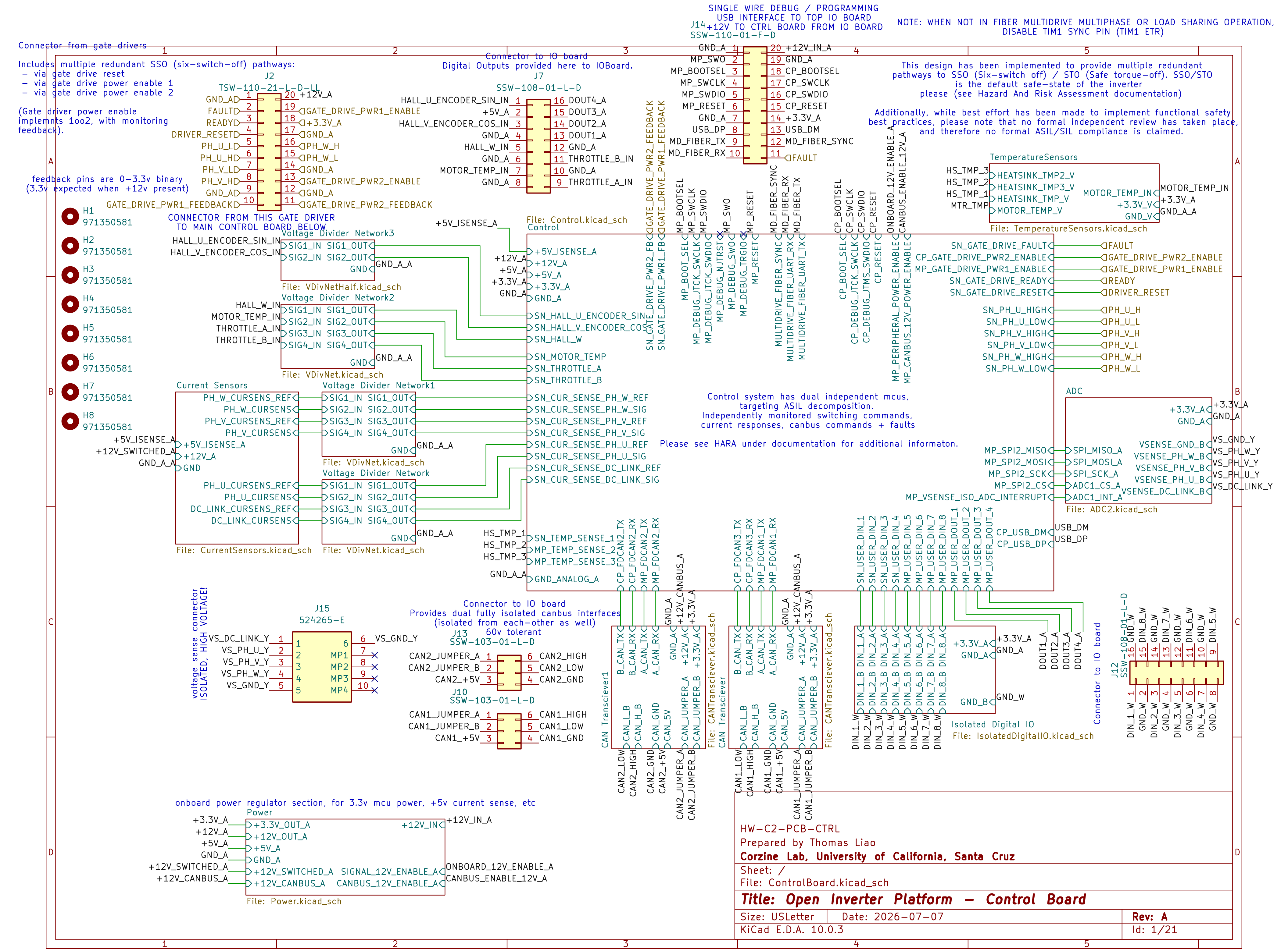


PCB ASSEMBLY: CONTROL BOARD

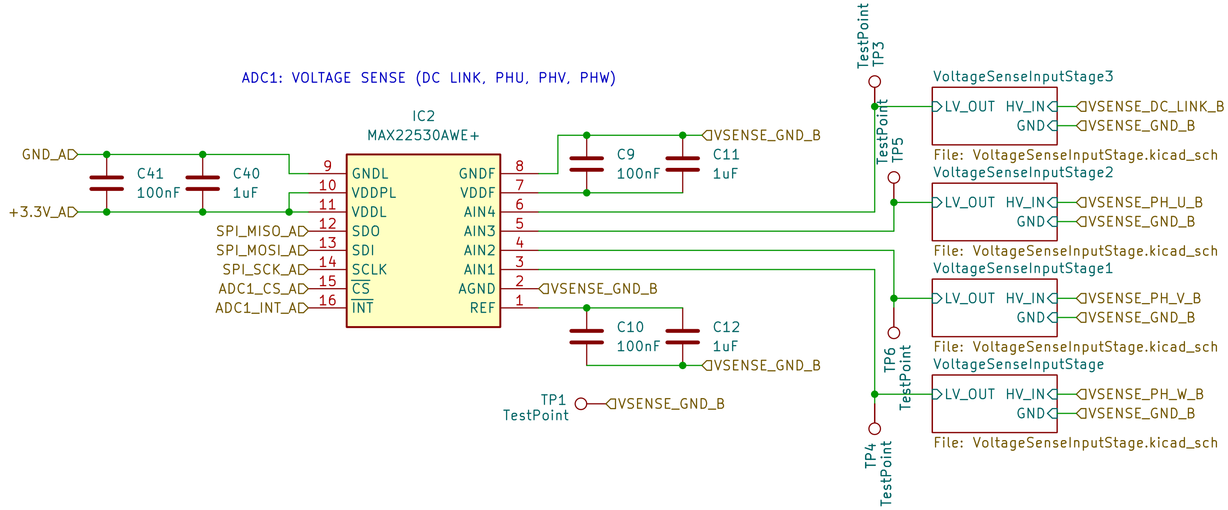
HW-C2-PCB-CTRL-A · printed circuit board — schematic and PCB layers







SINGLE WIRE DEBUG / PROGRAMMING
USB INTERFACE TO TOP IO BOARD
+12V TO CTRL BOARD FROM IO BOARD
J14
SSW-110-01-F-D
NOTE: WHEN NOT IN FIBER MULTIDRIVE MULTIPHASE OR LOAD SHARING OPERATION,
DISABLE TIM1 SYNC PIN (TIM1 ETR)
Connector from gate drivers
Includes multiple redundant SSO (six-switch-off) pathways:
- via gate drive reset
- via gate drive power enable 1
- via gate drive power enable 2
(Gate driver power enable implemnts 1oo2, with monitoring feedback).
feedback pins are 0-3.3v binary
(3.3v expected when +12v present)
J2
TSW-110-21-L-D-LL
CONNECTOR FROM THIS GATE DRIVER
TO MAIN CONTROL BOARD BELOW
Connector to IO board
Digital Outputs provided here to IOBoard.
J7
SSW-108-01-L-D
This design has been implemented to provide multiple redundant pathways to SSO (Six-switch off) / STO (Safe torque-off). SSO/STO is the default safe-state of the inverter
please (see Hazard And Risk Assessment documentation)
Additionally, while best effort has been made to implement functional safety best practices, please note that no formal independent review has taken place, and therefore no formal ASIL/SIL compliance is claimed.
TemperatureSensors
File: TemperatureSensors.kicad_sch
Control
File: Control.kicad_sch
Voltage Divider Network3
File: VDivNetHalf.kicad_sch
Voltage Divider Network2
File: VDivNet.kicad_sch
Current Sensors
File: CurrentSensors.kicad_sch
Voltage Divider Network1
File: VDivNet.kicad_sch
Voltage Divider Network
File: VDivNet.kicad_sch
Control system has dual independent mcus, targeting ASIL decomposition.
Independently monitored switching commands, current responses, canbus commands + faults
Please see HARA under documentation for additional informaton.
ADC
File: ADC2.kicad_sch
H1 971350581
H2 971350581
H3 971350581
H4 971350581
H5 971350581
H6 971350581
H7 971350581
H8 971350581
voltage sense connector
ISOLATED, HIGH VOLTAGE!
J15
524265-E
Connector to IO board
Provides dual fully isolated canbus interfaces
(isolated from each-other as well)
60v tolerant
J13
SSW-103-01-L-D
J10
SSW-103-01-L-D
CAN Transciever1
File: CANTransciever.kicad_sch
CAN Transciever
File: CANTransciever.kicad_sch
Isolated Digital IO
File: IsolatedDigitalIO.kicad_sch
Connector to IO board
J12
SSW-108-01-L-D
onboard power regulator section, for 3.3v mcu power, +5v current sense, etc
Power
File: Power.kicad_sch
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /
File: ControlBoard.kicad_sch
Title: Open Inverter Platform - Control Board
Size: USLetter
Date: 2026-07-07
KiCad E.D.A. 10.0.3
Rev: A
Id: 1/21

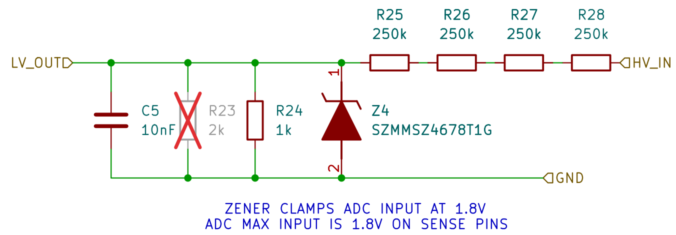


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/
File: ADC2.kicad_sch

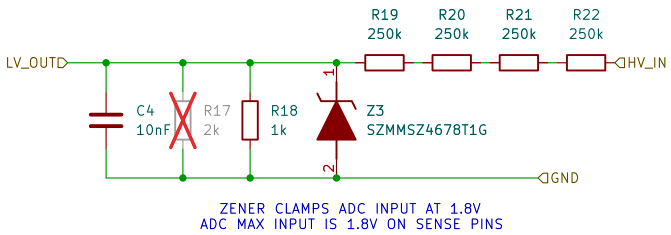
Title: Open Inverter Platform - Control Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 4/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/VoltageSenseInputStage/
File: VoltageSenseInputStage.kicad_sch
Title: Open Inverter Platform – Control Board

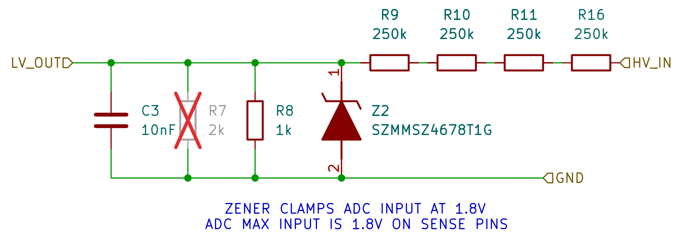
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 2/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/VoltageSenseInputStage1/
File: VoltageSenseInputStage.kicad_sch

Title: Open Inverter Platform - Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 3/21

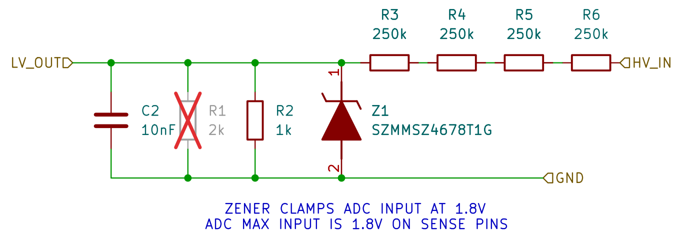


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage2/
File: VoltageSenseInputStage.kicad_sch

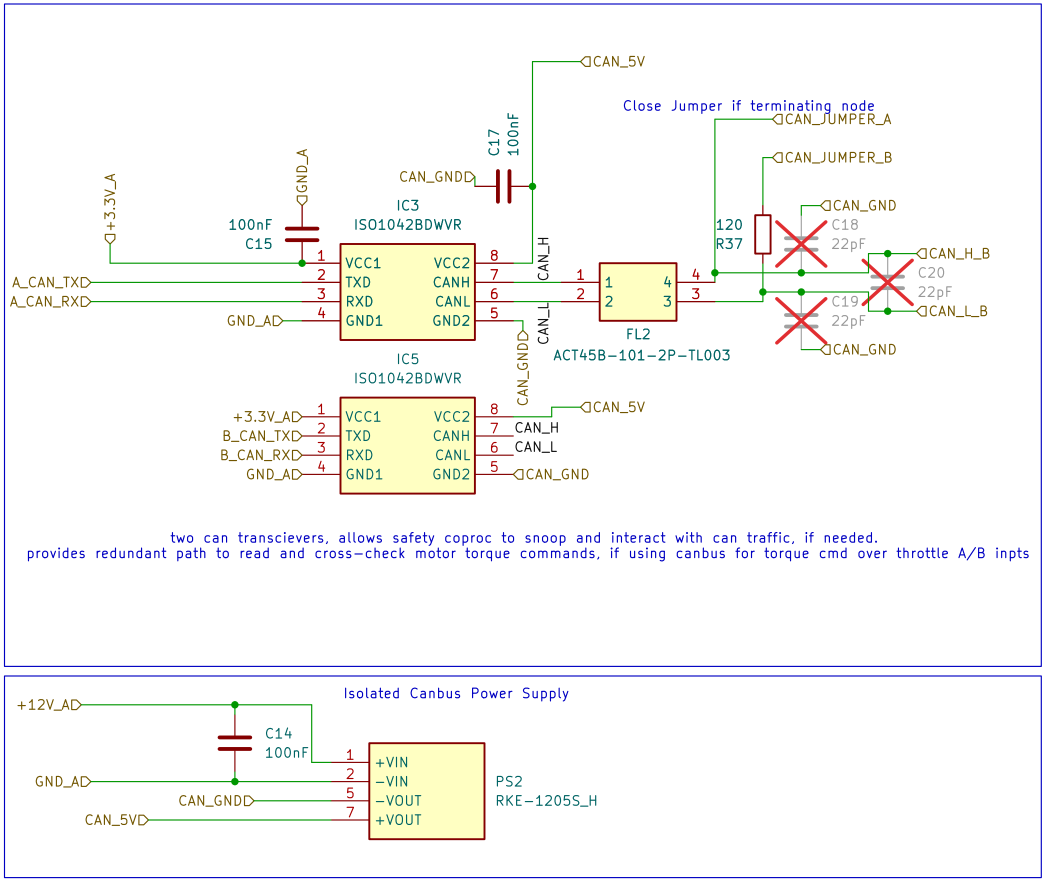
Title: Open Inverter Platform - Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 5/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/VoltageSenseInputStage3/
File: VoltageSenseInputStage.kicad_sch
Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 6/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

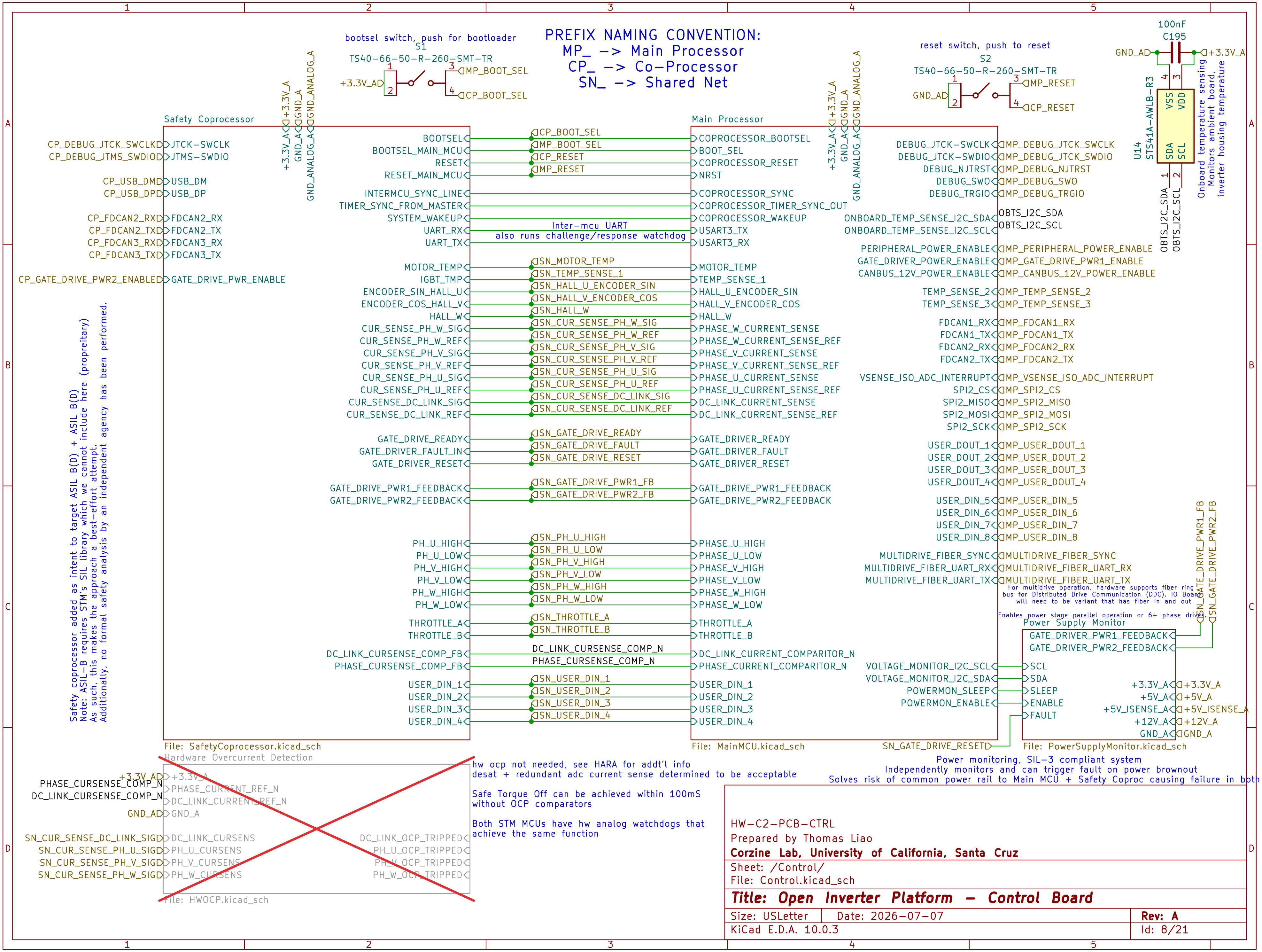
Corzine Lab, University of California, Santa Cruz

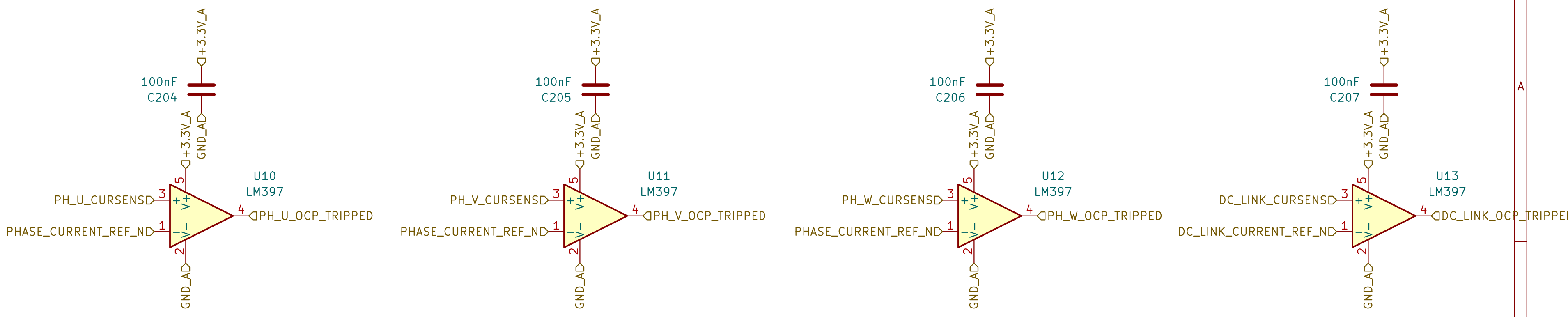
Sheet: /CAN Transciever1/

File: CANTransciever.kicad_sch

Title: Open Inverter Platform - Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 7/21


PREFIX NAMING CONVENTION:
MP_ -> Main Processor
CP_ -> Co-Processor
SN_ -> Shared Net
bootsel switch, push for bootloader
S1
TS40-66-50-R-260-SMT-TR
+3.3V_AD
MP_BOOT_SEL
CP_BOOT_SEL
reset switch, push to reset
S2
TS40-66-50-R-260-SMT-TR
GND_AD
MP_RESET
CP_RESET
100nF
C195
GND_AD
+3.3V_A
U14
STS41A-AWLB-R3
VSS
VDD
SDA
SCL
Onboard temperature sensing
Monitors ambient board,
inverter housing temperature
OBTS_I2C_SDA
OBTS_I2C_SCL
Safety Coprocessor
CP_DEBUG_JTCK_SWCLK
CP_DEBUG_JTMS_SWDIO
CP_USB_DM
CP_USB_DP
CP_FDCAN2_RX
CP_FDCAN2_TX
CP_FDCAN3_RX
CP_FDCAN3_TX
CP_GATE_DRIVE_PWR2_ENABLED
JTCK-SWCLK
JTMS-SWDIO
USB_DM
USB_DP
FDCAN2_RX
FDCAN2_TX
FDCAN3_RX
FDCAN3_TX
GATE_DRIVE_PWR_ENABLE
+3.3V_A
GND_A
GND_ANALOG_A
BOOTSEL
BOOTSEL_MAIN_MCU
RESET
RESET_MAIN_MCU
INTERMCU_SYNC_LINE
TIMER_SYNC_FROM_MASTER
SYSTEM_WAKEUP
UART_RX
UART_TX
MOTOR_TEMP
IGBT_TMP
ENCODER_SIN_HALL_U
ENCODER_COS_HALL_V
HALL_W
CUR_SENSE_PH_W_SIG
CUR_SENSE_PH_W_REF
CUR_SENSE_PH_V_SIG
CUR_SENSE_PH_V_REF
CUR_SENSE_PH_U_SIG
CUR_SENSE_PH_U_REF
CUR_SENSE_DC_LINK_SIG
CUR_SENSE_DC_LINK_REF
GATE_DRIVE_READY
GATE_DRIVER_FAULT_IN
GATE_DRIVER_RESET
GATE_DRIVE_PWR1_FEEDBACK
GATE_DRIVE_PWR2_FEEDBACK
PH_U_HIGH
PH_U_LOW
PH_V_HIGH
PH_V_LOW
PH_W_HIGH
PH_W_LOW
THROTTLE_A
THROTTLE_B
DC_LINK_CURSENSE_COMP_FB
PHASE_CURSENSE_COMP_FB
USER_DIN_1
USER_DIN_2
USER_DIN_3
USER_DIN_4
File: SafetyCoprocessor.kicad_sch
Inter-mcu UART
also runs challenge/response watchdog
SN_MOTOR_TEMP
SN_TEMP_SENSE_1
SN_HALL_U_ENCODER_SIN
SN_HALL_V_ENCODER_COS
SN_HALL_W
SN_CUR_SENSE_PH_W_SIG
SN_CUR_SENSE_PH_W_REF
SN_CUR_SENSE_PH_V_SIG
SN_CUR_SENSE_PH_V_REF
SN_CUR_SENSE_PH_U_SIG
SN_CUR_SENSE_PH_U_REF
SN_CUR_SENSE_DC_LINK_SIG
SN_CUR_SENSE_DC_LINK_REF
SN_GATE_DRIVE_READY
SN_GATE_DRIVE_FAULT
SN_GATE_DRIVE_RESET
SN_GATE_DRIVE_PWR1_FB
SN_GATE_DRIVE_PWR2_FB
SN_PH_U_HIGH
SN_PH_U_LOW
SN_PH_V_HIGH
SN_PH_V_LOW
SN_PH_W_HIGH
SN_PH_W_LOW
SN_THROTTLE_A
SN_THROTTLE_B
DC_LINK_CURSENSE_COMP_N
PHASE_CURSENSE_COMP_N
SN_USER_DIN_1
SN_USER_DIN_2
SN_USER_DIN_3
SN_USER_DIN_4
Main Processor
COPROCESSOR_BOOTSEL
BOOT_SEL
COPROCESSOR_RESET
NRST
COPROCESSOR_SYNC
COPROCESSOR_TIMER_SYNC_OUT
COPROCESSOR_WAKEUP
USART3_TX
USART3_RX
MOTOR_TEMP
TEMP_SENSE_1
HALL_U_ENCODER_SIN
HALL_V_ENCODER_COS
HALL_W
PHASE_W_CURRENT_SENSE
PHASE_W_CURRENT_SENSE_REF
PHASE_V_CURRENT_SENSE
PHASE_V_CURRENT_SENSE_REF
PHASE_U_CURRENT_SENSE
PHASE_U_CURRENT_SENSE_REF
DC_LINK_CURRENT_SENSE
DC_LINK_CURRENT_SENSE_REF
GATE_DRIVER_READY
GATE_DRIVER_FAULT
GATE_DRIVER_RESET
GATE_DRIVE_PWR1_FEEDBACK
GATE_DRIVE_PWR2_FEEDBACK
PHASE_U_HIGH
PHASE_U_LOW
PHASE_V_HIGH
PHASE_V_LOW
PHASE_W_HIGH
PHASE_W_LOW
THROTTLE_A
THROTTLE_B
DC_LINK_CURRENT_COMPARITOR_N
PHASE_CURRENT_COMPARITOR_N
USER_DIN_1
USER_DIN_2
USER_DIN_3
USER_DIN_4
File: MainMCU.kicad_sch
DEBUG_JTCK-SWCLK
DEBUG_JTCK-SWDIO
DEBUG_NJTRST
DEBUG_SWO
DEBUG_TRGIO
ONBOARD_TEMP_SENSE_I2C_SDA
ONBOARD_TEMP_SENSE_I2C_SCL
PERIPHERAL_POWER_ENABLE
GATE_DRIVER_POWER_ENABLE
CANBUS_12V_POWER_ENABLE
TEMP_SENSE_2
TEMP_SENSE_3
FDCAN1_RX
FDCAN1_TX
FDCAN2_RX
FDCAN2_TX
VSENSE_ISO_ADC_INTERRUPT
SPI2_CS
SPI2_MISO
SPI2_MOSI
SPI2_SCK
USER_DOUT_1
USER_DOUT_2
USER_DOUT_3
USER_DOUT_4
USER_DIN_5
USER_DIN_6
USER_DIN_7
USER_DIN_8
MULTIDRIVE_FIBER_SYNC
MULTIDRIVE_FIBER_UART_RX
MULTIDRIVE_FIBER_UART_TX
VOLTAGE_MONITOR_I2C_SCL
VOLTAGE_MONITOR_I2C_SDA
POWERMON_SLEEP
POWERMON_ENABLE
SN_GATE_DRIVE_RESET
MP_DEBUG_JTCK_SWCLK
MP_DEBUG_JTCK_SWDIO
MP_DEBUG_NJTRST
MP_DEBUG_SWO
MP_DEBUG_TRGIO
MP_PERIPHERAL_POWER_ENABLE
MP_GATE_DRIVE_PWR1_ENABLE
MP_CANBUS_12V_POWER_ENABLE
MP_TEMP_SENSE_2
MP_TEMP_SENSE_3
MP_FDCAN1_RX
MP_FDCAN1_TX
MP_FDCAN2_RX
MP_FDCAN2_TX
MP_VSENSE_ISO_ADC_INTERRUPT
MP_SPI2_CS
MP_SPI2_MISO
MP_SPI2_MOSI
MP_SPI2_SCK
MP_USER_DOUT_1
MP_USER_DOUT_2
MP_USER_DOUT_3
MP_USER_DOUT_4
MP_USER_DIN_5
MP_USER_DIN_6
MP_USER_DIN_7
MP_USER_DIN_8
MULTIDRIVE_FIBER_SYNC
MULTIDRIVE_FIBER_UART_RX
MULTIDRIVE_FIBER_UART_TX
For multidrive operation, hardware supports fiber ring bus for Distributed Drive Communication (DDC). IO Board will need to be variant that has fiber in and out
Enables power stage parallel operation or 6+ phase drives.
SN_GATE_DRIVE_PWR1_FB
SN_GATE_DRIVE_PWR2_FB
Power Supply Monitor
GATE_DRIVER_PWR1_FEEDBACK
GATE_DRIVER_PWR2_FEEDBACK
SCL
SDA
SLEEP
ENABLE
FAULT
+3.3V_A
+5V_A
+5V_ISENSE_A
+12V_A
GND_A
File: PowerSupplyMonitor.kicad_sch
Power monitoring, SIL-3 compliant system
Independently monitors and can trigger fault on power brownout
Solves risk of common power rail to Main MCU + Safety Coproc causing failure in both
Safety coprocessor added as intent to target ASIL B(D) + ASIL B(D)
Note: ASIL-B requires STM's SIL library which we cannot include here (proprietary)
As such, this makes the approach a best-effort attempt.
Additionally, no formal safety analysis by an independent agency has been performed.
Hardware Overcurrent Detection
+3.3V_AD
PHASE_CURSENSE_COMP_N
DC_LINK_CURSENSE_COMP_N
GND_AD
SN_CUR_SENSE_DC_LINK_SIG
SN_CUR_SENSE_PH_U_SIG
SN_CUR_SENSE_PH_V_SIG
SN_CUR_SENSE_PH_W_SIG
+3.3V_A
PHASE_CURRENT_REF_N
DC_LINK_CURRENT_REF_N
GND_A
DC_LINK_CURSENS
PH_U_CURSENS
PH_V_CURSENS
PH_W_CURSENS
DC_LINK_OCP_TRIPPED
PH_U_OCP_TRIPPED
PH_V_OCP_TRIPPED
PH_W_OCP_TRIPPED
File: HWOCP.kicad_sch
hw ocp not needed, see HARA for addt'l info
desat + redundant adc current sense determined to be acceptable
Safe Torque Off can be achieved within 100mS
without OCP comparators
Both STM MCUs have hw analog watchdogs that
achieve the same function
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/
File: Control.kicad_sch
Title: Open Inverter Platform - Control Board
Size: USLetter
Date: 2026-07-07
Rev: A
KiCad E.D.A. 10.0.3
Id: 8/21



HW–C2–PCB–CTRL

Prepared by Thomas Liao

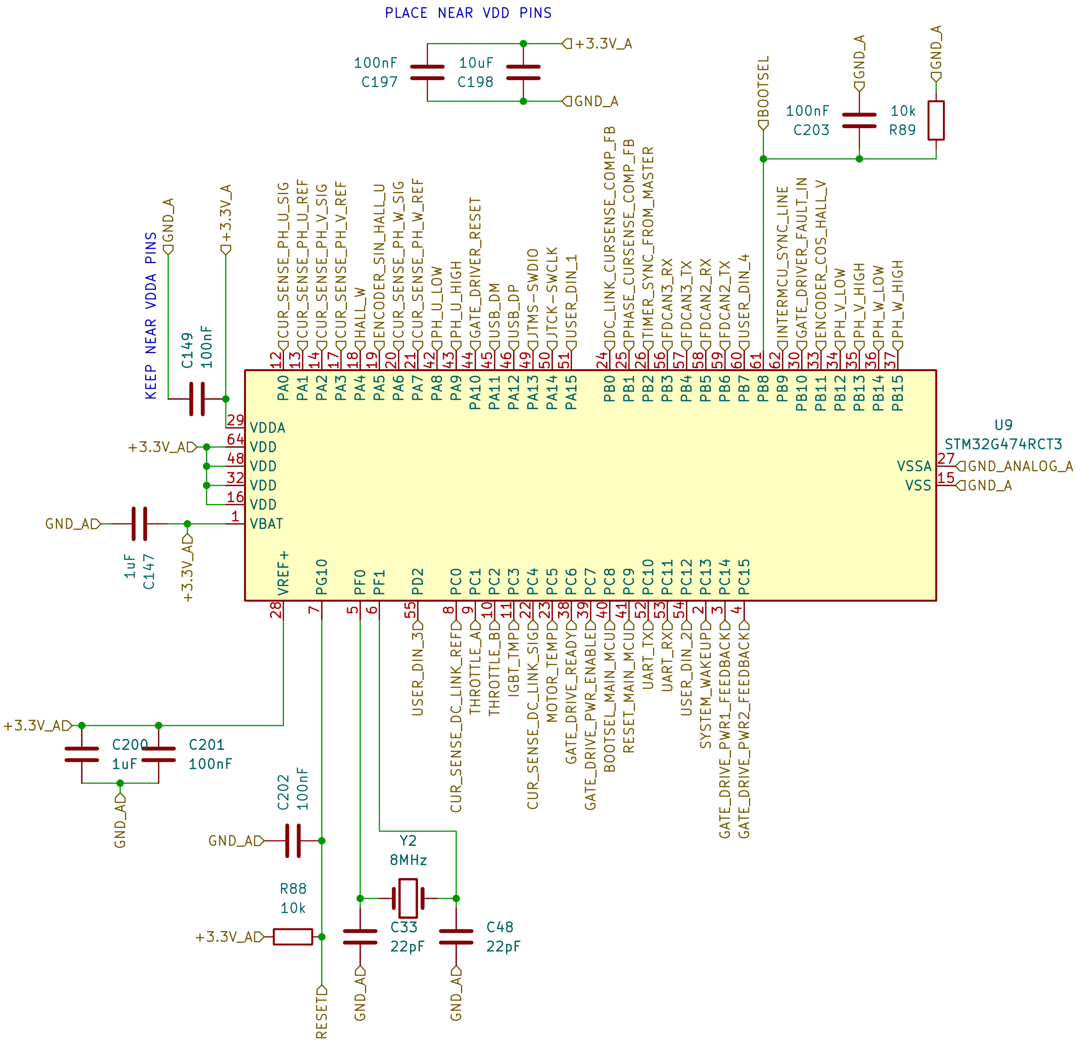
Corzine Lab, University of California, Santa Cruz

Sheet: /Control/Hardware Overcurrent Detection/

File: HWOCP.kicad_sch

Title: Open Inverter Platform – Control Board

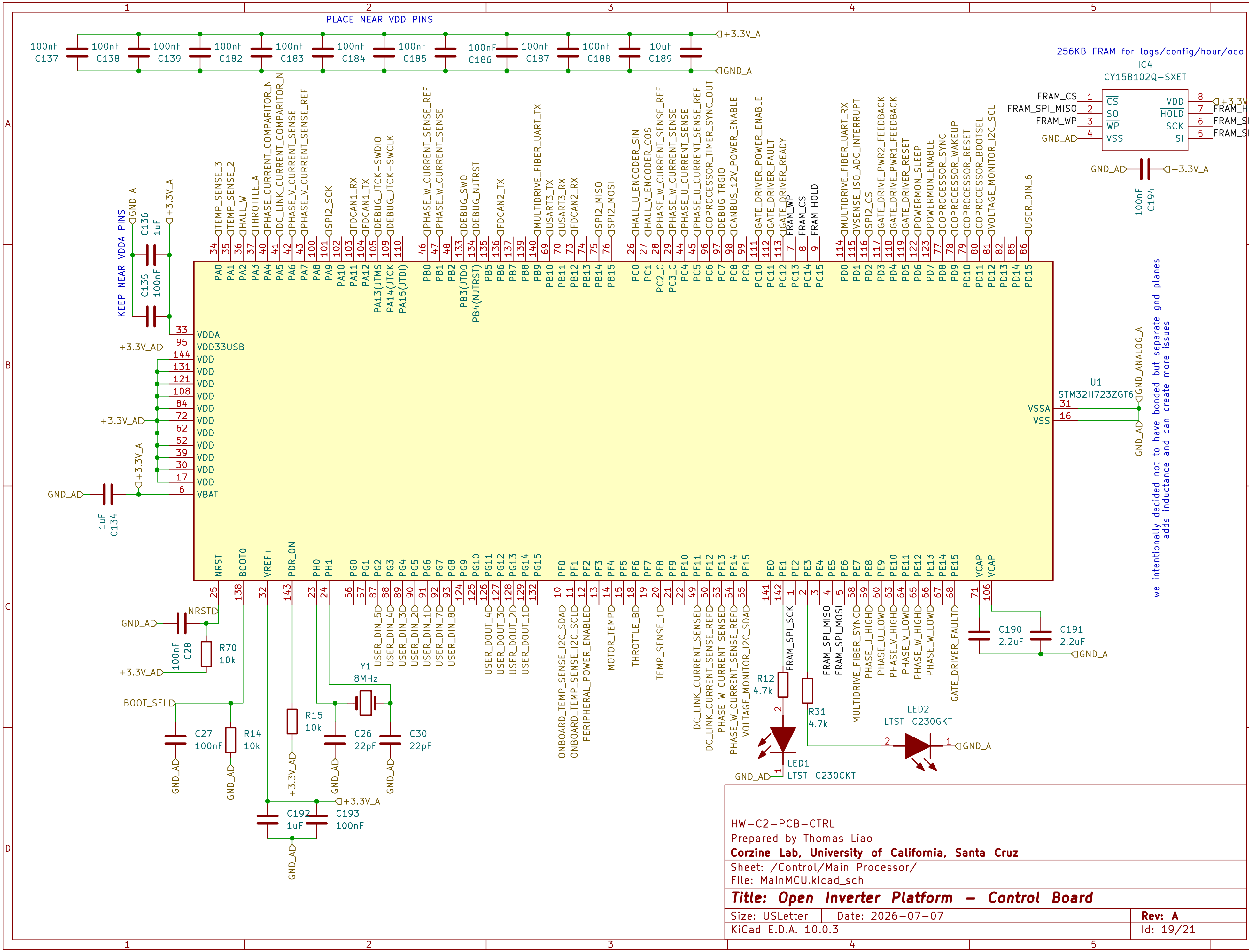
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 10/21

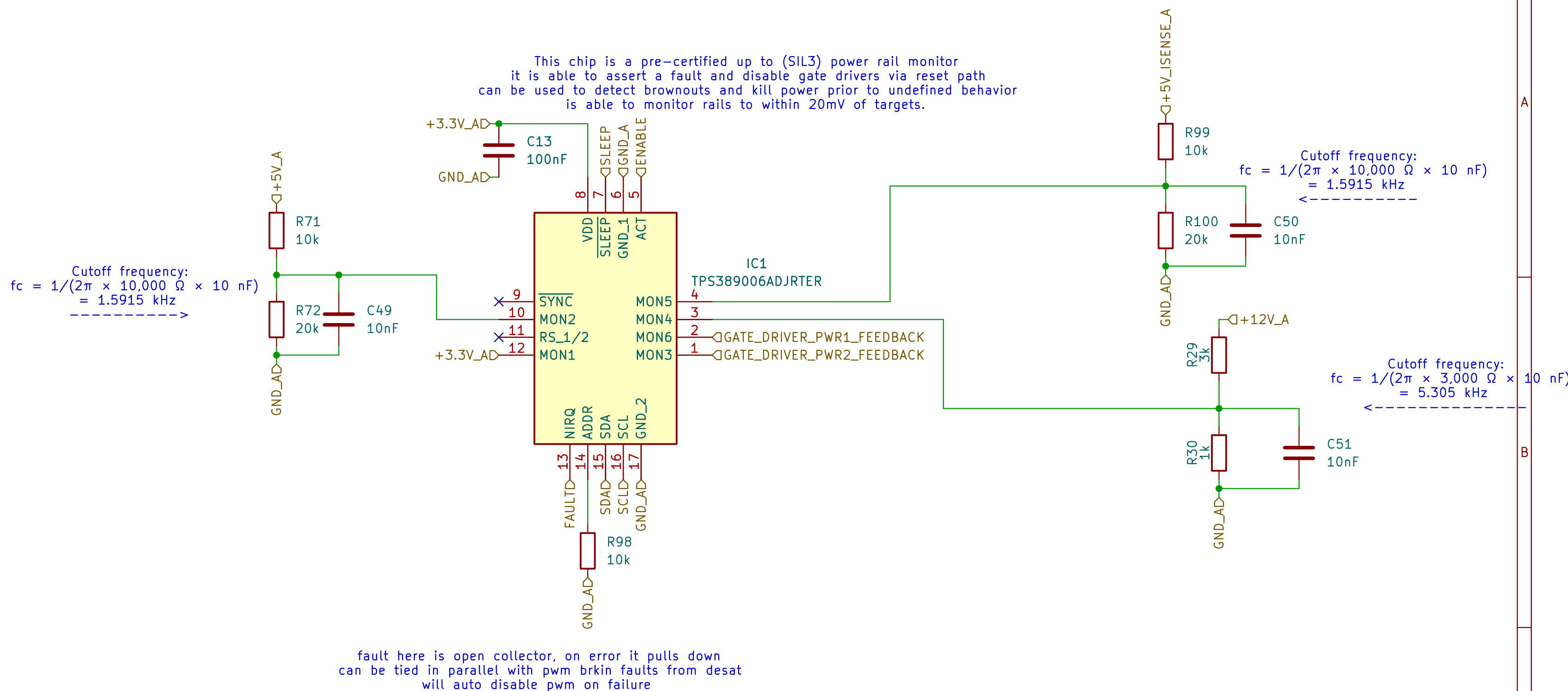


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/Safety Coprocessor/
File: SafetyCoprocessor.kicad_sch

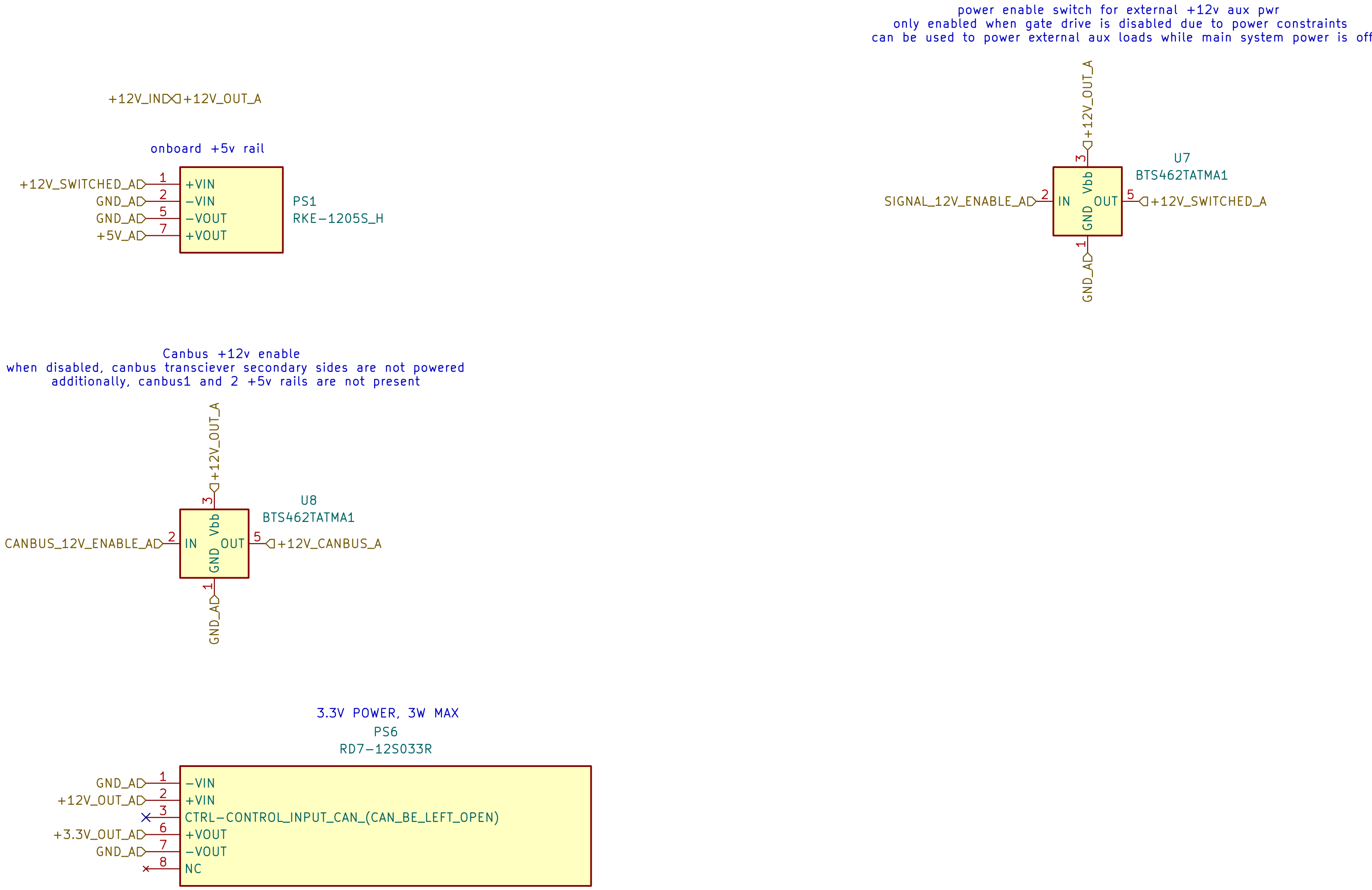
Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 16/21


PLACE NEAR VDD PINS
256KB FRAM for logs/config/hour/odo
IC4
CY15B102Q-SXET
KEEP NEAR VDDA PINS
U1
STM32H723ZGT6
we intentionally decided not to have bonded but separate gnd planes
adds inductance and can create more issues
Y1
8MHz
LED1
LTST-C230CKT
LED2
LTST-C230GKT
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/Main Processor/
File: MainMCU.kicad_sch
Title: Open Inverter Platform – Control Board
Size: USLetter
Date: 2026-07-07
Rev: A
KiCad E.D.A. 10.0.3
Id: 19/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/Power Supply Monitor/
File: PowerSupplyMonitor.kicad_sch
Title: Open Inverter Platform - Control Board
Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 20/21

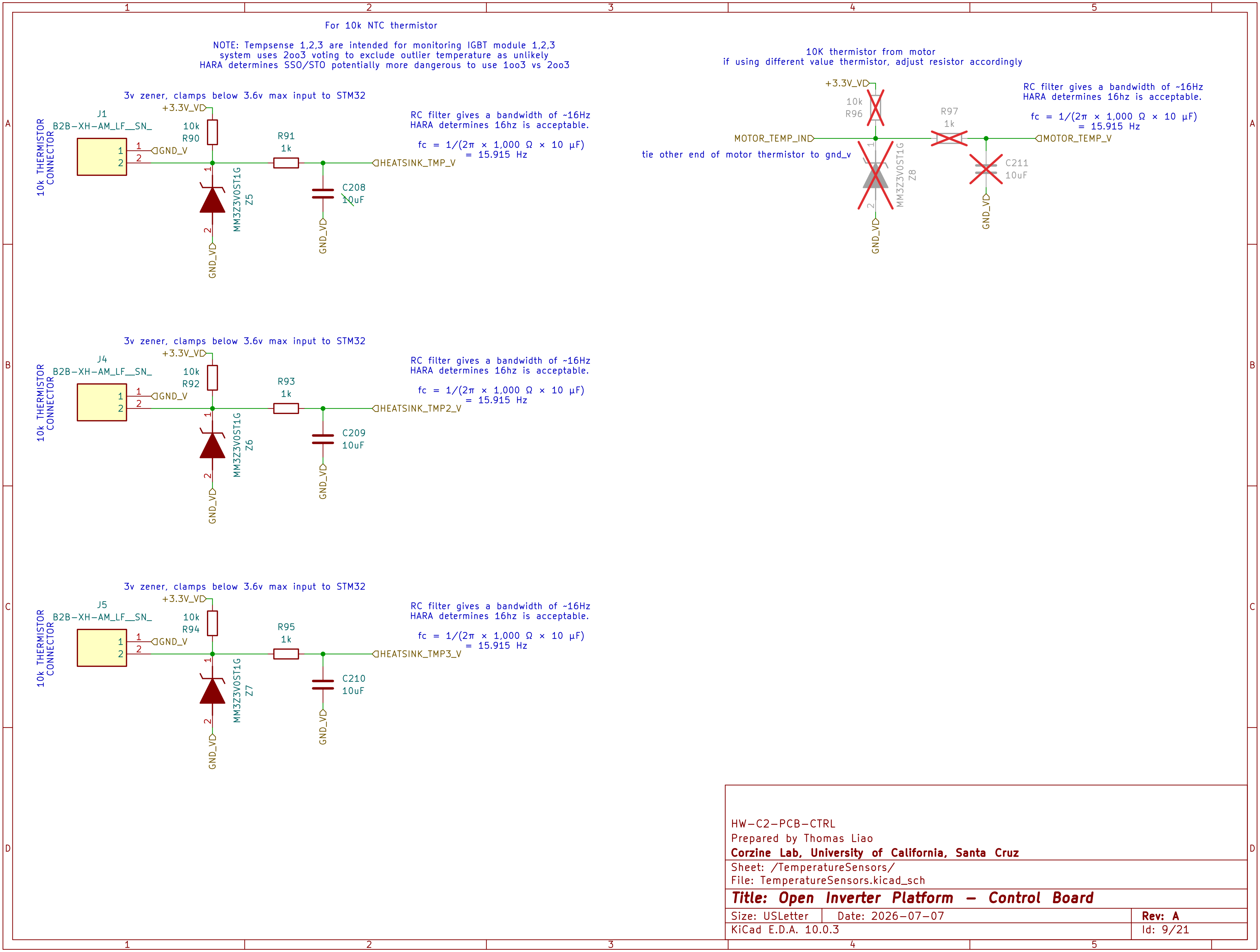


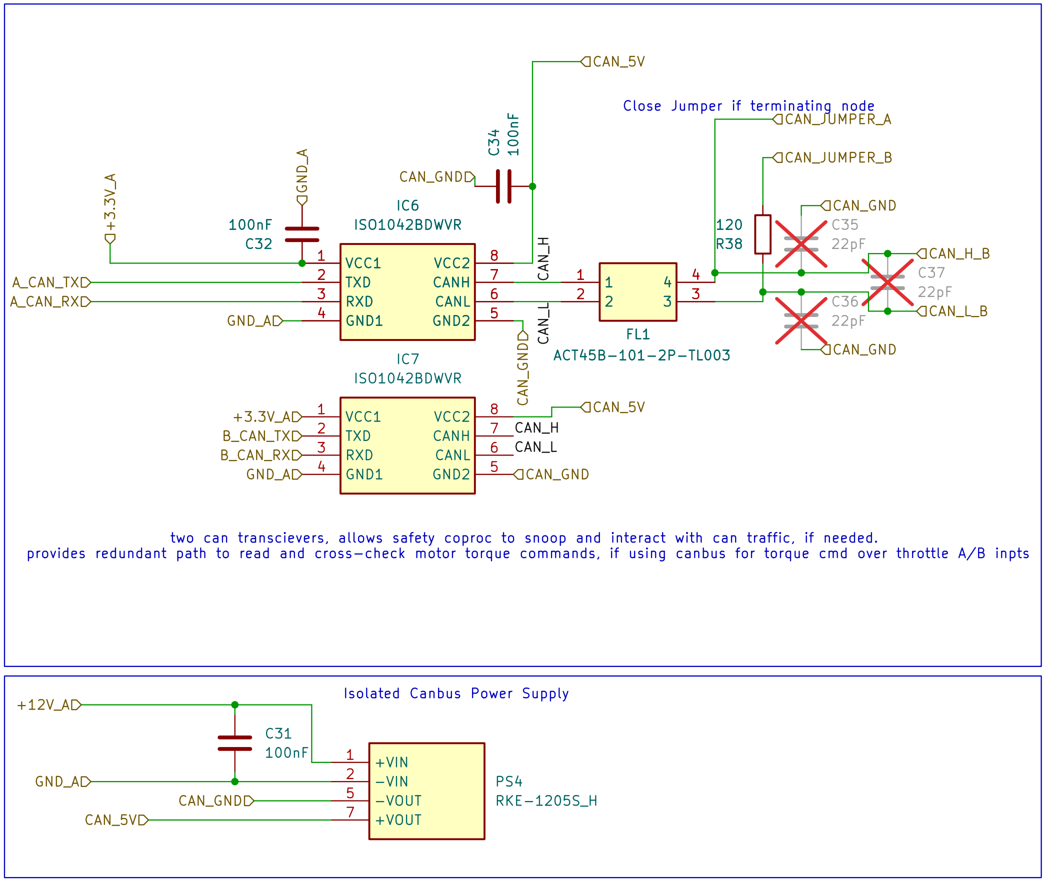
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Power/
File: Power.kicad_sch

Title: Open Inverter Platform - Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 9/21

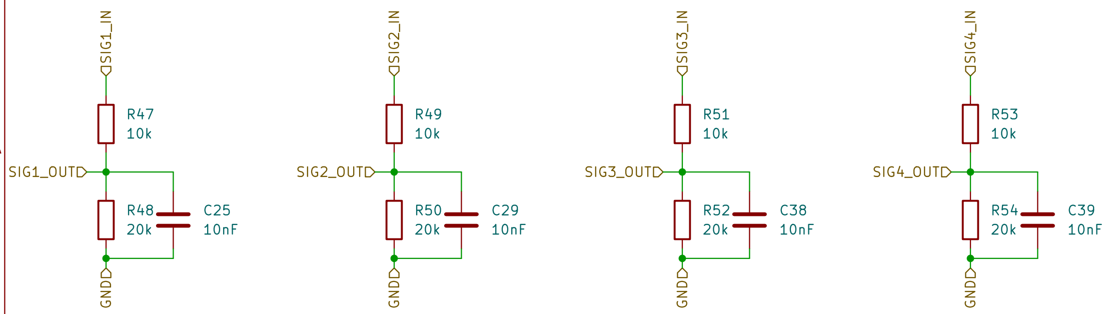

For 10k NTC thermistor
NOTE: Tempsense 1,2,3 are intended for monitoring IGBT module 1,2,3
system uses 2oo3 voting to exclude outlier temperature as unlikely
HARA determines SSO/STO potentially more dangerous to use 1oo3 vs 2oo3
3v zener, clamps below 3.6v max input to STM32
+3.3V_VD
J1
B2B-XH-AM_LF__SN_
10k THERMISTOR CONNECTOR
GND_V
10k
R90
R91
1k
HEATSINK_TMP_V
C208
10uF
MM3Z3V0ST1G
Z5
GND_VD
RC filter gives a bandwidth of ~16Hz
HARA determines 16hz is acceptable.
fc = 1/(2π × 1,000 Ω × 10 µF)
= 15.915 Hz
J4
R92
R93
HEATSINK_TMP2_V
C209
Z6
J5
R94
R95
HEATSINK_TMP3_V
C210
Z7
10K thermistor from motor
if using different value thermistor, adjust resistor accordingly
MOTOR_TEMP_IND
tie other end of motor thermistor to gnd_v
R96
R97
Z8
C211
MOTOR_TEMP_V
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /TemperatureSensors/
File: TemperatureSensors.kicad_sch
Title: Open Inverter Platform - Control Board
Size: USLetter
Date: 2026-07-07
Rev: A
KiCad E.D.A. 10.0.3
Id: 9/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /CAN Transciever/
File: CANTransciever.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 11/21

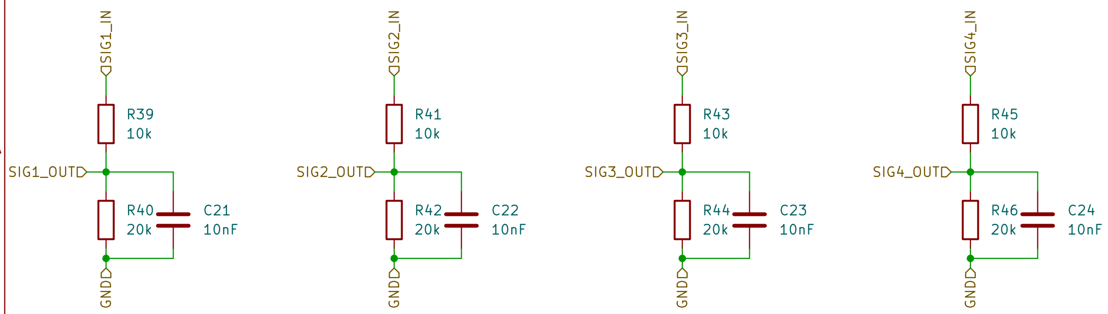


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network1/
File: VDivNet.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 12/21

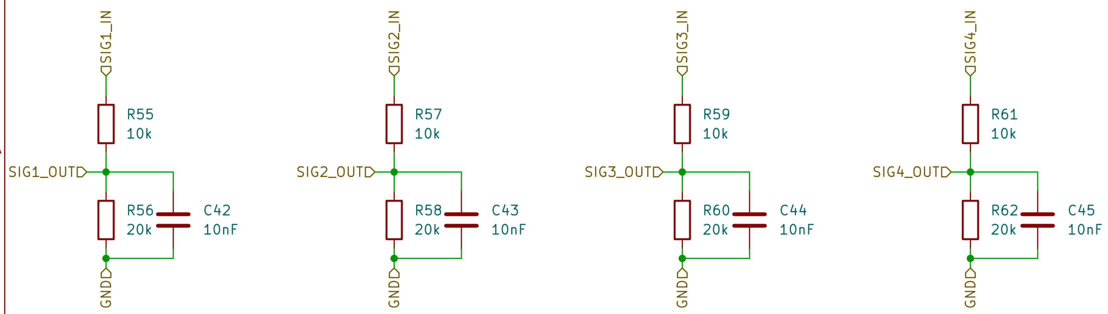


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network/
File: VDivNet.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 13/21

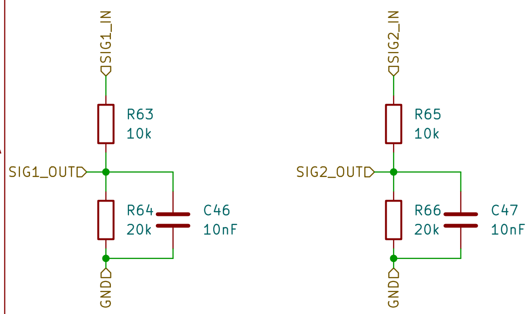


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network2/
File: VDivNet.kicad_sch

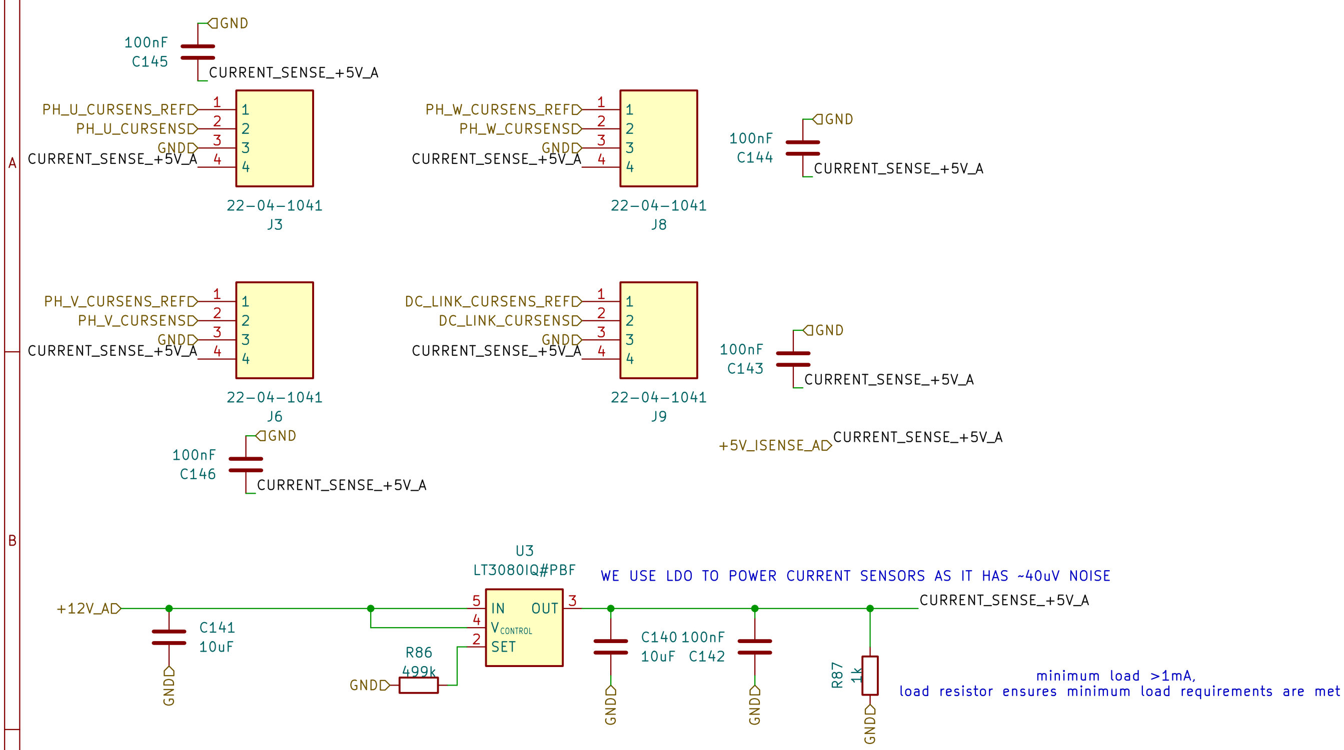
Title: Open Inverter Platform — Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 14/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Voltage Divider Network3/
File: VDivNetHalf.kicad_sch
Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 15/21

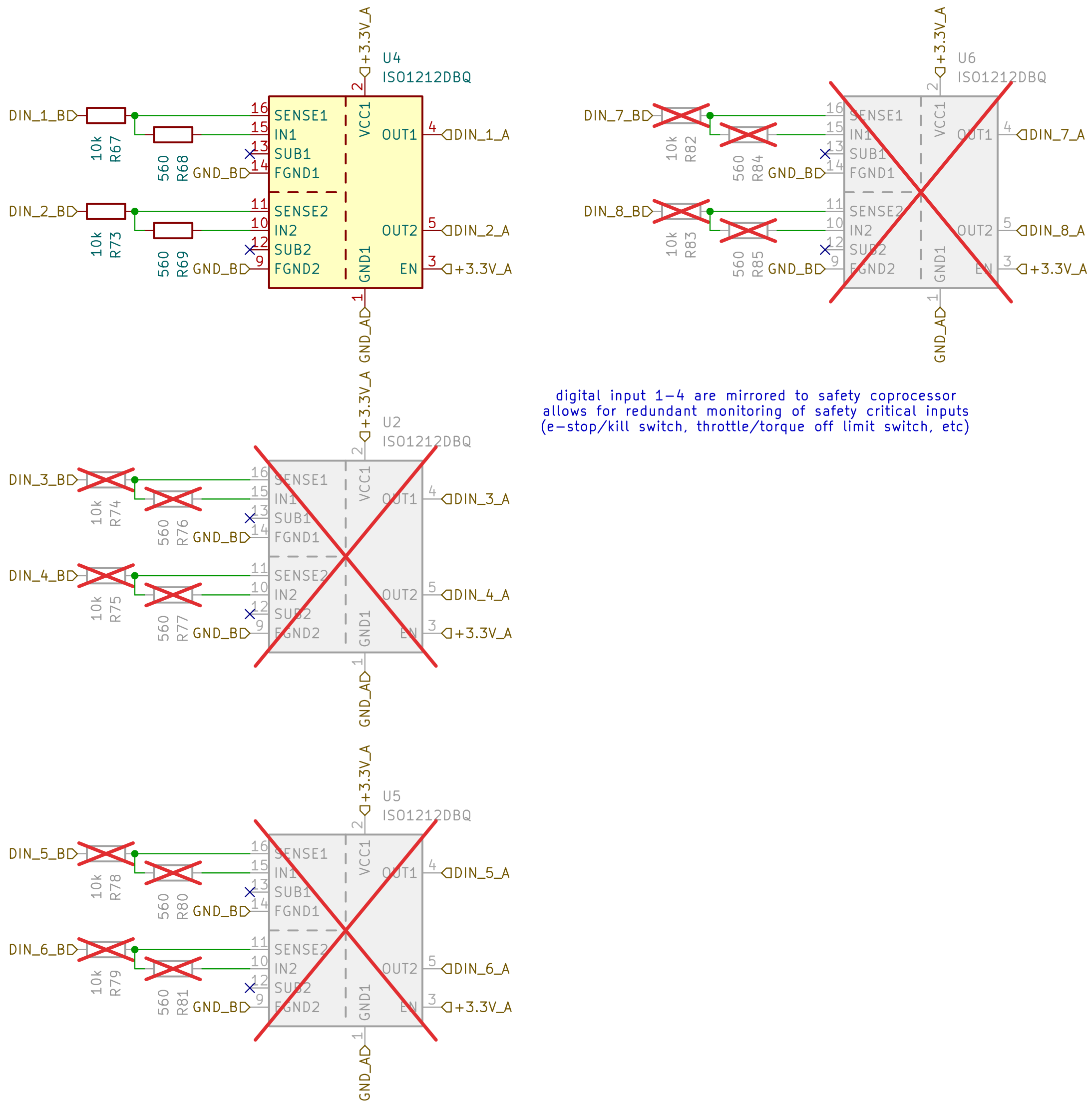


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Current Sensors/
File: CurrentSensors.kicad_sch

Title: Open Inverter Platform – Control Board

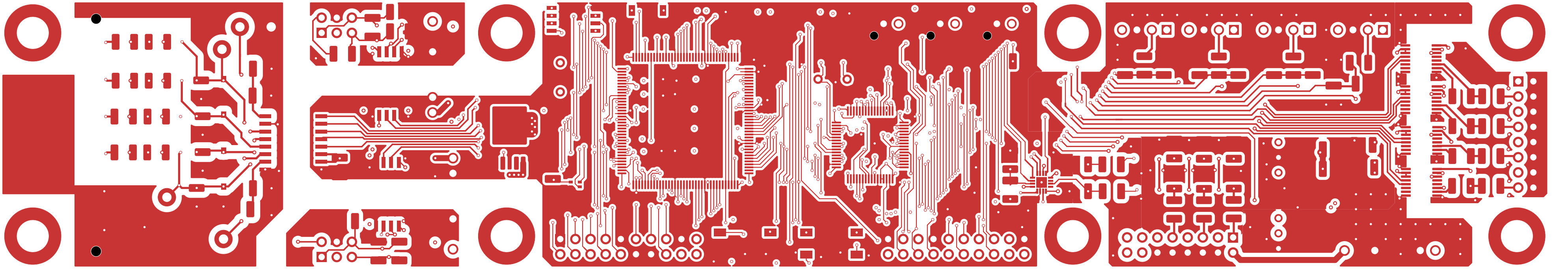
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 17/21

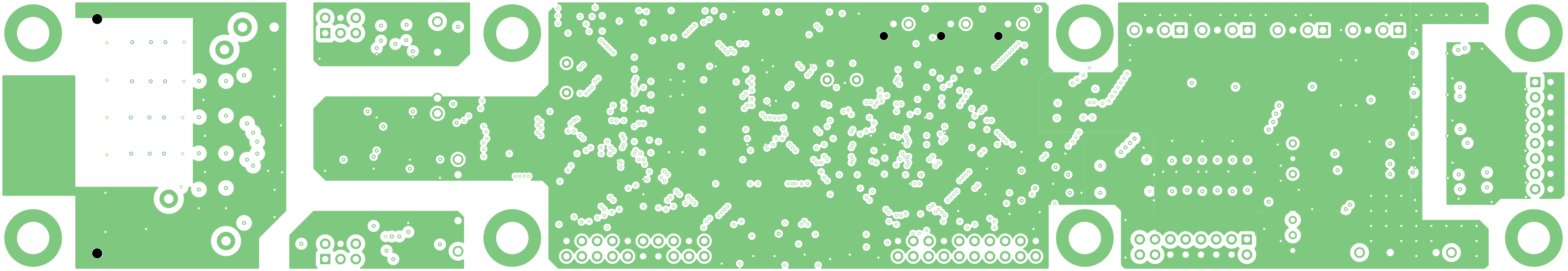


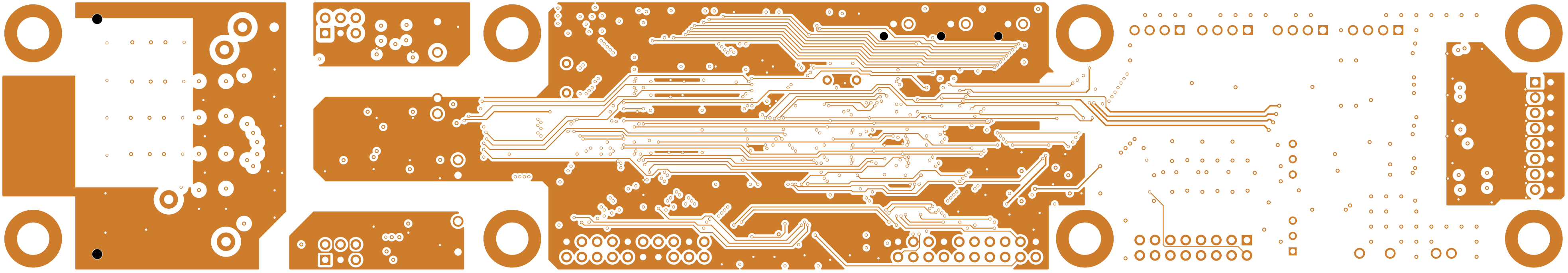
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Isolated Digital IO/
File: IsolatedDigitalIO.kicad_sch

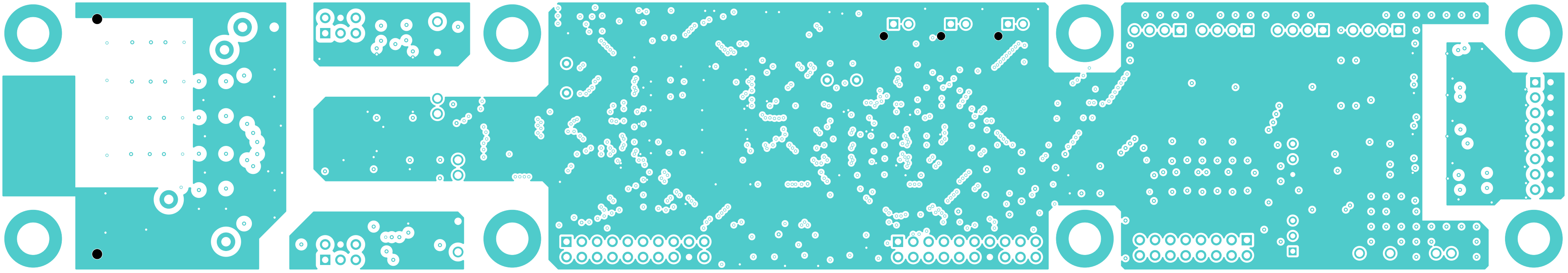
Title: Open Inverter Platform – Control Board

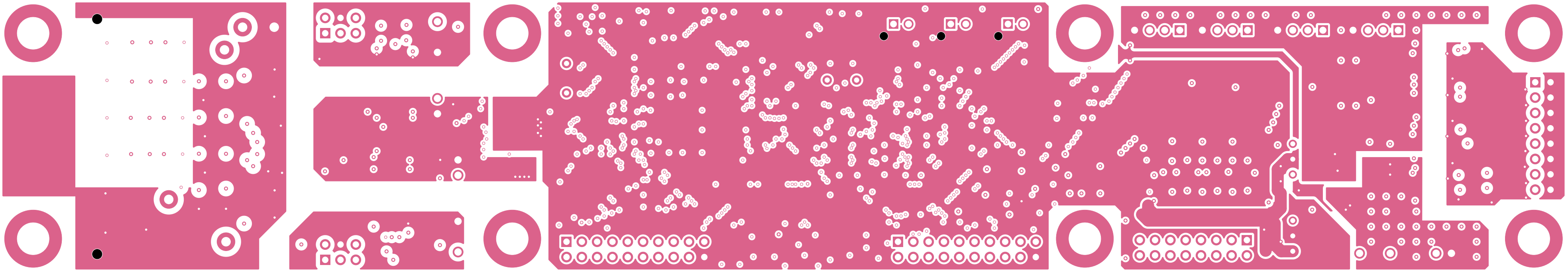
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 18/21

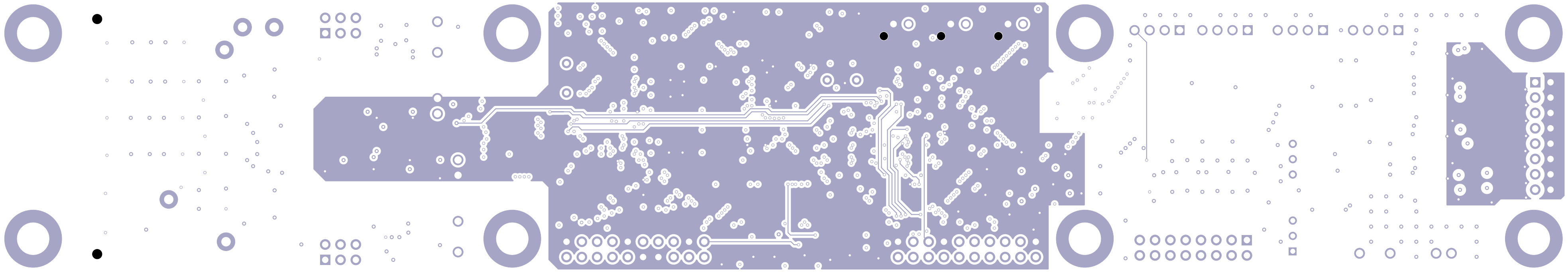


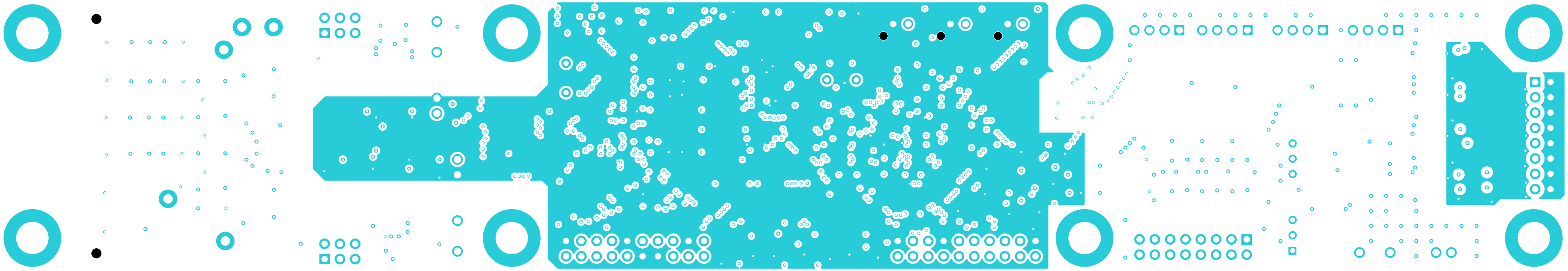


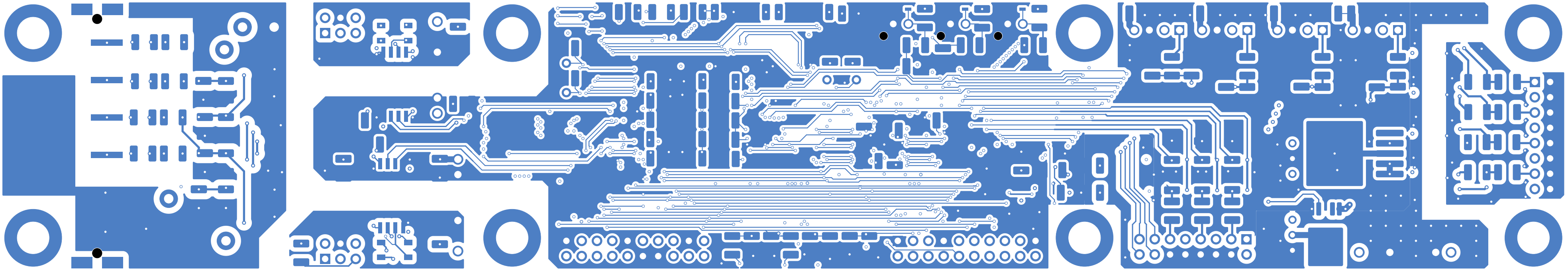


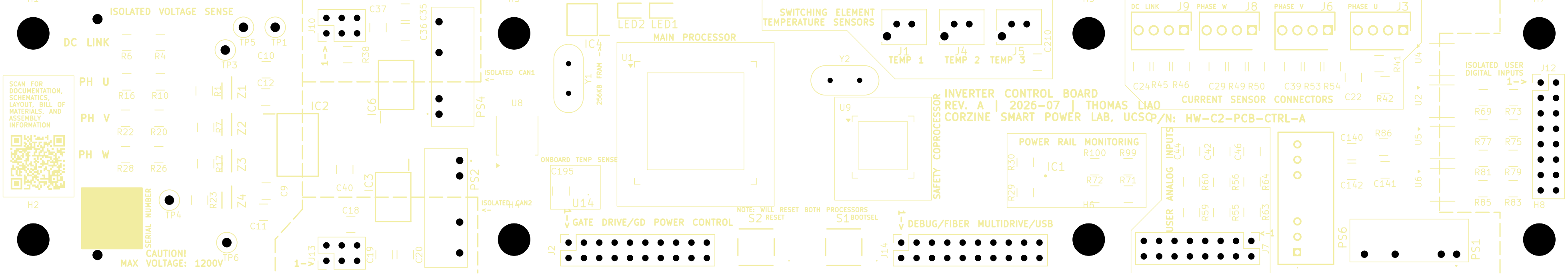


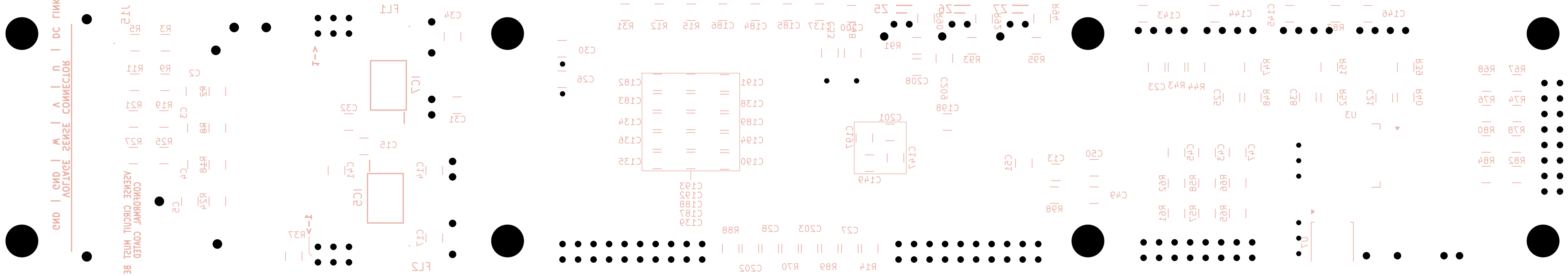









ISOLATED VOLTAGE SENSE
DC LINK
PH U
PH V
PH W
SCAN FOR DOCUMENTATION, SCHEMATICS, LAYOUT, BILL OF MATERIALS, AND ASSEMBLY INFORMATION
SERIAL NUMBER
CAUTION!
MAX VOLTAGE: 1200V
ISOLATED CAN1
ISOLATED CAN2
ONBOARD TEMP SENSE
256KB FRAM
MAIN PROCESSOR
SWITCHING ELEMENT TEMPERATURE SENSORS
TEMP 1
TEMP 2
TEMP 3
INVERTER CONTROL BOARD
REV. A | 2026-07 | THOMAS LIAO
CORZINE SMART POWER LAB, UCSC
P/N: HW-C2-PCB-CTRL-A
SAFETY COPROCESSOR
POWER RAIL MONITORING
DC LINK
PHASE W
PHASE V
PHASE U
CURRENT SENSOR CONNECTORS
USER ANALOG INPUTS
ISOLATED USER DIGITAL INPUTS
GATE DRIVE/GD POWER CONTROL
NOTE: WILL RESET BOTH PROCESSORS
RESET
BOOTSEL
DEBUG/FIBER MULTIDRIVE/USB

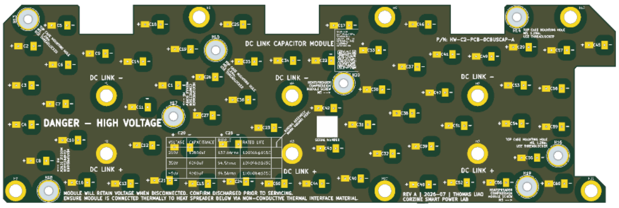


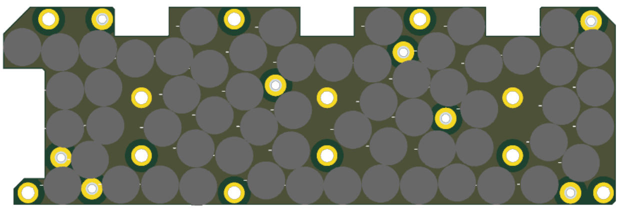
DRC — ControlBoard

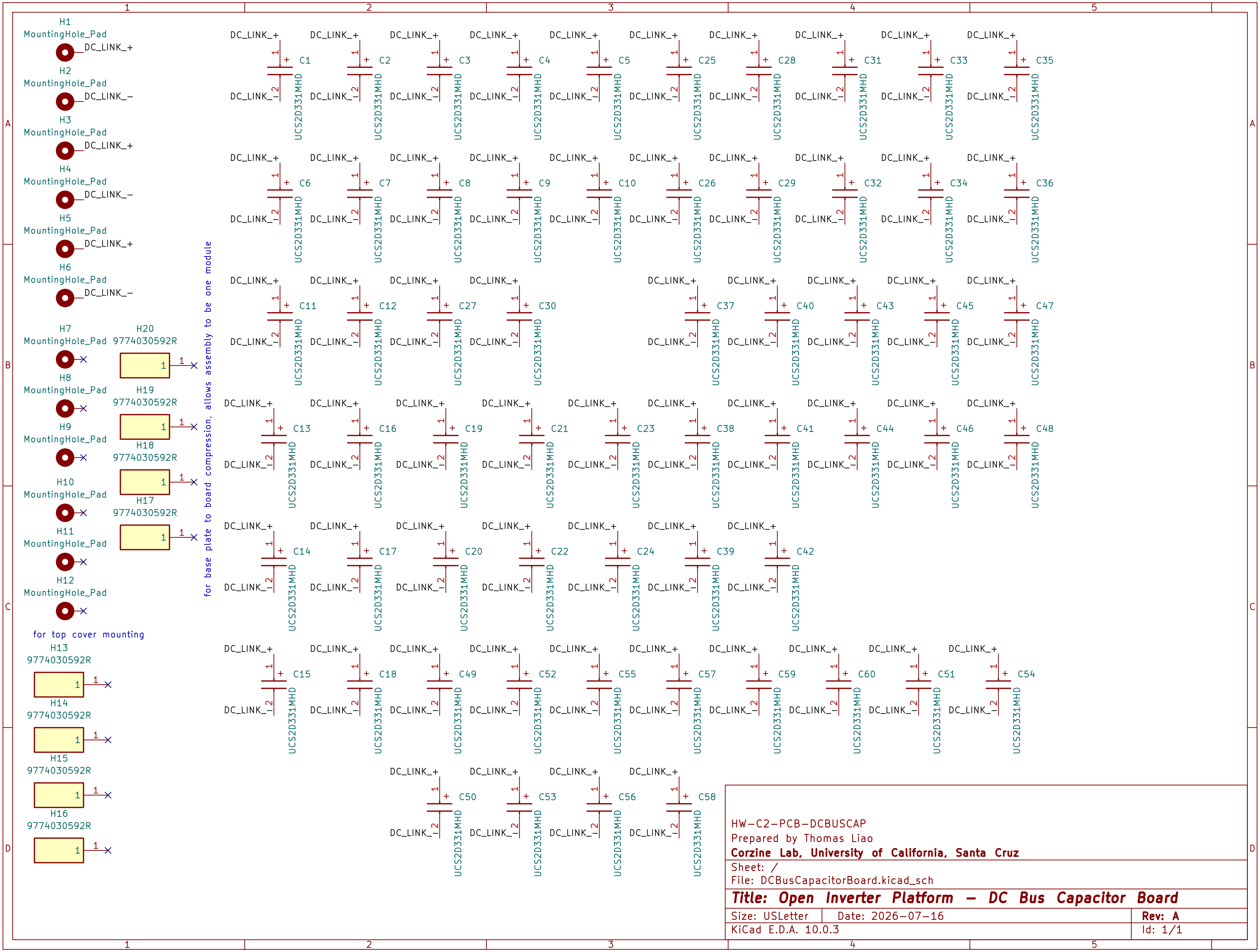
No DRC errors (warnings suppressed).

PCB ASSEMBLY: DC BUS CAPACITOR BOARD

HW-C2-PCB-DCBUSCAP-A · printed circuit board — schematic and PCB layers





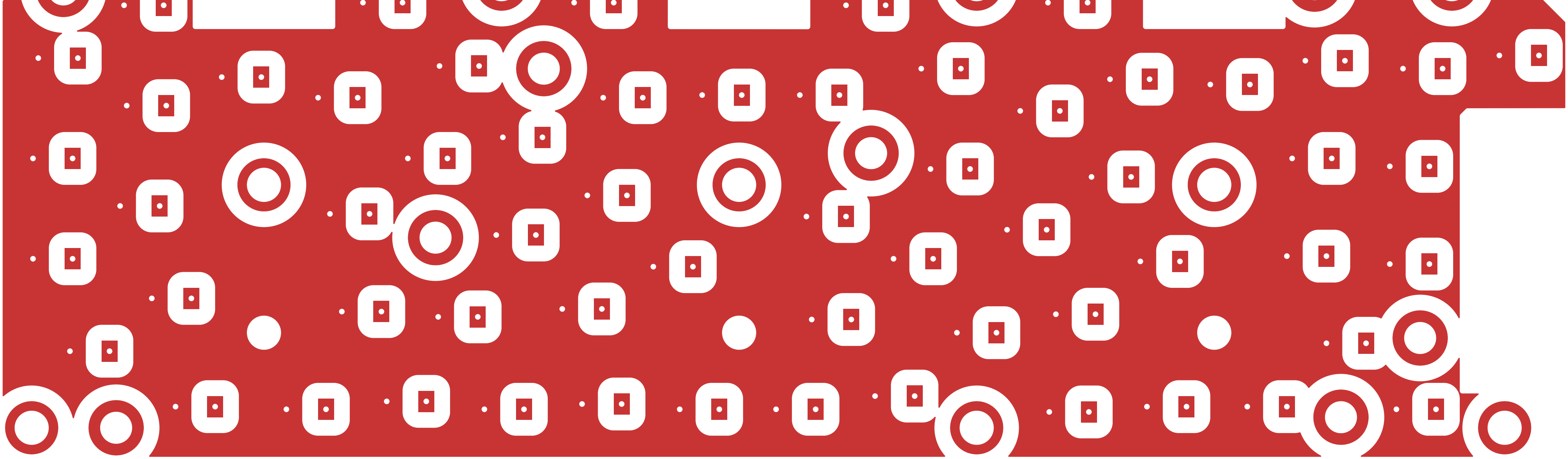


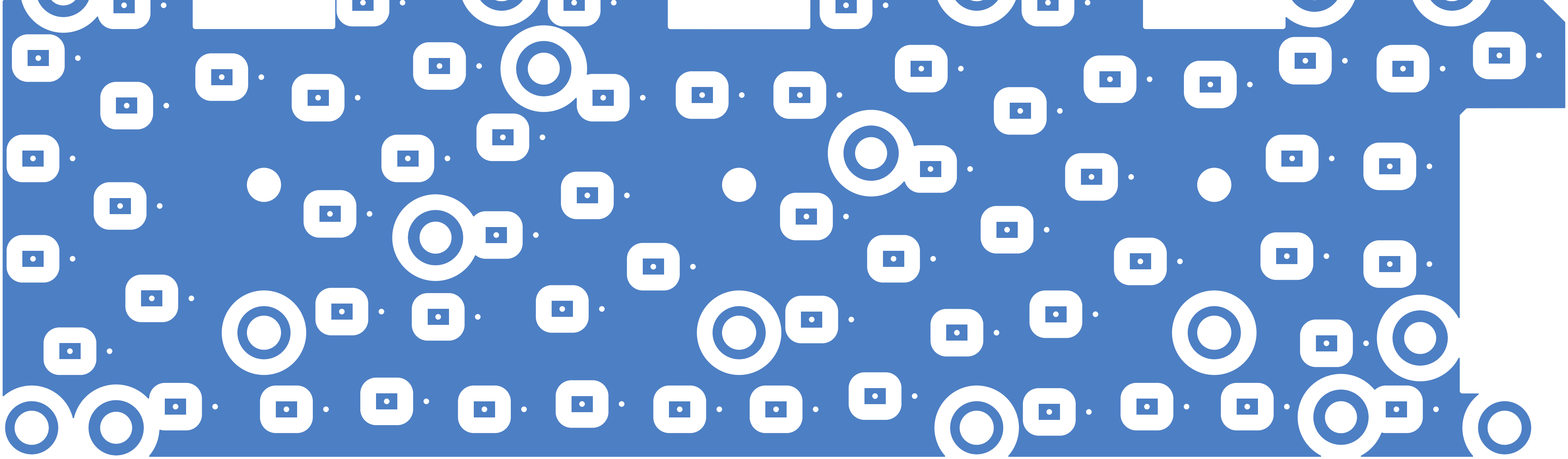
HW−C2−PCB−DCBUSCAP
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

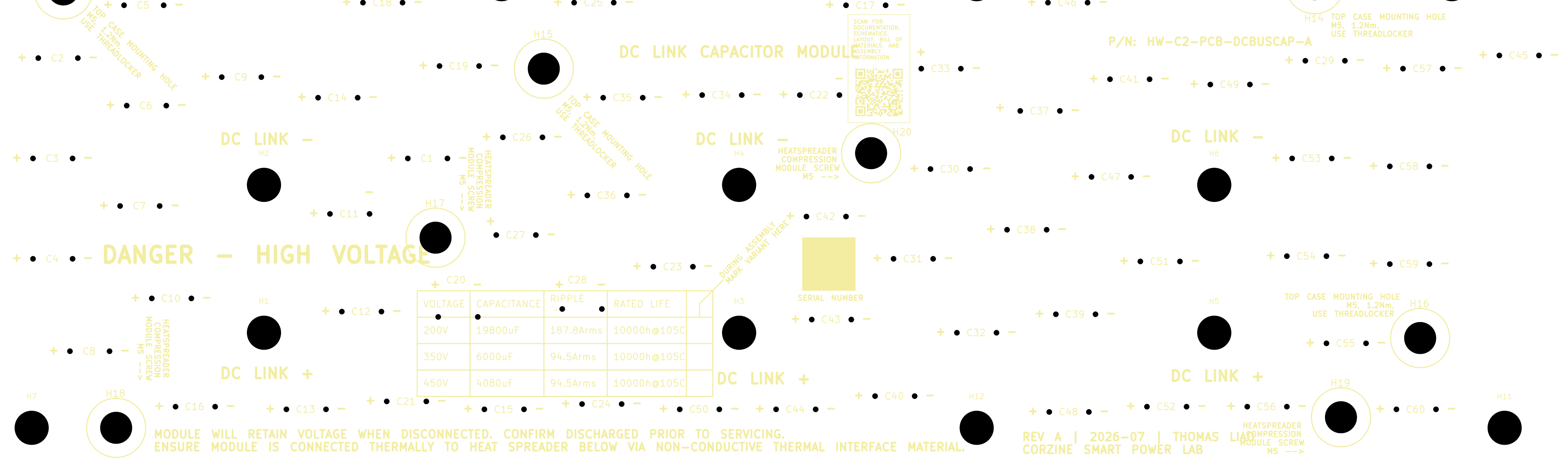
Sheet: /
File: DCBusCapacitorBoard.kicad_sch

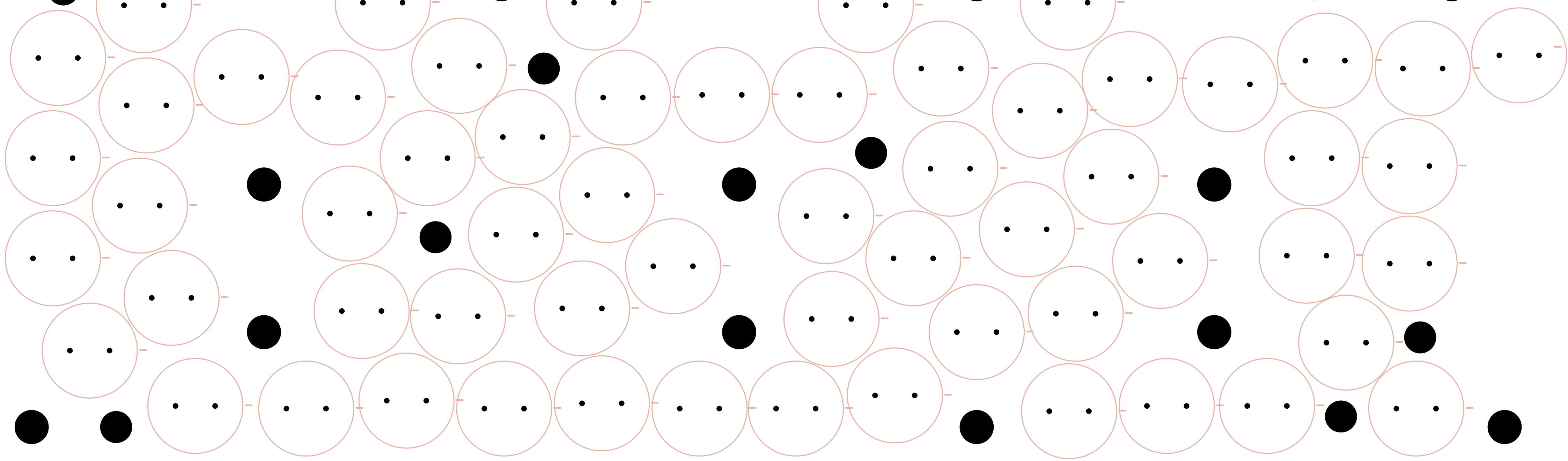
Title: Open Inverter Platform − DC Bus Capacitor Board

Size: USLetter	Date: 2026−07−16	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 1/1






DC LINK CAPACITOR MODULE
P/N: HW-C2-PCB-DCBUSCAP-A
SCAN FOR DOCUMENTATION, SCHEMATICS, LAYOUT, BILL OF MATERIALS, AND ASSEMBLY INFORMATION
DANGER - HIGH VOLTAGE
DC LINK -
DC LINK -
DC LINK -
DC LINK +
DC LINK +
DC LINK +
TOP CASE MOUNTING HOLE M5, 1.2Nm, USE THREADLOCKER
HEATSPREADER COMPRESSION MODULE SCREW M5 -->
DURING ASSEMBLY MARK VARIANT HERE
SERIAL NUMBER
VOLTAGE
CAPACITANCE
RIPPLE
RATED LIFE
200V
19800uF
187.8Arms
10000h@105C
350V
6000uF
94.5Arms
10000h@105C
450V
4080uF
94.5Arms
10000h@105C
MODULE WILL RETAIN VOLTAGE WHEN DISCONNECTED. CONFIRM DISCHARGED PRIOR TO SERVICING.
ENSURE MODULE IS CONNECTED THERMALLY TO HEAT SPREADER BELOW VIA NON-CONDUCTIVE THERMAL INTERFACE MATERIAL.
REV A | 2026-07 | THOMAS LIAO
CORZINE SMART POWER LAB

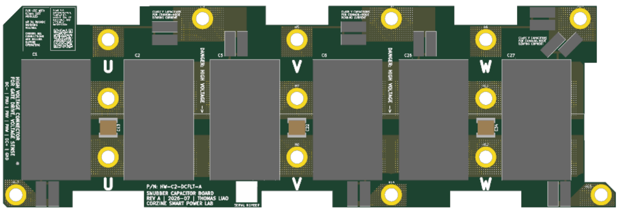


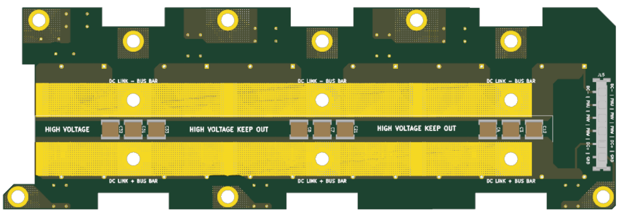
DRC — DCBusCapacitorBoard

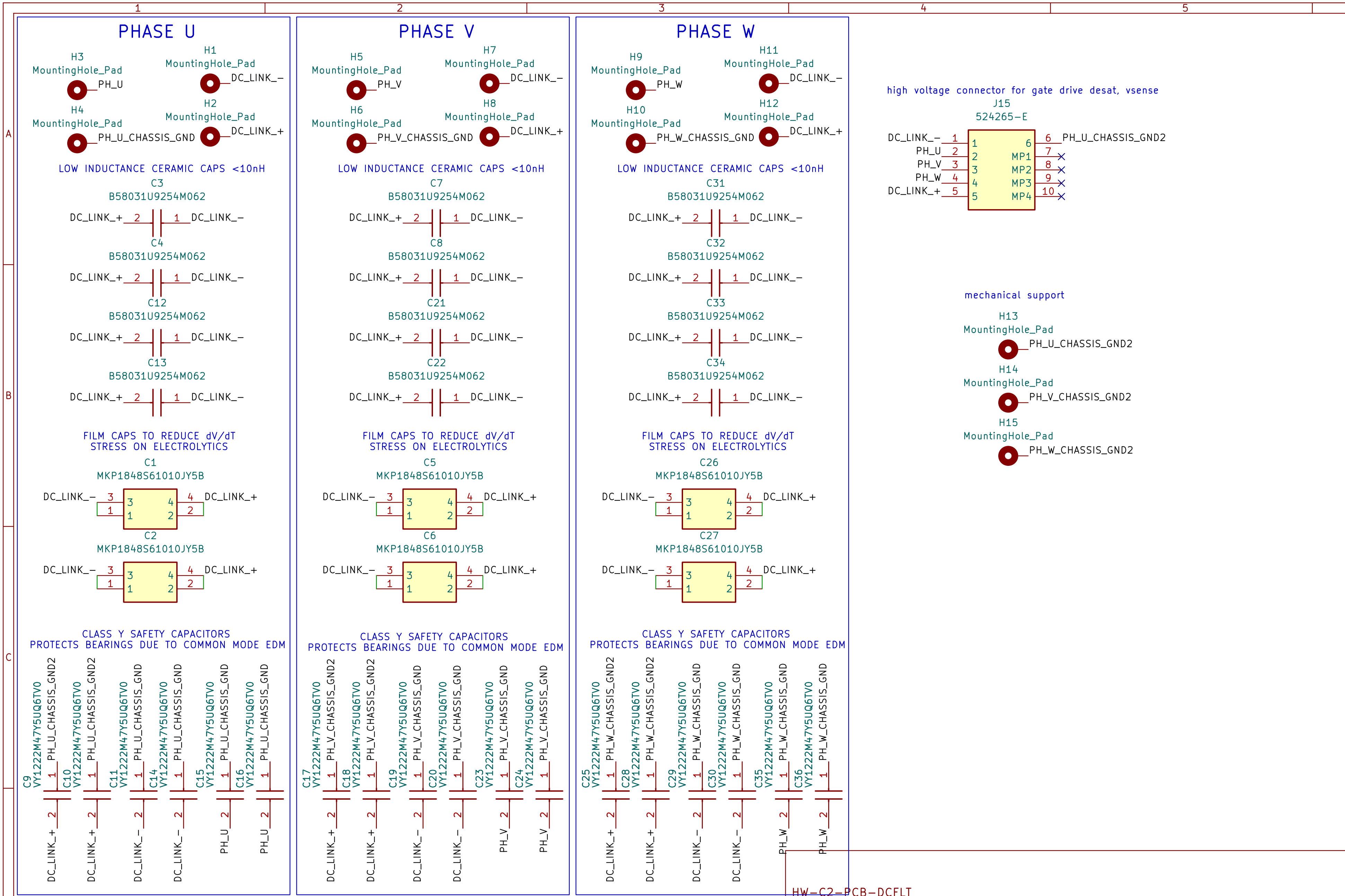
No DRC errors (warnings suppressed).

PCB ASSEMBLY: FILTER BOARD

HW-C2-PCB-DCFLT-A · printed circuit board — schematic and PCB layers

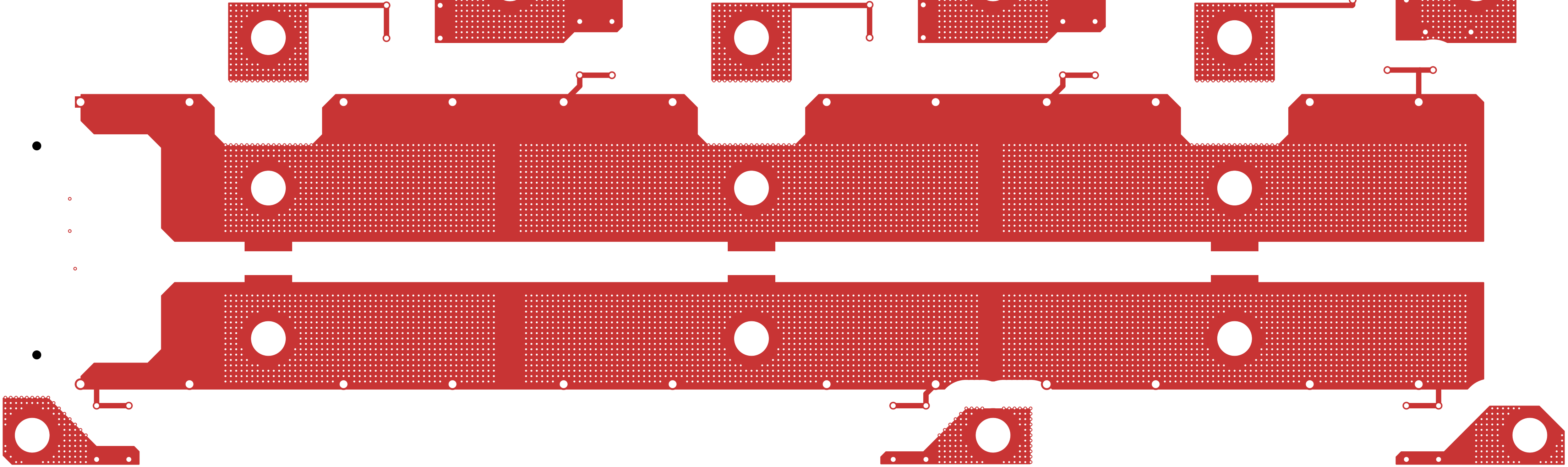


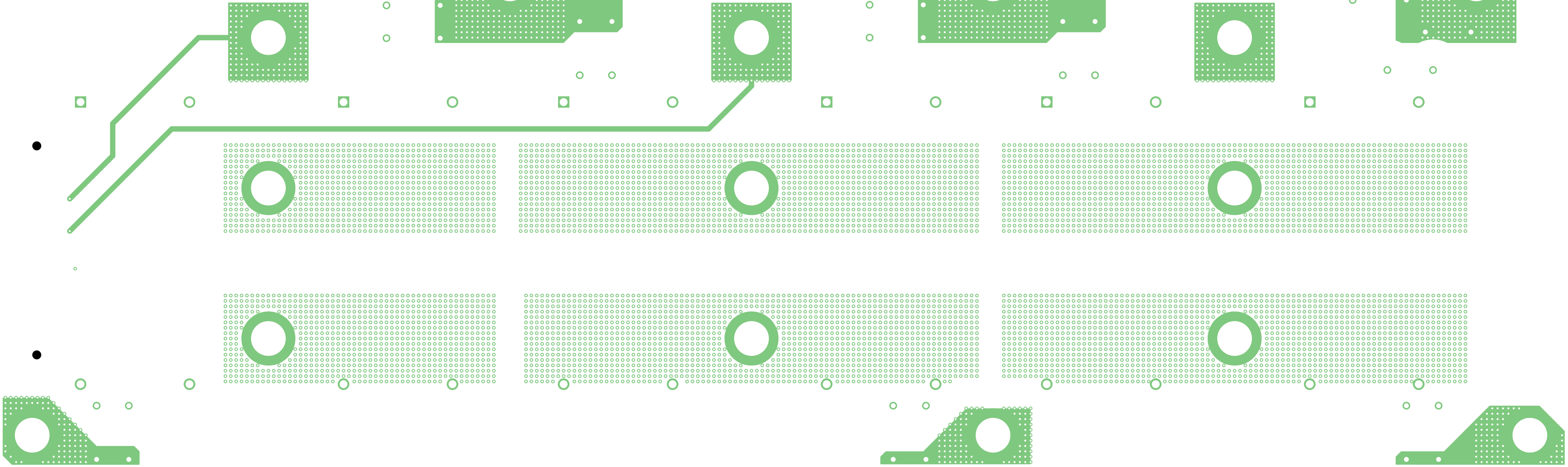


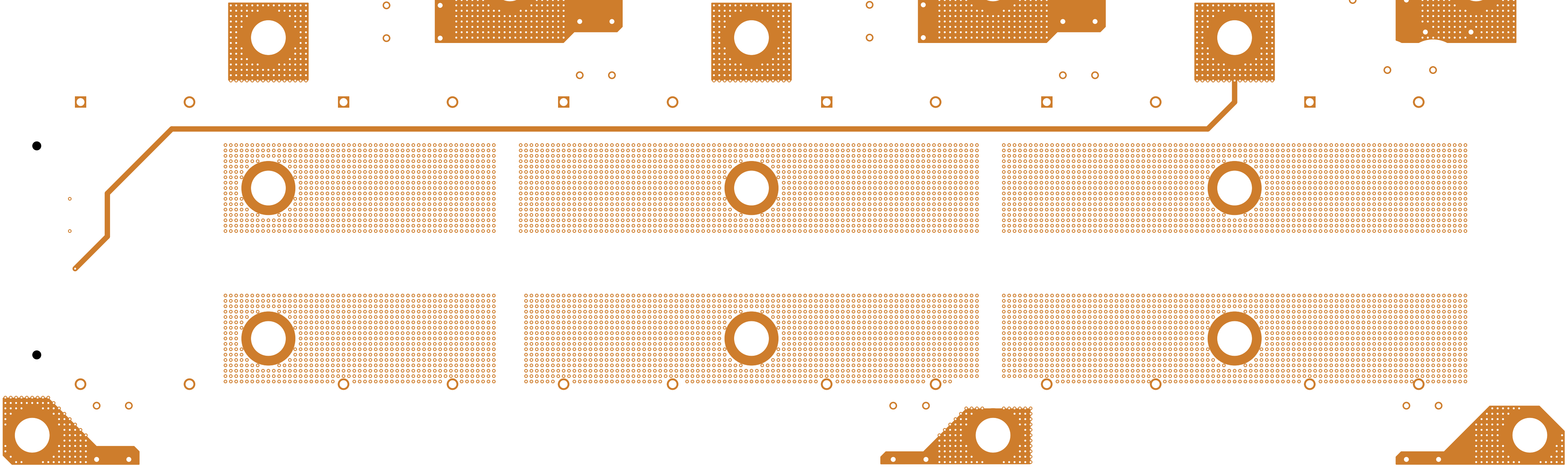


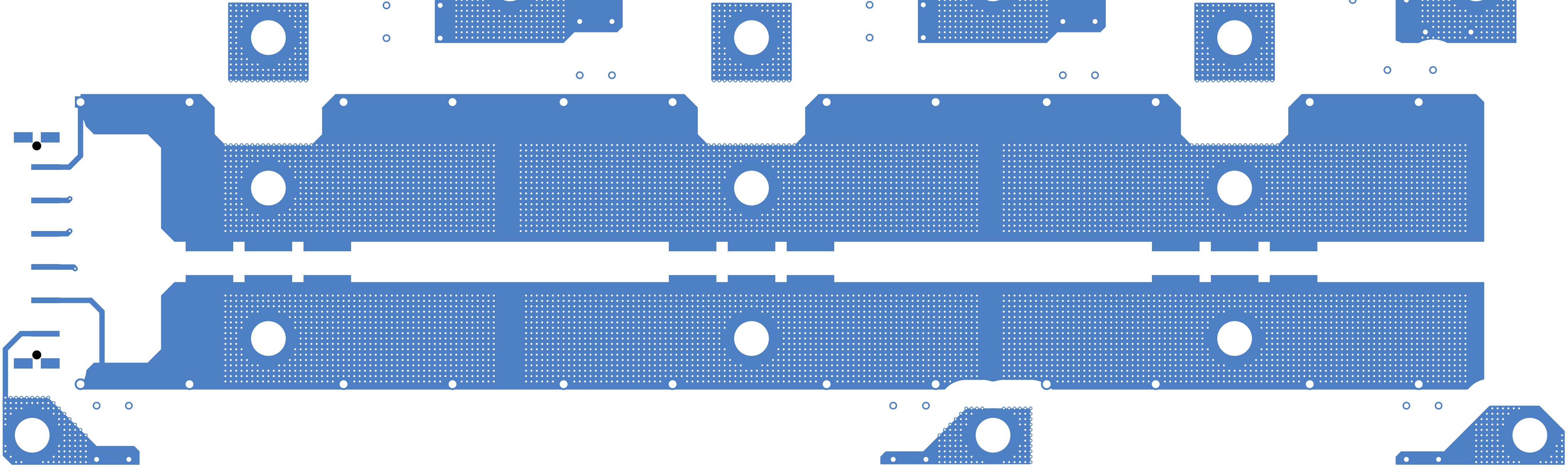
HW-C2-PCB-DCFLT
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /
File: DCBusFilter.kicad_sch
Title: Open Inverter Platform - Filter Board

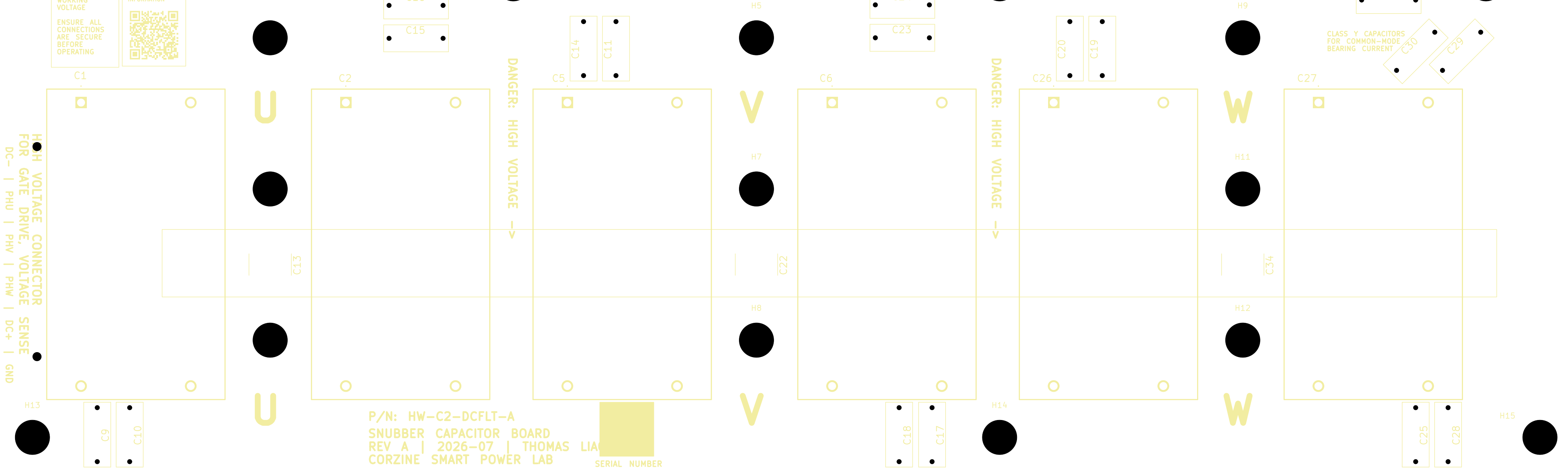
Size: USLetter	Date: 2026-07-16	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 1/1

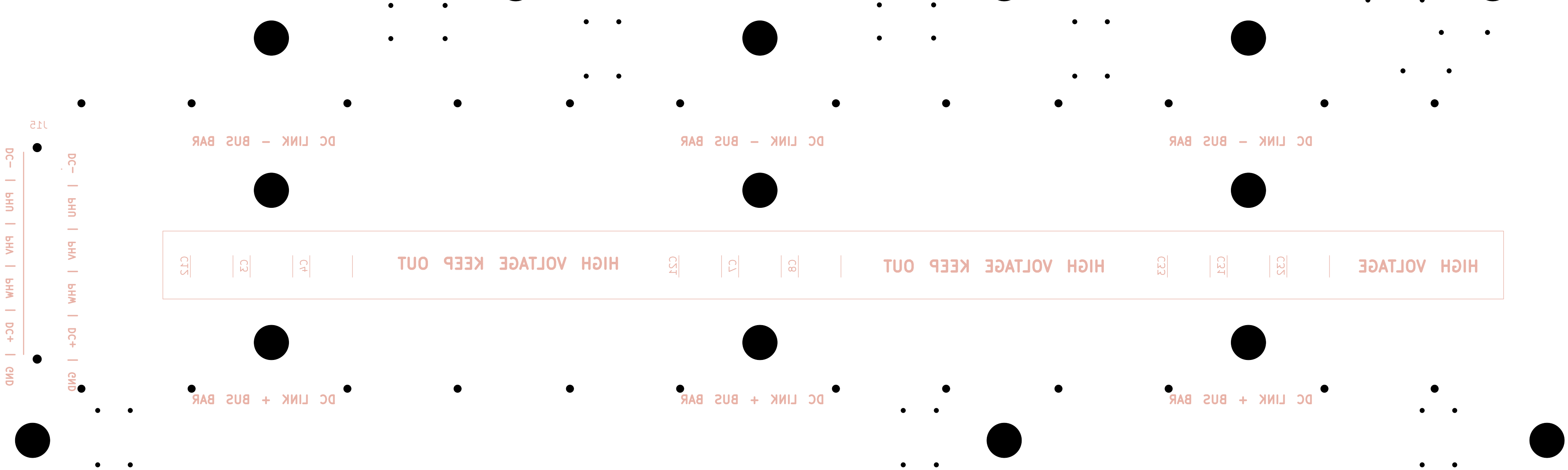









ENSURE ALL CONNECTIONS ARE SECURE BEFORE OPERATING
HIGH VOLTAGE CONNECTOR FOR GATE DRIVE, VOLTAGE SENSE
DC- | PHU | PHV | PHW | DC+ | GND
DANGER: HIGH VOLTAGE ->
DANGER: HIGH VOLTAGE ->
CLASS Y CAPACITORS FOR COMMON-MODE BEARING CURRENT
U
U
V
V
W
W
P/N: HW-C2-DCFLT-A
SNUBBER CAPACITOR BOARD
REV A | 2026-07 | THOMAS LIA
CORZINE SMART POWER LAB
SERIAL NUMBER

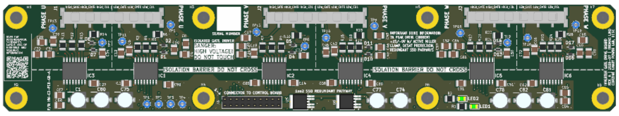

J15
DC– | PhU | PhV | PhW | DC+ | GND
DC LINK – BUS BAR
DC LINK – BUS BAR
DC LINK – BUS BAR
HIGH VOLTAGE KEEP OUT
HIGH VOLTAGE KEEP OUT
HIGH VOLTAGE
DC LINK + BUS BAR
DC LINK + BUS BAR
DC LINK + BUS BAR

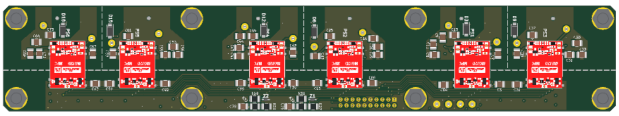
DRC — DCBusFilter

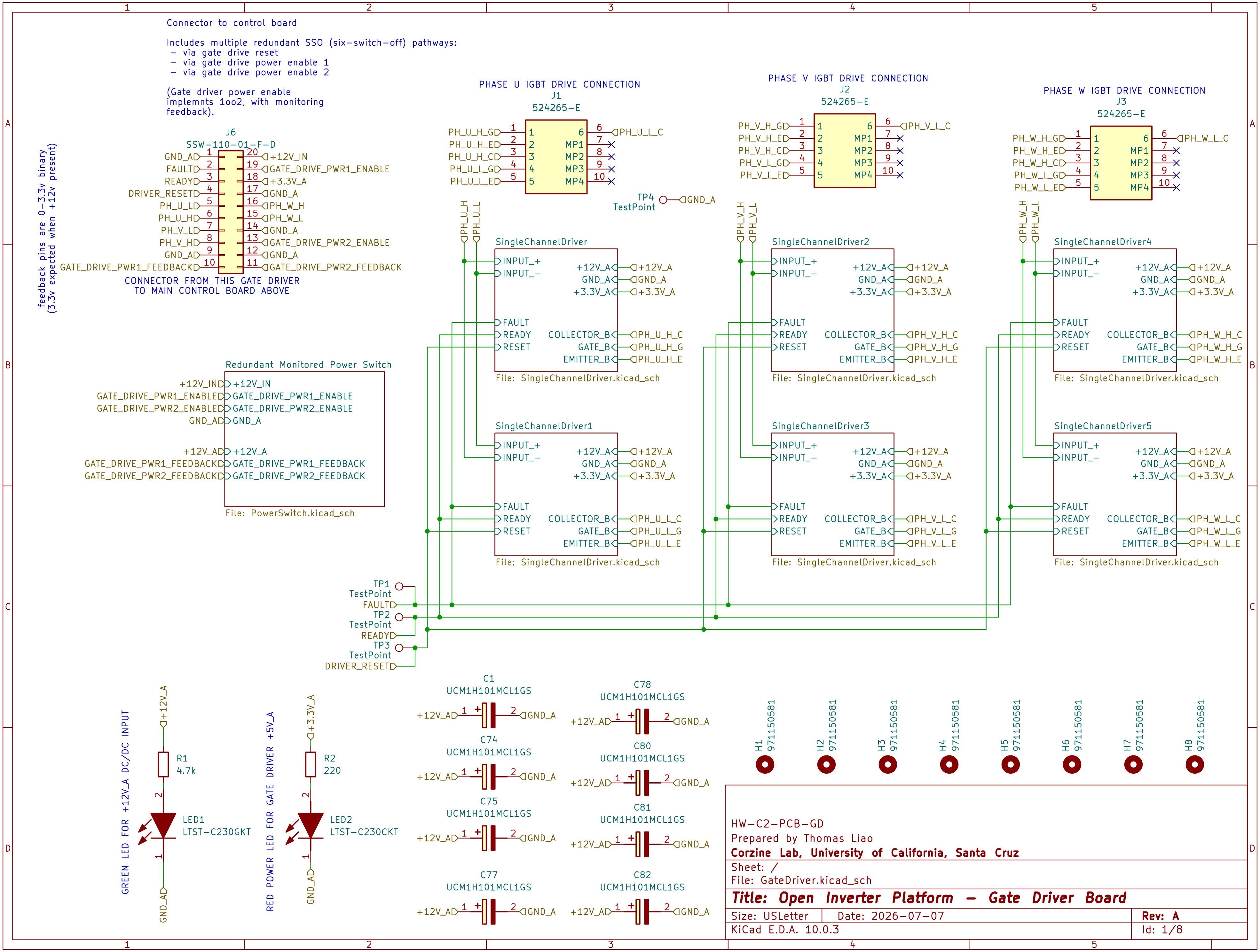
No DRC errors (warnings suppressed).

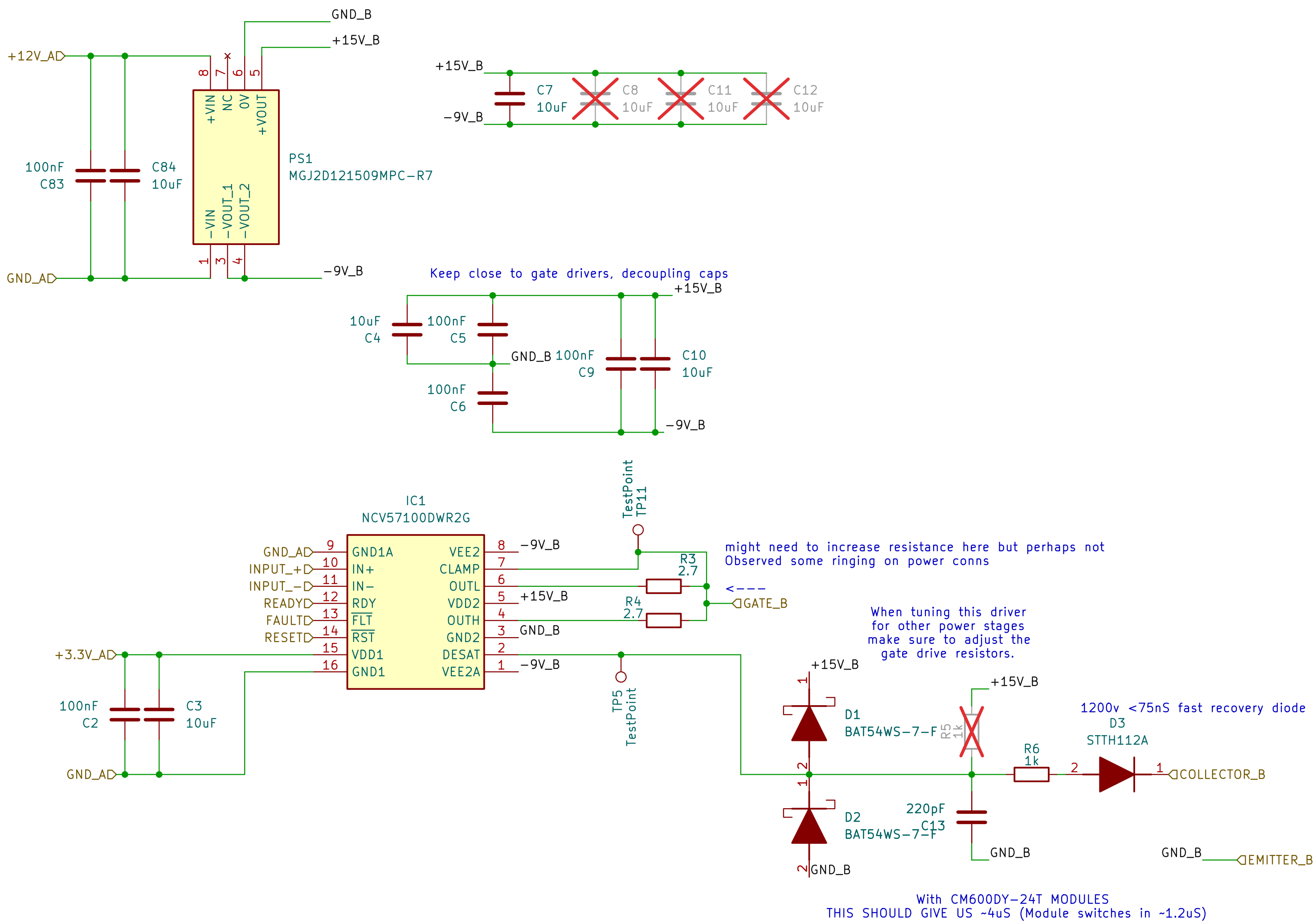
PCB ASSEMBLY: GATE DRIVER BOARD

HW-C2-PCB-GD-A · printed circuit board — schematic and PCB layers






Connector to control board
Includes multiple redundant SSO (six-switch-off) pathways:
– via gate drive reset
– via gate drive power enable 1
– via gate drive power enable 2
(Gate driver power enable implemnts 1oo2, with monitoring feedback).
feedback pins are 0-3.3v binary (3.3v expected when +12v present)
J6
SSW-110-01-F-D
GND_AD
FAULTD
READYD
DRIVER_RESETD
PH_U_LD
PH_U_HD
PH_V_LD
PH_V_HD
GND_AD
GATE_DRIVE_PWR1_FEEDBACKD
+12V_IN
GATE_DRIVE_PWR1_ENABLE
+3.3V_A
GND_A
PH_W_H
PH_W_L
GND_A
GATE_DRIVE_PWR2_ENABLE
GND_A
GATE_DRIVE_PWR2_FEEDBACK
CONNECTOR FROM THIS GATE DRIVER TO MAIN CONTROL BOARD ABOVE
PHASE U IGBT DRIVE CONNECTION
J1
524265-E
PH_U_H_GD
PH_U_H_ED
PH_U_H_CD
PH_U_L_GD
PH_U_L_ED
PH_U_L_C
PHASE V IGBT DRIVE CONNECTION
J2
524265-E
PH_V_H_GD
PH_V_H_ED
PH_V_H_CD
PH_V_L_GD
PH_V_L_ED
PH_V_L_C
PHASE W IGBT DRIVE CONNECTION
J3
524265-E
PH_W_H_GD
PH_W_H_ED
PH_W_H_CD
PH_W_L_GD
PH_W_L_ED
PH_W_L_C
TP4
TestPoint
GND_A
SingleChannelDriver
SingleChannelDriver1
SingleChannelDriver2
SingleChannelDriver3
SingleChannelDriver4
SingleChannelDriver5
INPUT_+
INPUT_–
FAULT
READY
RESET
+12V_A
GND_A
+3.3V_A
COLLECTOR_B
GATE_B
EMITTER_B
File: SingleChannelDriver.kicad_sch
Redundant Monitored Power Switch
+12V_IN
GATE_DRIVE_PWR1_ENABLE
GATE_DRIVE_PWR2_ENABLE
GND_A
+12V_A
GATE_DRIVE_PWR1_FEEDBACK
GATE_DRIVE_PWR2_FEEDBACK
File: PowerSwitch.kicad_sch
TP1
TestPoint
FAULTD
TP2
TestPoint
READYD
TP3
TestPoint
DRIVER_RESETD
GREEN LED FOR +12V_A DC/DC INPUT
+12V_A
R1
4.7k
LED1
LTST-C230GKT
GND_AD
RED POWER LED FOR GATE DRIVER +5V_A
+3.3V_A
R2
220
LED2
LTST-C230CKT
GND_AD
C1
UCM1H101MCL1GS
C74
UCM1H101MCL1GS
C75
UCM1H101MCL1GS
C77
UCM1H101MCL1GS
C78
UCM1H101MCL1GS
C80
UCM1H101MCL1GS
C81
UCM1H101MCL1GS
C82
UCM1H101MCL1GS
+12V_AD
GND_A
H1 971150581
H2 971150581
H3 971150581
H4 971150581
H5 971150581
H6 971150581
H7 971150581
H8 971150581
HW-C2-PCB-GD
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /
File: GateDriver.kicad_sch
Title: Open Inverter Platform – Gate Driver Board
Size: USLetter
Date: 2026-07-07
Rev: A
KiCad E.D.A. 10.0.3
Id: 1/8



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

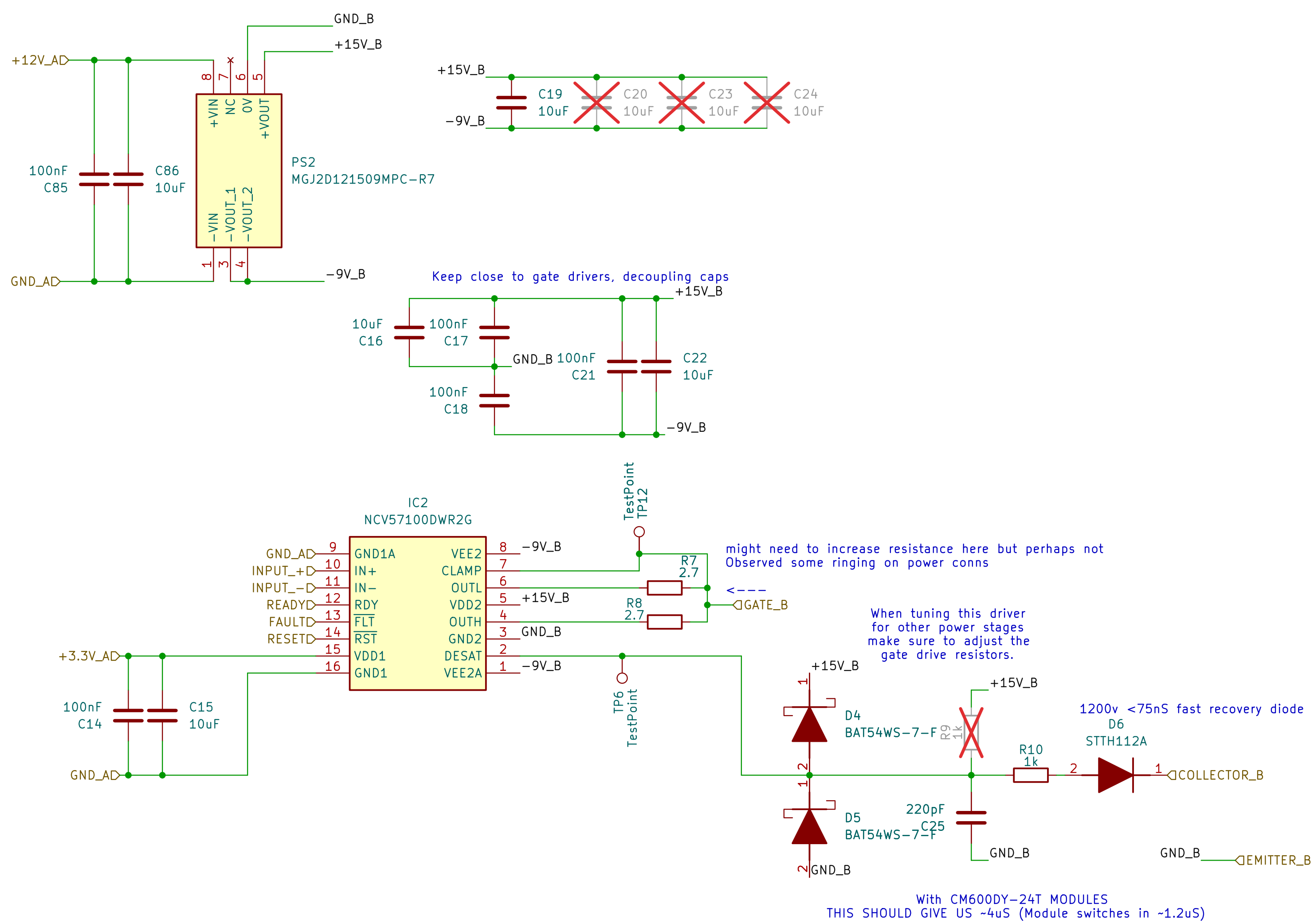
Sheet: /SingleChannelDriver1/

File: SingleChannelDriver.kicad_sch

Title: Open Inverter Platform - Gate Driver Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**

KiCad E.D.A. 10.0.3 | Id: 2/8



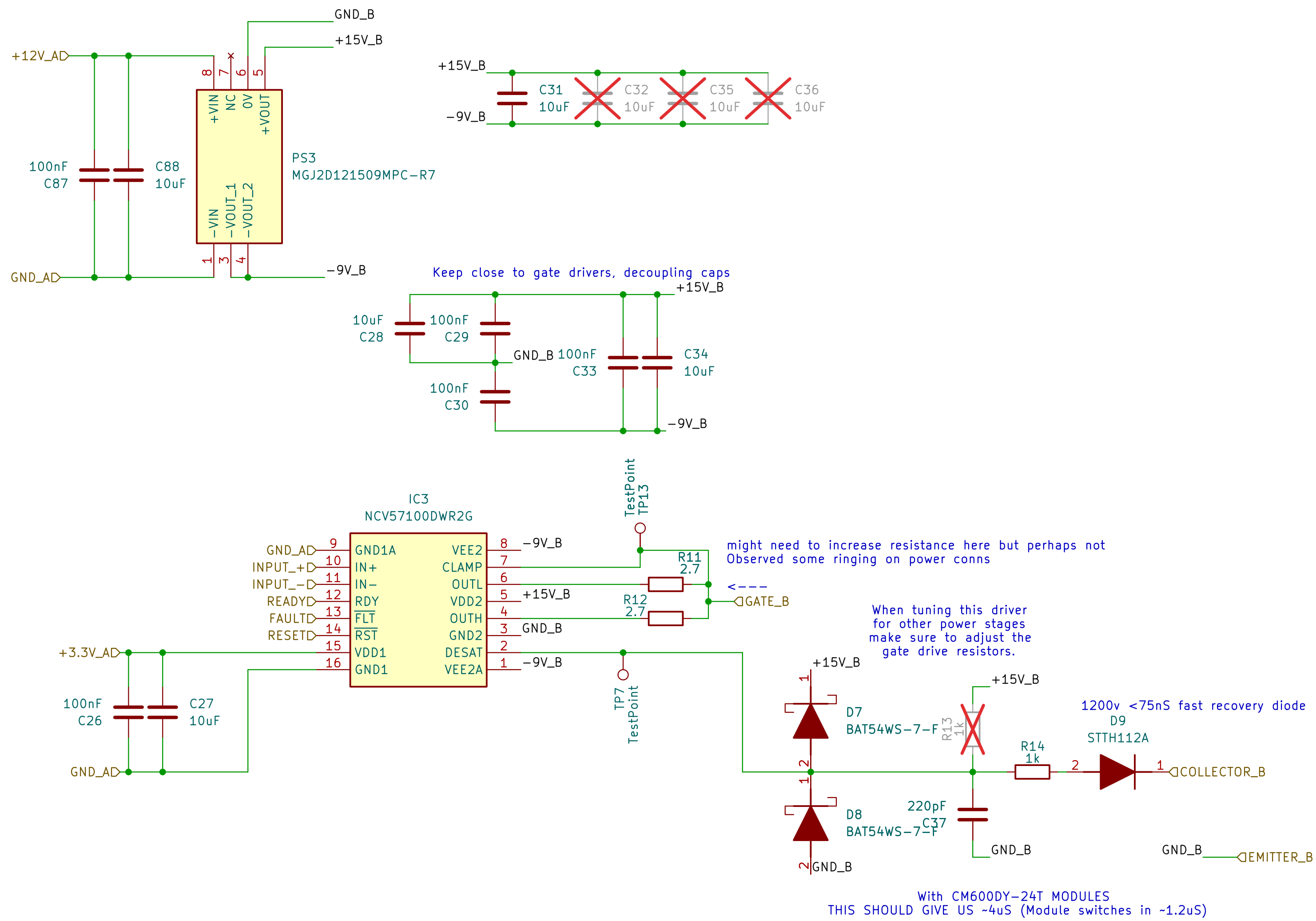
HW-C2-PCB-GD
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /SingleChannelDriver2/
File: SingleChannelDriver.kicad_sch

Title: Open Inverter Platform - Gate Driver Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**

KiCad E.D.A. 10.0.3 | Id: 3/8



HW-C2-PCB-GD

Prepared by Thomas Liao

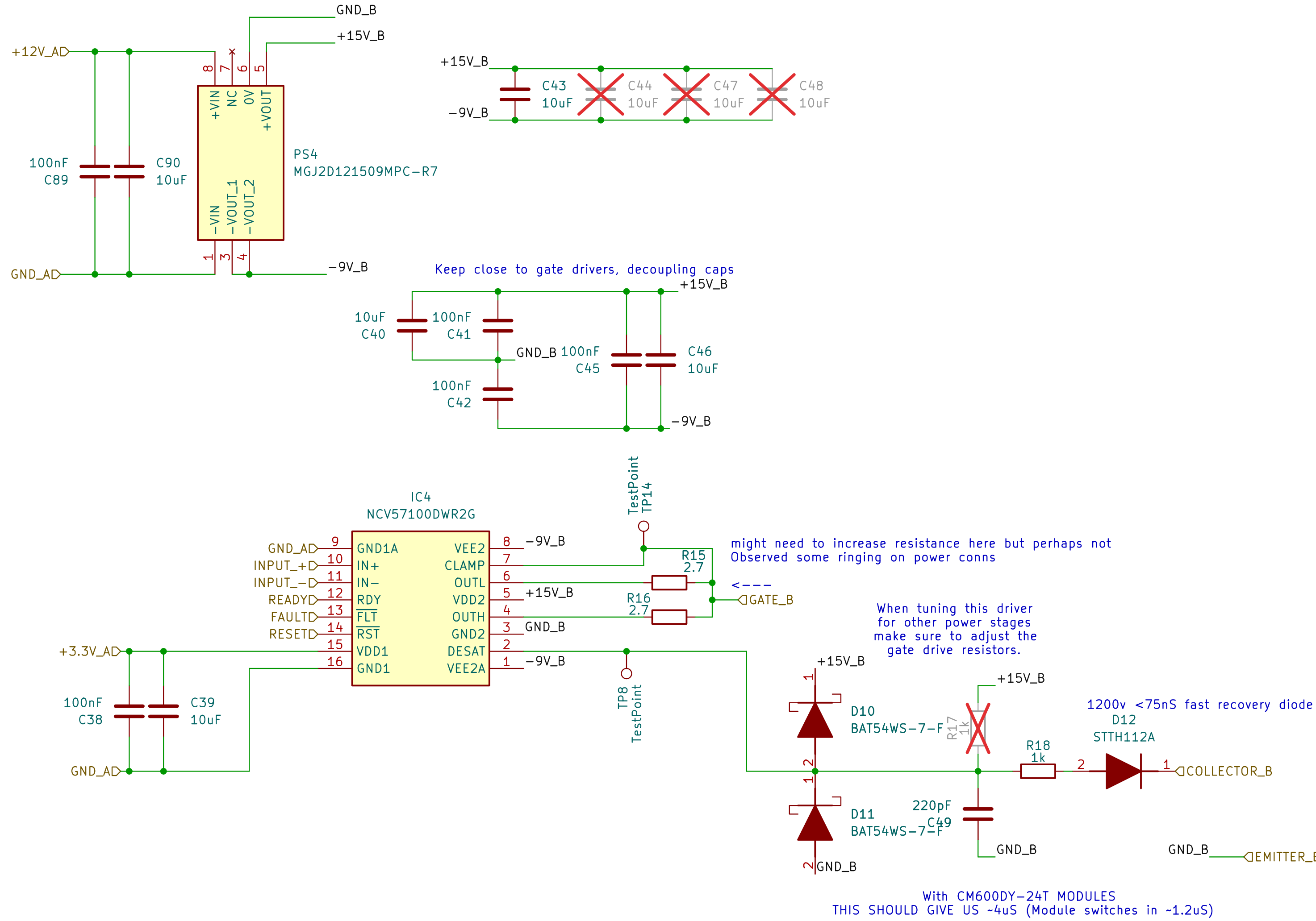
Corzine Lab, University of California, Santa Cruz

Sheet: /SingleChannelDriver/

File: SingleChannelDriver.kicad_sch

Title: Open Inverter Platform - Gate Driver Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 4/8



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

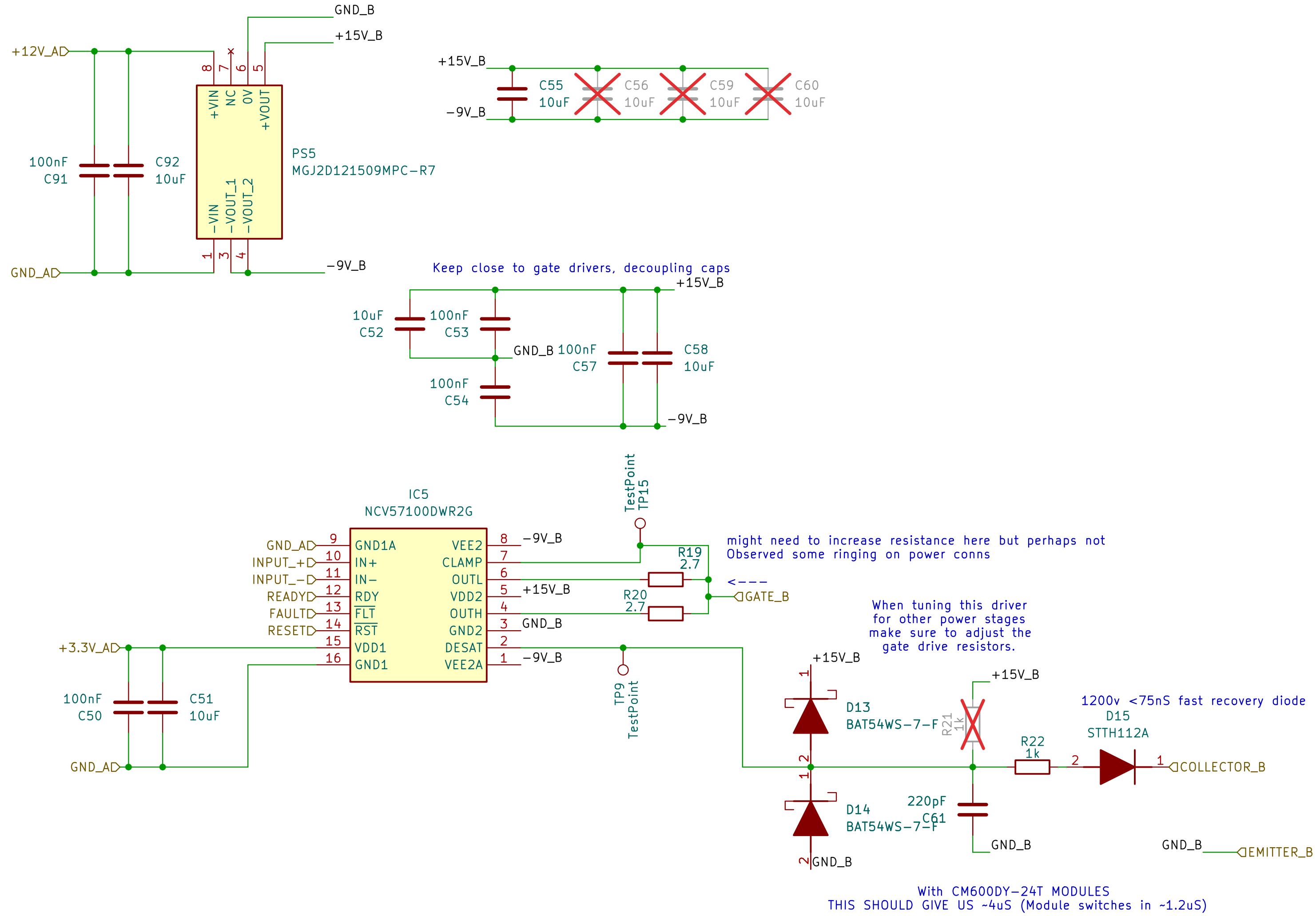
Sheet: /SingleChannelDriver3/

File: SingleChannelDriver.kicad_sch

Title: Open Inverter Platform - Gate Driver Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**

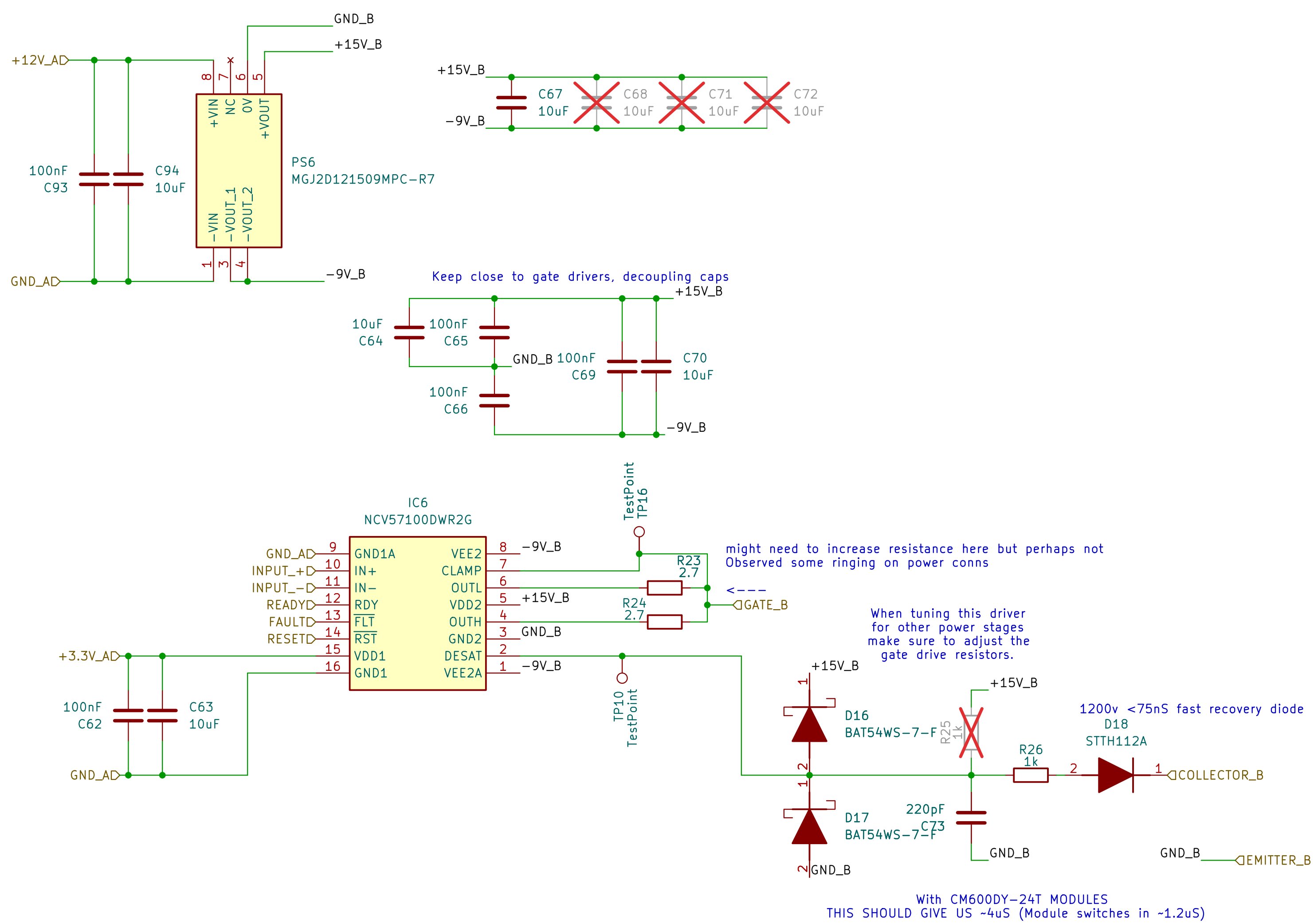
KiCad E.D.A. 10.0.3 | Id: 5/8



HW-C2-PCB-GD
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /SingleChannelDriver4/
File: SingleChannelDriver.kicad_sch

Title: Open Inverter Platform - Gate Driver Board

Size: USLetter	Date: 2026-07-07	Rev: A
KiCad E.D.A. 10.0.3		Id: 6/8



HW-C2-PCB-GD
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /SingleChannelDriver5/
File: SingleChannelDriver.kicad_sch

Title: Open Inverter Platform - Gate Driver Board

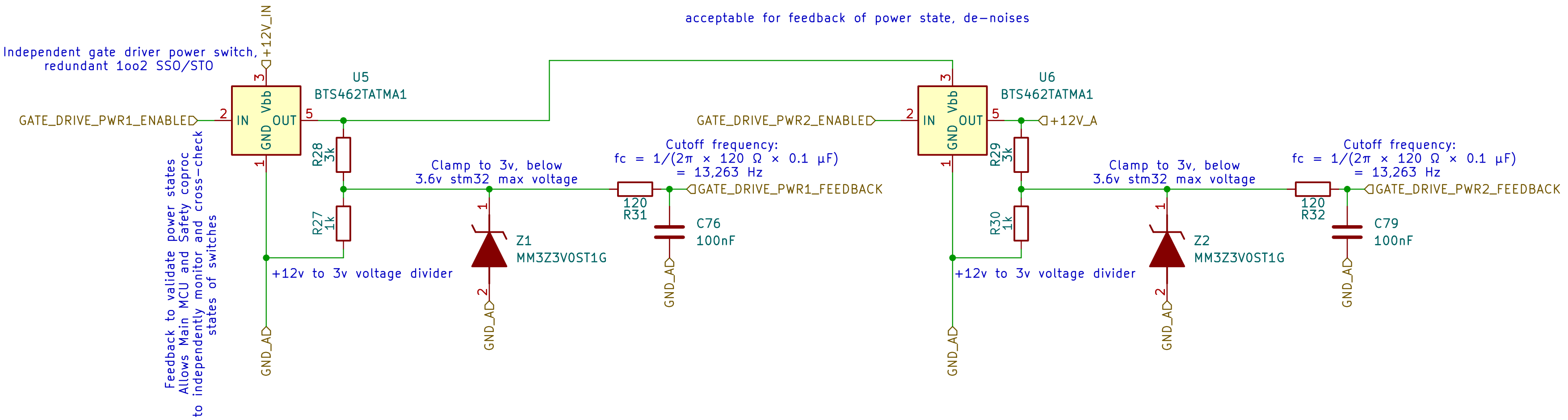
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 7/8

This circuit implements a redundent pathway for both mcus to implement Six Switch Open / Safe Torque Off in addition to regular pwm break and gate driver reset.

The following switches are independent systems with feedback monitoring. Each MCU cross monitors returned inputs, and can validate if the switch failed to respond as commanded - thus removing latent failures from this safety subsystem.

cutoff period (1/freq) ->
75.4 uS

acceptable for feedback of power state, de-noises

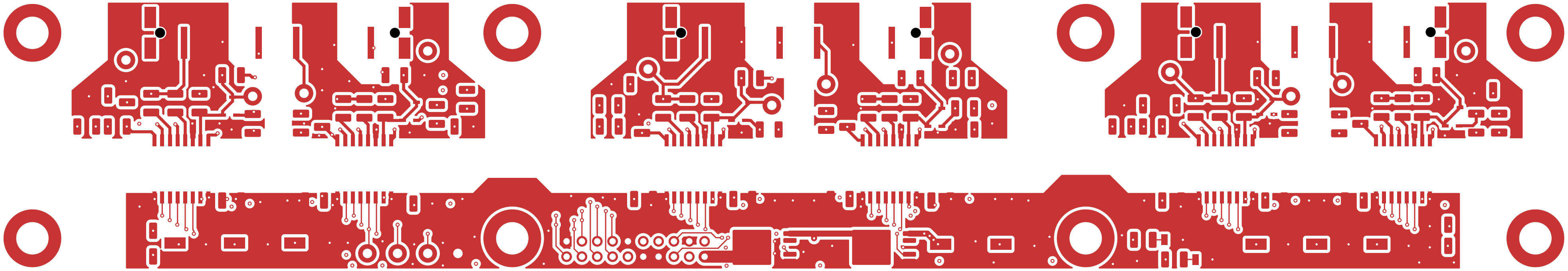


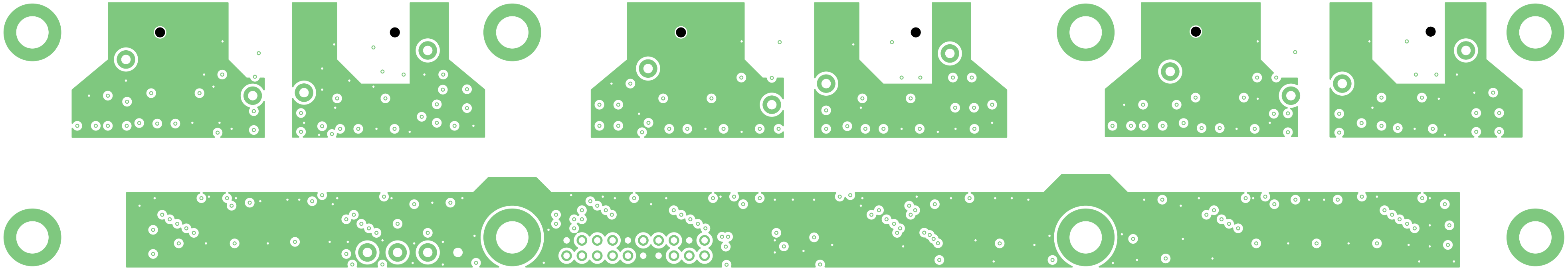
HW-C2-PCB-GD
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

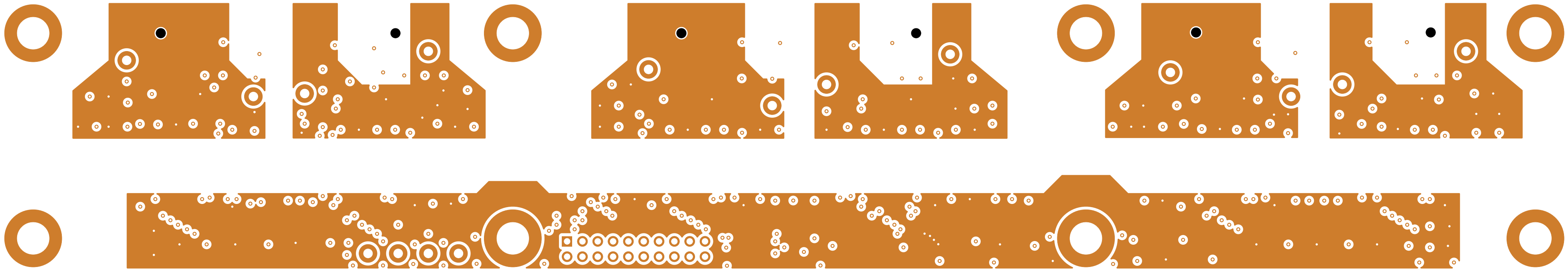
Sheet: /Redundant Monitored Power Switch/
File: PowerSwitch.kicad_sch

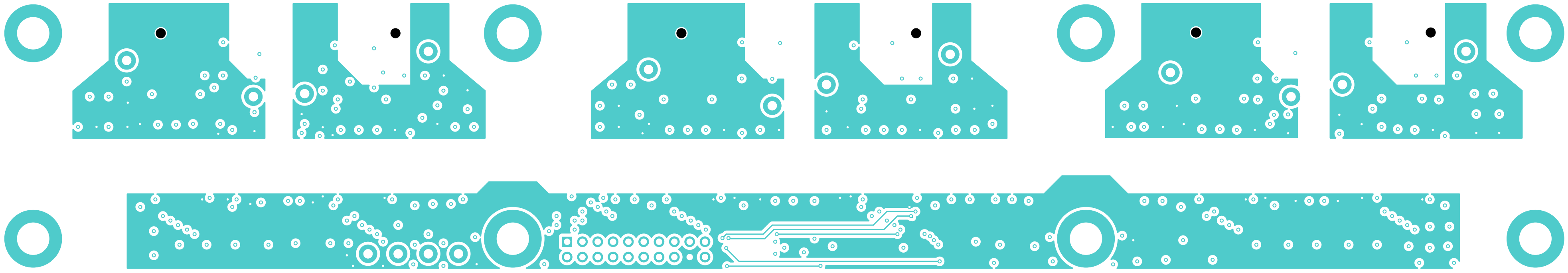
Title: Open Inverter Platform - Gate Driver Board

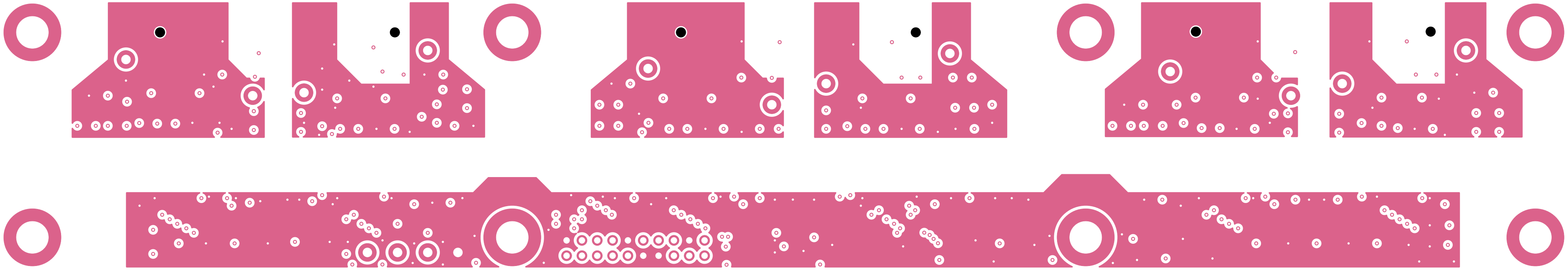
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 8/8

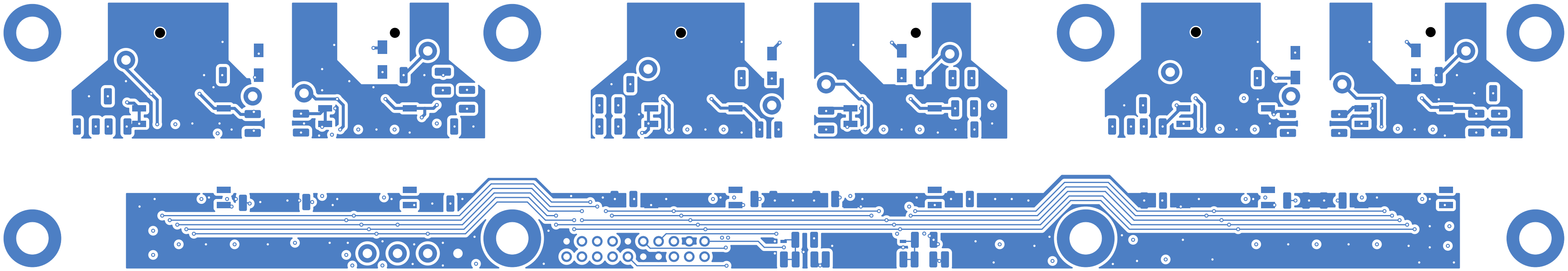


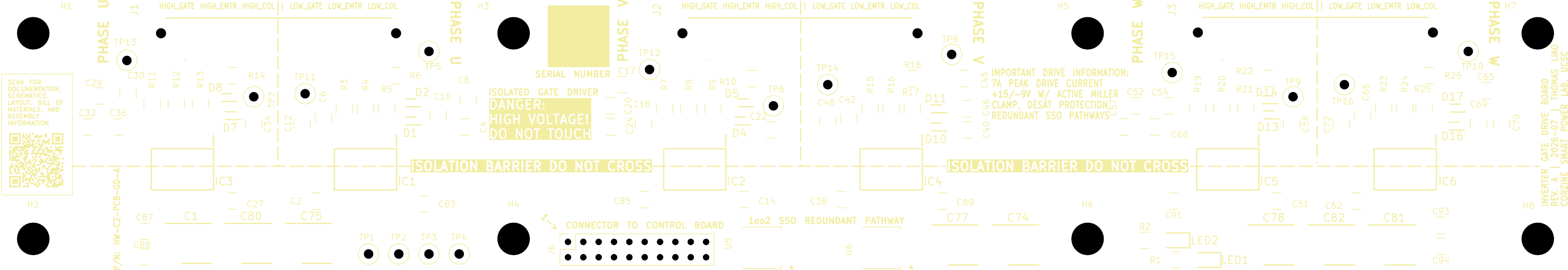




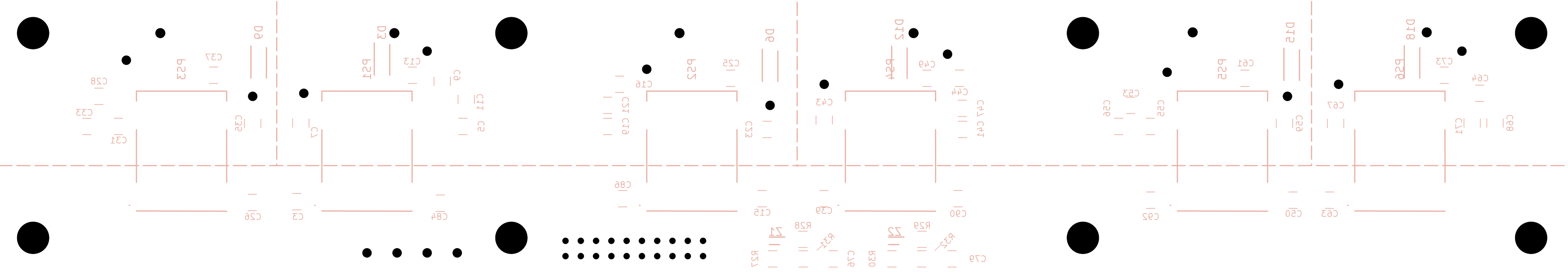








PHASE U
PHASE V
PHASE W
HIGH_GATE HIGH_EMTR HIGH_COL
LOW_GATE LOW_EMTR LOW_COL
SERIAL NUMBER
ISOLATED GATE DRIVER
DANGER:
HIGH VOLTAGE!
DO NOT TOUCH
IMPORTANT DRIVE INFORMATION:
7A PEAK DRIVE CURRENT
+15/-9V W/ ACTIVE MILLER
CLAMP, DESAT PROTECTION,
REDUNDANT SSO PATHWAYS
ISOLATION BARRIER DO NOT CROSS
ISOLATION BARRIER DO NOT CROSS
SCAN FOR DOCUMENTATION, SCHEMATICS, LAYOUT, BILL OF MATERIALS, AND ASSEMBLY INFORMATION
CONNECTOR TO CONTROL BOARD
1oo2 SSO REDUNDANT PATHWAY
P/N: HW-C2-PCB-GD-A
INVERTER GATE DRIVE BOARD
REV. A | 2026-07 | THOMAS LUO
CORZINE SMART POWER LAB, UCSC

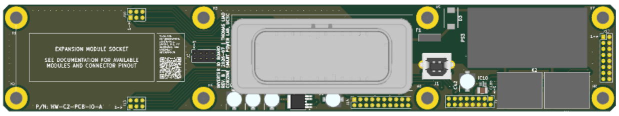


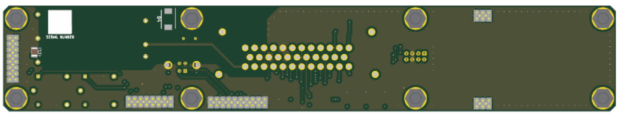
DRC — GateDriver

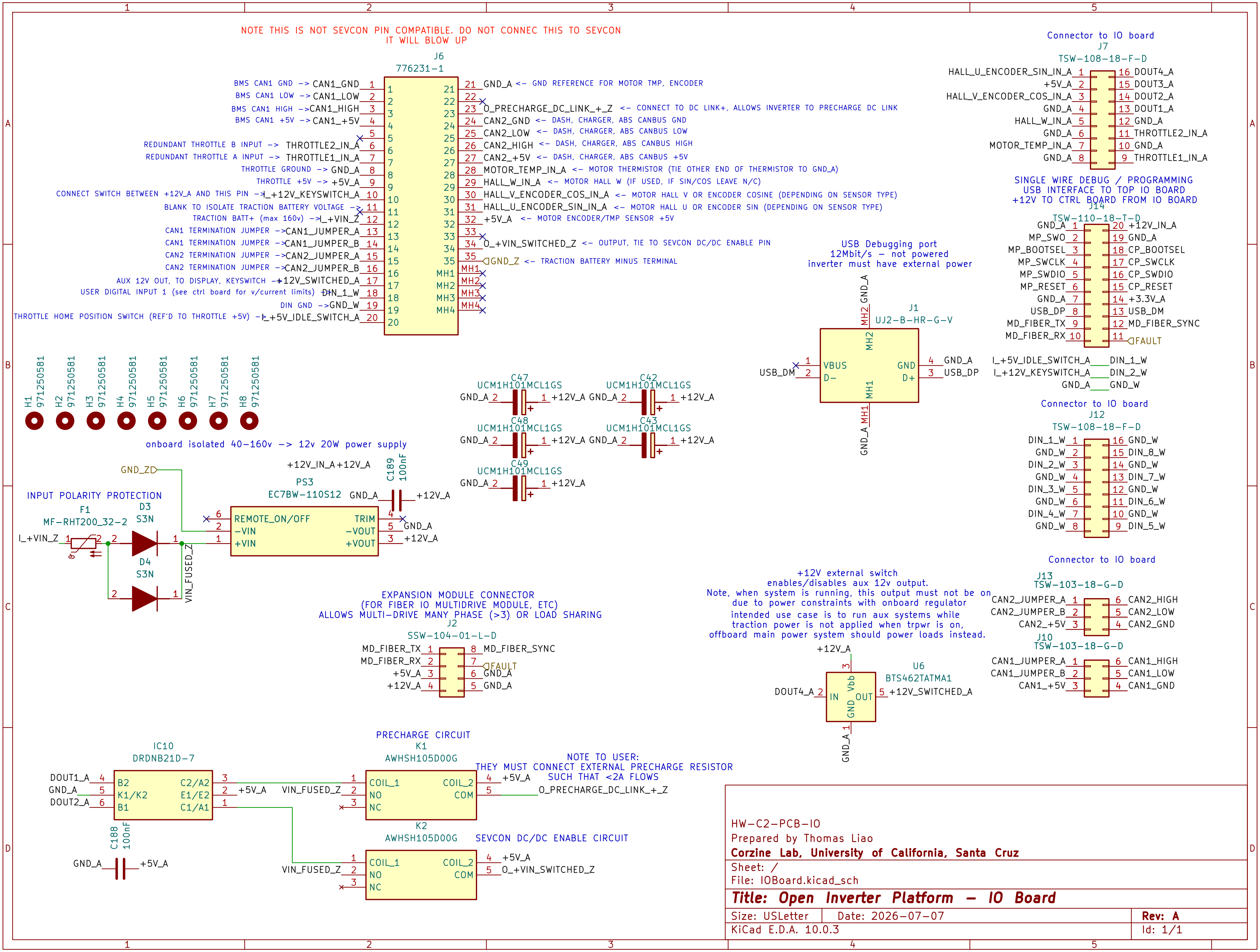
No DRC errors (warnings suppressed).

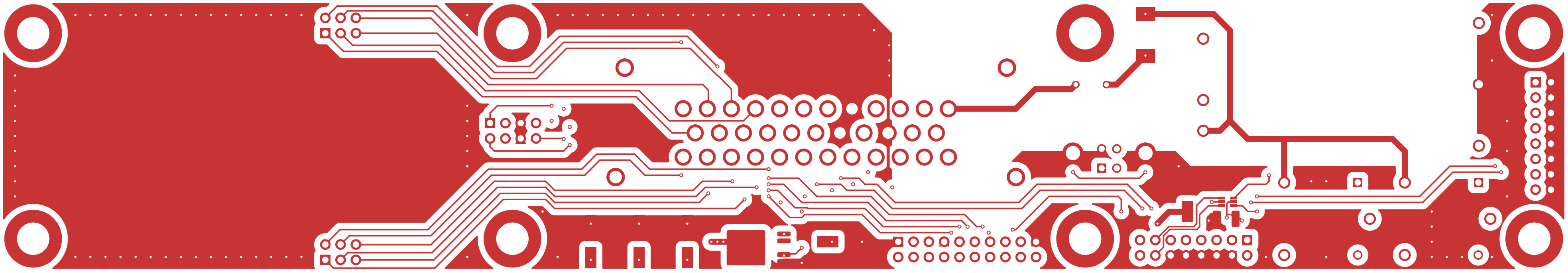
PCB ASSEMBLY: IO BOARD

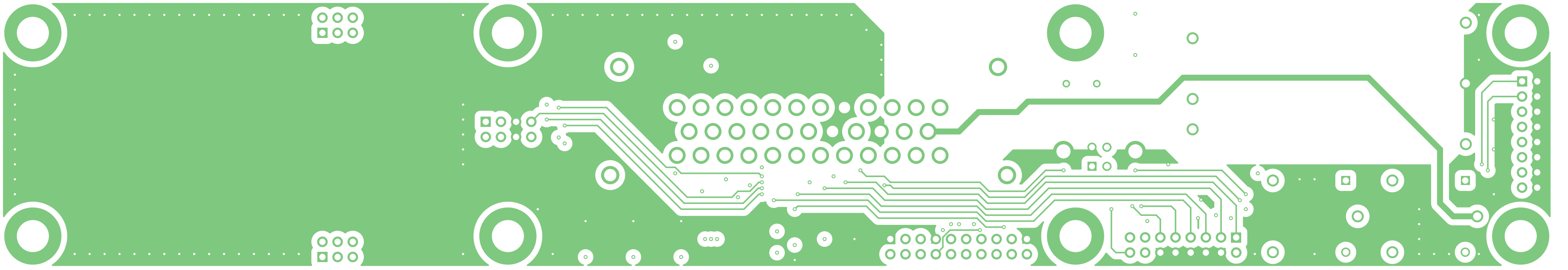
HW-C2-PCB-IO-A · printed circuit board — schematic and PCB layers

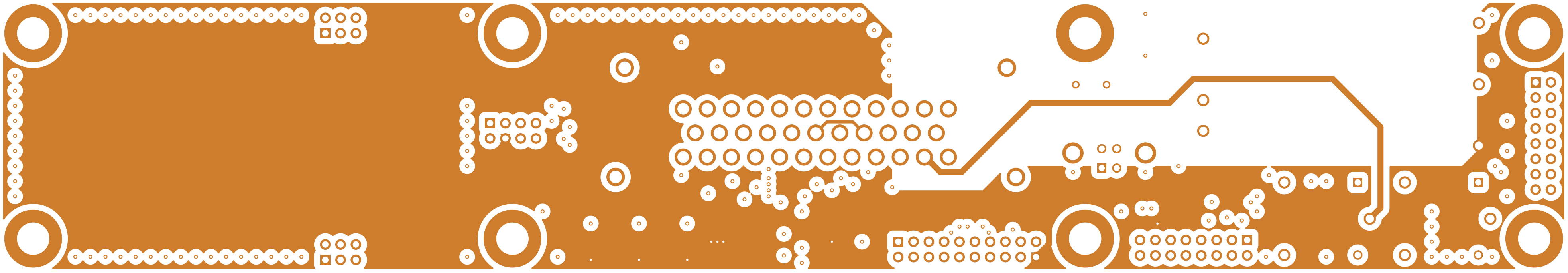


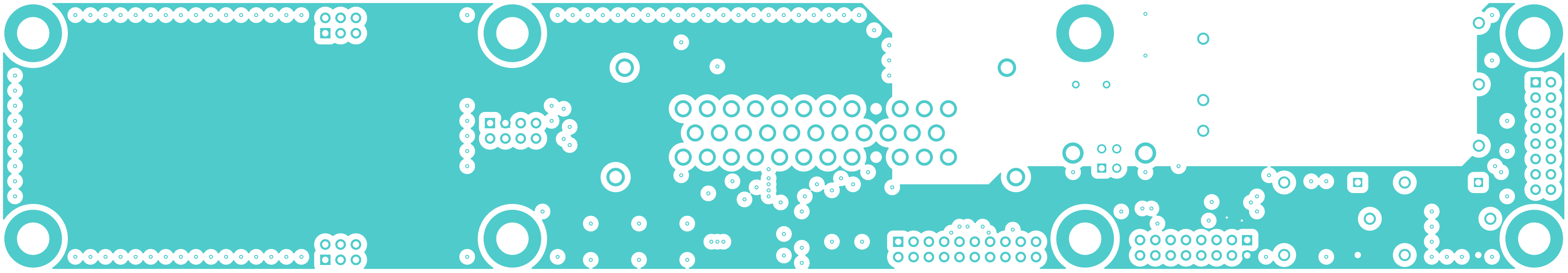


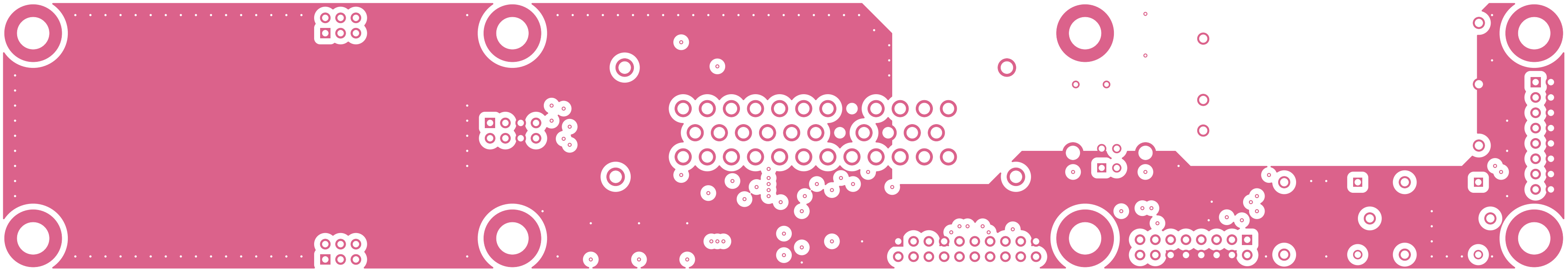

NOTE THIS IS NOT SEVCON PIN COMPATIBLE. DO NOT CONNEC THIS TO SEVCON
IT WILL BLOW UP
J6
776231-1
BMS CAN1 GND -> CAN1_GND
BMS CAN1 LOW -> CAN1_LOW
BMS CAN1 HIGH -> CAN1_HIGH
BMS CAN1 +5V -> CAN1_+5V
REDUNDANT THROTTLE B INPUT -> THROTTLE2_IN_A
REDUNDANT THROTTLE A INPUT -> THROTTLE1_IN_A
THROTTLE GROUND -> GND_A
THROTTLE +5V -> +5V_A
CONNECT SWITCH BETWEEN +12V_A AND THIS PIN -> I_+12V_KEYSWITCH_A
BLANK TO ISOLATE TRACTION BATTERY VOLTAGE ->
TRACTION BATT+ (max 160v) -> I_+VIN_Z
CAN1 TERMINATION JUMPER -> CAN1_JUMPER_A
CAN1 TERMINATION JUMPER -> CAN1_JUMPER_B
CAN2 TERMINATION JUMPER -> CAN2_JUMPER_A
CAN2 TERMINATION JUMPER -> CAN2_JUMPER_B
AUX 12V OUT, TO DISPLAY, KEYSWITCH -> +12V_SWITCHED_A
USER DIGITAL INPUT 1 (see ctrl board for v/current limits) -> DIN_1_W
DIN GND -> GND_W
THROTTLE HOME POSITION SWITCH (REF'D TO THROTTLE +5V) -> I_+5V_IDLE_SWITCH_A
GND_A <- GND REFERENCE FOR MOTOR TMP, ENCODER
O_PRECHARGE_DC_LINK_+_Z <- CONNECT TO DC LINK+, ALLOWS INVERTER TO PRECHARGE DC LINK
CAN2_GND <- DASH, CHARGER, ABS CANBUS GND
CAN2_LOW <- DASH, CHARGER, ABS CANBUS LOW
CAN2_HIGH <- DASH, CHARGER, ABS CANBUS HIGH
CAN2_+5V <- DASH, CHARGER, ABS CANBUS +5V
MOTOR_TEMP_IN_A <- MOTOR THERMISTOR (TIE OTHER END OF THERMISTOR TO GND_A)
HALL_W_IN_A <- MOTOR HALL W (IF USED, IF SIN/COS LEAVE N/C)
HALL_V_ENCODER_COS_IN_A <- MOTOR HALL V OR ENCODER COSINE (DEPENDING ON SENSOR TYPE)
HALL_U_ENCODER_SIN_IN_A <- MOTOR HALL U OR ENCODER SIN (DEPENDING ON SENSOR TYPE)
+5V_A <- MOTOR ENCODER/TMP SENSOR +5V
O_+VIN_SWITCHED_Z <- OUTPUT, TIE TO SEVCON DC/DC ENABLE PIN
GND_Z <- TRACTION BATTERY MINUS TERMINAL
Connector to IO board
J7
TSW-108-18-F-D
HALL_U_ENCODER_SIN_IN_A
+5V_A
HALL_V_ENCODER_COS_IN_A
GND_A
HALL_W_IN_A
GND_A
MOTOR_TEMP_IN_A
GND_A
DOUT4_A
DOUT3_A
DOUT2_A
DOUT1_A
GND_A
THROTTLE2_IN_A
GND_A
THROTTLE1_IN_A
SINGLE WIRE DEBUG / PROGRAMMING
USB INTERFACE TO TOP IO BOARD
+12V TO CTRL BOARD FROM IO BOARD
J14
TSW-110-18-T-D
GND_A
MP_SWO
MP_BOOTSEL
MP_SWCLK
MP_SWDIO
MP_RESET
GND_A
USB_DP
MD_FIBER_TX
MD_FIBER_RX
+12V_IN_A
GND_A
CP_BOOTSEL
CP_SWCLK
CP_SWDIO
CP_RESET
+3.3V_A
USB_DM
MD_FIBER_SYNC
FAULT
USB Debugging port
12Mbit/s - not powered
inverter must have external power
J1
UJ2-B-HR-G-V
VBUS
D-
GND
D+
USB_DM
GND_A
USB_DP
I_+5V_IDLE_SWITCH_A DIN_1_W
I_+12V_KEYSWITCH_A DIN_2_W
GND_A GND_W
Connector to IO board
J12
TSW-108-18-F-D
DIN_1_W
GND_W
DIN_2_W
GND_W
DIN_3_W
GND_W
DIN_4_W
GND_W
GND_W
DIN_8_W
GND_W
DIN_7_W
GND_W
DIN_6_W
GND_W
DIN_5_W
H1 971250581
H2 971250581
H3 971250581
H4 971250581
H5 971250581
H6 971250581
H7 971250581
H8 971250581
C47 UCM1H101MCL1GS
C42 UCM1H101MCL1GS
C48 UCM1H101MCL1GS
C43 UCM1H101MCL1GS
C49 UCM1H101MCL1GS
GND_A
+12V_A
onboard isolated 40-160v -> 12v 20W power supply
GND_ZD
+12V_IN_A +12V_A
C189 100nF
PS3
EC7BW-110S12
REMOTE_ON/OFF
-VIN
+VIN
TRIM
-VOUT
+VOUT
INPUT POLARITY PROTECTION
F1
MF-RHT200_32-2
D3 S3N
D4 S3N
I_+VIN_Z
VIN_FUSED_Z
EXPANSION MODULE CONNECTOR
(FOR FIBER IO MULTIDRIVE MODULE, ETC)
ALLOWS MULTI-DRIVE MANY PHASE (>3) OR LOAD SHARING
J2
SSW-104-01-L-D
MD_FIBER_TX
MD_FIBER_RX
+5V_A
+12V_A
MD_FIBER_SYNC
FAULT
GND_A
+12V external switch
enables/disables aux 12v output.
Note, when system is running, this output must not be on
due to power constraints with onboard regulator
intended use case is to run aux systems while
traction power is not applied when trpwr is on,
offboard main power system should power loads instead.
U6
BTS462TATMA1
IN
Vbb
GND
OUT
DOUT4_A
+12V_SWITCHED_A
Connector to IO board
J13
TSW-103-18-G-D
CAN2_JUMPER_A
CAN2_JUMPER_B
CAN2_+5V
CAN2_HIGH
CAN2_LOW
CAN2_GND
J10
TSW-103-18-G-D
CAN1_JUMPER_A
CAN1_JUMPER_B
CAN1_+5V
CAN1_HIGH
CAN1_LOW
CAN1_GND
IC10
DRDNB21D-7
B2
K1/K2
B1
C2/A2
E1/E2
C1/A1
DOUT1_A
GND_A
DOUT2_A
C188 100nF
PRECHARGE CIRCUIT
K1
AWHSH105D00G
COIL_1
NO
NC
COIL_2
COM
NOTE TO USER:
THEY MUST CONNECT EXTERNAL PRECHARGE RESISTOR
SUCH THAT <2A FLOWS
O_PRECHARGE_DC_LINK_+_Z
K2
AWHSH105D00G
SEVCON DC/DC ENABLE CIRCUIT
O_+VIN_SWITCHED_Z
HW-C2-PCB-IO
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /
File: IOBoard.kicad_sch
Title: Open Inverter Platform - IO Board
Size: USLetter
Date: 2026-07-07
Rev: A
KiCad E.D.A. 10.0.3
Id: 1/1

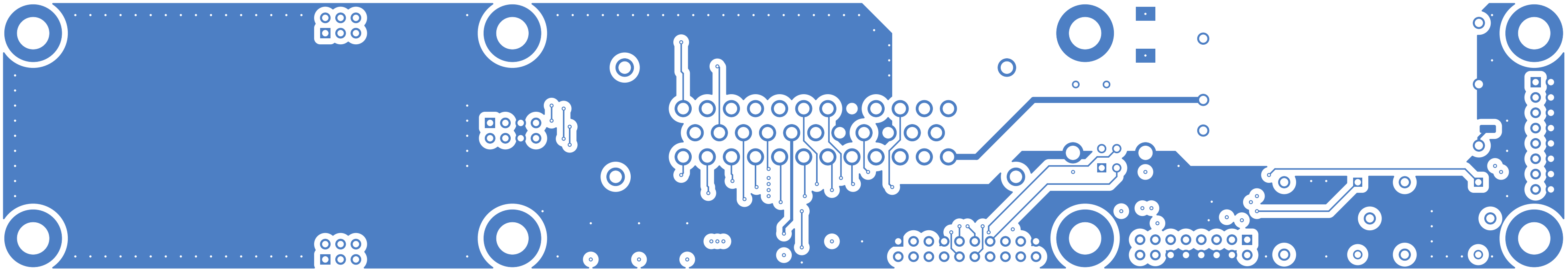


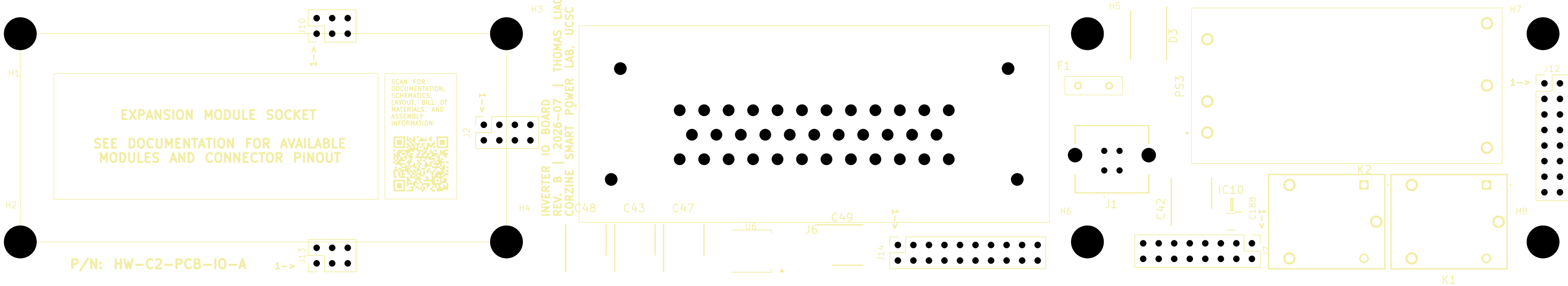




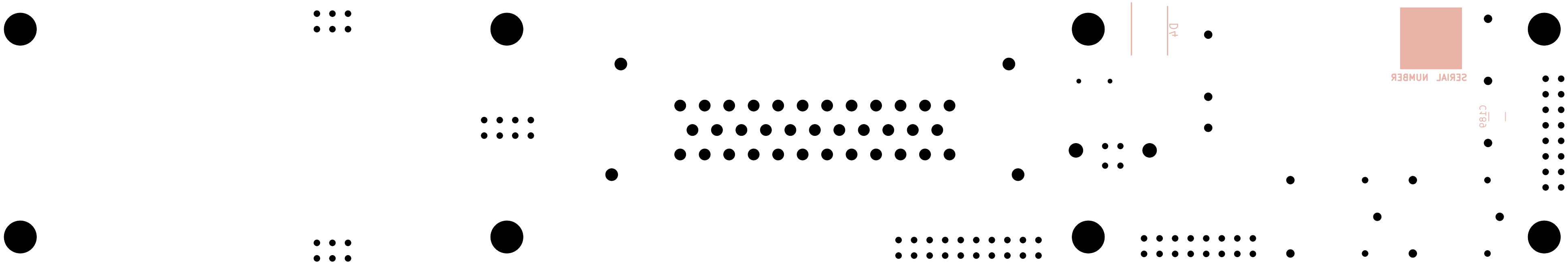








EXPANSION MODULE SOCKET
SEE DOCUMENTATION FOR AVAILABLE
MODULES AND CONNECTOR PINOUT
SCAN FOR DOCUMENTATION, SCHEMATICS, LAYOUT, BILL OF MATERIALS, AND ASSEMBLY INFORMATION
P/N: HW-C2-PCB-IO-A
INVERTER IO BOARD
REV. B | 2026-07 | THOMAS LIAO
CORZINE SMART POWER LAB, UCSC

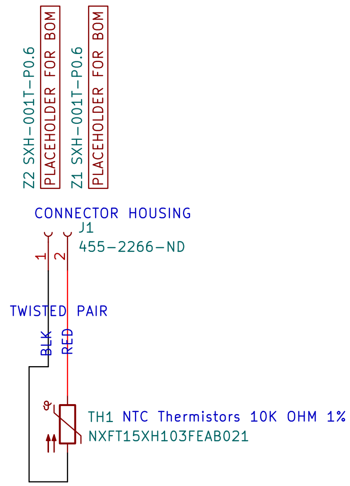


DRC — IOBoard

No DRC errors (warnings suppressed).

CAPACITOR TEMP SENSE WIRING HARNESS

HW-C2-WH-TSBCM-A · wiring harness — schematic



NOTE:
THERMISTOR SHOULD BE SECURELY PLACED INSIDE DC LINK CAPACITOR MODULE BETWEEN CAPACITORS AND ADHERED WITH THERMALLY CONDUCTIVE MATERIAL.

NOTE:
WHEN ASSEMBLING, USE HEAT-SHRINK TUBING AROUND THERMISTOR LEADS, AND ENSURE ELECTRICAL ISOLATION IS MAINTAINED BETWEN THERMISTOR AND DC LINK

Wiring Harness Note

All harnesses shall be constructed using #22 AWG stranded conductor, 1kV rated, PVC insulated wire (or cross-linked polyolefin where higher temperature rating is required). Conductors shall be color-coded per MIL-STD-681D (see chart below). Terminations shall be crimped using approved tooling with pull-test verification per IPC/WHMA-A-620, Chapter 19. Harnesses shall be bundled with lacing cord or expandable braided sleeving; tie-wraps are acceptable only in non-vibration environments per drawing callout. All harnesses require 100% continuity and 500V DC hipot testing prior to release. RoHS compliance required unless otherwise specified.

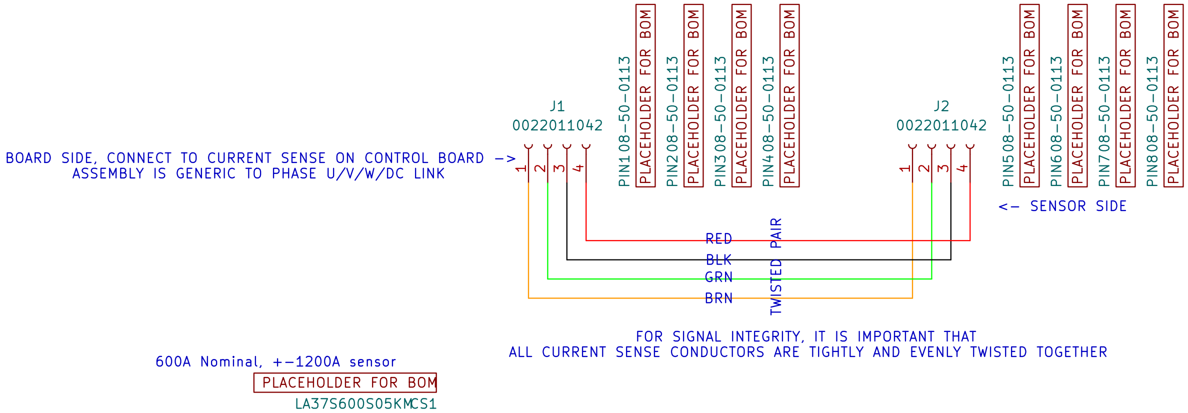
COLOR CODE CHART (MIL-STD-681D)

Cond#	Color	Abbr
0	Black	BLK
1	Brown	BRN
2	Red	RED
3	Orange	ORG
4	Yellow	YEL
5	Green	GRN
6	Blue	BLU
7	Violet	VIO
8	Gray	GRY
9	White	WHT

HW-C2-WH-TSBCM
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /
File: CapacitorTempSenseHarness.kicad_sch
Title: Open Inverter Platform – Capacitor Temp Sense Wiring Harness
Size: USLetter | Date: 2026-07-16 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 1/1

CURRENT SENSOR WIRING HARNESS

HW-C2-WH-CS-A · wiring harness — schematic



Wiring Harness Note

All harnesses shall be constructed using #22 AWG stranded conductor, 1kV rated, PVC insulated wire (or cross-linked polyolefin where higher temperature rating is required). Conductors shall be color-coded per MIL-STD-681D (see chart below). Terminations shall be crimped using approved tooling with pull-test verification per IPC/WHMA-A-620, Chapter 19. Harnesses shall be bundled with lacing cord or expandable braided sleeving; tie-wraps are acceptable only in non-vibration environments per drawing callout. All harnesses require 100% continuity and 500V DC hipot testing prior to release. RoHS compliance required unless otherwise specified.

COLOR CODE CHART (MIL-STD-681D)

Cond#	Color	Abbr
0	Black	BLK
1	Brown	BRN
2	Red	RED
3	Orange	ORG
4	Yellow	YEL
5	Green	GRN
6	Blue	BLU
7	Violet	VIO
8	Gray	GRY
9	White	WHT

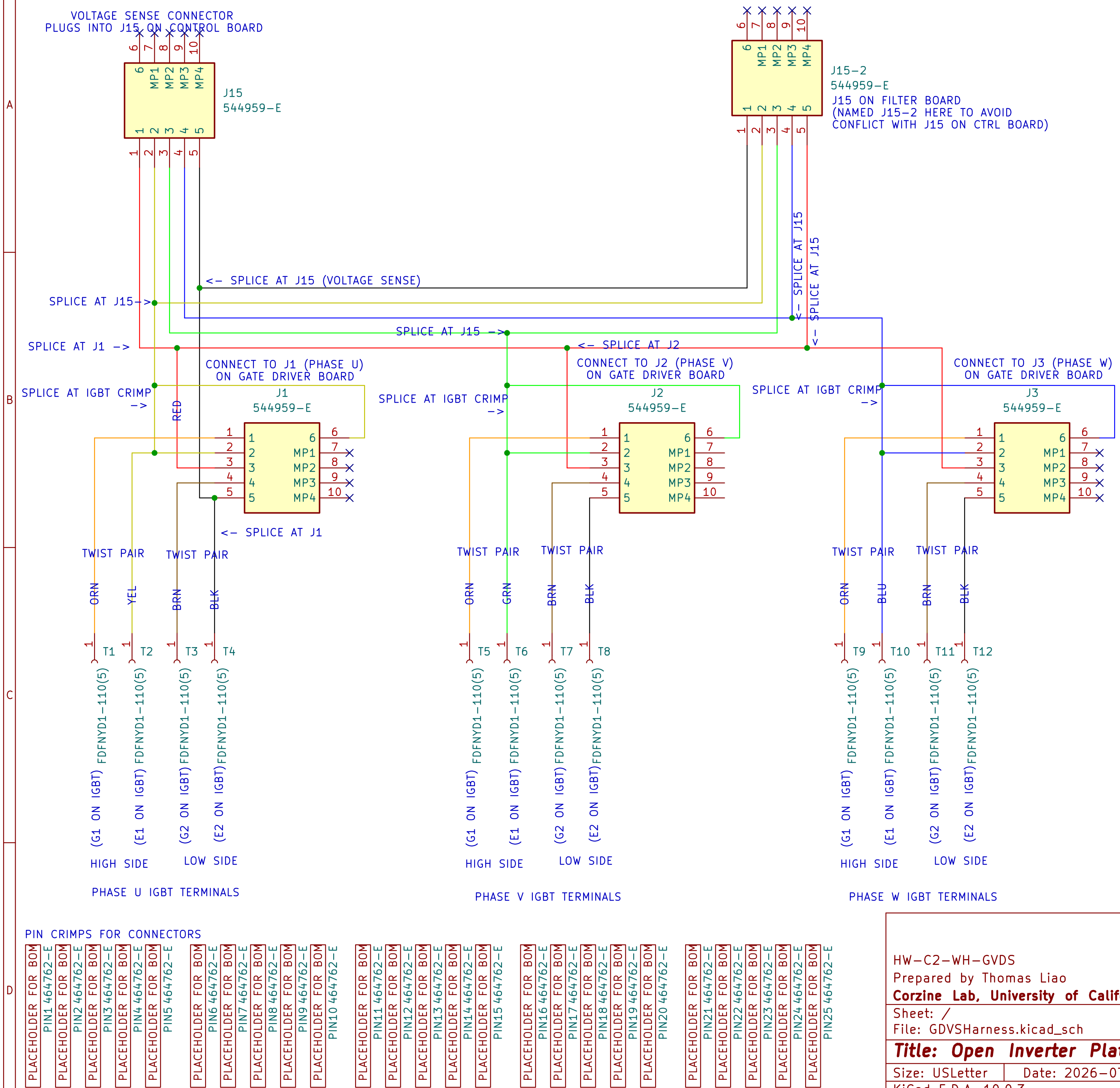
HW-C2-WH-CS
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /
File: CurrentSenseHarness.kicad_sch

Title: Open Inverter Platform – Current Sensor Wiring Harness

Size: USLetter | Date: 2026-07-16 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 1/1

GATE DRIVE WIRING HARNESS

HW-C2-WH-GDVS-A · wiring harness — schematic



Wiring Harness Note

All harnesses shall be constructed using #22 AWG stranded conductor, 1kV rated, PVC insulated wire (or cross-linked polyolefin where higher temperature rating is required). Conductors shall be color-coded per MIL-STD-681D (see chart below). Terminations shall be crimped using approved tooling with pull-test verification per IPC/WHMA-A-620, Chapter 19. Harnesses shall be bundled with lacing cord or expandable braided sleeving; tie-wraps are acceptable only in non-vibration environments per drawing callout. All harnesses require 100% continuity and 500V DC hipot testing prior to release. RoHS compliance required unless otherwise specified.

COLOR CODE CHART (MIL-STD-681D)

Cond#	Color	Abbr
0	Black	BLK
1	Brown	BRN
2	Red	RED
3	Orange	ORG
4	Yellow	YEL
5	Green	GRN
6	Blue	BLU
7	Violet	VIO
8	Gray	GRY
9	White	WHT

HW-C2-WH-GVDS
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

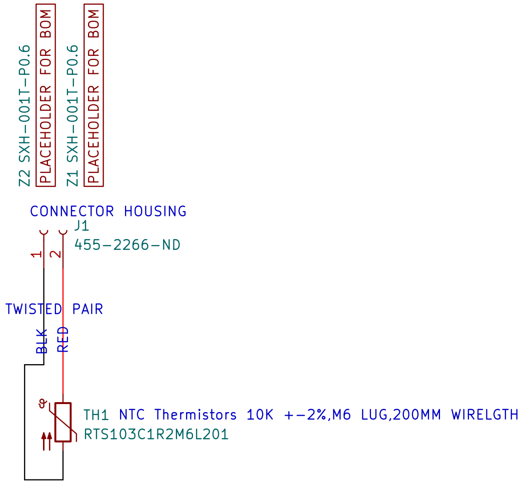
Sheet: /
File: GDVSHarness.kicad_sch

Title: Open Inverter Platform – Gate Drive Wiring Harness

Size: USLetter | Date: 2026-07-16 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 1/1

IGBT TEMPERATURE WIRING HARNESS

HW-C2-WH-TSIGBT-A · wiring harness — schematic



Wiring Harness Note

All harnesses shall be constructed using #22 AWG stranded conductor, 1kV rated, PVC insulated wire (or cross-linked polyolefin where higher temperature rating is required). Conductors shall be color-coded per MIL-STD-681D (see chart below). Terminations shall be crimped using approved tooling with pull-test verification per IPC/WHMA-A-620, Chapter 19. Harnesses shall be bundled with lacing cord or expandable braided sleeving; tie-wraps are acceptable only in non-vibration environments per drawing callout. All harnesses require 100% continuity and 500V DC hipot testing prior to release. RoHS compliance required unless otherwise specified.

COLOR CODE CHART (MIL-STD-681D)

Cond#	Color	Abbr
0	Black	BLK
1	Brown	BRN
2	Red	RED
3	Orange	ORG
4	Yellow	YEL
5	Green	GRN
6	Blue	BLU
7	Violet	VIO
8	Gray	GRY
9	White	WHT

HW-C2-WH-TSIGBT
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /
File: TempSenseHarness.kicad_sch

Title: Open Inverter Platform – IGBT Temperature Wiring Harness

Size: USLetter | Date: 2026-07-16 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 1/1

DESIGN DOCUMENTS

project documentation — safety, security, manual, analyses

THERMAL ANALYSIS — DC LINK MODULE STANDOFF HEAT PATH

HW-C2-DOC-DC-LINK-THERMAL-ANAL-A · design document — rev A

DESIGN DOCUMENT

Thermal Analysis — DC Link Module Standoff Heat Path

Part HW-C2-DOC-DC-LINK-THERMAL-ANAL-A · Rev A

University of California, Santa Cruz — Corzine Lab

Contents

Thermal Analysis — DC Link Module Standoff Heat Path 3
1. Methodology 3
2. Design Evolution 4
3. Final Design Calculation 4
4. Sensitivity & Margin 5
5. Recommendations 6
6. Assumptions & Limitations 6

Thermal Analysis — DC Link Module Standoff Heat Path

Heat load at rated ripple was calculated to be 40W across all capacitors.

1. Methodology

All calculations use one-dimensional steady-state thermal resistance:

$\Delta T = Q \times R_{th}$

where

$R_{th} = L / (k \times A)$

Total system resistance is the sum of three series components:

1. **Standoff conduction** — $R_{standoff} = L / (k_{standoff} \times A_{standoff} \times n)$
2. **Contact resistance** (both faces in series) — $R_{contact} = (2 \times \rho_{contact}) / (n \times A_{standoff})$
3. **Aluminium spreading** — $R_{spread} \approx (\ln(r_{cell} / r_{standoff}) - 0.5) / (2\pi \times k_{Al} \times t_{Al})$

Material properties

Material	Thermal conductivity k [W/(m·K)]
Aluminium (6063 / generic)	200
Brass (C36000)	120
Copper (C11000)	400
Carbon steel	50
18-8 Stainless steel	16

Contact resistivity values

Condition	Resistivity ρ_contact [m²·K/W]
Dry metal-to-metal	1.0×10^{-4}
With thermal paste / thin pad	5.0×10^{-5}

Geometry constants

- Heat-spreader plate: 3.18 mm (1/8 in) thick aluminium *(corrected from 4 mm to match the fabricated plate, HW-C2-PLT-CHSP-A; all spreading-resistance values below use 3.18 mm)*
- Standoff length: 55 mm (final design)
- Number of standoffs: 6 (final design)
- Standoff spacing: assumed ~100 mm centre-to-centre (spreading cell radius $r_{cell} \approx 50$ mm)

2. Design Evolution

2.1 Initial concepts (for reference)

Configuration	k [W/m·K]	Area [mm²]	R_standoff [K/W]	ΔT_standoff [°C]	Total ΔT (paste) [°C]
8 mm hex brass, hollow M5	120	35.8	2.13	85.4	123
10 mm hex Al, hollow M5	200	67.0	0.684	27.4	53.9
16 mm round Al, hollow M8	200	150.8	0.304	12.2	29.9
16 mm round Al, solid	200	201.1	0.228	9.1	25.8

All at 40 W, 6 standoffs, 55 mm long. Values rounded.

2.2 Selected design — 13 mm round aluminium spacers

Final part specification: - Outer diameter: 13.0 mm - Inner diameter (M6 clearance): 6.3 mm *(Note: M6 major diameter = 6.0 mm; 6.3 mm ID provides thread engagement or clearance depending on part type)* - Wall thickness: 3.35 mm - Length: 55 mm - Material: Aluminium - Thread: M6 × 1 (male-female or through-hole with bolt) - Quantity: 6

Cross-sectional area:

$A = \pi \times (r\_outer^2 - r\_inner^2) = \pi \times (6.5^2 - 3.15^2) \times 10^{-6} = 101.6 \times 10^{-6}\ m^2$

---

3. Final Design Calculation

3.1 Standoff conduction resistance

$R\_standoff = L / (k\_Al \times A \times n) = 0.055 / (200 \times 101.6 \times 10^{-6} \times 6) = 0.451$ K/W

$\Delta T\_standoff = 40 \times 0.451 =$ **18.1 °C**

3.2 Contact resistance (both faces)

With thermal paste:

$R\_contact = (2 \times 5.0 \times 10^{-5}) / (6 \times 101.6 \times 10^{-6}) = 0.164$ K/W

$\Delta T\_contact = 40 \times 0.164 =$ **6.6 °C**

Dry metal-to-metal:

$R\_contact = (2 \times 1.0 \times 10^{-4}) / (6 \times 101.6 \times 10^{-6}) = 0.328$ K/W

$\Delta T\_contact = 40 \times 0.328 =$ **13.1 °C**

3.3 Aluminium spreading resistance

$R\_spread = (\ln(50 / 6.5) - 0.5) / (2\pi \times 200 \times 0.00318) \approx 0.385$ K/W

$\Delta T\_spread = 40 \times 0.385 =$ **15.4 °C**

(Spreading resistance is independent of standoff material; it depends only on plate conductivity, thickness, and cell geometry.)

3.4 Total temperature rise

Condition	ΔT_total
With thermal paste	18.1 + 6.6 + 15.4 = **40.1 °C**
Dry metal-to-metal	18.1 + 13.1 + 15.4 = **46.6 °C**

3.5 Absolute temperatures (heatsink base = 40 °C)

Condition	Aluminium plate temperature
With thermal paste	**~80 °C**
Dry metal-to-metal	**~87 °C**

450 V capacitor-only upgrade: The 450 V upgrade is a single part-number swap to 60× Nichicon UCS2W680MHD 68 µF / 450 V capacitors (4.08 mF total). These parts are 5 mm shorter than the 200 V UCS2D331MHD, so the standoff length is reduced by 5 mm (from 55 mm to 50 mm). The same 13 mm OD aluminium standoff thermal path applies, with marginally lower conduction resistance due to the shorter length.

---

4. Sensitivity & Margin

4.1 Effect of standoff material

If stainless steel (18-8, k = 16 W/m·K) were used instead of aluminium:

$\Delta T\_standoff = 40 \times 0.055 / (16 \times 101.6 \times 10^{-6} \times 6) \approx 226$ °C

Total rise would exceed 245 °C. Stainless steel is **not acceptable** for this thermal path.

4.2 Effect of quantity

Standoff count	R_standoff [K/W]	ΔT_standoff [°C]	Total ΔT (paste) [°C]
4	0.677	27.1	52.3
6 (selected)	0.451	18.1	40
8	0.338	13.5	33.9

Six standoffs provides adequate margin; eight would be better but is not required at 40 W.

4.3 Effect of length

Length [mm]	ΔT_standoff [°C]	Total ΔT (paste) [°C]
30	9.9	32
55 (selected)	18.1	40
65	21.4	43

5. Recommendations

1. **Use aluminium standoffs/spacers only.** Do not substitute stainless or carbon steel.
2. **Apply thermal paste** (or a thin graphite / indium thermal pad) at both the plate-to-standoff and standoff-to-heatsink interfaces. This saves ~6.5 °C and improves long-term thermal stability.
3. **Ensure adequate clamping force** on the M6 bolts to minimize contact resistance. Target ~5–10 N·m on steel bolts into aluminium.
4. **Verify standoff placement** is reasonably distributed across the plate. Uneven distribution will increase local spreading resistance and hot-spot temperatures.
5. **If power increases above ~60 W**, consider upgrading to 8 standoffs or thicker-wall spacers (e.g., 16 mm OD / 6 mm ID).

6. Assumptions & Limitations

- One-dimensional conduction assumed; actual 3D spreading may vary ±20 %.
- Contact resistivity values are typical estimates; actual values depend on surface finish, flatness, and clamping pressure.
- Heat generation is assumed uniform across the aluminium plate. Localised hot spots will increase peak temperatures.
- Radiation and natural convection from the plate are neglected; in reality they provide additional heat rejection, so actual plate temperature may be slightly lower.
- Ambient / heatsink base temperature is assumed constant at 40 °C; if the heatsink warms up under load, the absolute plate temperature rises proportionally.

Prepared for DC link module thermal design review.

HAZARD ANALYSIS & RISK ASSESSMENT

HW-C2-DOC-HARA-A · design document — rev A

HAZARD ANALYSIS & RISK ASSESSMENT

Traction Inverter

MCUs	STM32H723ZG + STM32G474RCTx
Operating Temp	−40 °C to +85 °C
Version	4.1
Prepared by	Thomas Liao
Reviewed by	(not yet reviewed)
Date	July 13, 2026
Part	HW-C2-DOC-HARA-A · Rev A

University of California, Santa Cruz — Corzine Lab

Contents

1. Introduction . . . 4
    1.1 Compliance Statement . . . 4
    1.2 Scope . . . 5
    1.3 Reference Standards . . . 5
2. Item Definition . . . 6
    2.1 System Boundaries . . . 6
    2.2 Architecture Overview . . . 6
    2.3 Mitigation Strategy . . . 9
    2.4 Technical Parameters . . . 11
    2.5 Gate Drivers (Non-ASIL) . . . 12
    2.6 Onboard Power . . . 12
    2.7 Future Considerations . . . 13
3. Operational Situations . . . 13
4. Hazard Identification (HAZID) . . . 14
5. Risk Assessment & ASIL Assignment . . . 16
    ASIL Assignment . . . 16
6. Safety Goals . . . 18
7. Functional Safety Requirements . . . 19
8. Current Design Coverage . . . 20
9. Gap Analysis . . . 21
    9.1 Architecture Gaps . . . 21
    9.2 Component Gaps . . . 22
    9.2 Gap Summary . . . 22
10. Fault Injection Test Plan . . . 23
    10.1 Test Philosophy . . . 23
    10.2 Test Environment . . . 24
    10.3 Component-Level Tests . . . 25
    10.4 System-Level Tests . . . 43
    10.5 Integration Tests . . . 53
    10.6 Safety Goal Traceability . . . 60
    10.7 Coverage Justification . . . 61
    10.8 Pass/Fail Criteria . . . 63
    10.9 Known Test Limitations . . . 64
    10.10 Environmental and Stress Tests . . . 66
    10.11 Recommended Test Execution Order . . . 71
11. Implementation Roadmap . . . 76
    11.1 Phase 1: Safety-Critical Foundation (Immediate) . . . 76
    11.2 Phase 2: Graceful Degradation (Short Term) . . . 76
    11.3 Phase 3: Test Execution and Validation . . . 76
    11.4 Coprocessor Integration Validation . . . 77
References . . . 78

Document History 78

1. Introduction

1.1 Compliance Statement

This Is Not an ISO 26262 Compliance Claim

This document **applies ISO 26262:2018 methodology** (Part 3: Concept Phase) to identify hazards, assess risk via Severity/Exposure/Controllability classification, and derive Safety Goals with ASIL ratings. The following must be understood clearly:

- **ASIL ratings are targets** derived from the HARA process. They represent the assessed risk level of identified hazardous events, **not a claim that the design has been verified or certified to meet those ASILs**.
- **The system described is not ISO 26262 compliant.** Full compliance would require: complete product development per Parts 4-6, hardware architectural metrics (SPFM ≥ 90% for ASIL B, ≥ 97% for ASIL D; LFM ≥ 60% for ASIL B, ≥ 80% for ASIL D), dependent failure analysis (DFA), software tool qualification, verification and validation testing (including fault injection), configuration management, change control, and independent safety assessment. **None of these have been completed.**
- The hardware implements a **dual-MCU architecture**: STM32H723ZG main MCU + STM32G474RCTx safety coprocessor. The coprocessor provides independent ADC monitoring of all safety-critical signals, 1oo2 gate drive power kill, challenge/response watchdog, independent CAN bus snooping, and bidirectional NRST. This makes **ASIL D achievable for SG-01 and SG-13 via ASIL B(D) + ASIL B(D) decomposition**.
- The gate driver ICs (onsemi NCV57100) are **automotive-qualified (AEC-Q100)** but are **not ISO 26262 safety elements**. No safety manual, FMEDA, or ASIL claim is available from the manufacturer. Internal protections (DESAT, anti-shoot-through, UVLO) provide hardware-level risk reduction but cannot be claimed as ASIL-rated safety mechanisms without additional justification.
- **Software Test Library (STL) limitation:** This project uses ST's publicly available X-CUBE-CLASSB library (IEC 60730-1 Class B certified). The ISO 26262-certified Class D STL (X-CUBE-STL) requires an NDA and is not available for open-source use. The following ASIL process gaps result: no FMEDA/SPFM/LFM metrics derived for this hardware configuration; no ISO 26262 software tool qualification (compiler, static analysis); no fault injection testing campaign with coverage evidence; no independent safety assessment. ASIL decomposition using the Class B STL is a design and educational exercise only.
- The firmware is **architecturally complete**. Field-Oriented Control (FOC) is implemented. Safety mechanisms (tractive effort plausibility checking, fault ramp-down, boot CRC, challenge/response watchdog) are specified in this document; their implementation status is tracked in Sections 8 and 11.
- This HARA and its accompanying Fault Injection Test Plan are **living design-input documents** intended to guide development and establish a safety engineering baseline. They do not constitute a product safety case, compliance certification, or warranty of fitness for any purpose.

This is an **open-source aftermarket traction inverter** project. The documentation is published in the interest of transparency. Users bear full responsibility for evaluating the suitability of this design for their specific application, risk tolerance, and applicable regulatory requirements.

1.2 Scope

This document presents the Hazard Analysis and Risk Assessment (HARA) and Fault Injection Test Plan for a combined traction inverter and Vehicle Control Unit (VCU) designed to drive 3-phase PMSM traction motors from a DC link bus supply. The primary application focus is high-performance electric motorcycles (aftermarket replacement for Zero, Energica, and similar platforms), though the traction inverter may be used with any compatible DC link bus supply and 3-phase traction motor.

The DC link is supplied from a traction battery pack with Battery Management System (BMS), though the traction inverter itself is agnostic to the DC source type provided the voltage remains within the specified operating window. The analysis covers the traction inverter/VCU unit and its interfaces to external systems that affect safety.

The scope includes all hardware and software within the combined unit: the STM32H723ZG main MCU, the STM32G474RCTx safety coprocessor, six NCV57100 gate drivers, 3-phase IGBT power stage, current/voltage/temperature sensing, HVIL circuit, tractive effort control input processing, motor control algorithm, CAN communication, inter-MCU challenge/response watchdog, and fault handling. The Safety Coprocessor is **part of the current design** and provides independent monitoring, 1oo2 gate drive power kill, and ASIL B(D) decomposition.

Out of scope: The traction motor itself (external product), the rotor position sensor/encoder (part of the external motor), the Battery Management System (BMS), the IO board, the charger, and the vehicle display — these are external CAN nodes interfaced by the VCU but not designed or manufactured by this project. The CAN protocol definitions in this document are the VCU-side interface only.

> **ASIL Decomposition via Dual MCU**
>
> This design implements **ASIL B(D) decomposition** through two independent MCUs: the STM32H723ZG (main processor) and the STM32G474RCTx (safety coprocessor). Each MCU independently monitors all safety-critical signals. Either MCU can trigger safe state entry. The 1oo2 gate drive power kill (GATE_DRIVE_PWR1_ENABLE from main, GATE_DRIVE_PWR2_ENABLE from coprocessor) provides an independent supply shutdown path. Target ASIL D is achievable for SG-01 and SG-13 via ASIL B(D) + ASIL B(D) decomposition.

1.3 Reference Standards

Standard	Title	Application
ISO 26262-1:2018	*Road vehicles — Functional Safety — Part 1: Vocabulary*	Definitions and abbreviations
ISO 26262-3:2018	*Part 3: Concept phase*	HARA methodology, Safety Goals, FSR derivation
ISO 26262-4:2018	*Part 4: Product development at the system level*	Technical Safety Requirements, system design
ISO 26262-5:2018	*Part 5: Product development at the hardware level*	Hardware architectural metrics (SPFM, LFM)
ISO 26262-8:2018	*Part 8: Supporting processes*	Test planning, change management
ISO 26262-9:2018	*Part 9: ASIL-oriented and safety-oriented analyses*	ASIL decomposition, safety analysis

Standard	Title	Application
ISO 6469-3:2018	*Electrically propelled road vehicles — Safety specifications — Part 3: Electrical safety*	HV isolation requirements
EN 50155	*Railway applications — Electronic equipment used on rolling stock*	Onboard power supply qualification reference

2. Item Definition

2.1 System Boundaries

Table 1 — System Boundary Inclusions and Exclusions

Category	Description
In Scope	Combined traction inverter/VCU PCB, six onsemi NCV57100 isolated gate drivers (non-ASIL), 3-phase 2-level IGBT bridge, **STM32H723ZG main MCU + STM32G474RCTx safety coprocessor** (dual independent core), phase current sensing (3-phase + DC link), DC link bus voltage sensing, phase voltage sensing, IGBT temperature sensing (3 modules, 1 NTC per module), traction motor temperature sensing, traction motor encoder input, HVIL circuit, dual redundant tractive effort control input + end-travel limit switch, CAN1 (BMS), CAN2 (ABS, display, charger, IO board), precharge control, fault handling logic, 1oo2 gate drive power supply kill (GATE_DRIVE_PWR1_ENABLE + GATE_DRIVE_PWR2_ENABLE), inter-MCU challenge/response watchdog, CY15B102Q-SXET 256 KB FRAM (main MCU side)
Interfaced (external)	Traction battery pack with BMS (provides DC link, CAN1 heartbeat 5 s), ABS module (CAN2, independently powered), display/dash (CAN2), charger(s) (CAN2), IO board (CAN2, brake switch, kickstand switch, turn signal feedback, headlight feedback, 1 s heartbeat), 3-phase PMSM traction motor with encoder, 12 V onboard power (EC7BW-110S12 DC/DC from DC link)
Out of Scope	BMS internal cell protection and contactor control, ABS hydraulic/mechanical system, charger AC-side circuitry, vehicle chassis, traction motor construction, DC link source beyond electrical interface

2.2 Architecture Overview

The traction inverter/VCU is a combined unit that performs both motor control (inverter) and vehicle-level control (VCU) functions. The architecture is organized into three functional layers: the primary control path, the hardware protection layer, and the safe state actuation layer.

Table 2 — System Architecture

PRIMARY CONTROL PATH — STM32H723ZG MAIN MCU		
Core Processing 550 MHz Cortex-M7 ECC RAM Brown-out detect Internal WDT **Software Functions:** FOC motor control Tractive effort command processing Sensor acquisition & plausibility Fault detection & handling CAN communication (FDCAN1 + FDCAN2) Safe state management Gate drive power kill (Path 2a) **Coprocessor interface:** Inter-MCU UART Timer sync line Bidirectional NRST **Storage:** CY15B102Q-SXET 256 KB FRAM (SPI) for fault logs, configuration, hour meter, odometer. Hardware WP pin.	**Interface**	**Connected System**
	DC link (102–350 V)	Traction battery (or compatible DC source)
	3-Phase AC via NCV57100 x6	PMSM traction motor
	CAN1 (heartbeat 5 s)	BMS
	CAN2 (heartbeat 1 s)	IO board (brake, kickstand, signals)
	CAN2	ABS module, display, charger
	Analog (dual pot + limit switch)	Tractive effort control input
	Sin/Cos analog / Hall effect	Traction motor encoder
	EC7BW-110S12	12 V onboard power rail
	FLT (OR'd, active low)	Six NCV57100 fault outputs
HARDWARE PROTECTION LAYER — NCV57100 GATE DRIVERS (NON-ASIL)		

PRIMARY CONTROL PATH — STM32H723ZG MAIN MCU		
Six NCV57100 devices provide **local hardware protection** for each IGBT: DESAT short-circuit detection (<2 us), complementary anti-shoot-through inputs, UVLO, active Miller clamp, soft turn-off, gate active pull-down. These protections operate independently of the STM32 and provide the first line of defense against power stage faults. All six FLT outputs are OR'd together and fed to the STM32 fault input. **Note: these protections are not ASIL-rated but provide valuable hardware-level risk reduction.**		
INDEPENDENT SAFETY MONITOR — STM32G474RCTx SAFETY COPROCESSOR		
Core: 170 MHz Cortex-M4+FPU, 8 MHz crystal, shared +3.3 V rail (dedicated RD7-12S033R DC/DC converter), independent oscillator. **Independent ADC access:** All 4 current sense signals (phase U/V/W + DC link) + REF, all 3 heatsink temperature sensors + motor temp, all encoder signals (Hall U/V/W, Sin/Cos). **Independent gate drive monitoring:** GATE_DRIVE_FAULT (OR'd FLT), GATE_DRIVE_READY, GATE_DRIVE_RESET, GATE_DRIVE_PWR1_FEEDBACK, GATE_DRIVE_PWR2_FEEDBACK. All 6 PWM outputs monitored (PH_U/V/W_HIGH/LOW). **Independent CAN:** FDCAN2 + FDCAN3 — can snoop both CAN buses to cross-check torque commands and node heartbeats. **Inter-MCU communication:** Dedicated UART + timer sync line + bidirectional NRST cross-reset. **1oo2 gate drive power kill:** GATE_DRIVE_PWR2_ENABLE (coprocessor) in logical-OR with GATE_DRIVE_PWR1_ENABLE (main). Either MCU deasserting its enable kills all six gate drive supplies. Each enable has independent feedback (GATE_DRIVE_PWR1_FB, GATE_DRIVE_PWR2_FB). **Challenge/response watchdog:** Coprocessor issues challenge; main MCU must respond within window. Failure → coprocessor asserts NRST on main MCU → SSO. **Safe state authority:** Coprocessor can independently trigger SSO via gate drive power kill, gate drive RESET, or main MCU NRST.		
RAIL SUPERVISOR — TPS389006-Q1 6-CHANNEL WINDOW SUPERVISOR		
A TPS389006-Q1 six-channel window supervisor monitors the +3.3 V, +5 V, +12 V, and sensor +5 V rails plus both gate-drive power feedbacks (GATE_DRIVE_PWR1_FB, GATE_DRIVE_PWR2_FB). Window thresholds are I2C-configured by the main MCU at boot. On any out-of-window rail fault — including brownout of the +3.3 V rail shared by both MCUs — its NIRQ output asserts the shared GATE_DRIVER_FAULT line, which is monitored by both MCUs (the same net that carries the OR'd gate-driver FLT). The device is TI Functional Safety-Compliant and supports designs up to SIL 3 / ASIL D per TI. The boot-time I2C threshold configuration is a dependency to be covered by the pending DFA (LIMIT-08).		
SAFE STATE ACTUATION LAYER — SIX REDUNDANT SSO PATHWAYS		

PRIMARY CONTROL PATH — STM32H723ZG MAIN MCU		
Path 1 (hardware, <100 ns): TIM1 break input (TIM1_BKIN) → hardware clears MOE, all PWM outputs disabled. Triggered by OR'd gate driver FLT (DESAT/UVLO) or software fault. Independent of both CPU states after trigger. **Path 2a (active, ~10 us):** Main MCU → GATE_DRIVE_PWR1_ENABLE low → all six Murata MGJ2D121509MPC-R7 supplies shut down → NCV57100 UVLO → active pull-down → SSO. Feedback via GATE_DRIVE_PWR1_FB. **Path 2b (active, ~10 us):** Coprocessor → GATE_DRIVE_PWR2_ENABLE low → same supply shutdown. Independent of Path 2a. Feedback via GATE_DRIVE_PWR2_FB. Either Path 2a or 2b alone achieves SSO (1oo2). The ~10 us figures are actuation-only; time-to-SSO on the power-kill paths additionally depends on gate-bias rail decay to the NCV57100 UVLO threshold and is under characterization. **Path 3 (hardware, passive):** Loss of shared 3.3V rail → NCV57100 VDD lost → internal active pull-down → SSO. Automatic, no software intervention. **Path 4 (active, <1 us):** Either MCU → DRIVER_RESET asserted → all NCV57100 RESET inputs → outputs immediately disabled (hard turn-off via OUTL active pull-down; on the NCV57100, soft turn-off exists only on the DESAT path) → SSO. Both MCUs share the RESET line (either can assert). The TPS389006-Q1 rail supervisor is an additional fault source on this pathway: on any monitored-rail fault, including brownout of the +3.3 V rail shared by both MCUs, it asserts the shared GATE_DRIVER_FAULT line, signaling both MCUs to assert DRIVER_RESET. **Path 5 (active, ~100 ms):** Coprocessor challenge/response watchdog failure → coprocessor asserts main MCU NRST → system reset → SSO during boot. Main MCU WDT timeout as backup. **Path 6 (active, <10 us):** Coprocessor detects critical fault independently → asserts GATE_DRIVE_PWR2_ENABLE low + GATE_DRIVE_RESET → SSO without relying on main MCU.		

2.3 Mitigation Strategy

The following table explains how each hazard class is mitigated by the architecture, which mechanisms cover which failure modes, and where the known limitations are.

Table 3 — Hazard Mitigation Strategy

Hazard Category	Failure Mode	Mitigation Mechanism	Limitation
Unintended tractive effort (H-01, H-15)	Throttle sensor fault (open, short, drift)	Dual redundant pots with >5% discrepancy check (FSR-01) on both MCUs; throttle limit switch override (FSR-18); rate limiter (FSR-03)	Both MCUs independently read throttle; either detecting discrepancy triggers SSO via independent power kill
Unintended tractive effort (H-01, H-15)	Software error in tractive effort calculation	Torque command vs. measured current plausibility (FSR-02); boot CRC (FSR-19); ECC RAM (FSR-20); windowed WDT (FSR-15)	Plausibility check is software-based; common-cause with main control possible

Hazard Category	Failure Mode	Mitigation Mechanism	Limitation
Unintended tractive effort (H-01, H-15)	MCU latch-up / runaway	Windowed watchdog ≤50 ms (FSR-15); breakpoint HW PWM disable (FSR-14)	WDT is on-chip; common-cause with MCU failure possible
Loss of tractive effort (H-03, H-03a)	Fault-triggered safe state entry	Torque ramp-down at ≤200 Nm/s before SSO (FSR-05); gradual removal prevents destabilization	Implemented via controlled deceleration profile
Loss of tractive effort (H-03, H-03a)	External system loss (CAN, BMS)	CAN heartbeat timeouts with safe defaults (FSR-17); graceful degradation	IO board loss → zero tractive effort (safe but abrupt if rider not expecting)
Over-torque (H-06)	Excessive tractive effort command	Software torque limit LUT (FSR-07); torque command plausibility (FSR-02); coprocessor independent current monitoring with cross-check	DESAT handles hard short-circuit (<2 us). Regular overcurrent detected within 100 ms by dual-MCU integrated monitoring — sufficient for safe state off without hardware damage. No separate HW OCP comparator required.
Over-torque (H-06)	Short-circuit / shoot-through	NCV57100 DESAT (<2 us) (FSR-13); complementary anti-shoot-through inputs (FSR-12); active Miller clamp	Gate drivers are non-ASIL; coprocessor independently monitors all 6 PWM output pairs for deadtime violations and stuck-on/stuck-off
Over-temperature (H-07)	IGBT thermal runaway	3 IGBT modules, 1 NTC sensor per module, 2-out-of-3 voting with 100 °C hard cap (FSR-08); progressive derating; critical threshold → SSO	2oo3 voter implemented; stuck-high sensor can cause unnecessary derating (known trade-off: safety over availability)
Encoder loss (H-08)	Loss of rotor position feedback	Encoder timeout detection <100 ms (FSR-09); immediate SSO on loss	Single encoder (no redundancy); bounded sensorless fallback if implemented
DC link overvoltage (H-10)	Regen-induced bus rise	Isolated ADC monitoring (FSR-11); regen disable at warning threshold; SSO at critical threshold	None significant
HV isolation (H-09, H-11)	HV exposure / contactor weld	HVIL continuous monitoring (FSR-10); interruption → PWM disable + contactor open request	Contactor control is in BMS domain; VCU can only request
Safe state failure (H-13, H-14)	Cannot reach SSO; latched tractive effort	Six redundant SSO pathways (Path 2a and Path 2b are redundant channels of one 1oo2 power-kill pathway): Path 1 = TIM1_BKIN hardware (<100 ns); Path 2a/2b = 1oo2 gate-drive power kill (GATE_DRIVE_PWR1_ENABLE / GATE_DRIVE_PWR2_ENABLE); Path 3 = shared 3.3 V rail loss → NCV57100 pull-down; Path 4 = GATE_DRIVE_RESET; Path 5 = coprocessor watchdog → NRST; Path 6 = coprocessor independent fault trigger. WDT reset (FSR-15); POST before PWM enable (FSR-16).	1oo2 power kill: either GATE_DRIVE_PWR1_ENABLE or GATE_DRIVE_PWR2_ENABLE going low achieves SSO. Each has independent feedback (GATE_DRIVE_PWR1_FB, GATE_DRIVE_PWR2_FB). Coprocessor provides fully independent safe state actuation. Six pathways provide extensive redundancy against any single-point failure.

Hazard Category	Failure Mode	Mitigation Mechanism	Limitation
Gate driver fault (H-16, H-17)	DESAT/ UVLO not detected; PWM deadtime violation	OR'd FLT input to STM32 (FSR-13); DESAT self-test at POST (FSR-16); complementary inputs (FSR-12)	OR'd FLT monitored by both MCUs; coprocessor additionally monitors the combined READY signal and all 6 PWM outputs for independent fault diagnosis

> **How to read this table:** Each row maps a hazard category to the specific failure modes that could cause it, the mitigation mechanisms in the current architecture that address those failure modes, and the known limitations of those mitigations. The left column is the **what could go wrong**; the middle column is the **how we prevent or detect it**; the right column is the **why this might not be enough**. The limitations are addressed in Section 9 (Gap Analysis).

2.4 Technical Parameters

Table 4 — Key Technical Parameters

Parameter	Value / Range
Vehicle type	Electric 2-wheel motorcycle (aftermarket)
Target platforms	Zero (102 V nom), Energica (<320 V), similar
Max traction motor power	100 kW peak (200+ kW possible with derating)
Max vehicle speed	150 mph (240 km/h)
Vehicle kerb weight	Up to 600 lbs (272 kg)
Traction motor type	3-phase PMSM, FOC controlled
DC link voltage range	48 V – 350 V (nominal 102 V – 320 V)
DC link source	Traction battery with BMS (or compatible DC source)
Main MCU	STM32H723ZG, 550 MHz Cortex-M7, ECC RAM, FDCAN1/2, HRTIM
Gate driver	onsemi NCV57100 x6 (AEC-Q100, non-ASIL)
Isolation	>5 kV_{rms} (reinforced) per channel
Power stage	3-phase 2-level IGBT bridge
Onboard DC/DC	Cincon EC7BW-110S12 (railway grade, EN 50155)
Fail-safe default	Six-switch-open (SSO) via NCV57100 active pull-down
Gate kill paths	Six redundant SSO pathways (TIM1_BKIN, 1oo2 gate-drive power kill, shared 3.3 V rail loss, GATE_DRIVE_RESET, coprocessor NRST, coprocessor independent trigger)
FLT outputs	All six NCV57100 FLT OR'd to fault input read by both MCUs
Regenerative braking	One-pedal (tractive effort control rollback), not wheel-lock capable

2.5 Gate Drivers (Non-ASIL)

Six **onsemi NCV57100** isolated high-current IGBT gate drivers. Automotive-qualified per AEC-Q100. These devices are **not ISO 26262 safety elements** — no safety manual, FMEDA, or ASIL claim is available.

Table 5 — NCV57100 Safety-Relevant Features

Feature	Specification	Safety Role	ASIL Credit
Reinforced isolation	>5 kV_{rms}, 1200 V working	HV-to-logic barrier	Hardware only
Complementary inputs (IN+/IN−)	Internal anti-shoot-through logic	Prevents HS+LS simultaneous ON	None (not ASIL-rated)
DESAT protection	V_{TH} = 6.5 V, prog. blanking	Short-circuit detection every ON cycle	None (not ASIL-rated)
Soft turn-off	Controlled slope on DESAT	Limits di/dt overvoltage	None (not ASIL-rated)
Active Miller clamp	Internal N-FET	Prevents dV/dt turn-on	None (not ASIL-rated)
Gate active pull-down	OUT pulled low on fault/UVLO	Ensures IGBT OFF	None (not ASIL-rated)
UVLO	V_{UVLO+} = 12.2 V, V_{UVLO-} = 11.3 V	No operation at low gate drive	None (not ASIL-rated)
Negative gate drive	VEE2 to −9 V	Robust OFF-state	Hardware only
FLT output	Open-drain, active low	Fault reporting to MCU	OR'd; monitored by both MCUs

> **Non-ASIL Gate Driver Implications:** The NCV57100 internal protections provide valuable hardware-level risk reduction but cannot be counted toward ASIL metrics. The OR'd FLT output is monitored by **both** the main MCU (via TIM1_BKIN) and the coprocessor (via independent GPIO). Either MCU detecting FLT can trigger SSO. The coprocessor additionally monitors the combined NCV57100 READY output and can detect a stuck-active FLT line through its independent PWM output monitoring (all 6 phase high/low signals). This dual monitoring closes the single-point FLT path gap.

2.6 Onboard Power

Table 6 — EC7BW-110S12 Parameters

Parameter	Specification
Input voltage	40–160 VDC (4:1 range)
Output	12 V / 1.67 A (20 W)
I/O isolation	1500 VDC
Protection	OTP, OCP, OVP, UVLO
Qualification	EN 50155, EN 45545-2, IEC/EN/UL 62368-1
Temperature	−40 °C to +105 °C case

2.7 Future Considerations

> **Safety Coprocessor — STM32G474RCTx (Implemented)**
>
> The Safety Coprocessor is a **STM32G474RCTx** (170 MHz Cortex-M4+FPU, 3x FDCAN, advanced motor-control timers) that operates as an **independent safety monitor** alongside the main STM32H723ZG. It is part of the current design and provides:
>
> - **Independent gate driver supply kill** (GATE_DRIVE_PWR2_ENABLE, Path 2b) — 1oo2 with main MCU's GATE_DRIVE_PWR1_ENABLE. Independent feedback via GATE_DRIVE_PWR2_FB.
> - **Independent ADC monitoring** of all current sensors, temperature sensors, and encoder signals via voltage divider networks
> - **Challenge-response watchdog** with the main STM32 via inter-MCU UART
> - **Independent PWM output monitoring** — all 6 phase high/low signals monitored for deadtime violations, stuck-on, stuck-off
> - **Independent gate driver FLT monitoring** via OR'd fault line + combined READY signal
> - **Independent CAN bus snooping** via FDCAN2 + FDCAN3 — cross-checks torque commands and heartbeat timing
> - **Bidirectional NRST** — coprocessor can reset main MCU; main MCU can reset coprocessor
>
> The coprocessor enables **target ASIL D claims for SG-01 and SG-13 via ASIL B(D) + ASIL B(D) decomposition**. Both MCUs must independently agree that operation is safe; either can trigger SSO.

3. Operational Situations

Table 7 — Operational Situations for HARA

ID	Operational Situation	Description	Speed
OS-01	Stationary, system active	Vehicle stopped, key-on, rider present, in gear	0 mph
OS-02	Creep / low speed	Parking lot, driveway, traffic jam	0–10 mph
OS-03	Urban driving	City streets, intersections, moderate traffic	10–45 mph
OS-04	Rural / arterial road	Secondary roads, moderate curves	45–65 mph
OS-05	Highway cruising	Multi-lane highway, straight, dense traffic	65–85 mph
OS-06	High speed highway	High-speed lane, limited maneuvering space	85–150 mph
OS-07	Hard acceleration	Wide open throttle, max tractive effort request	Variable
OS-08	Regenerative braking	Tractive effort control rollback, one-pedal regen active	Variable
OS-09	Combined braking	Mechanical brake + regen blend	Variable
OS-10	Cornering / lean	Banked turn, leaned over, on tractive effort	Variable
OS-10a	**Cornering at limit / racetrack**	**High lean angle, maximum traction demand, rider relying on tractive effort to hold line**	**Variable (high)**
OS-11	Power-on sequence	Precharge, system initialization	0 mph
OS-12	Power-off / shutdown	Key-off, DC link discharge, contactor open	0 mph

ID	Operational Situation	Description	Speed
OS-13	Rain / wet road	Reduced traction, any speed	Variable
OS-14	High ambient temperature	Desert operation, >40 °C ambient	Variable
OS-15	Low traction surface	Gravel, sand, wet leaves, painted lines	Variable

> **OS-10a (Cornering at Limit):** This situation is distinct from OS-10 (general cornering) because on a racetrack or spirited canyon road, the rider is at high lean angle with the rear tire at or near its traction limit. The rider is actively using tractive effort to modulate the cornering line. A sudden complete loss of tractive effort in this situation is not merely an inconvenience — it causes the motorcycle to stand up and run wide, potentially off the track/road surface. Recovery is extremely difficult at full lean. This situation elevates the controllability rating for loss-of-power hazards from C2 to **C3**.

Environmental Conditions

Table 8 — Environmental Operating Conditions

Parameter	Range
Ambient temperature	−20 °C to +50 °C
Humidity	0% to 100% (condensing)
Road surface	Dry, wet, standing water, gravel, sand
Altitude	0–3,000 m MSL
Vibration / shock	Motorcycle-mounted, high vibration

4. Hazard Identification (HAZID)

Hazards are identified via systematic Functional Hazard Analysis (FHA) combining top-down FMEA perspective, expert judgment on power electronics and motorcycle dynamics, and ISO 26262 hazard category checklists. All hazards are stated at the **vehicle level** (harm to occupants or other road users).

Table 9 — Identified Hazards

ID	Malfunctioning Behavior	Vehicle-Level Hazard	Affected Road Users
H-01	Unintended positive tractive effort production not requested by rider	Unintended acceleration, loss of vehicle control	Rider, passengers, other road users
H-02	Unintended reverse tractive effort	Unexpected rearward motion, tip-over	Rider, nearby pedestrians
H-03	Sudden loss of tractive effort (inability to produce requested tractive effort)	Unexpected loss of drive, rear collision risk	Rider, following vehicles
H-03a	**Loss of tractive effort during cornering at lean**	**Motorcycle stands up and runs wide off road/track; loss of directional control at full lean**	**Rider**

ID	Malfunctioning Behavior	Vehicle-Level Hazard	Affected Road Users
H-04	Inability to produce requested regenerative braking tractive effort	Extended stopping distance	Rider, following vehicles
H-05	Unintended regenerative braking tractive effort (uncommanded)	Unexpected deceleration, loss of stability	Rider, following vehicles
H-06	Excessive tractive effort exceeding design limits	Wheel spin, loss of traction, component damage	Rider
H-07	Failure to limit tractive effort during over-temperature	Fire, thermal damage, burn injury	Rider, nearby persons
H-08	Motor overspeed due to loss of rotor position feedback	Loss of FOC control, unpredictable tractive effort	Rider
H-09	HV electrical isolation failure	Electric shock, potentially fatal	Rider, service personnel
H-10	DC link bus overvoltage not detected / not limited	Component rupture, arc flash, fire	Rider, nearby persons
H-11	HV contactor welding / inability to disconnect	HV always present, fire/shock risk	Rider, emergency responders
H-12	IGBT shoot-through (HS and LS simultaneously ON)	Immediate DC link short, fire, catastrophic failure	Rider
H-13	Failure to execute safe state (SSO) on detected fault	Hazardous operation continues despite fault	Rider
H-14	Corrupted or latched tractive effort command held indefinitely	Persistent unintended acceleration or braking	Rider
H-15	Incorrect tractive effort command due to software error	Non-requested tractive effort, unexpected drivability	Rider
H-16	Gate driver fault (DESAT/UVLO/TSD) not acted upon	Phase loss, imbalance, unexpected tractive effort	Rider
H-17	PWM deadtime violation or stuck-on not detected upstream	Shoot-through, DC link short, fire	Rider

H-03a: Loss of Tractive Effort During Cornering

This hazard was added after review of motorcycle dynamics. When a motorcycle is at significant lean angle, the rider uses tractive effort to maintain the cornering line. The rear tire is loaded and generating lateral force. A sudden complete loss of tractive effort removes the lateral force component that balances the turn, causing the bike to **stand up and run wide**. Unlike a 4-wheeled vehicle where loss of power merely reduces speed, on a leaned motorcycle this causes an immediate and involuntary change in trajectory. On a racetrack this means leaving the track surface; on a mountain road this means crossing the centerline or leaving the roadway. Recovery requires significant rider skill and sufficient road/runoff area, neither of which may be available. This hazard elevates the controllability of loss-of-tractive-effort events during cornering to **C3**.

5. Risk Assessment & ASIL Assignment

Severity (ISO 26262-3 Table 1)

Class	Description
S1	Light and moderate injuries
S2	Severe, life-threatening (survival probable)
S3	Life-threatening to fatal (survival uncertain)

Exposure (ISO 26262-3 Table 2)

Class	Description
E1	Very low probability (<1% of time)
E2	Low (1–10% of time)
E3	Medium (10–50% of time)
E4	High (>50% of time)

Controllability (ISO 26262-3 Table 3)

Class	Description
C1	Simply controllable
C2	Normally controllable (requires attention/skill)
C3	Difficult to control or uncontrollable

> **Motorcycle Controllability Consideration:** Motorcycles are inherently less stable than 4-wheeled vehicles. Unintended tractive effort at the rear wheel is especially difficult to control due to: (1) only two contact patches, (2) rider must manage roll/pitch/yaw simultaneously, (3) high CG relative to wheelbase, (4) single driven wheel with limited traction reserve. Unintended acceleration/braking at highway speed is therefore **C3** rather than C2. Additionally, loss of tractive effort **during cornering at lean** (OS-10a) is **C3** because the rider has no ability to recover trajectory at full lean.

ASIL Assignment

Table 10 — Hazard Risk Assessment & ASIL Assignment

Hazard	Worst OS	S	E	C	Target ASIL	Achievable Now	Rationale
H-01	OS-06 (85–150 mph)	S3	E3	C3	D	D	Unintended tractive effort at high speed causes loss of control. E3 reflects highway/track use. Achievable ASIL D via dual-MCU ASIL B(D) decomposition: coprocessor provides independent throttle ADC monitoring, independent CAN snooping, and 1oo2 power kill.

Hazard	Worst OS	S	E	C	Target ASIL	Achievable Now	Rationale
H-02	OS-01 (stationary)	S2	E2	C2	B	B	Rearward tip-over at standstill. Rider can brace.
H-03	OS-06 (highway)	S3	E3	C2	C	C	Sudden loss of tractive effort at highway speed. E3 reflects highway/track proportion of total operating time. Rear collision risk.
H-03a	OS-10a (corner at limit)	S3	E3	C3	C	C	Loss of tractive effort mid-corner causes bike to stand up and run wide. Extremely difficult to recover at full lean.
H-04	OS-09 (braking)	S2	E3	C2	A	A	Loss of regen; friction brakes remain as backup.
H-05	OS-13 (wet road)	S3	E3	C3	C	C	Unexpected deceleration on wet surface during cornering.
H-06	OS-06 (highway WOT)	S3	E3	C3	C	C	Wheel spin at speed. Dual-MCU independent current monitoring detects overcurrent within 100 ms. DESAT handles hard short-circuit (<2 us).
H-07	OS-14 (desert)	S3	E2	C2	B	B	IGBT thermal runaway. 3x redundant temp sensing.
H-08	OS-06 (downhill)	S3	E2	C3	C	C	Loss of encoder at speed. Single physical encoder is an external constraint; both MCUs independently monitor the same encoder signal for plausibility.
H-09	OS-01 (crash/ service)	S3	E1	C2	A	A	HV shock. HVIL + reinforced isolation.
H-10	OS-08 (regen downhill)	S3	E2	C2	B	B	DC link overvoltage from regen.
H-11	OS-01 (post-crash)	S3	E1	C3	B	B	HV present after crash endangers responders.
H-12	OS-07 (hard accel)	S3	E2	C3	C	C	Shoot-through. NCV57100 anti-shoot-through is HW non-ASIL. Coprocessor independently monitors all 6 PWM output pairs for deadtime violations and stuck-on/stuck-off.
H-13	OS-06 (fault at speed)	S3	E3	C3	D	D	Six redundant SSO pathways (Path 1: TIM1_BKIN; Path 2: 1oo2 power kill via GATE_DRIVE_PWR1_ENABLE / GATE_DRIVE_PWR2_ENABLE; Path 3: 3.3 V rail loss; Path 4: GATE_DRIVE_RESET; Path 5: coprocessor watchdog/NRST; Path 6: coprocessor independent fault trigger).
H-14	OS-06 (highway)	S3	E3	C3	C	C	Stuck tractive effort command. WDT catches CPU halt.
H-15	OS-06 (highway)	S3	E3	C3	D	D	Software error producing incorrect tractive effort. Coprocessor provides independent verification: cross-checks commanded torque vs. measured currents, monitors CAN torque commands, and can trigger independent SSO via GATE_DRIVE_PWR2_ENABLE.

Hazard	Worst OS	S	E	C	Target ASIL	Achievable Now	Rationale
H-16	OS-07 (hard accel)	S3	E2	C3	C	C	Gate driver fault not acted upon. Both MCUs independently monitor the OR'd FLT line. Coprocessor additionally monitors the combined READY signal and PWM outputs to detect stuck-active FLT.
H-17	OS-07 (hard accel)	S3	E3	C3	C	C	PWM fault upstream of gate driver. Coprocessor independently monitors all 6 PWM output pairs (PH_U/V/W_HIGH/LOW) for deadtime violations, stuck-on, and stuck-off.

6. Safety Goals

Table 11 — Safety Goals

SG	Safety Goal	Target	Achievable	Hazards	Safe State
SG-01	Prevent unintended positive tractive effort when rider not requesting propulsion. Unintended tractive effort ≤10 Nm for ≤200 ms before safe state.	D	D	H-01, H-15	SSO, zero tractive effort
SG-02	Prevent unintended reverse tractive effort. Reverse rejected → safe state.	B	B	H-02	SSO, zero tractive effort
SG-03	Safe degradation on loss of tractive effort. Tractive effort cut-off rate-limited to ≤200 Nm/s.	C	C	H-03, H-03a	Zero tractive effort (gradual)
SG-04	Ensure regenerative braking availability. Friction brakes remain on loss.	A	A	H-04	Zero regen (gradual)
SG-05	Prevent unintended regenerative braking. Uncommanded >10 Nm → safe state within 200 ms.	C	C	H-05	SSO, zero tractive effort
SG-06	Limit max tractive effort to calibrated max. Over-torque >110% → safe state within 100 ms.	C	C	H-06	SSO, zero tractive effort
SG-07	Detect over-temperature, progressively derate. Critical temp → safe state.	B	B	H-07	SSO, monitoring
SG-08	Detect loss of rotor position → safe state within 100 ms.	C	C	H-08	SSO, zero tractive effort
SG-09	Maintain HV isolation (>500 Ohm/V). Isolation fault → safe state.	A	A	H-09	SSO, contactor open
SG-10	Detect DC link bus overvoltage, limit regen. Critical OV → safe state.	B	B	H-10	SSO, regen disable
SG-11	Detect HVIL interruption, request contactor open within 100 ms.	B	B	H-11	SSO, HV disconnect
SG-12	Prevent IGBT shoot-through. Risk → PWM disable <10 us via hardware.	C	C	H-12	SSO, HW PWM kill
SG-13	Achieve safe state (SSO) within 200 ms of any fault. Safe state entry independent of main control loop.	D	D	H-13, H-14	SSO, HW-enforced

SG	Safety Goal	Target	Achievable	Hazards	Safe State
SG-14	Detect any NCV57100 fault (DESAT, UVLO, TSD) and transition to safe state.	C	C	H-16	SSO, FLT detect
SG-15	Detect PWM deadtime violations and stuck-on faults. Deadtime collapse or stuck-high >100 us → safe state.	C	C	H-17	SSO, PWM kill

7. Functional Safety Requirements

Table 12 — Functional Safety Requirements (FSRs)

FSR	Requirement	Tgt	Now	SG
FSR-01	Read dual redundant tractive effort control pots on both MCUs. Discrepancy >5% or either OOR detected by either MCU → safe state.	D	D	SG-01
FSR-02	Tractive effort command plausibility: commanded current reference vs. measured phase currents. Deviation >20% for >100 ms → safe state.	D	C	SG-01, SG-06
FSR-03	Tractive effort control input rate-limited. Max tractive effort rate: 500 Nm/s (calibration parameter).	D	C	SG-01
FSR-04	Reject reverse tractive effort when speed >0. Reverse only when stationary AND explicitly selected.	B	B	SG-02
FSR-05	On fault, ramp down tractive effort at ≤200 Nm/s before SSO. Abrupt removal prohibited unless <50 ms safety timing requires faster response.	C	C	SG-03
FSR-06	Monitor regenerative braking tractive effort continuously. Uncommanded >10 Nm for >200 ms → safe state.	C	C	SG-05
FSR-07	Max tractive effort limit: software LUT on both MCUs with cross-check. Dual-MCU independent current monitoring detects overcurrent within 100 ms. DESAT handles hard short-circuit (<2 us).	C	C	SG-06
FSR-08	One NTC temp sensor per IGBT module (3 total), 2-out-of-3 voting with 100 °C hard cap. Critical threshold in ECC memory.	B	B	SG-07
FSR-09	Encoder loss → safe state within 100 ms. Sensorless fallback if implemented: ≤2 s, ≤30% tractive effort.	C	C	SG-08
FSR-10	HVIL monitored continuously. Interruption → immediate PWM disable + BMS contactor open request within 50 ms.	B	B	SG-09, SG-11
FSR-11	DC link bus voltage monitored with isolated ADC. OV threshold → immediate regen disable. Critical OV → SSO within 50 ms.	B	B	SG-10
FSR-12	NCV57100 complementary inputs (IN+/IN−) prevent HS+LS simultaneous conduction. Power-on self-test confirms.	C	A	SG-12
FSR-13	NCV57100 DESAT detection active on all six IGBTs. DESAT event → local PWM disable within <2 us, independent of MCU.	C	A	SG-12, SG-14
FSR-14	STM32 breakpoint input → HW PWM disable within <10 us on any critical fault. Independent of main CPU execution.	D	C	SG-13
FSR-15	Independent windowed watchdog. Failure to service → automatic PWM disable + system reset. Timeout ≤50 ms.	D	C	SG-13, SG-01

FSR	Requirement	Tgt	Now	SG
FSR-16	Power-on self-test (POST): ADC references, current sensor offsets, gate driver communications, encoder signal, HVIL continuity, watchdog, NCV57100 DESAT self-test. All must pass before PWM enable.	D	C	SG-13
FSR-17	IO board CAN heartbeat 1 s timeout. Loss → safe-state defaults (brake pressed, kickstand down) and tractive effort restricted to zero.	C	C	SG-01
FSR-18	Tractive effort control limit switch monitored independently of analog pots. Activation overrides any analog value and commands zero tractive effort.	D	C	SG-01
FSR-19	Boot-time CRC-32 over safety-critical code and calibration data. Mismatch → PWM enable prevented, fault logged.	D	C	SG-01, SG-13
FSR-20	ECC RAM for all safety-critical variables. Single-bit errors corrected (SECDED). Double-bit errors → safe state.	D	C	SG-15
FSR-21	Monitor DC link bus undervoltage. UV → tractive effort derate, critical UV → safe state. Prevents overcurrent due to insufficient DC link voltage.	B	B	SG-03, SG-10

8. Current Design Coverage

Table 13 — Honest Assessment of Design vs. FSRs

FSR	Tgt	Now	Status	Existing / Planned	Gap
FSR-01	D	D	Covered	Dual pots on both MCUs; independent voters	Achieved via dual-MCU ASIL B(D) decomposition
FSR-02	D	D	Planned	Commanded vs. measured cross-check on both MCUs	Not yet implemented in firmware (Phase 2)
FSR-03	D	C	Covered	Rate limiter implemented	Part of SG-01 decomposition; main MCU handles rate limiting
FSR-04	B	B	Planned	Reverse interlock: speed >100 rpm → reverse clamped	Not yet implemented in firmware (GAP-SW-02)
FSR-05	C	C	Planned	Direct SSO on fault today	Ramp-down ≤200 Nm/s to be implemented (GAP-HW-02)
FSR-06	C	C	Planned	Uncommanded regen monitor on both MCUs	Not yet implemented in firmware (Phase 2)
FSR-07	C	C	Covered	Dual-MCU independent current monitoring	100 ms detection sufficient; DESAT for hard shorts
FSR-08	B	B	Planned	Three redundant IGBT temp sensors (hardware)	2oo3 voter software not yet implemented (Phase 2)
FSR-09	C	C	Limited	Single encoder (external constraint)	Immediate SSO on loss; bounded sensorless if used
FSR-10	B	B	Covered	HVIL circuit implemented	Verify ≤50 ms E2E
FSR-11	B	B	Covered	DC link isolated ADC	Define OV thresholds
FSR-12	C	A	Covered	NCV57100 complementary inputs	Non-ASIL; no credit claimed
FSR-13	C	A	Covered	NCV57100 DESAT on all six	Non-ASIL; no credit claimed

FSR	Tgt	Now	Status	Existing / Planned	Gap
FSR-14	D	D	Covered	Breakpoint input to HW PWM disable + 1oo2 gate drive power kill	Six redundant SSO pathways; either MCU achieves SSO independently
FSR-15	D	C	Covered	Independent watchdog on STM32	On-chip WDT subject to common cause
FSR-16	D	C	Partial	NCV57100 DESAT self-test exists	Expand to all safety functions
FSR-17	C	C	Planned	1-second heartbeat timeout confirmed	Safe defaults on CAN loss
FSR-18	D	C	Covered	Limit switch input implemented	Verify override precedence
FSR-19	D	C	To implement	STM32 flash CRC peripheral available	Add boot-time CRC check
FSR-20	D	C	Covered	STM32H723 has ECC RAM	Enable + DED handler to safe state
FSR-21	B	B	Planned	DC link ADC can measure UV	Define UV thresholds and response

9. Gap Analysis

9.1 Architecture Gaps

GAP-ARCH-01: Single-Core MCU Without Independent Monitor (P0)

Issue: All safety mechanisms execute on one STM32H723. A single MCU fault (clock failure, latch-up, supply collapse) can disable all safety functions simultaneously. The independent watchdog is on-chip and subject to common-cause failure.

Impact: SG-01 and SG-13 target ASIL D, achievable via dual-MCU ASIL B(D) + ASIL B(D) decomposition (Table 11). The remaining gap is the formal ASIL D claim, pending the Dependent Failure Analysis (LIMIT-08).

Mitigation path: Dual-MCU ASIL B(D) decomposition — implemented. Main MCU + coprocessor each achieve ASIL B(D); combined via 1oo2 voter on safe state actuation.

GAP-ARCH-02: Power Kill Without Feedback Monitoring (P1, was P0)

Issue: The GATE_DRIVE_PWR1_ENABLE (main MCU) and GATE_DRIVE_PWR2_ENABLE (coprocessor) paths provide a 1oo2 active SSO mechanism. **Feedback:** GATE_DRIVE_PWR1_FB and GATE_DRIVE_PWR2_FB provide independent per-supply status. When the NCV57100 detects VDD_UVLO (loss of gate drive supply), it asserts FLT (active low), which both MCUs can read. The 1oo2 architecture means a stuck-high GATE_DRIVE_PWR1_ENABLE is not a single-point failure — the coprocessor can still achieve SSO via GATE_DRIVE_PWR2_ENABLE (Path 2b), GATE_DRIVE_RESET (Path 4), or NRST (Path 5). The shared 3.3V rail provides a passive SSO path (NCV57100 pull-down on VDD loss, Path 3).

Impact: SG-13 targets ASIL D via ASIL B(D) decomposition. Six SSO pathways exist (see Section 2.2). The 1oo2 power kill with independent feedback provides diagnostic coverage for the supply kill path.

Mitigation path: Implemented — GATE_DRIVE_PWR1_FB (main MCU) and GATE_DRIVE_PWR2_FB (coprocessor) provide independent per-supply feedback. Six SSO pathways (TIM1_BKIN; 1oo2 power kill via GATE_DRIVE_PWR1_ENABLE / GATE_DRIVE_PWR2_ENABLE; 3.3 V rail loss;

GATE_DRIVE_RESET; coprocessor watchdog/NRST; coprocessor independent fault trigger) provide extensive redundancy. Verify feedback paths in C-17, C-26, C-27, S-10, and S-11.

GAP-ARCH-03: Non-ASIL Gate Driver Protections (P1)

Issue: NCV57100 internal protections not ASIL-rated. OR'd FLT output is a single shared wire (single point, not diagnosed), though it is now read by both MCUs.

Impact: SG-12, SG-14, SG-15 limited. ASIL credit requires external monitoring.

Mitigation path: Coprocessor FLT/READY monitoring and PWM integrity check.

9.2 Component Gaps

GAP-HW-01: Hardware Overcurrent Comparator — CLOSED (Not Required)

The schematic includes LM397 comparators for phase and DC link overcurrent detection, but these are **not required for safe operation**. Hard short-circuit protection is handled by NCV57100 DESAT (<2 us). Regular overcurrent (non-DESAT) is detected within **100 ms** by the dual-MCU integrated monitoring: both the main STM32H723 and the coprocessor STM32G474 independently sample all four current sense channels at high rate. Either MCU detecting overcurrent triggers SSO via its independent gate drive power kill path. This 100 ms detection time is faster than the thermal time constant of the IGBT module and therefore cannot cause permanent damage. The LM397 comparators are retained on the PCB as a redundant monitoring layer but are not relied upon for the safety case. **Closed — integrated dual-MCU monitoring is sufficient.**

GAP-HW-02: No Tractive Effort Ramp-Down (P1)

Direct SSO on fault. Implement software ramp at ≤200 Nm/s before SSO.

GAP-SW-01: No Boot CRC (P1)

FSR-19: Boot CRC verification is not yet implemented. Add a boot-time CRC-32 using the STM32 CRC peripheral to validate safety-critical code and calibration data before PWM enable.

GAP-SW-02: No Reverse Interlock (P1)

Software interlock: speed >100 rpm forward → negative tractive effort clamped to zero.

GAP-SW-03: Sensorless Fallback Policy Undefined (P1)

Define: immediate SSO on encoder loss, or bounded sensorless (≤2 s, ≤30% tractive effort).

9.2 Gap Summary

Table 14 — Gap Mitigation Priority

Gap	Priority	Mitigation	Effort
Dual MCU with coprocessor	**RESOLVED**	STM32G474RCTx implemented — 1oo2 power kill, independent ADC, challenge/response watchdog, independent CAN snoop	Complete

Gap	Priority	Mitigation	Effort
HW overcurrent comparator	**CLOSED**	Dual-MCU integrated monitoring detects overcurrent within 100 ms — sufficient for SSO without damage. LM397 retained as non-safety redundant layer.	N/A
Power kill feedback monitoring	**P1**	GATE_DRIVE_PWR1_FB and GATE_DRIVE_PWR2_FB provide independent per-supply feedback. Verify in S-16 through S-19.	Low
Non-ASIL gate driver credit	**P1**	Coprocessor monitors FLT, READY, all 6 PWM outputs. Verify cross-check logic in C-14, S-16.	Medium
Tractive effort ramp-down	**P1**	Software	Medium
Boot CRC	**P1**	STM32 CRC peripheral	Low
Reverse interlock	**P1**	Software	Low
Sensorless policy	**P1**	Software + documentation	Low
No DFA (ISO 26262-9)	**P0**	Dependent Failure Analysis	Medium
No EMI/EMC pre-compliance	**P1**	CISPR 25 pre-compliance test	Medium

Note: With the dual-MCU architecture (STM32H723 + STM32G474 coprocessor), ASIL D is achievable for SG-01 and SG-13 via ASIL B(D) decomposition. The four hazards previously limited to ASIL A (H-06, H-16, H-17, and H-13 safe state failure) are now fully covered: H-06 by dual-MCU independent current monitoring (100 ms detection + independent power kill), H-16 by coprocessor FLT/READY/PWM monitoring, H-17 by coprocessor independent deadtime monitoring, and H-13 by six redundant SSO pathways. GAP-HW-01 (HW OCP comparator) is closed — dual-MCU integrated monitoring is sufficient. No hazards remain below their target ASIL.

10. Fault Injection Test Plan

10.1 Test Philosophy

The purpose of this Fault Injection Test Plan is to provide a comprehensive, traceable methodology for verifying that the safety mechanisms implemented in the traction inverter/VCU detect faults and transition to the defined safe state (SSO) within the required time budgets. The test plan defines **99 tests** organized into four categories:

- **Component-Level Tests (C-01 to C-50):** Individual hardware component validation — inject faults into a single sensor, input, or protection circuit and verify detection and safe state entry. Includes supply faults (brownout, rail shorts), phase faults (open, short-to-short, short-to-DC), gate driver validation, clock failures, GPIO stuck-at, ADC drift, SPI, CAN bus off, watchdog starvation, thermal runaway, pre-charge, Miller clamp, propagation delay, deadtime, isolation.
- **System-Level Tests (S-01 to S-19):** Full-system fault scenarios — simulate real-world fault conditions with the complete hardware and software running closed-loop control and verify end-to-end response. Includes thermal camera survey, full-load regen, startup/shutdown sequencing, and power dip ride-through.
- **Integration-Level Tests (I-01 to I-18):** External interface and communication fault scenarios — simulate failures in connected systems (BMS, IO board, CAN bus) and verify safe degradation. Includes CAN fuzzing, bus load testing, multi-node faults, and wiring faults.

- **Environmental Tests (E-01 to E-12):** Environmental and stress validation — vibration, thermal shock, humidity, EMI immunity, ESD, and water ingress. These are type tests for production qualification.

Each test case is designed with the following principles:

1. **Every Safety Goal (SG-01 through SG-15) is covered by at least one test case.**
2. **Every Functional Safety Requirement (FSR-01 through FSR-21) is covered by at least one test case.**
3. **Every identified hazard (H-01 through H-17, including H-03a) is covered by at least one test case.**
4. **Response time requirements are verified** where measurable (e.g., <10 us HW PWM disable, ≤50 ms WDT timeout, ≤200 ms safe state entry).
5. **Fault injection is realistic** — faults represent credible failure modes observed in traction power electronics systems (stuck sensor, shorted wiring, open connection, corrupted data, supply anomaly, etc.).
6. **Tests are independently executable** where possible to allow incremental validation as software matures.

10.2 Test Environment

Table 15 — Test Equipment and Setup

Item	Description	Purpose
Test article	Traction inverter/VCU PCB with STM32H723, six NCV57100, IGBT power stage, all sensors	Device under test (DUT)
DC link source	Programmable DC power supply 0–400 V, ≥50 A or traction battery with contactor control	DC link for low-power testing; traction battery for high-power
Test motor	Low-voltage PMSM on dyno or equivalent inertia load	Closed-loop traction motor control testing
Dyno	Motor dynamometer with torque/speed measurement and programmable load	Full-load characterization and fault testing under load
Fault injection hardware	Switchable resistor networks, relay cards, programmable loads, short-circuit switches	Physically inject faults into sensors, wiring, gate signals
Oscilloscope	4-channel, ≥100 MHz, with current probes and HV differential probes	Timing verification of PWM disable, DESAT response, gate signals
CAN interface	PCAN or equivalent USB-CAN adapter	Monitor CAN traffic, simulate BMS/IO board messages
RTE (Real Time Examiner)	Host tool connected via CAN or debug interface	Monitor internal variables, tractive effort commands, fault status
Thermal chamber	Environmental chamber −20 °C to +60 °C	Overtemperature derating and thermal fault testing
Isolation tester	Megohmmeter / insulation resistance tester	HV isolation measurement
Thermal camera	Infrared camera, 320x240 minimum, <50 mK NETD	Full-load thermal survey (S-06), hotspot identification
Vibration table	Random/sinusoidal vibration table, 5–2000 Hz, ≥10 g	Environmental vibration testing (E-01 to E-03)
EMC chamber	ALSE or anechoic chamber with RF amplifiers	Radiated EMI immunity (E-08), conducted BCI (E-09)
ESD gun	ESD simulator per ISO 10605	Contact and air discharge testing (E-10, E-11)
Humidity chamber	Environmental chamber with humidity control	High humidity operation (E-07)

Item	Description	Purpose
Water spray rig	IPX4/IPX5/IPX6 spray nozzles per IEC 60529	Water ingress validation (E-12)
Common-mode probe	High-bandwidth differential probe, ≥100 MHz	Bearing voltage / common-mode measurement (C-36)
Shaft voltage probe	Carbon brush or capacitive coupling to motor shaft	Bearing voltage ratio measurement (C-36)

Safety During Testing

All high-power fault injection tests (tests involving >48 V DC link or >10 A phase current) must be conducted with:

- Personnel trained in HV electrical safety (NFPA 70E or equivalent)
- Appropriate PPE (HV-rated gloves, face shield, non-conductive tools)
- Emergency stop button within arm's reach of all test stations
- Fire extinguisher (Class C) present
- Test area marked and restricted to authorized personnel only
- Pre-charge/discharge protocol followed for all DC link connections

10.3 Component-Level Tests

C-01: Tractive Effort Control Potentiometer 1 Open Circuit

Objective: Verify FSR-01 — Dual tractive effort control plausibility detects open circuit on primary potentiometer.

Covered: SG-01 (ASIL D), FSR-01, H-01

Procedure:

1. Connect tractive effort control assembly with both pots functional. System at key-on, traction motor not running.
2. Apply 50% tractive effort request via both pots (matched within 2%).
3. While maintaining input, open-circuit potentiometer 1 signal wire (disconnect from PCB).
4. Observe system response via RTE and oscilloscope on PWM outputs.
5. Repeat with traction motor spinning at 50% rated speed under dyno load.

Pass Criteria: System detects pot 1 out-of-range/discrepancy >5% within <100 ms and transitions to safe state (SSO). No tractive effort produced after fault. Fault logged with correct code.

Fail Criteria: Tractive effort continues after fault, fault detection >200 ms, or incorrect fault code logged.

Why this covers the requirement: An open circuit on one potentiometer is a credible wiring failure. The dual-pot plausibility check (FSR-01) must detect the discrepancy and enter safe state before any unintended tractive effort can be commanded. This directly addresses H-01 (unintended tractive effort) by ensuring a single sensor fault cannot cause an incorrect tractive effort request.

C-02: Tractive Effort Control Potentiometer 2 Shorted to +5 V

Objective: Verify FSR-01 — Dual tractive effort control plausibility detects short-to-rail on secondary potentiometer.

Covered: SG-01 (ASIL D), FSR-01, H-01

Procedure:

1. Connect tractive effort control with both pots functional. System at key-on.
2. Apply 25% tractive effort request via pot 1, short pot 2 to +5 V reference (simulating wiring short).
3. Observe response via RTE and scope.
4. Repeat with traction motor running at 25% speed under load.

Pass Criteria: Discrepancy >5% detected within <100 ms. Safe state entered. Fault logged.

Why this covers the requirement: Short-to-rail is a common wiring fault (chafed harness). Pot 2 at 100% while pot 1 at 25% creates a 75% discrepancy that must be caught. This is a distinct failure mode from C-01 (open circuit) and verifies the plausibility check works for both high and low anomalies.

C-03: Tractive Effort Control Potentiometer 1 Shorted to Ground

Objective: Verify FSR-01 — Short-to-ground detection on primary pot.

Covered: SG-01 (ASIL D), FSR-01, H-01

Procedure: Short pot 1 signal to ground while pot 2 at 50%. Observe response. Repeat under motor load.

Pass Criteria: Discrepancy detected <100 ms. Safe state. Fault logged.

Why this covers the requirement: Ground short is distinct from open and +5 V short (different ADC reading). Verifies the plausibility check is robust to all three common wiring fault modes.

C-04: Tractive Effort Control Potentiometer Drift (Gradual Divergence)

Objective: Verify FSR-01 detects gradual potentiometer mismatch (simulating sensor degradation).

Covered: SG-01 (ASIL D), FSR-01, H-01

Procedure: Use programmable resistors to gradually increase offset between pot 1 and pot 2 from 0% to 10% over 10 seconds while traction motor running at constant speed. Observe when fault triggers.

Pass Criteria: Fault triggers when discrepancy exceeds 5% threshold. No false trips below threshold.

Why this covers the requirement: Gradual sensor drift (wear, temperature aging) must be distinguished from normal variation. Verifies threshold is calibrated correctly and not overly sensitive (nuisance trips) or insensitive (missed faults).

C-05: Tractive Effort Control Limit Switch Activation at Speed

Objective: Verify FSR-18 — Limit switch independently commands zero tractive effort, overriding analog values.

Covered: SG-01 (ASIL D), FSR-18, H-01

Procedure:

1. Traction motor running at 50% rated speed under dyno load with 50% tractive effort request applied.
2. Activate tractive effort control limit switch (mechanical end-stop) while maintaining potentiometer position.
3. Observe tractive effort command and PWM output via RTE and scope.

Pass Criteria: Tractive effort command drops to zero within <50 ms of limit switch activation, regardless of potentiometer position. PWM disabled or zero duty. Fault logged.

Why this covers the requirement: The limit switch is an independent hardware path to zero tractive effort. Even if both potentiometers fail or software miscalculates, the mechanical switch must unconditionally override. This is a last-resort protection against a stuck control cable or sensor malfunction.

C-06: Phase Current Sensor Offset Drift

Objective: Verify FSR-02 — Tractive effort command plausibility detects current sensor offset error.

Covered: SG-01 (ASIL D), SG-06 (ASIL C), FSR-02, H-01, H-06

Procedure:

1. Traction motor running at 50% rated tractive effort under dyno load.
2. Inject DC offset (+20% of rated current) into one phase current sensor via external bias circuit.
3. Observe commanded tractive effort vs. measured current. Note when plausibility check triggers.
4. Repeat with −20% offset.

Pass Criteria: Plausibility deviation >20% detected within <100 ms. Safe state entered.

Why this covers the requirement: Current sensor offset drift (hall sensor temperature drift, ADC reference drift) causes the MCU to misread actual current. FSR-02 requires comparing commanded current reference against measured phase currents. This test verifies the end-to-end plausibility chain catches sensor errors that would lead to incorrect tractive effort.

C-07: DC Link Current Sensor Open Circuit

Objective: Verify FSR-02 — Loss of DC link current sensor detection.

Covered: SG-01 (ASIL D), FSR-02, FSR-07, H-01, H-06

Procedure: Open-circuit DC link current sensor signal wire while traction motor running at 25% load. Observe plausibility check and safe state entry.

Pass Criteria: Sensor out-of-range detected. If sensor has reference signal: reference mismatch detected. Safe state within <200 ms.

Why this covers the requirement: The DC link sensor provides the sum current check against phase currents. Its loss removes a plausibility layer. This verifies the system handles the failure gracefully.

C-08: IGBT Overtemperature (Simulated)

Objective: Verify FSR-08 — 2-out-of-3 temperature voting and progressive derating.

Covered: SG-07 (ASIL B), FSR-08, H-07

Procedure:

1. Use programmable resistor or heat gun to elevate one IGBT temp sensor reading above warning threshold (e.g., 100 °C) while other two sensors remain normal.
2. Observe tractive effort derating behavior via RTE.
3. Elevate second sensor to warning threshold. Verify further derating.
4. Elevate all three sensors above critical threshold (e.g., 130 °C). Verify safe state entry.
5. Test 1-of-3 failure: one sensor stuck at 150 °C (implausible) while others normal. Verify 2oo3 voter rejects outlier and continues operation.

Pass Criteria: Progressive derate at warning thresholds. Safe state at critical threshold. 2oo3 voter correctly rejects single implausible sensor. Response times: derate <500 ms, safe state <1 s from critical threshold crossing.

Why this covers the requirement: Thermal runaway of IGBTs is a credible failure mode under high load or blocked cooling. FSR-08 requires 2-out-of-3 voting to tolerate one sensor failure. This test verifies both the temperature response curve and the voting logic.

C-09: Traction Motor Encoder Signal Loss (Quadrature)

Objective: Verify FSR-09 — Encoder loss detection and safe state entry.

Covered: SG-08 (ASIL C), FSR-09, H-08

Procedure:

1. Traction motor spinning at 50% rated speed under dyno load (closed-loop FOC).
2. Disconnect encoder A or B channel (open circuit).
3. Observe FOC behavior and safe state entry via RTE and scope.
4. Repeat with encoder supply disconnected (both channels lost).
5. If sensorless fallback implemented: verify bounded operation (≤2 s, ≤30% tractive effort) before SSO.

Pass Criteria: Encoder loss detected within <100 ms. Safe state entered. If sensorless fallback: bounded to ≤2 s and ≤30% tractive effort, then SSO. Fault logged with correct code.

Why this covers the requirement: Loss of position feedback at speed causes FOC to lose synchronization, producing unpredictable tractive effort. This is H-08. FSR-09 requires immediate safe state because the single encoder has no redundancy. The test verifies the timeout and response.

C-10: DC Link Bus Overvoltage (Simulated)

Objective: Verify FSR-11 — DC link bus overvoltage detection and regen disable.

Covered: SG-10 (ASIL B), FSR-11, H-10

Procedure:

1. System running with DC link at nominal voltage. Traction motor spinning with regen active.
2. Gradually increase DC link bus voltage via programmable supply to overvoltage warning threshold.
3. Verify regen tractive effort is disabled / reduced.
4. Continue increasing to critical overvoltage threshold.
5. Verify safe state entry (SSO) within <50 ms of critical threshold crossing.

Pass Criteria: Regen disabled at warning threshold. SSO at critical threshold within <50 ms. No overvoltage damage to DC link capacitors or semiconductors.

Why this covers the requirement: Overvoltage occurs during hard regen on long downhills or when BMS cell balancing fails. FSR-11 requires two thresholds: a warning that disables regen (preventing voltage rise) and a critical that forces SSO (preventing capacitor rupture). Both must be verified.

C-11: DC Link Bus Undervoltage (Simulated)

Objective: Verify FSR-21 — DC link bus undervoltage detection and response.

Covered: SG-03 (ASIL C), SG-10 (ASIL B), FSR-21, H-03, H-03a

Procedure: Gradually reduce DC link bus voltage while traction motor running at 50% load. Observe tractive effort derating and safe state entry at critical UV threshold.

Pass Criteria: Tractive effort derated progressively as bus voltage drops. Safe state at critical UV. No overcurrent event due to insufficient DC link voltage.

Why this covers the requirement: Insufficient DC link voltage causes the current controller to saturate, potentially commanding excessive duty cycle that leads to overcurrent when voltage recovers. FSR-21 prevents this by derating and entering safe state before saturation occurs.

C-12: HVIL Interruption

Objective: Verify FSR-10 — HVIL loop interruption detection and response.

Covered: SG-09 (ASIL A), SG-11 (ASIL B), FSR-10, H-09, H-11

Procedure:

1. System at key-on with HVIL loop intact.
2. Open HVIL loop (disconnect HVIL connector or cut loop wire).
3. Measure time from HVIL open to PWM disable and contactor open request on CAN1.

Pass Criteria: PWM disabled within <50 ms. Contactor open request transmitted on CAN1 within <100 ms. Fault logged.

Why this covers the requirement: HVIL interruption indicates a connector disconnection or safety interlock trigger (e.g., crash sensor). FSR-10 requires immediate PWM disable and BMS notification. Both the local response (PWM) and external notification (CAN) must be verified.

C-13: Watchdog Timeout (STM32 Failure to Service)

Objective: Verify FSR-15 — Independent watchdog detects MCU runaway and forces reset.

Covered: SG-13 (ASIL D), SG-01 (ASIL D), FSR-15, H-13, H-14

Procedure:

1. Traction motor running at 50% load. System operating normally.
2. Via debug interface, halt CPU execution (simulating code runaway or infinite loop).
3. Measure time from CPU halt to PWM disable and system reset.
4. Alternatively, use a firmware build that intentionally stops servicing the watchdog after a trigger condition.

Pass Criteria: PWM disabled and system reset occurs within watchdog timeout period (≤50 ms). No tractive effort produced during or after reset until POST completes and key cycle occurs.

Why this covers the requirement: A CPU runaway (e.g., pointer corruption causing infinite loop, stack overflow) can leave PWM running at a constant duty cycle, producing H-14 (latched tractive effort). FSR-15 requires the watchdog to catch this and force safe state. The halt-via-debug method is a realistic fault injection technique for this failure mode.

C-14: STM32 Breakpoint Input (HW PWM Disable)

Objective: Verify FSR-14 — Hardware breakpoint input disables all PWM within <10 us, independent of software.

Covered: SG-13 (ASIL D), FSR-14, H-13

Procedure:

1. Traction motor running at 50% load with active PWM.
2. Assert breakpoint input pin (pull to active level via external switch).
3. Measure time from breakpoint assertion to all six PWM outputs going low, using oscilloscope.
4. Verify PWM remains disabled while breakpoint held.
5. De-assert breakpoint and verify PWM does NOT resume without system reset and POST.

Pass Criteria: All PWM outputs disabled within <10 us of breakpoint assertion. Independent of CPU state (test with CPU halted via debugger to prove hardware-only path). PWM does not auto-resume.

Why this covers the requirement: The breakpoint input is the fastest and most reliable safe-state path. It must work even if the CPU is completely non-functional. Measuring with the CPU halted proves the path is truly hardware-independent. The <10 us requirement addresses shoot-through and other time-critical faults.

C-15: Gate Driver DESAT (Simulated Short Circuit)

Objective: Verify FSR-13 — NCV57100 DESAT protection detects and responds to simulated short circuit.

Covered: SG-12 (ASIL C), SG-14 (ASIL C), FSR-13, H-12, H-16

Procedure:

1. System at key-on with DC link present. Gate drivers active.
2. Use a low-inductance pulse generator to inject a brief overcurrent pulse into one phase, or use a DESAT test fixture that momentarily pulls the DESAT pin above threshold.
3. Measure FLT output response time and PWM disable via scope.
4. Verify soft turn-off slope (not hard shutoff that causes voltage spike).
5. Verify FLT is latched and readable by STM32.

Pass Criteria: DESAT detected and PWM disabled within <2 us. Soft turn-off observed. FLT asserted and latched. System enters safe state.

Why this covers the requirement: DESAT is the primary short-circuit protection in the gate driver. While the NCV57100 is non-ASIL, verifying its response time and behavior is essential for the overall safety argument. This test also forms the basis for the DESAT self-test (C-16).

C-16: Gate Driver DESAT Self-Test at Power-On

Objective: Verify FSR-16 — DESAT self-test confirms protection circuit functional before PWM enable.

Covered: SG-12 (ASIL C), SG-14 (ASIL C), FSR-16, H-12, H-16

Procedure: Power cycle the system. Observe the DESAT self-test sequence (brief test pulse to verify DESAT detection circuit responds). Verify PWM is not enabled until all six gate drivers pass self-test. Introduce a fault in one DESAT circuit (e.g., open DESAT diode) and verify system refuses to enable PWM.

Pass Criteria: Self-test completes on all six channels. PWM enable gated by self-test pass. With DESAT fault injected: system remains in safe state, fault logged, PWM never enabled.

Why this covers the requirement: A failed DESAT circuit (open DESAT diode, broken connection) is a latent fault that would prevent short-circuit detection. The power-on self-test must catch this before the system can operate. This is essential for latent fault coverage.

C-17: Gate Driver UVLO (Simulated Low Supply)

Objective: Verify gate driver UVLO prevents operation at insufficient gate drive voltage.

Covered: SG-12 (ASIL C), FSR-12, H-12

Procedure: Gradually reduce the +15 V gate driver supply voltage while system is operating. Observe the point at which UVLO triggers and PWM is disabled.

Pass Criteria: UVLO triggers at $V_{UVLO-} \approx 11.3$ V. All PWM disabled. FLT asserted. Safe state entered.

Why this covers the requirement: Insufficient gate drive voltage causes the IGBT to operate in the linear region, leading to excessive conduction loss and thermal destruction. UVLO prevents this by forcing gate OFF when supply is inadequate.

C-18: ECC RAM Single-Bit Error Injection

Objective: Verify FSR-20 — ECC RAM corrects single-bit errors and allows continued operation.

Covered: SG-13 (ASIL D), SG-15 (ASIL C), FSR-20

Procedure: Use STM32's built-in ECC error injection capability (if available) or debug interface to flip a single bit in a safety-critical RAM location (e.g., tractive effort limit variable). Observe system behavior via RTE.

Pass Criteria: Single-bit error detected and corrected. System continues operating. Error logged. No unsafe state entered.

Why this covers the requirement: Cosmic radiation and EMI can corrupt RAM. SECDED ECC must handle single-bit errors transparently. This verifies the error correction path works without causing false trips.

C-19: ECC RAM Double-Bit Error Injection

Objective: Verify FSR-20 — Double-bit errors trigger safe state (uncorrectable).

Covered: SG-13 (ASIL D), FSR-20

Procedure: Flip two bits in a safety-critical RAM location. Observe system response.

Pass Criteria: Double-bit error detected. Safe state entered immediately. Fatal error logged. System requires reset.

Why this covers the requirement: Double-bit errors are uncorrectable. The system must treat them as critical faults and enter safe state, because the corrupted data could be a safety-critical variable (tractive effort limit, fault threshold, etc.).

C-20: Boot CRC Mismatch

Objective: Verify FSR-19 — Boot CRC prevents operation with corrupted firmware.

Covered: SG-01 (ASIL D), SG-13 (ASIL D), FSR-19, H-15

Procedure: Deliberately corrupt one byte in the safety-critical code flash region (using debugger or flash programming tool). Power cycle and observe boot behavior.

Pass Criteria: CRC-32 mismatch detected at boot. PWM enable prevented. Fault logged. System remains in safe state.

Why this covers the requirement: Corrupted firmware (flash wear, programming error, EMI during write) could modify the tractive effort mapping, safety thresholds, or fault handling logic. FSR-19 prevents the system from operating with untrusted code. This directly addresses H-15 (incorrect tractive effort from software error).

C-21: STM32 Supply Brownout — Gradual Vdd Drop

Objective: Verify brownout detection (BOR) triggers safe state before MCU operation becomes unreliable.

Covered: SG-13 (ASIL D), FSR-14, FSR-15, H-13, H-14

Procedure:

1. System operating at 50% tractive effort. STM32 Vdd = 3.3 V nominal.
2. Using programmable power supply, ramp Vdd down gradually at 10 mV/s.
3. Observe BOR threshold (typical 2.7 V for level 2). Verify reset/assertion before Vdd reaches 2.5 V.
4. Verify PWM disabled at BOR assertion. Verify safe state entered.
5. Repeat with faster ramp (100 mV/s) and slower ramp (1 mV/s).
6. Repeat at different operating points: idle, full load, regen.

Pass Criteria: BOR asserts at consistent threshold (±50 mV). PWM disabled before Vdd < 2.5 V. No erratic PWM behavior observed during decay. Safe state entered reliably. Tractive effort zero before MCU becomes unreliable.

Why this covers the requirement: A declining supply (failing DC/DC, loose connection, corroded terminal) can cause the MCU to operate in an undefined region where it may output random PWM patterns. The BOR must catch this before erratic behavior occurs. This test validates the brownout threshold and response across multiple slew rates.

C-22: Brownout Recovery — Power Dip Ride-Through

Objective: Verify system behavior during brief power dips and recovery.

Covered: SG-13 (ASIL D), FSR-16, H-13

Procedure:

1. System operating at 50% tractive effort.
2. Apply brief Vdd dip: 3.3 V → 2.8 V for 10 ms (above BOR threshold). Verify system continues operating.
3. Apply brief Vdd dip: 3.3 V → 2.6 V for 5 ms (below BOR threshold). Verify BOR reset triggers.
4. Apply Vdd interruption: 3.3 V → 0 V for 100 ms → 3.3 V. Verify full POST sequence executes on recovery.
5. Repeat with dip on +12 V rail (gate drive supply) while monitoring gate driver UVLO.

Pass Criteria: Dips above BOR: uninterrupted operation. Dips below BOR: clean reset, PWM disabled, POST runs on recovery. +12 V dip: NCV57100 UVLO triggers, PWM disabled. All cases: safe state maintained during anomaly.

Why this covers the requirement: Real-world power disturbances (load dump, starter motor engagement, loose battery connection) cause brief voltage dips. The system must either ride through cleanly or reset cleanly — never enter an undefined operational state. This validates both the brownout detector and the gate driver UVLO independence.

C-23: +12 V Logic Rail Short to Ground

Objective: Verify system response when +12 V logic rail is shorted to ground.

Covered: SG-13 (ASIL D), FSR-14, FSR-15, H-13

Procedure:

1. System operating at 50% tractive effort. +12 V rail monitored.
2. Apply controlled short to ground on +12 V rail via current-limited test fixture (<5 A limit to protect wiring).
3. Observe: STM32 supply (3.3 V via RD7-12S033R DC/DC converter), gate driver supplies (via isolated DC/DC), CAN transceivers.
4. Verify which subsystems lose power and which remain operational.
5. Remove short. Verify recovery behavior.

Pass Criteria: If +12 V collapse causes 3.3 V dropout: BOR triggers, safe state entered. Gate driver isolated supplies must remain operational (isolated from logic rail) or UVLO triggers safe state. No erratic PWM during collapse. Short removal requires key cycle to resume.

Why this covers the requirement: A +12 V short can occur from chafed wiring, failed downstream load, or moisture ingress. The test validates that the system fails safely — either the brownout catches it, or the gate driver UVLO catches it, or both. The isolated gate drive supplies should remain up (they have their own isolated DC/DC), but if they droop, the NCV57100 UVLO provides independent protection.

C-24: +5 V Sensor Rail Short to Ground

Objective: Verify system response when +5 V sensor supply is shorted.

Covered: SG-01 (ASIL D), SG-13 (ASIL D), FSR-01, FSR-09, H-01, H-08

Procedure:

1. System operating. +5 V rail powers throttle pots, current sensor references, encoder.
2. Apply controlled short on +5 V rail (current-limited to <2 A).
3. Observe throttle pot readings (should go to zero / OOR). Observe current sensor reference loss.
4. Verify FSR-01 catches pot OOR. Verify FSR-09 catches encoder signal loss (if encoder powered from +5 V).
5. Verify safe state entered before any incorrect tractive effort is produced.

Pass Criteria: +5 V short detected via sensor OOR. Safe state entered within <200 ms. No tractive effort produced based on invalid sensor data. Fault logged with distinct DTC.

Why this covers the requirement: The +5 V rail powers critical sensors. A short causes all sensors to read zero or OOR. The system must recognize this as a fault condition, not interpret zero readings as valid "zero throttle" and "zero current." This tests the sensor OOR detection paths.

C-25: +3.3 V MCU Supply Short to Ground

Objective: Verify safe state behavior on MCU supply collapse.

Covered: SG-13 (ASIL D), FSR-14, FSR-15, H-13

Procedure:

1. System operating at 50% tractive effort. 3.3 V rail monitored.
2. Apply controlled short on 3.3 V rail (current-limited).
3. Observe BOR behavior. If BOR does not trigger (short faster than BOR response), observe gate driver behavior independently.
4. Verify breakpoint PWM disable triggers as supply collapses.
5. Verify watchdog triggers if CPU stalls.

Pass Criteria: At least one safe-state path triggers (BOR, watchdog, or breakpoint). PWM disabled. No sustained erratic operation. System requires key cycle after short removal.

Why this covers the requirement: The 3.3 V short is the worst-case supply fault because it directly affects the MCU. Multiple independent safe-state paths must exist. This test validates that at least one path works even under the most severe supply fault.

C-26: Gate Driver +15 V Supply Short

Objective: Verify NCV57100 UVLO response when +15 V gate drive supply is shorted.

Covered: SG-12 (ASIL C), SG-14 (ASIL C), FSR-12, FSR-13, H-12, H-16

Procedure:

1. System operating. Individual gate driver +15 V supplies monitored (six isolated supplies).
2. Short +15 V supply of one gate driver (U-phase high-side) to ground via current-limited fixture.
3. Observe NCV57100 UVLO trigger. Measure UVLO threshold (typical V_{UVLO-} = 11.3 V).
4. Verify FLT asserted. Verify PWM disabled for that phase.
5. Verify system enters safe state (not just disabling one phase, which creates imbalance).
6. Repeat for all six gate driver positions.

Pass Criteria: UVLO triggers at V_{UVLO-} ≈ 11.3 V (±0.5 V). FLT asserted. PWM disabled. Safe state entered. All six positions show consistent behavior. No IGBT damage.

Why this covers the requirement: Each gate driver has its own isolated +15 V supply. A short in one supply (failed DC/DC, broken wire) causes that driver's UVLO to trigger. The FLT output signals the fault. The system must enter safe state because operating with one disabled phase creates severe current imbalance and vibration.

C-27: Gate Driver −9 V Supply Short

Objective: Verify NCV57100 response when negative gate drive supply is shorted.

Covered: SG-12 (ASIL C), FSR-12, H-12

Procedure: Same as C-26 but short −9 V supply to ground for each gate driver position.

Pass Criteria: Negative supply loss detected. Gate cannot be properly turned off (Miller current has no return path). FLT asserted. Safe state entered. No shoot-through.

Why this covers the requirement: The −9 V supply ensures the IGBT stays off during dv/dt transitions (active Miller clamp). Loss of negative rail means the Miller current charges the gate capacitance through the Miller capacitor, potentially turning the IGBT on unexpectedly. The NCV57100 must detect this condition and force safe state.

C-28: Phase U Open Circuit (Motor Disconnect)

Objective: Verify system detects and responds to an open-circuited motor phase.

Covered: SG-08 (ASIL C), FSR-09, H-08

Procedure:

1. System operating at 30% tractive effort on dyno. Closed-loop FOC active.
2. Open Phase U connection (using contactor or relay in series with motor lead).
3. Observe current sensor readings: Phase U current drops to zero while V and W show imbalance.

4. Verify FSR-02 (tractive effort plausibility) detects commanded vs. actual current mismatch.
5. Verify safe state entered within <200 ms.
6. Repeat at multiple operating points: low speed, high speed, regen.

Pass Criteria: Open phase detected via current imbalance. Safe state entered within <200 ms. No sustained single-phase operation (which would cause severe vibration and potential motor damage). Fault logged with specific DTC.

Why this covers the requirement: An open phase (loose terminal, broken wire, burnt connector) causes the motor to operate single-phase. FOC cannot maintain control with one phase open — the current feedback is missing for that phase. The plausibility check (FSR-02) must catch the mismatch between commanded and actual current. Operating single-phase causes severe torque ripple, vibration, and potential bearing damage.

C-29: Phase V Open Circuit

Objective: Same as C-28 for Phase V.

Covered: SG-08 (ASIL C), FSR-09, H-08

Procedure: Identical to C-28, open Phase V instead of Phase U.

Pass Criteria: Same as C-28.

Why this covers the requirement: All three phases must be independently validated. The detection mechanism may behave differently depending on which phase is open due to FOC coordinate transformation dependencies.

C-30: Phase W Open Circuit

Objective: Same as C-28 for Phase W.

Covered: SG-08 (ASIL C), FSR-09, H-08

Procedure: Identical to C-28, open Phase W instead of Phase U.

Pass Criteria: Same as C-28.

Why this covers the requirement: All three phases must be independently validated. The detection mechanism may behave differently depending on which phase is open due to FOC coordinate transformation dependencies.

C-31: Phase-to-Phase Short (U-V)

Objective: Verify DESAT and overcurrent protection respond to phase-to-phase short circuit.

Covered: SG-12 (ASIL C), SG-14 (ASIL C), FSR-13, H-12, H-16

Procedure:

1. System at key-on with DC link energized. Low-voltage test recommended (reduced DC link to 50 V) to limit fault energy.
2. Apply phase-to-phase short between U and V using low-inductance contactor.
3. Measure DESAT response time (<2 us target). Measure peak current.
4. Verify soft turn-off (not hard shutoff). Verify no IGBT damage.
5. Verify FLT latched. Verify system safe state.
6. Inspect IGBTs after test (no degradation).

Pass Criteria: DESAT triggers within <2 us. Peak current limited by DC link inductance. Soft turn-off observed. No IGBT damage. FLT latched. Safe state entered. System requires reset.

Why this covers the requirement: Phase-to-phase short is the most severe fault the inverter can experience (H-12). The NCV57100 DESAT is the primary protection. This test validates the full short-circuit protection chain: DESAT detection → soft turn-off → FLT assertion → safe state. **WARNING:** This test must be conducted at reduced DC link voltage (50 V or less) with appropriate safety barriers and energy-limiting precautions. Even at reduced voltage, fault energy can be destructive if protection fails.

C-32: Phase-to-Phase Short (V-W and U-W)

Objective: Same as C-31 for V-W and U-W phase combinations.

Covered: SG-12 (ASIL C), SG-14 (ASIL C), FSR-13, H-12

Procedure: Identical to C-31 but short V-W and U-W combinations separately. Use reduced DC link voltage.

Pass Criteria: Same as C-31 for all phase combinations.

Why this covers the requirement: All three phase-pair combinations must be independently validated. The DESAT protection must work regardless of which phases are shorted.

C-33: Phase-to-DC-Positive Short

Objective: Verify protection response when a motor phase shorts to DC link positive rail.

Covered: SG-12 (ASIL C), FSR-13, H-12

Procedure:

1. System at key-on. DC link at reduced voltage (50 V).
2. Apply short from Phase U to DC+ using low-inductance contactor.
3. If the high-side IGBT of Phase U is ON: DESAT should trigger immediately.
4. If the low-side IGBT is ON: shoot-through path created through Phase U winding to DC+. Overcurrent protection must trigger.
5. Verify safe state. Inspect hardware.

Pass Criteria: Protection triggers within <10 us. No hardware damage at reduced voltage. Safe state entered. Fault logged.

Why this covers the requirement: This fault simulates insulation failure in the motor winding (turn-to-ground fault at DC+ potential). The current path depends on which switch is conducting. Either DESAT or the overcurrent comparator must catch it.

C-34: Phase-to-DC-Negative Short

Objective: Same as C-33 for phase shorted to DC link negative rail.

Covered: SG-12 (ASIL C), FSR-13, H-12

Procedure: Identical to C-33 but short Phase U to DC-. Test all three phases.

Pass Criteria: Same as C-33.

Why this covers the requirement: Motor winding ground fault to DC- is a common failure mode (winding insulation degrades, moisture ingress, mechanical damage). All three phases must be independently validated.

C-35: DC Link Capacitor Temperature Monitoring

Objective: Verify DC link capacitor temperature sensing and thermal protection.

Covered: SG-07 (ASIL B), FSR-08 (extended), H-07

Procedure:

1. Install temperature sensor(s) on DC link capacitor bank (thermistor or RTD).
2. Operate system at various load levels while monitoring capacitor temperature.
3. Apply elevated ambient temperature or reduced cooling to raise capacitor temp.
4. Verify warning threshold triggers power derate. Verify critical threshold triggers safe state.
5. Verify temperature reading is stable and accurate (compare with external thermometer).
6. Verify sensor open/short detection (if applicable).

Pass Criteria: Temperature reading accurate (±5°C). Warning derate at T_{warn} (e.g., 70°C). Critical safe state at T_{crit} (e.g., 85°C). Sensor fault detected and safe state entered.

Why this covers the requirement: DC link capacitor lifetime decreases exponentially with temperature (Arrhenius law: lifetime halves every 10°C rise). Capacitor overtemperature can lead to electrolyte venting (explosion for electrolytic) or capacitance loss (film capacitors). Monitoring capacitor temperature extends service life and prevents catastrophic failure. **Note:** This may require adding a temperature sensor to the capacitor bank if not already present.

C-36: Bearing Current / Class Y Capacitor Effectiveness

Objective: Validate that Class Y safety capacitors effectively shunt common-mode bearing currents to ground, protecting motor bearings from electrical discharge machining (EDM).

Covered: SG-07 (ASIL B, extended), H-07 (indirect — bearing damage leads to mechanical failure)

Procedure:

1. System operating at rated speed and load. Motor shaft grounded through insulated coupling on dyno.
2. Measure common-mode voltage at motor terminals using high-bandwidth differential probe.
3. Measure shaft voltage (voltage between motor shaft and ground) using carbon brush or capacitive coupling.
4. Calculate bearing voltage ratio (BVR = V_{shaft} / $V_{common\text{-}mode}$). Target: BVR < 0.1 with Class Y caps installed.
5. Temporarily remove Class Y capacitors. Verify BVR increases significantly (>0.3).
6. Reinstall capacitors. Verify BVR returns to low value.

Pass Criteria: With Class Y capacitors: BVR < 0.1. Shaft voltage < 1 V peak (below bearing oil film breakdown threshold of ~5–15 V). Without capacitors: BVR > 0.3 (confirms capacitors are effective). No bearing damage after extended operation.

Why this covers the requirement: PWM inverters generate high dv/dt common-mode voltage that capacitively couples to the motor shaft. When shaft voltage exceeds the bearing lubricant's dielectric strength (~5–15 V), electrical discharge occurs through the bearing, pitting the raceways. This is called Electrical Discharge Machining (EDM). Class Y safety capacitors (rated for line-to-ground use, fail-safe) provide a low-impedance path for common-mode current to flow to ground, bypassing the bearings. This test validates the capacitor effectiveness by measuring the bearing voltage ratio. This is a long-term reliability test, not an immediate safety test, but bearing failure at high speed is a safety concern.

C-37: STM32 HSE Clock Failure

Objective: Verify system response when the external high-speed crystal (HSE) fails.

Covered: SG-13 (ASIL D), FSR-15, H-13, H-14

Procedure:

1. System operating normally with HSE as clock source.
2. Disable HSE (remove crystal drive signal or switch off oscillator circuit).
3. Verify STM32 CSS (Clock Security System) detects HSE failure and switches to HSI (internal RC).
4. Verify system enters safe state (operating on HSI at reduced accuracy is not acceptable for motor control).
5. Verify fault logged. Verify system does not attempt full-load operation on HSI.

Pass Criteria: CSS detects HSE loss within <1 ms. System enters safe state. Fault logged. No continued motor control on degraded clock.

Why this covers the requirement: HSE failure (crystal crack, cold solder joint, EMI damage) causes the MCU to lose its accurate timebase. FOC requires precise PWM timing; operating on the internal HSI RC oscillator (±1% accuracy vs. ±30 ppm for crystal) causes current control degradation and potential instability. The CSS must detect this and force safe state.

C-38: STM32 PLL Unlock

Objective: Verify safe state when PLL loses lock (unstable system clock).

Covered: SG-13 (ASIL D), FSR-15, H-13

Procedure:

1. System operating with PLL providing 550 MHz system clock.
2. Introduce Vdd noise or temperature transient that causes PLL to lose lock.
3. Alternatively, use debug interface to momentarily change PLL configuration to unstable values.
4. Verify system detects PLL unlock (if monitoring available) or watchdog triggers due to timing chaos.
5. Verify safe state entered.

Pass Criteria: PLL unlock detected or watchdog triggers within <50 ms. Safe state entered. System resets and runs POST.

Why this covers the requirement: PLL unlock causes the system clock frequency to drift unpredictably. All timing-dependent functions (PWM, ADC sampling, control loop) become erratic. If explicit PLL lock monitoring is not implemented, the watchdog serves as the fallback.

C-39: Flash Bit Rot / Corruption Detection

Objective: Verify boot CRC (FSR-19) detects flash corruption from bit rot, EMI, or wear.

Covered: SG-01 (ASIL D), SG-13 (ASIL D), FSR-19, H-15

Procedure:

1. Corrupt a single bit in safety-critical flash region (using debugger).
2. Corrupt multiple bits in the same word.
3. Corrupt calibration data (torque LUT, temperature thresholds).
4. Corrupt the CRC checksum itself (but leave code intact).
5. Power cycle after each corruption. Verify boot behavior.

Pass Criteria: All corruption scenarios detected at boot. PWM enable prevented. Fault logged with distinct DTC for code corruption vs. calibration corruption. System remains in safe state until reprogrammed.

Why this covers the requirement: Flash memory is subject to: (1) bit rot from cosmic radiation and charge leakage over time, (2) corruption from EMI during write operations, (3) wear from repeated erase/program cycles.

FSR-19 prevents the system from operating with corrupted firmware that could modify safety-critical behavior. This test validates the CRC catches all realistic corruption patterns.

C-40: GPIO Stuck-At Fault Simulation

Objective: Verify system responds safely when critical GPIO pins are stuck-at-1 or stuck-at-0.

Covered: SG-01 (ASIL D), SG-13 (ASIL D), FSR-14, H-13, H-14

Procedure:

1. System operating normally. Identify critical GPIOs: PWM enable, breakpoint, HVIL, throttle ADC inputs, FLT input.
2. Use debug interface to force each critical GPIO to stuck-at-1, then stuck-at-0.
3. PWM enable stuck high: verify breakpoint override still works.
4. PWM enable stuck low: verify no PWM output (safe but non-functional).
5. Breakpoint input stuck at non-fault level: verify watchdog still provides safe-state path.
6. HIL input stuck: verify other safety paths are not compromised.

Pass Criteria: No stuck-at fault creates an unsafe condition. At least one independent safe-state path remains functional for each fault. With six SSO pathways and 1oo2 power kill, no single-point fault can prevent safe state entry.

Why this covers the requirement: GPIO stuck-at faults occur from: broken traces, damaged I/O cells, solder bridges, connector faults. The analysis identifies which stuck-at faults are safe and which create dangerous single-point failures. This guides hardware redesign (add pull-ups/pull-downs, redundant paths) and documents acceptable vs. unacceptable fault modes.

C-41: ADC Reference Voltage Drift

Objective: Verify ADC reference stability and detect reference drift that causes sensor misreading.

Covered: SG-01 (ASIL D), FSR-02, H-01, H-06

Procedure:

1. System operating. Apply known precision voltage to one ADC channel (calibration reference).
2. Monitor ADC reading of precision reference over temperature sweep (25°C to 85°C).
3. Verify ADC reading stays within ±1% of expected value.
4. If STM32 has internal Vrefint channel: monitor Vrefint reading as proxy for ADC reference stability.
5. Apply external Vref drift (using programmable reference) and verify FSR-02 (plausibility) catches resulting sensor errors.

Pass Criteria: ADC reference stable within ±1% over temperature. Vrefint reading in expected range. If Vref drifts beyond ±2%: plausibility check detects sensor mismatch and enters safe state.

Why this covers the requirement: ADC reference drift (from temperature, aging, or supply noise) causes ALL analog sensor readings to scale proportionally. This is dangerous because the throttle and current sensors will read incorrectly in a correlated way. The plausibility check (FSR-02) is the primary defense — it compares commanded vs. measured current, which should reveal the scaling error.

C-42: SPI Communication Fault (MAX22530 Isolated ADC)

Objective: Verify system handles SPI communication failures with the isolated voltage measurement ADC.

Covered: SG-10 (ASIL B), FSR-11, H-10

Procedure:

1. System operating with MAX22530 providing DC link voltage readings.
2. Inject SPI faults: SCLK stuck, MISO open, CSN stuck high, corrupted MOSI data.
3. Verify DC link voltage reading becomes invalid or stale.
4. Verify system detects loss of valid DC link data and enters safe state.
5. Verify DC link overvoltage protection still functions via independent path (if any).

Pass Criteria: SPI fault detected within <200 ms. Safe state entered. No operation with invalid DC link measurement. Fault logged.

Why this covers the requirement: The MAX22530 provides isolated DC link and phase voltage measurements via SPI. An SPI fault (broken trace, connector issue, EMI) means the system loses DC link voltage visibility. Operating without DC link voltage data risks overvoltage events going undetected. The system must detect the communication loss and enter safe state.

C-43: CAN Bus Off State and Recovery

Objective: Verify system handles CAN bus-off state correctly (TEC > 255).

Covered: SG-01 (ASIL D), FSR-17, H-03

Procedure:

1. System operating with active CAN communication.
2. Inject CAN errors at high rate to force TEC (Transmit Error Counter) above 255, triggering bus-off.
3. Verify system detects bus-off state.
4. Verify safe-state defaults applied (as if heartbeat lost).
5. Stop error injection. Verify CAN peripheral recovers per ISO 11898-1 (128 occurrences of 11 consecutive recessive bits).
6. Verify system resumes normal operation only after key cycle (not automatic).

Pass Criteria: Bus-off detected. Safe state entered. Recovery follows ISO 11898-1 protocol. System does not auto-resume without key cycle.

Why this covers the requirement: CAN bus-off occurs during severe bus faults (shorted CAN_H/CAN_L, failed transceiver on another node, broken termination). The system must not continue operating as if CAN is healthy. The bus-off state is the CAN controller's built-in self-protection. The test validates that the VCU treats bus-off as a communication loss and applies safe defaults.

C-44: Watchdog Starvation Under CPU Load

Objective: Verify watchdog triggers when CPU is overloaded and cannot service WDT in time.

Covered: SG-13 (ASIL D), FSR-15, H-13, H-14

Procedure:

1. System operating normally with WDT timeout = 50 ms.
2. Introduce CPU load spike: heavy interrupt load, DMA saturation, or floating-point-intensive task.
3. Verify watchdog service is missed. Verify WDT triggers reset.
4. Verify safe state (SSO) is maintained during reset and POST.
5. Verify system does not produce tractive effort during the reset window.

Pass Criteria: WDT triggers within specified window. System resets. POST executes. No tractive effort during reset. Safe state maintained throughout.

Why this covers the requirement: A software bug (infinite loop, priority inversion, stack overflow) or EMI-induced code execution fault can prevent the WDT from being serviced. The WDT is the last line of defense against runaway software. This test validates that the WDT is not accidentally masked by normal CPU load (windowed watchdog helps here) and that it triggers reliably when the CPU is truly stuck.

C-45: IGBT Thermal Runaway Profile

Objective: Verify thermal protection catches IGBT thermal runaway before junction temperature exceeds $T_{j,max}$.

Covered: SG-07 (ASIL B), FSR-08, H-07

Procedure:

1. System operating at overload condition (e.g., 120% rated current with reduced cooling).
2. Monitor all three IGBT temperature sensors and compare against junction temperature estimation (from $V_{CE(sat)}$ measurement or thermal model).
3. Verify progressive derate occurs as temperature rises: 100% → 80% → 50% → 20%.
4. Continue until critical temperature reached. Verify SSO entered.
5. Verify junction temperature never exceeds $T_{j,max}$ (typically 150°C or 175°C).

Pass Criteria: Progressive derate observed. SSO at critical temperature. $T_j < T_{j,max}$ at all times. 2oo3 voter works correctly (test with one sensor reading artificially low to simulate sensor fault).

Why this covers the requirement: H-07 is thermal runaway leading to fire. The 2oo3 voter must correctly identify the true temperature even if one sensor is faulty. The progressive derate gives the rider time to react (reduce load, find shade) before SSO. The test validates the full thermal protection chain under realistic overload conditions.

C-46: Pre-Charge and Inrush Current Validation

Objective: Verify pre-charge sequence limits inrush current into DC link capacitors.

Covered: SG-10 (ASIL B), FSR-11, H-10, H-12

Procedure:

1. DC link capacitors fully discharged.
2. Initiate key-on sequence. Monitor DC link voltage and inrush current.
3. Verify pre-charge resistor (or active pre-charge circuit) limits inrush to acceptable level (<50 A peak for <100 ms).
4. Verify pre-charge completes before contactor closes (voltage > 95% of battery voltage).
5. Verify main contactor only closes after pre-charge complete.
6. Verify failed pre-charge (open resistor, shorted capacitor) is detected and prevents contactor closure.

Pass Criteria: Inrush current < specified limit. Pre-charge completes within timeout (<3 s). Contactor only closes after pre-charge verified. Failed pre-charge detected and safe state entered. No sparking/welding on contactor closure.

Why this covers the requirement: Closing the main contactor into discharged capacitors creates massive inrush current (hundreds of amps) that welds contacts, damages capacitors, and trips BMS protection. The pre-charge sequence limits this current to a safe level. A failed pre-charge (open resistor) means the contactor would close into a short-circuit-like condition. This test validates the entire pre-charge control logic.

C-47: Active Miller Clamp Validation

Objective: Verify NCV57100 active Miller clamp prevents dv/dt-induced false turn-on.

Covered: SG-12 (ASIL C), FSR-12, H-12

Procedure:

1. System operating at rated DC link voltage. High-side IGBT of Phase U switching at maximum dv/dt.
2. Monitor low-side gate voltage of Phase U during high-side turn-on (the dv/dt couples through Miller capacitance).
3. Verify gate voltage spike is clamped to <2 V by active Miller clamp.
4. Compare with Miller clamp disabled (if test fixture allows): verify gate spike is significantly higher without clamp.
5. Repeat at maximum DC link voltage and maximum temperature.

Pass Criteria: Miller-induced gate spike < $V_{GE(th)}$ (typically <2 V). No false turn-on observed. Active Miller clamp current within specification.

Why this covers the requirement: When the high-side IGBT turns on, the rapid dv/dt across the half-bridge creates a current through the Miller capacitance (C_{GC}) that can charge the low-side gate capacitance. If the gate voltage exceeds $V_{GE(th)}$, the low-side IGBT turns on briefly, creating shoot-through. The active Miller clamp provides a low-impedance path to shunt this current, preventing false turn-on. This is especially critical at high DC link voltage where dv/dt is maximum.

C-48: Gate Driver Propagation Delay Matching

Objective: Verify propagation delay matching across all six gate drivers to prevent skew-induced shoot-through.

Covered: SG-12 (ASIL C), FSR-12, H-12

Procedure:

1. Apply identical PWM test signal to all six gate driver inputs.
2. Measure propagation delay (input edge to output edge) for each gate driver using oscilloscope.
3. Record turn-on delay ($t_{d,on}$) and turn-off delay ($t_{d,off}$) for all six.
4. Calculate delay mismatch: $\max(t_d) - \min(t_d)$ across all six drivers.
5. Verify mismatch < deadtime setting (typically 1–2 us).

Pass Criteria: Propagation delay variation < 100 ns between all six drivers. Delay mismatch < deadtime margin. No driver significantly slower/faster than others.

Why this covers the requirement: Propagation delay mismatch between gate drivers can cause one phase to switch before another, creating instantaneous voltage imbalance and potential shoot-through in the phase that switches earliest. For complementary switching (one phase on, another off), delay mismatch can create a window where both are on. This test ensures the hardware switching is well-matched and the deadtime margin is adequate.

C-49: PWM Deadtime Verification

Objective: Verify deadtime is always present between complementary switching edges.

Covered: SG-12 (ASIL C), FSR-12, H-17

Procedure:

1. Generate PWM at various duty cycles (5%, 50%, 95%) and switching frequencies.
2. Measure high-side and low-side gate signals of one phase on oscilloscope.
3. Verify deadtime (both gates off) is present between every switching transition.

4. Measure deadtime duration. Verify within programmed value (±10%).
5. Verify no deadtime collapse at minimum or maximum duty cycle.
6. Verify complementary breakpoint disable produces simultaneous OFF (not ON).

Pass Criteria: Deadtime present on every transition. Duration = programmed value ±10%. No deadtime collapse at any operating point. Breakpoint disable = both OFF simultaneously.

Why this covers the requirement: H-17 is PWM deadtime violation causing shoot-through. TIM1 generates deadtime in hardware, but this test verifies it is actually present on the gate signals (not just in the timer registers). Deadtime collapse at extreme duty cycles is a known timer bug risk. This is a production test that should be run on every unit.

C-50: Isolation Barrier Degradation (HV)

Objective: Verify reinforced isolation barriers maintain integrity under operating stress.

Covered: SG-09 (ASIL A), FSR-10, H-09

Procedure:

1. Measure isolation resistance between HV DC link and logic ground using 500 V megohmmeter.
2. Verify >500 Ohm/V per ISO 6469-3 (e.g., >100 kOhm for 200 V system, >250 kOhm for 500 V system).
3. Apply HiPot test at 2 kV AC (or 2.8 kV DC) for 60 seconds across isolation barrier.
4. Verify no breakdown, no corona, leakage current < 5 mA.
5. Repeat after thermal aging (85°C soak for 24 hours).

Pass Criteria: Isolation resistance > 500 Ohm/V. HiPot passes without breakdown. Post-thermal isolation resistance > 80% of initial value.

Why this covers the requirement: H-09 is HV electrical shock from isolation failure. Isolation barriers (gate driver isolation, DC/DC isolation, CAN isolation) degrade over time from thermal cycling, humidity, and electrical stress. The HiPot test validates the barrier can withstand the rated working voltage plus safety margin. This is a type test (not run on every unit) but must pass before any HV operation.

10.4 System-Level Tests

System-level tests exercise complete end-to-end fault scenarios with all hardware and software running in closed-loop FOC control on a dyno or test rig. These tests validate that the integrated system responds correctly to realistic fault conditions.

S-01: Unintended Tractive Effort from Throttle Fault

Objective: Verify SG-01, FSR-01, FSR-03, FSR-18 — System rejects unintended tractive effort when throttle input is implausible.

Covered: SG-01 (ASIL D), H-01, FSR-01, FSR-03, FSR-18

Setup: Traction motor on dyno. System in normal operating mode at key-on. DC link bus supply at nominal voltage.

Procedure:

1. Apply 50% tractive effort request via both throttle pots (matched). Verify system produces corresponding tractive effort.
2. Inject pot 1 short to +5 V (simulating wiring fault). Pot 2 remains at 50%.

3. Observe system response: should detect >5% discrepancy between pots.
4. Repeat with pot 1 short to GND, pot 1 open circuit, pot 2 drifting (slow ramp offset).
5. Verify limit switch override: apply 50% throttle on both pots, activate limit switch. Verify tractive effort commands to zero regardless of pot values.

Pass Criteria: All fault injections detected within <100 ms. Tractive effort ramped to zero at ≤500 Nm/s (FSR-03). Safe state entered. Fault logged with correct DTC. Limit switch override has absolute priority over all pot values.

Why this covers the requirement: This is the highest-severity hazard (H-01). The test validates the complete chain from sensor fault through detection to safe state entry. The dual-pot redundancy with discrepancy check (FSR-01) is the primary defense. The limit switch (FSR-18) provides the independent override. The rate limiter (FSR-03) prevents a step change if one pot fails high and the other is trusted.

S-02: Unintended Reverse Tractive Effort

Objective: Verify SG-02, FSR-04 — Reverse tractive effort is rejected when speed > 0.

Covered: SG-02 (ASIL B), H-02, FSR-04

Setup: Traction motor spinning forward on dyno (simulated 30 mph equivalent).

Procedure:

1. Rotate traction motor forward at >100 rpm.
2. Command reverse tractive effort (negative torque request from rider interface).
3. Verify system clamps negative tractive effort to zero.
4. Bring motor to stop (<10 rpm). Command reverse tractive effort with reverse gear explicitly selected. Verify reverse tractive effort is permitted.
5. Bring motor to stop. Command reverse tractive effort WITHOUT reverse gear selected. Verify request is rejected.

Pass Criteria: Reverse tractive effort rejected when speed > 100 rpm. Reverse permitted only when stationary AND reverse explicitly selected. All other cases → zero tractive effort, fault logged.

Why this covers the requirement: H-02 is rearward tip-over at standstill or low speed. FSR-04 prevents reverse tractive effort at any forward speed. The two-condition interlock (stationary + selected) prevents inadvertent reverse from a software glitch or sensor fault.

S-03: Sudden Loss of Tractive Effort (Highway)

Objective: Verify SG-03, FSR-05 — Tractive effort removed gradually to prevent rear-end collision hazard.

Covered: SG-03 (ASIL C), H-03, FSR-05

Setup: Traction motor on dyno at 80% load (simulating highway cruise). System in normal closed-loop control.

Procedure:

1. Stabilize system at 80% rated tractive effort. Record baseline.
2. Trigger a non-critical fault that requires safe state entry (e.g., minor temperature excursion, non-critical CAN timeout).
3. Measure tractive effort ramp-down rate using dyno torque sensor.
4. Repeat with critical fault (encoder loss, DESAT) that demands immediate SSO.

Pass Criteria: Non-critical faults: tractive effort ramps down at ≤200 Nm/s before SSO. Critical faults: SSO permitted within <50 ms if safety timing demands (FSR-05 exception). Dyno shows smooth torque decay, no step change.

Why this covers the requirement: H-03 is sudden loss of tractive effort at highway speed. The primary danger is the following vehicle colliding with the motorcycle. A step torque removal causes rapid deceleration that following traffic may not anticipate. FSR-05 requires gradual removal (≤200 Nm/s) so the rider can react (activate hazard lights, move to shoulder) and following vehicles have time to adjust. This test directly measures the ramp rate to validate it meets the requirement.

S-04: Loss of Tractive Effort During Cornering at Lean

Objective: Verify SG-03, FSR-05 — Gradual tractive effort removal during cornering does not cause the motorcycle to stand up and run wide.

Covered: SG-03 (ASIL C), H-03a, FSR-05

Setup: This test validates the **rate limiter** behavior. Full cornering validation requires vehicle-level testing or motorcycle dynamics simulation (see Known Limitations, Section 10.9). The dyno test validates the controllable portion: torque ramp rate.

Procedure:

1. Stabilize system at 60% rated tractive effort on dyno.
2. Inject a fault requiring tractive effort removal while motor is loaded.
3. Measure actual torque ramp-down rate. Verify ≤200 Nm/s.
4. Simulate repeated fault/recovery cycles to ensure ramp-down is consistent.

Pass Criteria: Torque ramp-down rate never exceeds 200 Nm/s. No step changes observed. SSO only after ramp completes (unless <50 ms safety exception applies).

Why this covers the requirement (Motorcycle Dynamics Rationale): H-03a is distinct from H-03. When a motorcycle is cornering at lean angle, the rider balances lateral acceleration (cornering force) against gravity and tire friction. The rear tire contact patch provides both lateral cornering force and longitudinal propulsive force. At full lean on a racetrack, the tire is operating at or near its friction limit.

When tractive effort is suddenly removed (step to zero), the longitudinal force component at the rear contact patch collapses instantly. This has two destabilizing effects: (1) the motorcycle chassis experiences a forward weight transfer (dive) that reduces rear tire normal force, which reduces available lateral friction; (2) the loss of rearward propulsive force causes the motorcycle to **stand up** (reduce lean angle) because the lateral force requirement changes with the reduced longitudinal slip. The rider, committed to a cornering line at full lean, now finds the motorcycle running wide toward the outside of the turn (toward oncoming traffic, barriers, or track edge).

Unlike a four-wheeled vehicle where loss of drive simply causes deceleration, a leaned motorcycle's trajectory is **coupled to its propulsive state**. The rider has zero margin to recover: both hands are committed to handlebar control, the body is positioned for the corner, and the tire is already at its friction limit. There is no "catching" a run-wide event at 40-degree lean and 80 mph.

FSR-05 limits the ramp-down rate to ≤200 Nm/s precisely to keep the change in longitudinal force gradual enough that the rider can modulate lean angle progressively. This test validates the ramp rate on the dyno. The vehicle-level effect (does the bike run wide?) can only be validated by track testing with an experienced rider, which is noted as a limitation.

S-05: Uncommanded Regenerative Braking

Objective: Verify SG-05, FSR-06 — Uncommanded regenerative braking tractive effort is detected and rejected.

Covered: SG-05 (ASIL C), H-05, FSR-06

Setup: Traction motor on dyno in motoring mode (simulating downhill or coasting with no rider brake request).

Procedure:

1. Spin traction motor at 40 mph equivalent. Set rider brake request to zero (no regen requested).
2. Inject a software fault causing negative Id current (regen) to be commanded.
3. Verify FSR-06 monitors the magnitude of regenerative braking tractive effort.
4. Observe detection threshold at 10 Nm and response time.

Pass Criteria: Uncommanded regen >10 Nm detected within <200 ms. Safe state entered. Friction brakes remain available (independent system).

Why this covers the requirement: H-05 is unexpected deceleration on a wet or slippery surface during cornering. The one-pedal regen strategy is intentionally mild to prevent wheel lock. However, a software fault could command excessive regen. FSR-06 provides the independent monitor. This test validates that the monitor catches software-induced uncommanded regen.

S-06: Full Load Continuous Operation with Thermal Camera Survey

Objective: Identify all thermal hotspots under sustained full-load operation using infrared thermography.

Covered: SG-07 (ASIL B), FSR-08, H-07 (extended to all components)

Setup: Traction motor on dyno. Infrared thermal camera (320x240 minimum, <50 mK NETD). Emissivity-calibrated targets on key surfaces. Ambient temperature 25°C with controlled airflow.

Procedure:

1. Apply thermal emissivity tape or paint ($\varepsilon = 0.95$) to: IGBT modules, DC link capacitors, gate driver PCBs, current sensors, DC/DC converter, STM32 MCU, CAN transceivers, busbars, contactors/relays, motor connection terminals.
2. Begin operation at 25% rated load. Run for 15 minutes. Capture thermal image.
3. Increase to 50% load. Run 15 minutes. Capture thermal image.
4. Increase to 75% load. Run 15 minutes. Capture thermal image.
5. Increase to 100% rated load. Run for minimum 60 minutes (or until thermal steady-state). Capture thermal images every 10 minutes.
6. Record maximum temperature of every component. Flag any component exceeding 80% of its rated maximum temperature.
7. Identify unexpected hotspots (poor solder joints, undersized traces, concentrated current paths).

Pass Criteria: All component temperatures < 80% of rated maximum at 100% load steady-state. No unexpected hotspots >10°C above surrounding area. IGBT T_{case} < 100°C. DC link capacitor T < 70°C. Busbars < 80°C.

Why this covers the requirement: Thermal design validation cannot rely solely on temperature sensors at a few points. A thermal camera reveals: poor solder joints (higher resistance = localized heating), current crowding in traces, inadequate heatsink contact, unexpected coupling between hot components. Full-load steady-state is the worst-case thermal scenario. Identifying hotspots during bench testing prevents field failures. This test should be run on the first production-representative unit and repeated after any hardware revision.

S-07: Thermal Cycling — IGBT and DC Link Capacitor

Objective: Validate thermal protection and mechanical integrity under rapid temperature changes.

Covered: SG-07 (ASIL B), FSR-08, FSR-11, H-07, H-10

Setup: Traction motor on dyno. Temperature chamber or local heating/cooling for IGBT and capacitor. Thermal camera monitoring.

Procedure:

1. Cold start from −20°C (or minimum expected ambient). Verify system starts and operates correctly.
2. Ramp to full load while monitoring IGBT temperature rise rate.
3. Verify temperature sensors read correctly at low temperature (check for frozen calibration).
4. Rapid transition: full load hot (>80°C) → immediate shutdown → immediate restart. Verify no false fault from thermal shock.
5. Repeat 10 thermal cycles: cold → full load hot → cooldown. Monitor for solder joint degradation.

Pass Criteria: System starts at −20°C. All temperature readings accurate across range. No false faults during thermal shock. No measurable increase in thermal resistance after 10 cycles.

Why this covers the requirement: Thermal cycling causes mechanical stress from CTE (coefficient of thermal expansion) mismatch: solder joints, wire bonds, substrate attachments. Repeated cycling can crack solder joints (increasing thermal resistance) or delaminate substrates. The 10-cycle test is a screening test; the real validation comes from extended thermal cycling (500+ cycles) which is noted as a future environmental test.

S-08: Regenerative Braking at Maximum Power

Objective: Verify regen system handles maximum regenerative braking power without overvoltage or instability.

Covered: SG-05 (ASIL C), SG-10 (ASIL B), FSR-06, FSR-11, H-05, H-10

Setup: Traction motor driven by external dyno at maximum rated speed. DC link bus supply with overvoltage protection. Battery simulator or resistive load bank capable of absorbing max regen power.

Procedure:

1. Spin traction motor at rated speed via external dyno.
2. Apply maximum regenerative braking torque command.
3. Monitor DC link voltage during regen. Verify it stays below warning threshold.
4. Gradually increase regen torque until DC link reaches warning threshold. Verify regen is limited.
5. If battery simulator has limited absorption: verify system limits regen to what the battery can accept.
6. Verify smooth torque transition into and out of regen (no oscillation).
7. Repeat at multiple speeds including field-weakening region.

Pass Criteria: DC link voltage stable during max regen. No overvoltage events. Regen smoothly limited at battery capacity. No torque oscillation. All temperature limits respected.

Why this covers the requirement: Maximum regen power is a worst-case scenario for DC link overvoltage (H-10) and potentially for uncommanded regen (H-05) if the control loop oscillates. This test validates the complete regen control chain: torque command → Id current control → DC link voltage feedback → overvoltage limiter. Field-weakening region regen is particularly challenging because the back-EMF exceeds DC link voltage.

S-09: Field Weakening Region Operation

Objective: Verify stable operation in field-weakening region (speed above base speed).

Covered: SG-06 (ASIL C), FSR-07, H-06

Setup: Traction motor on dyno capable of exceeding base speed.

Procedure:

1. Operate at base speed with rated torque.
2. Increase speed above base speed. Verify control transitions to field weakening.
3. Verify torque is correctly derated above base speed ($T \propto 1/\omega$).

4. Operate at maximum field-weakening speed for 10 minutes.
5. Verify current magnitude does not exceed rated value (even though torque is lower).
6. Verify thermal limits respected at all speeds.

Pass Criteria: Smooth transition to field weakening. Correct torque derate. No current overshoot. Stable operation at max speed. Temperatures within limits.

Why this covers the requirement: Field weakening requires injecting negative Id current to oppose the permanent magnet flux. Incorrect Id control can cause overcurrent (H-06) or loss of current control. This is a performance validation test that indirectly verifies current control stability at operating extremes.

S-10: Startup Sequence Validation

Objective: Verify correct sequence from key-on to ready-to-drive.

Covered: SG-13 (ASIL D), FSR-16, H-13

Setup: Complete system with DC link bus supply, BMS simulator, IO board simulator.

Procedure:

1. System fully powered down. DC link discharged.
2. Turn key to ON. Verify sequence: (a) STM32 boots, (b) POST executes, (c) pre-charge starts, (d) DC link charges to >95%, (e) contactor closes, (f) gate driver self-test, (g) encoder validated, (h) CAN communication established, (i) READY indication.
3. Measure time for each phase. Total time key-on to ready < 5 s.
4. Verify tractive effort is NOT available until READY state.
5. Inject fault during each phase (e.g., open HVIL during pre-charge) and verify sequence aborts to safe state.

Pass Criteria: Sequence executes in correct order every time. Tractive effort only available in READY state. Fault at any phase aborts to safe state. Total startup time < 5 s.

Why this covers the requirement: The startup sequence is the most safety-critical operational phase because the system transitions from unpowered to high-voltage active. Any fault during startup (HVIL open, failed POST, bad encoder) must prevent the system from reaching READY. This test validates the state machine and all transition guards.

S-11: Shutdown Sequence Validation

Objective: Verify safe shutdown from any operating state.

Covered: SG-13 (ASIL D), FSR-05, H-13

Setup: System operating at various load levels on dyno.

Procedure:

1. Operate at 100% load. Turn key OFF. Verify: (a) tractive effort ramps to zero at ≤200 Nm/s, (b) PWM disabled, (c) contactor opens, (d) DC link discharged via the external discharge resistor per service procedure (no onboard bleeder — the bank remains at bus voltage for hours after shutdown), (e) gate driver supplies disabled.
2. Repeat at 50% load, 25% load, regen mode.
3. Verify abrupt key-off (emergency shutdown) still results in safe state (may bypass ramp-down for <50 ms safety).
4. Verify system cannot be restarted without full key cycle (OFF → ON).

Pass Criteria: Safe shutdown from all operating points. Tractive effort ramped (not stepped) where timing allows. PWM disabled before contactor opens. DC link discharged via the external discharge resistor per service procedure (no onboard bleeder; bank remains at bus voltage until discharged).

Why this covers the requirement: Shutdown is the second most safety-critical phase. An incorrect sequence (contactor opens before PWM disable = arcing, DC link not discharged = shock hazard) creates immediate danger. The ramp-down requirement (FSR-05) applies here too — the rider expects gradual deceleration when turning the key off.

S-12: Key-Cycle Stress Test

Objective: Verify reliability of startup/shutdown cycling over many repetitions.

Covered: SG-13 (ASIL D), FSR-16, H-13

Setup: Automated key-switch cycling. System on dyno at no-load.

Procedure:

1. Automated key cycle: ON (5 s) → OFF (5 s) → repeat.
2. Run 1000 cycles.
3. Log every POST result, every startup time, every fault.
4. After 1000 cycles: run full functional test (C-01 through C-05). Compare against baseline.

Pass Criteria: 1000 cycles with zero failures. All POST results pass. Startup time variation < 10%. No degradation in functional test results.

Why this covers the requirement: Key cycling causes wear on: contactor (inrush every cycle), DC link capacitors (charge/discharge), gate driver supplies (startup stress), connectors (thermal cycling), flash memory (if parameters are written at shutdown). This is a life-test screening that catches infant mortality failures.

S-13: Power Dip Ride-Through (Coast-Through)

Objective: Verify system survives brief DC link power interruptions without unsafe behavior.

Covered: SG-03 (ASIL C), FSR-05, FSR-11, H-03

Setup: System on dyno. Programmable DC link supply with interruption capability.

Procedure:

1. System operating at 50% load.
2. Interrupt DC link for 1 ms. Verify system rides through (DC link capacitors maintain voltage).
3. Interrupt DC link for 10 ms. Verify system detects undervoltage and enters safe state gracefully.
4. Interrupt DC link for 100 ms. Verify safe state entered. Verify no restart without key cycle.
5. Repeat with load at 25%, 75%, 100%.
6. Repeat with interruption during regen (current flowing into DC link).

Pass Criteria: Short dips (<5 ms): ride-through. Longer dips: safe state with gradual ramp-down. No erratic behavior at any dip duration. DC link capacitor energy sufficient for safe shutdown sequencing.

Why this covers the requirement: Real-world power interruptions occur from: loose battery terminals, vibration-induced connector movement, BMS contactor bounce, load dump events. The DC link capacitors provide ride-through energy for brief interruptions. This test validates the capacitor sizing and the undervoltage response (FSR-21) at the boundary between ride-through and safe-state.

S-14: Reverse Tractive Effort at Standstill

Objective: Verify reverse tractive effort functions correctly when explicitly requested at standstill.

Covered: SG-02 (ASIL B), FSR-04, H-02

Setup: Traction motor on dyno (can rotate freely in both directions).

Procedure:

1. Motor stationary (<10 rpm). Reverse gear selected.
2. Apply 25% reverse tractive effort. Verify motor rotates backward at expected speed.
3. Apply 50% reverse tractive effort. Verify stable operation.
4. Deselect reverse while moving backward. Verify tractive effort ramps to zero.
5. Select reverse while motor is spinning forward (coasting). Verify request rejected until speed < 10 rpm.

Pass Criteria: Reverse only permitted when stationary AND selected. Tractive effort correctly proportional to throttle. Rejection of reverse while moving forward. Smooth transitions.

Why this covers the requirement: This is the positive test for reverse functionality (complement to S-02 which tests the negative case). Reverse is a legitimate operating mode but must be strictly controlled to prevent H-02 (unintended reverse).

S-15: Overspeed Protection

Objective: Verify system limits maximum motor speed to prevent mechanical overspeed damage.

Covered: SG-06 (ASIL C), FSR-07, H-06

Setup: Traction motor on dyno capable of exceeding rated maximum speed.

Procedure:

1. Command acceleration toward rated maximum speed.
2. Verify speed limit is enforced (tractive effort reduced to zero at speed limit).
3. Attempt to exceed speed limit with external dyno assistance. Verify system resists overspeed (regen if needed).
4. Verify encoder-derived speed agrees with back-EMF-derived speed (plausibility).
5. Verify overspeed fault is logged.

Pass Criteria: Speed does not exceed programmed limit by >5%. Tractive effort reduced (not suddenly cut) as speed approaches limit. Overspeed fault logged. No mechanical damage.

Why this covers the requirement: Mechanical overspeed of the traction motor can cause rotor burst (permanent magnet disintegration, centrifugal failure). The speed limit is a hard limit that must be enforced even with external forcing (downhill, towed).

S-16: Modulation Scheme Transition During Acceleration — Torque Blip

Objective: Verify modulation scheme transitions do not produce perceptible torque disturbances during acceleration. Transition gating logic must inhibit switches during high di/dt.

Covered: SG-03 (ASIL C), FSR-05, H-03, H-03a

Setup: Traction motor on dynamometer with high-bandwidth torque transducer (≥10 kHz bandwidth). DC link at nominal voltage. Automatic modulation map enabled with at least 3 adjacent schemes (e.g., ARSVPWM → SVPWM → SHEPWM).

Procedure:

1. Command smooth acceleration from standstill through the speed range that triggers automatic scheme transitions.
2. Record torque transducer output at ≥20 kSPS during each scheme transition.
3. Measure peak torque deviation from pre-transition mean during the crossfade window.
4. Repeat with manual scheme switch commanded via RTE during hard acceleration (>80% torque request).
5. Verify the gating logic rejects or delays the manual switch during high di/dt.
6. Repeat at multiple operating temperatures (25°C, 60°C, 85°C ambient).

Pass Criteria: Automatic transitions produce peak torque deviation <2% of rated torque. Manual switches during hard acceleration are either gated (delayed until di/dt falls) or produce <5% deviation with crossfade. No fault trips during any transition. No audible tonal change at transition point.

Fail Criteria: Torque deviation >5% on automatic transition; ungated manual switch produces perceptible torque blip; false fault trip during transition.

Why this covers the requirement: H-03 and H-03a (sudden loss of tractive effort) are triggered by any abrupt torque change. Modulation transitions change the harmonic structure of the voltage output, which can momentarily disturb the current controller. The bumpless crossfade and di/dt gating are the safety mechanisms that prevent this disturbance from becoming a hazard.

S-17: Modulation Scheme Transition During Regenerative Braking — Torque Blip

Objective: Verify modulation scheme transitions do not produce perceptible torque disturbances during regenerative braking. Uncommanded torque disturbance during regen must be detected by FSR-06.

Covered: SG-05 (ASIL C), FSR-06, H-05, H-03

Setup: Traction motor on dynamometer in speed-controlled mode driving the DUT into regen. High-bandwidth torque transducer. Automatic modulation map enabled.

Procedure:

1. Establish steady-state regenerative braking at 50% rated regen torque.
2. Command a scheme switch via RTE during active regen (high di/dt in d-axis current).
3. Record torque transducer output during the transition.
4. Ramp dyno speed through the automatic map boundary that triggers a scheme change during regen.
5. Verify automatic transition behavior at the regen/acceleration crossover point (zero torque crossing).
6. Verify FSR-06 (uncommanded regen monitor) does not false-trip during legitimate scheme transitions.

Pass Criteria: Torque deviation during regen transition <3% of rated torque. No false trip of uncommanded regen monitor (FSR-06) during legitimate transitions. Transition at zero-torque crossing is clean (no direction ambiguity).

Fail Criteria: Torque blip >5% during regen transition; FSR-06 false trip during legitimate scheme change; torque direction ambiguity at zero crossing.

Why this covers the requirement: H-05 (uncommanded regenerative braking) covers any unexpected deceleration. A badly managed modulation transition during regen can produce a torque blip that feels like uncommanded braking. FSR-06 must distinguish between legitimate control transitions and true uncommanded regen faults.

S-18: Hysteresis at Modulation Scheme Boundary — No Jitter

Objective: Verify hysteresis prevents rapid back-and-forth switching when operating conditions oscillate near a scheme boundary.

Covered: SG-03 (ASIL C), H-03

Setup: Traction motor on dynamometer. Automatic modulation map with at least one speed-based boundary (e.g., SVPWM at <80% base speed, SHEPWM at >80%). Hysteresis band configurable (default 5%).

Procedure:

1. Operate motor at a speed just below the SVPWM→SHEPWM boundary (e.g., 78% base speed with boundary at 80%).
2. Introduce small speed oscillation (±2% of base speed, 1 Hz) via dyno to simulate road speed variation.
3. Count scheme switches over a 60-second interval.
4. Repeat with speed just above the boundary (82%).
5. Repeat with hysteresis band reduced to 1% and then increased to 10%. Verify switching count scales with hysteresis width.
6. Verify no audible jitter (clicking, tonal change) at the boundary under any condition.

Pass Criteria: With default 5% hysteresis, zero scheme switches during ±2% speed oscillation at the boundary. With 1% hysteresis, occasional switches are acceptable but must not exceed 1 per 10 seconds. No audible jitter under any test condition.

Fail Criteria: Rapid switching (>1 per second) at boundary with default hysteresis; audible jitter; torque ripple correlating with switching frequency.

Why this covers the requirement: Operating at a scheme boundary is a common real-world scenario (cruising near a speed threshold). Without hysteresis, every small speed variation would trigger a scheme switch, producing a continuous stream of torque micro-disturbances. This is a degradation of H-03 (sudden loss of tractive effort) — not a single event, but a chronic instability that erodes rider confidence and could cause loss of control.

S-19: Full Modulation Map Traversal — End-to-End

Objective: Verify the complete automatic modulation map functions correctly across all speed and torque regions with no dead zones, incorrect selections, or fault trips.

Covered: SG-03 (ASIL C), H-03, H-03a, H-05

Setup: Traction motor on dynamometer. Full automatic modulation map programmed with all planned schemes. High-bandwidth torque transducer.

Procedure:

1. Define a modulation map covering the full speed range (0 to field-weakening max) and torque range (zero to rated, both motoring and regen).
2. Command a slow speed sweep (1% base speed per second) from standstill to maximum speed at constant 50% torque.
3. Record active scheme, torque, and DC link voltage throughout the sweep.
4. Verify correct scheme is selected in each region per the map.
5. Verify all transitions are clean (no fault trips, no >2% torque deviation).
6. Repeat sweep at 25% torque and 75% torque.
7. Repeat in reverse direction (deceleration) to verify hysteresis is direction-independent.

Pass Criteria: Correct scheme selected in every region. All transitions clean. No fault trips. Torque deviation <2% at every transition. Map behavior is identical in acceleration and deceleration directions.

Fail Criteria: Wrong scheme selected in any region; transition causes fault trip; torque deviation >5% at any transition; different behavior accelerating vs. decelerating.

Why this covers the requirement: This is the integration test for the entire multi-modulation subsystem. Individual transition tests (S-16, S-17) validate specific boundary crossings; this test validates that all boundaries work together correctly in a realistic driving profile. Any map programming error, incorrect boundary definition, or interaction between adjacent transitions will be caught.

10.5 Integration Tests

Integration tests validate the interaction between the traction inverter/VCU and external systems connected via CAN bus and discrete I/O. These tests require the VCU connected to representative CAN nodes (or CAN simulation tools).

I-01: CAN1 (BMS) Heartbeat Loss

Objective: Verify FSR-17 — BMS CAN loss triggers safe degradation.

Covered: SG-01 (ASIL D), H-03, FSR-17

Setup: VCU connected to CAN bus with BMS simulator. System in normal operation producing tractive effort.

Procedure:

1. Establish normal CAN1 communication with BMS simulator sending heartbeat every 5 s.
2. Stop BMS heartbeat (simulate BMS failure or CAN1 bus fault).
3. Measure time from last heartbeat to safe state entry.
4. Verify safe-state defaults applied: tractive effort restricted to zero.

Pass Criteria: Safe state entered within 5 s + margin (<6 s total). Tractive effort ramped to zero (FSR-05). Fault logged. System does not attempt to operate without BMS data. No erroneous tractive effort produced.

Why this covers the requirement: The BMS provides cell voltage, temperature, and fault status. Operating without BMS data risks over-discharge, over-temperature, or use of a battery in fault condition. FSR-17 defines the 5-second timeout (external to VCU domain). This test validates the VCU correctly responds to loss of this critical external data source.

I-02: CAN2 (IO Board) Heartbeat Loss

Objective: Verify FSR-17 — IO board CAN loss triggers safe-state defaults.

Covered: SG-01 (ASIL D), H-03, FSR-17

Setup: VCU connected to CAN2 with IO board simulator. System in normal operation.

Procedure:

1. IO board simulator sending heartbeat every 1 s with brake=off, kickstand=up, valid state.
2. Stop IO board heartbeat.
3. Measure time from last heartbeat to safe state entry.
4. Verify safe-state defaults: brake=pressed, kickstand=down, all external inputs invalid.

Pass Criteria: Safe state entered within <1.5 s. Tractive effort zero. Fault logged. Safe-state defaults correctly applied (brake pressed prevents any tractive effort, kickstand down prevents any tractive effort).

Why this covers the requirement: The IO board provides brake switch, kickstand switch, and other safety-critical inputs. Loss of this node means the VCU cannot verify these safety conditions. FSR-17 requires assuming the worst-case safe defaults. This is more aggressive than BMS loss (1 s vs. 5 s) because the IO board provides real-time safety interlocks.

I-03: Simultaneous Throttle + Brake Request

Objective: Verify SG-01, FSR-01, FSR-18 — Brake request always overrides throttle.

Covered: SG-01 (ASIL D), H-01, FSR-01, FSR-18

Setup: VCU with IO board simulator. System in normal operation.

Procedure:

1. Apply 75% tractive effort via throttle pots (both valid, matched).
2. While maintaining throttle, set IO board brake=pressed.
3. Verify tractive effort immediately commands to zero.
4. Release brake. Verify tractive effort returns to rider-requested value (ramped, not step).
5. Repeat with kickstand=down instead of brake.

Pass Criteria: Brake or kickstand signal immediately overrides throttle. Tractive effort commands to zero within <100 ms. On release, tractive effort ramps back up at ≤500 Nm/s (no step). Fault logged for kickstand-down-at-speed condition.

Why this covers the requirement: This is the fundamental rider override. In any fault scenario where the throttle is stuck on, the rider's brake lever is the last-resort safety mechanism. The test validates that the brake input (via IO board CAN) unconditionally overrides any tractive effort command. The kickstand-down-at-speed is a special case that also demands zero tractive effort (rider likely did not intend to ride with kickstand down).

I-04: HVIL Interruption During Operation

Objective: Verify SG-09, SG-11, FSR-10 — HVIL interruption triggers immediate HV disconnect sequence.

Covered: SG-09 (ASIL A), SG-11 (ASIL B), H-09, H-11, FSR-10

Setup: VCU with HVIL loop simulator. System in normal operation producing tractive effort. Contactor control observable.

Procedure:

1. System operating at 50% tractive effort. HVIL loop closed (normal).
2. Open HVIL loop (simulate connector disconnect, crash sensor, or interlock fault).
3. Measure time from HVIL open to PWM disable.
4. Verify contactor open request sent to BMS within <50 ms.
5. Verify system does not re-enable PWM until HVIL is restored AND key cycle occurs.

Pass Criteria: PWM disabled within <50 ms of HVIL interruption. Contactor open request sent. Safe state entered. No tractive effort produced after HVIL open. Latch requires key cycle to clear (prevents automatic restart in unsafe condition).

Why this covers the requirement: HVIL is the primary HV safety interlock. H-09 (HV isolation failure) and H-11 (contactor welding) are both addressed by FSR-10. A crash or connector fault must immediately disconnect HV. The 50 ms timing requirement ensures the response is fast enough to prevent shock or fire in a post-crash scenario. The latch prevents the system from automatically re-energizing HV while the fault condition persists.

I-05: DC Link Overvoltage (Regen Event)

Objective: Verify SG-10, FSR-11 — DC link bus overvoltage is detected and limited.

Covered: SG-10 (ASIL B), H-10, FSR-11

Setup: VCU with DC link bus supply. System in regenerative braking operation (traction motor being driven externally to simulate downhill).

Procedure:

1. System in regen mode, DC link at nominal voltage.
2. Raise DC link bus voltage gradually (using external supply) toward warning threshold.
3. Verify regen is disabled at warning threshold (prevent further voltage rise).
4. Continue raising DC link to critical overvoltage threshold.
5. Verify SSO entered within <50 ms.

Pass Criteria: Regen disabled at warning threshold. SSO at critical threshold within <50 ms. Both events logged with distinct DTCs. No IGBT damage at any point.

Why this covers the requirement: H-10 is DC link bus overvoltage from regenerative braking (especially on a fully charged battery or during a long downhill). Excessive DC link voltage can rupture capacitors and IGBTs, creating arc flash and fire. FSR-11 provides two thresholds: warning (disable regen, prevent further rise) and critical (immediate SSO). This test validates both thresholds and response times.

I-06: Kickstand Down at Speed

Objective: Verify system responds safely to kickstand-down signal while vehicle is moving.

Covered: SG-01 (ASIL D), H-01 (indirect), FSR-17

Setup: VCU with IO board simulator. System producing tractive effort.

Procedure:

1. System operating at 50% tractive effort. IO board reports kickstand=up, speed >10 mph equivalent.
2. IO board reports kickstand=down (simulating mechanical failure or sensor fault).
3. Verify tractive effort commands to zero immediately.
4. Verify fault logged (kickstand down at speed).
5. Verify tractive effort remains zero even if throttle is applied while kickstand=down.

Pass Criteria: Tractive effort zero within <200 ms of kickstand-down signal. Fault logged. Tractive effort remains inhibited while kickstand=down. Safe state persists until key cycle.

Why this covers the requirement: Riding with the kickstand down is a dangerous condition (can catch on pavement, causing loss of control). The IO board provides the kickstand state via CAN2. This test validates that the VCU correctly interprets this input as a safety interlock and prevents any tractive effort when the kickstand is reported down, regardless of throttle position.

I-07: Simultaneous Multiple CAN Node Loss

Objective: Verify graceful degradation when multiple external systems fail simultaneously.

Covered: SG-01, SG-03, FSR-17

Setup: VCU connected to both CAN1 (BMS) and CAN2 (IO board, display, charger) simulators.

Procedure:

1. System operating normally. All CAN nodes active.
2. Simultaneously stop all CAN2 node heartbeats (IO board + display + charger + ABS).
3. Verify system enters safe state based on CAN2 timeout (1 s).
4. Restore CAN2. Verify system returns to normal operation after key cycle (does not auto-resume).

5. Simultaneously stop both CAN1 (BMS) and CAN2 (IO board) heartbeats.
6. Verify system enters safe state (whichever timeout fires first: CAN2 at 1 s or CAN1 at 5 s).

Pass Criteria: Safe state entered on first heartbeat timeout to fire. No tractive effort produced during any CAN-loss scenario. System does not auto-recover without key cycle. All fault conditions logged with appropriate DTCs.

Why this covers the requirement: Multiple simultaneous failures can occur in a crash or severe EMI event. The system must degrade to the safest possible state, not attempt to continue operating with partial external data. The test validates that the most restrictive timeout dominates and that the system requires a key cycle to resume (prevents oscillation between fault and operation).

I-08: CAN Bus Fuzzing — Random Frame Injection

Objective: Verify system ignores or gracefully handles random/corrupted CAN frames.

Covered: SG-01 (ASIL D), FSR-17, H-01, H-15

Setup: VCU on CAN bus with fuzzing tool (CANoe, PCAN, or custom tool). System in normal operation.

Procedure:

1. Generate random CAN frames at 10% bus load: random IDs (including IDs used by BMS and IO board), random data lengths (0–8 bytes), random payload.
2. Verify VCU does not malfunction, produce tractive effort, or enter safe state from random frames.
3. Increase to 50% bus load with random frames. Verify normal operation continues.
4. Inject frames with correct IDs but corrupted CRC. Verify frames are rejected by CAN controller.
5. Inject frames with valid BMS ID but payload that would indicate impossible values (e.g., cell voltage = 0xFFFF, temperature = 200°C). Verify sanity checks reject impossible values.

Pass Criteria: No malfunction from random frames. Valid heartbeat frames still processed correctly. Impossible values rejected. Bus load < 100% maintained.

Why this covers the requirement: CAN bus noise, failed nodes, or malicious injection can send unexpected frames. The VCU must be robust against parsing errors that could lead to incorrect tractive effort (H-15). This validates input validation and parser robustness.

I-09: CAN Bus Load Test (95% Utilization)

Objective: Verify system operates correctly under maximum CAN bus load.

Covered: SG-01 (ASIL D), FSR-17, H-03

Setup: CAN bus loaded to 95% utilization with background traffic. VCU operating normally.

Procedure:

1. Load CAN bus to 95% with frames from multiple simulated nodes.
2. Verify VCU transmits its frames without excessive delay (< 10 ms jitter).
3. Verify VCU receives BMS and IO board frames without dropping critical messages.
4. Verify heartbeat timeout still functions correctly (no false timeouts from bus congestion).
5. Inject a fault requiring safe state. Verify safe state entry is not delayed by bus load.

Pass Criteria: System operates normally at 95% bus load. No dropped critical messages. Heartbeat timeout accurate within ±10%. Safe state entry not delayed.

Why this covers the requirement: In a multi-node CAN network, bus load can spike during fault conditions (all nodes sending error frames, diagnostics). The VCU must continue receiving its critical messages even under congestion. This test validates CAN message prioritization and receive buffer sizing.

I-10: CAN Bus Off Recovery

Objective: Verify system correctly recovers from CAN bus-off state.

Covered: SG-01 (ASIL D), FSR-17, H-03

Setup: VCU on CAN bus with fault injection capability.

Procedure:

1. Inject errors on CAN bus to force VCU into bus-off state (TEC > 255).
2. Verify system enters safe state and applies safe defaults.
3. Stop error injection. Allow bus to go idle (11 consecutive recessive bits × 128).
4. Verify VCU CAN peripheral recovers to error-active state.
5. Verify VCU does NOT automatically resume operation (requires key cycle).
6. Perform key cycle. Verify system resumes normal operation.

Pass Criteria: Bus-off detected. Safe state entered. Recovery to error-active per ISO 11898-1. No auto-resume. Key cycle returns to normal operation.

Why this covers the requirement: Bus-off is a severe CAN fault. The recovery protocol (128 × 11 recessive bits) is defined by ISO 11898-1. The VCU must follow this protocol and only resume after explicit user action (key cycle), not automatically.

I-11: Invalid/Corrupted CAN Frame Injection

Objective: Verify system handles frames with valid IDs but semantically invalid data.

Covered: SG-01 (ASIL D), SG-10 (ASIL B), FSR-17, H-15

Setup: VCU on CAN bus with BMS and IO board simulators.

Procedure:

1. BMS simulator sends frame with valid ID but impossible cell voltage (>5 V or <0 V). Verify VCU rejects value and uses last-known-good or safe default.
2. BMS sends temperature = −50°C (impossible for operating battery). Verify rejected.
3. IO board sends brake=pressed AND throttle=100% simultaneously. Verify brake wins (tractive effort = 0).
4. IO board sends kickstand=down AND speed=100 mph. Verify zero tractive effort and fault logged.
5. Send frames with correct ID but DLC mismatch (e.g., DLC=8 for a 4-byte message). Verify parser handles correctly.

Pass Criteria: All invalid frames rejected. Safe defaults used when valid data unavailable. No tractive effort from invalid inputs. DLC mismatch handled safely.

Why this covers the requirement: Semantically invalid data (from a failed node, corrupted memory, or software bug) is more dangerous than random data because it passes CRC and ID filters. The VCU must have range checks, plausibility checks, and consistency checks on all received data.

I-12: Display Node Failure

Objective: Verify system operates safely when the display/dash node fails.

Covered: SG-01 (ASIL D), FSR-17, H-03

Setup: VCU on CAN2 with display simulator. System operating.

Procedure:

1. System operating normally. Display showing speed, SOC, fault status.
2. Stop display heartbeat.
3. Verify VCU continues operating (display is non-safety-critical).
4. Verify VCU logs display loss as non-critical fault.
5. Verify tractive effort is NOT affected by display loss.

Pass Criteria: Display loss does NOT affect tractive effort or safety functions. Fault logged. System continues operating. Display recovery does not require key cycle.

Why this covers the requirement: The display is a non-safety-critical node. Its loss must not affect propulsion. This test validates that the heartbeat timeout correctly distinguishes safety-critical nodes (BMS, IO board) from informational nodes (display).

I-13: Charger Node Interaction

Objective: Verify correct interaction with charger during charging and prevent drive-away-while-charging.

Covered: SG-01 (ASIL D), FSR-04, H-01

Setup: VCU on CAN2 with charger simulator.

Procedure:

1. Charger connected and active (charging in progress). Verify VCU detects charging state.
2. Attempt to command tractive effort while charging. Verify request rejected.
3. Verify interlock: charging active → tractive effort permanently disabled.
4. Charger completes charging (sends completion message). Verify VCU exits charging state.
5. Verify tractive effort available after charger disconnect confirmation.
6. Inject fault: charger stuck "active" after physical disconnect. Verify VCU has timeout to clear charging state.

Pass Criteria: No tractive effort during charging. Clean exit from charging state. Timeout for stuck charger message. Drive-away-prevention interlock functional.

Why this covers the requirement: Driving away while the charging cable is connected is a severe safety hazard (could damage charger, cable, or vehicle). The charger-active signal must unconditionally disable tractive effort. This test validates the interlock and its timeout behavior.

I-14: ABS Node Coordination

Objective: Verify correct interaction with ABS module during anti-lock braking events.

Covered: SG-05 (ASIL C), FSR-06, H-05

Setup: VCU on CAN2 with ABS simulator.

Procedure:

1. System in regenerative braking. ABS simulator sends "ABS active" message.
2. Verify VCU immediately reduces or disables regenerative braking (ABS needs wheel slip control).
3. Verify friction brakes remain fully available (independent of regen).
4. ABS sends "ABS inactive." Verify regen smoothly resumes.

5. Verify no tractive effort produced during ABS active period.

Pass Criteria: Regen disabled/reduced within <50 ms of ABS active. Smooth resume when ABS inactive. No interference with friction brake operation.

Why this covers the requirement: During ABS events, the ABS controller modulates brake pressure to prevent wheel lock. Regenerative braking (which applies torque directly to the wheel) can interfere with ABS slip control. The VCU must yield to ABS during active events. This is a coordination requirement, not a fault case.

I-15: BMS Fault Propagation

Objective: Verify VCU correctly responds to BMS fault messages.

Covered: SG-01 (ASIL D), SG-10 (ASIL B), FSR-17, H-01, H-10

Setup: VCU on CAN1 with BMS simulator. System operating.

Procedure:

1. BMS sends warning-level fault: cell imbalance > 50 mV. Verify VCU logs warning, derates tractive effort.
2. BMS sends critical fault: cell overvoltage (>4.25 V). Verify VCU immediately disables regen and enters safe state.
3. BMS sends critical fault: cell undervoltage (<2.5 V). Verify VCU immediately enters safe state.
4. BMS sends critical fault: over-temperature (>60°C). Verify VCU derates or enters safe state.
5. BMS sends contactor weld detection. Verify VCU logs fatal fault and prevents restart.

Pass Criteria: Warning faults: log + derate. Critical faults: safe state within <200 ms. Contactor weld: fatal fault + lockout. All responses appropriate to fault severity.

Why this covers the requirement: The BMS is the authoritative source for battery safety. The VCU must correctly interpret and act on BMS fault messages with appropriate severity. A BMS critical fault must never be treated as a warning.

I-16: Multi-Node Simultaneous Fault

Objective: Verify system handles simultaneous faults on multiple CAN nodes.

Covered: SG-01 (ASIL D), SG-03 (ASIL C), SG-13 (ASIL D), FSR-17, H-01, H-03, H-13

Setup: VCU on both CAN buses with all node simulators.

Procedure:

1. System operating at 50% tractive effort.
2. Simultaneously: BMS sends critical fault AND IO board stops heartbeat AND ABS sends active.
3. Verify VCU enters safe state (most restrictive response).
4. Verify safe state entry is not delayed by processing multiple faults.
5. Verify all three faults are logged with correct DTCs.
6. Repeat with different combinations of simultaneous faults.

Pass Criteria: Safe state entered within single worst-case timeout (not cumulative). All faults logged. No priority inversion (critical fault not delayed by less critical processing).

Why this covers the requirement: In a crash or severe EMI event, multiple faults can occur simultaneously. The fault handling must be deterministic: always take the most restrictive action, never get confused by multiple inputs, never delay the safe state entry.

I-17: CAN Bus Wiring Fault (Short and Open)

Objective: Verify system response to physical CAN bus wiring faults.

Covered: SG-01 (ASIL D), FSR-17, H-03

Setup: VCU with physical access to CAN bus wiring.

Procedure:

1. System operating normally.
2. Short CAN_H to CAN_L. Verify bus-off within error limit. Verify safe state.
3. Restore wiring. Verify bus-off recovery.
4. Open CAN_H (cut wire). Verify error passive then bus-off. Verify safe state.
5. Restore wiring. Verify recovery.
6. Short CAN_H to +12 V. Verify transceiver protection functions. Verify safe state.
7. Short CAN_L to ground. Verify safe state.

Pass Criteria: All wiring faults detected. Safe state entered within timeout. No transceiver damage from shorts. Recovery after wiring restored (key cycle required).

Why this covers the requirement: Physical CAN wiring faults (chafing, connector corrosion, crash damage) are common in automotive environments. The ISO1042 transceiver has protection against these faults, but the system must still detect the communication loss and enter safe state.

I-18: Wake/Sleep Cycle Test

Objective: Verify system handles wake-from-sleep and sleep-entry correctly.

Covered: SG-13 (ASIL D), FSR-16, H-13

Setup: VCU with sleep/wake capability (if implemented).

Procedure:

1. System in normal operation. Trigger sleep condition (key off, no activity timeout).
2. Verify graceful shutdown sequence before sleep entry.
3. Verify quiescent current in sleep mode (< 1 mA typical).
4. Trigger wake (key on, CAN wake-up frame, charger connection).
5. Verify full startup sequence executes (POST, pre-charge, etc.).
6. Verify system is in known good state after wake (no stale data from before sleep).

Pass Criteria: Sleep entry: graceful shutdown. Sleep current < limit. Wake: full POST executes. No stale data. System behavior identical to cold start.

Why this covers the requirement: If sleep/wake is implemented, wake must be treated as a fresh start — not a continuation of the previous session. Stale data (old fault status, old sensor values) must be discarded. This prevents a fault that occurred before sleep from being masked after wake.

10.6 Safety Goal Traceability

The following matrix maps each safety goal to the test cases that provide evidence of coverage. A test may cover multiple safety goals; a safety goal may require multiple tests.

Table 16 — Safety Goal to Test Case Traceability

SG	Target	Safety Goal	Covering Tests
SG-01	D	Prevent unintended positive tractive effort	C-01–C-06, C-13, C-14, C-20–C-25, C-39–C-41, C-44, S-01, S-02, I-01–I-03, I-06–I-08, I-11, I-15–I-17, E-08, E-09
SG-02	B	Prevent unintended reverse tractive effort	S-02, S-14, C-13
SG-03	C	Safe degradation on loss of tractive effort (≤200 Nm/s)	C-22, S-03, S-04, S-10–S-13, I-01, I-02, I-07
SG-04	A	Ensure regen availability (friction brakes backup)	C-11 (indirect: system preserves brake function independently)
SG-05	C	Prevent unintended regenerative braking	S-05, S-08, I-14
SG-06	C	Limit max tractive effort to calibrated max	C-07, C-08, C-13, C-41, S-09, S-15
SG-07	B	Detect over-temperature, progressively derate	C-09, C-10, C-35, C-45, S-06, S-07, E-04
SG-08	C	Detect loss of rotor position → safe state	C-11, C-28, C-29, C-30
SG-09	A	Maintain HV isolation	C-50, I-04, E-07, E-12
SG-10	B	Detect DC link bus overvoltage	C-12, C-46, I-05, S-08, S-13
SG-11	B	Detect HVIL interruption	I-04
SG-12	C	Prevent IGBT shoot-through	C-15, C-16, C-17, C-31–C-34, C-47–C-49
SG-13	D	Achieve safe state within 200 ms	C-13, C-14, C-15, C-16, C-17, C-19–C-25, C-37, C-38, C-40, C-43, C-44, S-01–S-03, S-10–S-12, I-04, E-01–E-03, E-10, E-11
SG-14	C	Detect gate driver fault	C-15, C-16, C-17, C-26, C-27, C-31–C-34
SG-15	C	Detect PWM deadtime violations	C-15, C-16, C-49 (DESAT indirectly covers stuck-on)

10.7 Coverage Justification

10.7.1 FSR Coverage Matrix

Table 17 — Functional Safety Requirement Coverage by Test Cases

FSR	Requirement Summary	Covering Tests
FSR-01	Dual throttle pot discrepancy >5% → safe state	C-01–C-06, S-01, I-03
FSR-02	Tractive effort command vs. measured current plausibility	C-07, C-08, C-28, C-29, C-30, C-41, S-01
FSR-03	Tractive effort rate limit ≤500 Nm/s	S-01, S-03
FSR-04	Reverse interlock (stationary + selected)	S-02, S-14, I-13
FSR-05	Fault ramp-down ≤200 Nm/s before SSO	S-03, S-04, S-11, S-13
FSR-06	Uncommanded regen >10 Nm → safe state	S-05, S-08, I-14

FSR	Requirement Summary	Covering Tests
FSR-07	Max tractive effort limit (SW LUT on both MCUs + dual-MCU current monitoring)	C-07, C-08, S-09, S-15
FSR-08	3 modules, 1 NTC per module, 2oo3 voting with 100 °C hard cap	C-09, C-10, C-35, C-45, S-06, S-07
FSR-09	Encoder loss → safe state <100 ms	C-11, C-28, C-29, C-30
FSR-10	HVIL interruption → PWM disable + contactor request <50 ms	I-04, C-50
FSR-11	DC link OV monitor: regen disable / SSO	C-12, C-46, I-05, S-08, S-13
FSR-12	Gate driver complementary inputs, anti-shoot-through	C-15, C-16, C-47–C-49
FSR-13	NCV57100 DESAT <2 us PWM disable	C-15, C-16, C-31–C-34
FSR-14	Breakpoint HW PWM disable <10 us	C-13, C-14, C-23–C-25
FSR-15	Windowed watchdog ≤50 ms	C-13, C-14, C-37, C-38, C-44
FSR-16	Power-on self-test (POST)	C-16, C-20, S-10, S-12
FSR-17	CAN heartbeat timeout with safe defaults	I-01, I-02, I-07, I-08–I-11, I-17
FSR-18	Throttle limit switch independent override	C-06, S-01, I-03
FSR-19	Boot CRC-32	C-20, C-39
FSR-20	ECC RAM: SECDED + double-bit → safe state	C-18, C-19
FSR-21	DC link undervoltage derate / safe state	C-12, C-22, S-13

10.7.2 Hazard Coverage Matrix

Table 18 — Hazard Coverage by Test Cases

Hazard	Description	Covering Tests
H-01	Unintended positive tractive effort	C-01–C-06, C-13, C-14, C-20, C-23–C-25, C-39–C-41, C-44, S-01, I-03, I-06–I-08, I-11, I-15–I-17, E-08, E-09
H-02	Unintended reverse tractive effort	S-02, S-14, C-13, I-13
H-03	Sudden loss of tractive effort	C-22, S-03, S-10, S-11, S-13, I-01, I-02, I-07, I-17
H-03a	Loss of tractive effort during cornering	S-04 (dyno portion)
H-04	Inability to produce regen	C-11, C-30 (indirect)
H-05	Uncommanded regenerative braking	S-05, S-08, I-14
H-06	Excessive tractive effort	C-07, C-08, C-41, S-01, S-09, S-15
H-07	Over-temperature	C-09, C-10, C-35, C-45, S-06, S-07, E-04, E-07
H-08	Motor overspeed / encoder loss	C-11, C-28, C-29, C-30, S-15
H-09	HV isolation failure	C-50, I-04, E-07, E-12
H-10	DC link bus overvoltage	C-12, C-46, I-05, S-08, S-13
H-11	HV contactor welding	I-04, C-46

Hazard	Description	Covering Tests
H-12	IGBT shoot-through	C-15, C-16, C-31–C-34, C-47–C-49
H-13	Failure to execute safe state	C-13, C-14, C-15, C-21–C-25, C-37, C-38, C-40, C-43, C-44, I-04, S-10, S-12, E-01–E-03, E-10, E-11
H-14	Corrupted/latched tractive effort	C-13, C-14, C-19, C-20, C-37, C-38, C-39, C-44
H-15	Incorrect tractive effort from software error	C-07, C-08, C-20, C-39, C-40, S-01, I-08, I-11
H-16	Gate driver fault not acted upon	C-15, C-16, C-17, C-26, C-27, C-31–C-34
H-17	PWM deadtime violation / stuck-on	C-15, C-16, C-49

10.8 Pass/Fail Criteria

All test cases in this plan use the following standardized pass/fail definitions. A test is **PASSED** only when all criteria are met. A test is **FAILED** if any criterion is not met.

Table 19 — Test Pass/Fail Criteria Definitions

Criterion	Pass Definition	Fail Definition
Safe State Entry	System enters defined safe state (SSO, zero tractive effort) within specified time limit. PWM outputs disabled. No tractive effort produced.	Safe state not entered within time limit. Tractive effort continues. PWM remains active.
Detection Time	Fault detected and logged within specified ms limit. DTC recorded with correct code.	Detection exceeds time limit. Incorrect DTC. No DTC recorded.
Torque Ramp Rate	Measured torque change rate ≤ specified Nm/s. Smooth, monotonic decay. No step changes.	Torque rate exceeds limit. Step change observed. Oscillation or non-monotonic behavior.
HW Response	Hardware protection (DESAT, breakpoint, watchdog) responds within specified us/ms limit, independent of software state.	HW response exceeds limit. Response depends on software execution. HW does not trigger.
Latching	Fault condition remains latched until key cycle or explicit reset. No automatic restart into operation.	System auto-recovers without reset. Fault condition clears spontaneously. Intermittent operation.
Logging	Fault recorded in non-volatile memory. DTC code matches fault type. Timestamp recorded.	Fault not logged. Incorrect DTC. Volatile-only storage (lost on power cycle).
No Degradation	System behavior outside the fault path is unaffected. Other safety functions remain operational.	Cascade failure: one fault causes unrelated safety function to fail. Side effects on non-fault paths.

10.8.1 Overall Test Plan Assessment

The test plan as a whole is assessed against the following criteria:

Table 20 — Overall Test Plan Assessment Criteria

Assessment	Definition
ALL TESTS PASSED	Every test case in Sections 10.3–10.5 achieves PASS on all criteria. All safety goals have at least one passing test. All FSRs have at least one passing test. The system is considered validated for its achievable ASIL level (ASIL D for SG-01, SG-13; ASIL B for SG-02, SG-07; ASIL C for SG-06, SG-12, SG-14, SG-15).
PASSED WITH EXCEPTIONS	All critical tests (covering P0 gaps) pass. Minor tests may fail with documented workarounds. Exceptions do not affect safety of the achievable ASIL claims. Mitigation plans documented for each exception.
FAILED — NOT ROADWORTHY	Any P0-gap test fails (throttle monitoring, safe state entry, watchdog response). System must not be operated on public roads until resolved. Hardware or software modifications required.
INCONCLUSIVE	Tests could not be executed due to equipment limitations or test environment constraints. Results insufficient for safety validation. Additional testing required before operation.

10.9 Known Test Limitations

This test plan is honest about its limitations. The following conditions cannot be fully validated by the test cases described and require additional verification methods.

LIMIT-01: Vehicle-Level Dynamics Testing (H-03a Cornering)

Limitation: Test S-04 validates the torque ramp rate on a dyno but cannot replicate the full motorcycle-cornering-dynamics scenario. The interaction between tractive effort removal, tire friction envelope, chassis geometry, rider position, and lean angle can only be validated on a real motorcycle with an experienced test rider on a controlled track.

Mitigation: (1) Dyno validates the controllable portion (ramp rate ≤200 Nm/s). (2) Track testing with graduated lean angles and known fault injection confirms no stand-up/run-wide behavior. (3) Rider feedback confirms controllability. This is essential before racetrack use.

LIMIT-02: Environmental Stress (Temperature, Vibration, EMI)

Limitation: Component and system tests are conducted at ambient temperature on a bench. The test plan does not include environmental stress screening (thermal chamber, vibration table, EMC chamber) that would be required for production qualification.

Mitigation: (1) Railway-grade DC/DC (EC7BW-110S12) qualified to EN 50155 provides confidence in power supply robustness. (2) Gate drivers automotive-qualified (AEC-Q100) for temperature. (3) EMC and environmental testing recommended before production but not required for prototype safety validation.

LIMIT-03: Long-Term Aging and Wear

Limitation: Tests are conducted on relatively new hardware. Long-term effects (capacitor aging, contactor wear, solder fatigue, connector fretting) are not covered.

Mitigation: (1) POST (FSR-16) catches degradation in critical protection circuits. (2) Regular inspection and maintenance intervals defined in service documentation. (3) BMS monitors cell degradation (outside VCU scope).

LIMIT-04: Single-Encoder Limitation (H-08)

Limitation: The system has a single encoder. There is no redundant rotor position sensor. All FSR-09 testing validates that the system *detects* encoder loss and enters safe state, but it cannot *prevent* loss of control during the detection window (up to 100 ms).

Mitigation: (1) Keep detection window as short as possible (<50 ms target). (2) Sensorless fallback if implemented: bounded to ≤2 s and ≤30% tractive effort. (3) Document limitation: rider must be prepared for occasional safe-state entry due to encoder-related faults.

LIMIT-05: Common-Cause MCU Failure (with Shared 3.3V Rail)

Limitation: The main STM32H723 and NCV57100 gate driver logic side share the same 3.3V supply rail — there is no independent 3.3V regulator for the gate drivers. However, this shared rail provides a *passive* safe-state path: total loss of 3.3V → NCV57100 VDD lost → internal active pull-down forces all IGBT gates to 0V → SSO, with no software involvement (Path 3). The dual-MCU architecture mitigates common-cause failures: the coprocessor has its own oscillator, independent ADC channels, and independent power kill (GATE_DRIVE_PWR2_ENABLE). A common-cause affecting the main MCU (clock failure, supply collapse, latch-up) does not disable the coprocessor's independent monitoring and safe state actuation.

Mitigation: (1) Six redundant SSO pathways (Path 2a/2b are redundant channels of one 1oo2 power-kill pathway): Path 1 = TIM1_BKIN hardware (<100 ns); Path 2a/2b = GATE_DRIVE_PWR1_ENABLE (main, with GATE_DRIVE_PWR1_FB) / GATE_DRIVE_PWR2_ENABLE (coprocessor, with GATE_DRIVE_PWR2_FB); Path 3 = shared 3.3 V passive pull-down; Path 4 = GATE_DRIVE_RESET (either MCU, <1 us); Path 5 = coprocessor watchdog → NRST (~100 ms); Path 6 = coprocessor independent fault trigger. (2) 1oo2 power kill: either GATE_DRIVE_PWR1_ENABLE or GATE_DRIVE_PWR2_ENABLE low achieves SSO. (3) Dual-MCU architecture enables ASIL D via ASIL B(D) + ASIL B(D) decomposition.

LIMIT-06: Gate Driver Non-ASIL Status

Limitation: The NCV57100 gate drivers are qualified to AEC-Q100, not ASIL. Internal DESAT, UVLO, and anti-shoot-through protections are not independently monitored. Tests C-15 through C-17 validate these protections function but do not provide ASIL credit.

Mitigation: (1) No ASIL credit claimed for gate driver internal protections (Table 13 reflects ASIL C achievable for SG-12, SG-14). (2) Coprocessor PWM monitoring (future) enables ASIL C claim. (3) OR'd FLT input provides basic monitoring with known single-point limitation.

LIMIT-07: Software Test Library — Class B vs. Class D

Limitation: ST's publicly available X-CUBE-CLASSB library is certified to IEC 60730-1 Annex H (Class B) only. The ISO 26262-assessed Class D STL (X-CUBE-STL) requires a manufacturer NDA and is not available for open-source projects. The following ASIL process requirements are therefore not met:

- No FMEDA / SPFM/LFM metrics derived for this specific STM32H723ZG hardware configuration
- No ISO 26262 software tool qualification (compiler, static analysis, test tools)
- No fault injection testing campaign with structural coverage evidence
- No independent safety assessment or third-party audit

This project implements ASIL decomposition as a **design and educational exercise only**. A production system targeting ISO 26262 compliance would require re-implementation under an ISO 26262-compliant development process, use of a certified STL (e.g., X-CUBE-STL under NDA from ST), or use of a pre-certified MCU + STL combination from an alternative supplier.

Mitigation: (1) X-CUBE-CLASSB provides CPU register test, RAM March-C test, and flash CRC — adequate for POST but not ASIL-traceable. (2) All software safety mechanisms are independently tested

> via fault injection (99 tests). (3) Document this gap explicitly: achievable ASIL is limited by STL certification status.
>
> **LIMIT-08: No Dependent Failure Analysis (DFA)**
>
> **Limitation:** ISO 26262-9 requires a Dependent Failure Analysis for ASIL C and above. No DFA has been performed. This means common-cause and cascading failures between the six SSO paths (e.g., a PCB delamination affecting multiple traces; a single ground bounce affecting both break inputs) are not formally analyzed. Without DFA, the independence claim for ASIL B(D) decomposition cannot be fully substantiated.
>
> **Mitigation:** (1) Physical separation of TIM1_BKIN, GATE_DRIVE_PWR1_ENABLE, and GATE_DRIVE_PWR2_ENABLE routing on the PCB. (2) Independent ground references where possible. (3) The coprocessor's independent oscillator reduces common-cause with the main MCU; the +3.3 V rail shared by both MCUs is supervised by the TPS389006-Q1, which detects brownout and asserts the shared GATE_DRIVER_FAULT line monitored by both MCUs. (4) DFA to be performed before any formal ASIL D audit. (5) Documented as a blocker for formal ASIL D claim.
>
> **LIMIT-09: No EMI/EMC Pre-Compliance Assessment**
>
> **Limitation:** The fault injection test plan includes environmental EMI tests (E-08 radiated immunity, E-09 conducted immunity, E-10/E-11 ESD), but no pre-compliance EMC assessment or design margin analysis has been performed. The inverter will be installed in unknown vehicle electromagnetic environments (ignition noise, fuel pump transients, radio transmitters, cell phones). EMI-induced ADC noise, CAN errors, or false triggering of the TIM1 break input are real risks that have not been quantified.
>
> **Mitigation:** (1) Environmental tests E-08 and E-09 provide a baseline when equipment is available. (2) CISPR 25 pre-compliance testing recommended before any field deployment. (3) PCB design includes best-practice grounding, shielding, and filtering. (4) Documented as an open risk for aftermarket installation.

10.10 Environmental and Stress Tests

Environmental tests validate the hardware's ability to withstand the physical and electromagnetic conditions encountered in motorcycle operation. These tests require specialized equipment (thermal chamber, vibration table, EMC chamber) and are typically conducted as type tests on a representative unit rather than on every production unit. They are included here as a reference test plan for when equipment becomes available.

10.10.1 Vibration Tests

E-01: Random Vibration — Operating

Objective: Verify no mechanical failure or electrical intermittent under random vibration profile.

Covered: SG-13 (ASIL D), FSR-16, H-13 (mechanical failure mode)

Setup: Vibration table with random vibration capability. DUT mounted in production-intent mounting orientation. Accelerometers on DUT.

Procedure:

1. Apply random vibration profile per ISO 16750-3, Test VII — Passenger car, sprung masses: 10–1000 Hz, 5.0 g RMS, 8 hours per axis (X, Y, Z).

2. Operate system at 50% tractive effort during vibration (motor running on dyno, electronics vibrating).
3. Monitor for: intermittent connections, CAN errors, sensor dropouts, reset events, unexpected safe state entries.
4. After test: visual inspection for cracked solder joints, loose fasteners, connector back-out, component detachment.
5. Run functional test (C-01 through C-05) post-vibration and compare to baseline.

Pass Criteria: No intermittent faults during vibration. No CAN errors above baseline. No resets. Post-vibration functional test passes. No visible mechanical damage.

Why this covers the requirement: Motorcycle operation subjects electronics to severe vibration from engine/motor imbalance, road irregularities, and wheel imbalance. The random vibration profile simulates the broadband energy of real road conditions. Operating during vibration (rather than just powered) exercises electrical connections under mechanical stress.

E-02: Sinusoidal Vibration Sweep — Resonance Search

Objective: Identify mechanical resonances that could cause fatigue failure.

Covered: SG-13 (ASIL D), H-13 (mechanical)

Setup: Sine vibration table. DUT with accelerometers at critical locations (PCB center, large components, connector backshells).

Procedure:

1. Sweep 5–500 Hz at 1.0 g peak, 1 octave/minute.
2. Record transfer function (response/input) at each accelerometer.
3. Identify resonance frequencies (Q > 10 considered high risk).
4. Dwell at each resonance frequency for 10 minutes at 1.0 g.
5. Repeat for all three axes.

Pass Criteria: No resonances with Q > 20. No failures after dwell. PCB resonant frequency > 200 Hz (above primary road vibration energy). Post-test functional verification passes.

Why this covers the requirement: Resonance causes amplification of vibration at specific frequencies. A PCB resonating at 80 Hz (common wheel imbalance frequency on motorcycles) can experience 10–20x amplification, causing solder fatigue in hours instead of years. Identifying resonances allows redesign (stiffening, damping, relocation) before production.

E-03: Mechanical Shock

Objective: Verify survival of pothole/curb-impact shock events.

Covered: SG-13 (ASIL D), H-13 (mechanical)

Setup: Shock test table. DUT in production mounting.

Procedure:

1. Apply half-sine shock: 50 g, 11 ms duration, 3 shocks per direction (±X, ±Y, ±Z).
2. Monitor for electrical intermittent during shock (system powered but not operating).
3. Post-shock: visual inspection and functional test.

Pass Criteria: No electrical intermittent. No mechanical damage. Post-shock functional test passes.

Why this covers the requirement: Pothole impacts and curb strikes create severe shock loads. Large components (DC link capacitors, heatsinks) have the highest inertia and experience the greatest stress. Connector back-out is a common shock failure mode.

10.10.2 Thermal Tests

E-04: High Temperature Soak

Objective: Verify correct operation at maximum expected ambient temperature.

Covered: SG-07 (ASIL B), FSR-08, H-07

Setup: Thermal chamber. DUT operating on dyno.

Procedure:

1. Ramp ambient to +60°C (desert parking + solar load).
2. Stabilize for 1 hour (DUT at thermal equilibrium).
3. Operate at 100% load for 2 hours.
4. Monitor all temperatures. Verify no thermal protection false trips.
5. Verify all electronic functions operate correctly at temperature.

Pass Criteria: All functions operate correctly. IGBT temperature < $T_{j,max}$ − 20°C margin. Capacitor temperature < rated max − 10°C. No thermal derate required within 30 minutes.

Why this covers the requirement: Desert operation with solar load can raise under-seat ambient to +60°C or higher. All semiconductor devices must operate within their safe operating area. Thermal protection thresholds must not false-trip at high ambient but must still protect against genuine over-temperature.

E-05: Low Temperature Soak

Objective: Verify correct operation at minimum expected ambient temperature.

Covered: SG-13 (ASIL D), FSR-16, H-13

Setup: Thermal chamber at −20°C.

Procedure:

1. Soak DUT at −20°C for 4 hours (cold-soaked, not operating).
2. Attempt key-on and startup. Verify system starts successfully.
3. Verify pre-charge functions (capacitor ESR higher at cold, may affect time constant).
4. Operate at 50% load for 30 minutes.
5. Monitor for: slow switching (higher IGBT $V_{CE(sat)}$ at cold), sluggish gate driver response, stiff cables/ connectors.

Pass Criteria: Successful cold start. Pre-charge completes within timeout. Stable operation. No false faults from cold-soaked components.

Why this covers the requirement: Cold temperature affects: capacitor ESR (electrolytic caps especially), battery capacity (reduced cranking power), cable flexibility (stiff HV cables stress connectors), semiconductor parameters (higher V_{th}, slower switching). The system must start and operate in winter conditions.

E-06: Thermal Shock

Objective: Verify survival of rapid temperature transitions.

Covered: SG-13 (ASIL D), H-13 (mechanical)

Setup: Two-zone thermal shock chamber or liquid thermal shock bath.

Procedure:

1. 20 cycles: −20°C (30 min) → +60°C (30 min). Transition time < 1 minute.
2. DUT non-operating during shock cycles.
3. After 20 cycles: visual inspection and functional test.

Pass Criteria: No condensation inside enclosure. No cracked solder joints. No connector seal breach. Functional test passes.

Why this covers the requirement: Thermal shock causes rapid CTE mismatch stress. Large components (capacitors, transformers, power modules) experience the highest stress. Condensation from temperature crossing the dew point is a risk for HV electronics. This is a type test for production qualification.

E-07: High Humidity Operation

Objective: Verify no performance degradation or insulation breakdown in high humidity.

Covered: SG-09 (ASIL A), FSR-10, H-09

Setup: Humidity chamber. 85% RH at 40°C.

Procedure:

1. Operate DUT at 50% load for 48 hours at 85% RH, 40°C.
2. Monitor isolation resistance (HV to logic ground) every 4 hours.
3. Verify no isolation degradation.
4. Post-test: HiPot test at rated voltage to confirm insulation integrity.

Pass Criteria: Isolation resistance stable throughout test. HiPot passes post-test. No corrosion visible on PCB or connectors.

Why this covers the requirement: Humidity reduces surface resistivity of PCBs and insulation materials. In extreme cases, condensation can create conductive paths between HV and logic. The test validates enclosure sealing and conformal coating effectiveness.

10.10.3 Electromagnetic Compatibility Tests

E-08: Radiated EMI Immunity (ALSE)

Objective: Verify system operates correctly under radiated electromagnetic interference.

Covered: SG-01 (ASIL D), SG-13 (ASIL D), FSR-15, H-01, H-13, H-15

Setup: ALSE (Absorber-Lined Shielded Enclosure) or anechoic chamber. RF signal generator and amplifiers. Bicone and log-periodic antennas.

Procedure:

1. System operating at 50% tractive effort on dyno (inside chamber or with motor outside via feedthrough).
2. Apply radiated EMI: 20 MHz – 6 GHz, 100 V/m, AM 80% 1 kHz, all polarizations (horizontal, vertical).
3. Dwell at 1% frequency steps for minimum 2 seconds each.
4. Monitor for: resets, CAN errors, sensor dropouts, unexpected tractive effort changes, safe state entries.

5. Pay special attention to GSM/Cellular bands (700 MHz, 850 MHz, 1.9 GHz), WiFi (2.4 GHz, 5 GHz), and ISM bands.

Pass Criteria: No Class A deviations (loss of function, unintended tractive effort). Class B deviations (temporary performance degradation) permitted if self-recovering. No safe state entries unless justified.

Why this covers the requirement: Cellular phone towers, emergency services radio, and amateur radio can create strong RF fields. The STM32 and analog sensors are susceptible to RF interference that can cause: ADC reading errors (appearing as throttle or current faults), CPU resets, CAN bit errors. This is a critical test for safety validation.

E-09: Conducted EMI Immunity (BCI)

Objective: Verify immunity to conducted interference on harnesses.

Covered: SG-01 (ASIL D), SG-13 (ASIL D), FSR-15, H-01, H-15

Setup: BCI (Bulk Current Injection) probe and calibration fixture. DUT with full harness.

Procedure:

1. Inject RF current onto each harness line: 1 MHz – 400 MHz, up to 100 mA (per ISO 11452-4).
2. Test all harnesses: CAN bus, throttle wiring, encoder cable, HVIL, +12 V, gate driver connections.
3. Monitor for same deviations as E-08.

Pass Criteria: Same as E-08. No Class A deviations on any harness.

Why this covers the requirement: Conducted coupling is often more severe than radiated for low-frequency interference (AM radio, CB radio). Long harnesses act as antennas. The BCI test injects current directly onto the harness to simulate coupling from nearby transmitters.

E-10: ESD — Contact Discharge

Objective: Verify system survives electrostatic discharge to accessible surfaces.

Covered: SG-13 (ASIL D), FSR-15, H-13

Setup: ESD gun per ISO 10605. DUT fully assembled in production enclosure.

Procedure:

1. Apply contact discharge: ±4 kV, ±6 kV, ±8 kV to all accessible metal surfaces: connector shells, mounting bolts, heatsink, enclosure seams.
2. 10 discharges per point per voltage level.
3. System operating during test.
4. Monitor for resets, safe state entries, fault logging.

Pass Criteria: No Class A deviations at ±4 kV and ±6 kV. At ±8 kV: Class B permitted (self-recovering). No permanent damage.

Why this covers the requirement: ESD occurs during rider contact (mounting/dismounting), fueling/charging, and maintenance. The discharge can couple into signals through connector shells and enclosure seams. The test validates enclosure shielding and ESD protection devices.

E-11: ESD — Air Discharge

Objective: Verify system survives air discharge to insulated surfaces.

Covered: SG-13 (ASIL D), FSR-15, H-13

Setup: ESD gun. DUT in production enclosure.

Procedure:

1. Apply air discharge: ±4 kV, ±8 kV, ±15 kV to all accessible insulated surfaces: plastic covers, labels, connectors with plastic housings.
2. 10 discharges per point per voltage level.
3. Monitor for same deviations as E-10.

Pass Criteria: Same as E-10 with adjusted levels for air discharge.

Why this covers the requirement: Air discharge has a slower rise time but can find paths into the enclosure through seams and ventilation. The plastic enclosure is the primary defense; this test validates its effectiveness.

E-12: Water Ingress / IP Rating Validation

Objective: Verify enclosure sealing against water ingress.

Covered: SG-09 (ASIL A), FSR-10, H-09, H-11

Setup: Water spray rig per IEC 60529. DUT in production enclosure, mounted in production orientation.

Procedure:

1. **IPX4 (splashing water):** Spray from all directions for 10 minutes. Inspect interior for water entry.
2. **IPX5 (water jets):** 12.5 L/min nozzle, 3 m distance, all directions, 3 minutes. Inspect.
3. **IPX6 (powerful jets):** 100 L/min nozzle, 3 m distance, 3 minutes. Inspect.
4. After each test: insulation resistance measurement (HV to ground). HiPot test.
5. Post-water: operate system and verify no moisture-induced faults.

Pass Criteria: No water inside enclosure at target IP rating. Insulation resistance > 500 Ohm/V. HiPot passes. System operates correctly post-test. Target: minimum IP54 (IP65 preferred for under-seat motorcycle installation).

Why this covers the requirement: Motorcycle electronics are exposed to rain, wheel spray, and pressure washing. Water ingress causes: short circuits (HV to logic), corrosion (connector degradation), insulation failure (shock hazard). The target IP rating must be validated, not just designed.

10.11 Recommended Test Execution Order

The 99 defined tests (50 component, 19 system, 18 integration, 12 environmental) are organized into an execution sequence designed to detect simple, non-destructive faults early while deferring potentially destructive tests until basic functionality is confirmed. This approach follows the principle of **progressive validation**: each test group builds confidence before moving to the next, more aggressive group. A failure in an early group prevents advancing to later groups until resolved, avoiding expensive damage from compound failures.

10.11.1 Execution Sequence

Table 21 — Test Execution Order and Justification

Order	Group	Tests	Risk Level	Justification
1	Power-On Self-Test	C-16 (DESAT POST) C-20 (Boot CRC)	None	Validate that POST and boot checks function before any HV is applied or PWM is enabled. These are purely diagnostic — no load, no HV, no movement. If POST fails, the unit must not be energized further.
2	Supply Integrity	C-21 (Brownout) C-22 (Brownout recovery) C-46 (Pre-charge) C-41 (ADC ref drift)	Very Low	Validate power supply behavior before energizing the inverter. Brownout tests use no load. Pre-charge validates sequencing logic. ADC reference validates sensor measurement foundation. All can be done with reduced DC link voltage.
3	Gate Driver Integrity	C-47 (Miller clamp) C-48 (Prop delay) C-49 (Deadtime) C-17 (UVLO) C-26 (+15V short) C-27 (−9V short)	Low	Gate driver tests validate the PWM output stage without connecting a motor. Miller clamp, propagation delay, and deadtime are bench tests with scope only. UVLO and supply short tests validate independent protection. No motor load = no movement risk.
4	Isolation & HV Safety	C-50 (Isolation) I-04 (HVIL) C-10 (IGBT temp sensors) C-35 (DC link cap temp)	Low-Medium	HiPot validates isolation before HV is applied repeatedly. HVIL validates the safety interlock. Temperature sensor tests validate monitoring before thermal stress. DC link at operational voltage for first time.
5	Sensor Validation	C-01 to C-06 (Throttle) C-07, C-08 (Current sensor) C-09 (Temp sensors) C-11 (Encoder) C-12 (DC link voltage) C-28, C-29, C-30 (Phase open) C-42 (SPI ADC)	Medium	All sensors validated with motor connected but at low load. Throttle, current, temperature, encoder, voltage — the complete feedback chain. Phase open tests validate continuity. Low power first to catch sensor faults before full power.
6	Control Loop Validation	C-13 (WDT) C-14 (Breakpoint) C-37 (Clock failure) C-38 (PLL unlock) C-43 (CAN bus off) C-44 (WDT starvation) S-10 (Startup) S-11 (Shutdown)	Medium	Validate the fundamental safety mechanisms with motor running at low-to-medium load. Watchdog, breakpoint, clock, and PLL tests verify safe-state paths. Startup/shutdown sequencing validated. CAN fault handling at operational load.

Order	Group	Tests	Risk Level	Justification
7	Software Integrity	C-18, C-19 (ECC RAM) C-39 (Flash bit rot) C-40 (GPIO stuck-at) C-15, C-16 (DESAT + self-test) I-08 (CAN fuzzing) I-09 (CAN bus load) I-10 (Bus off recovery) I-11 (Invalid frames)	Medium	Software fault injection with motor at medium load. ECC RAM, flash corruption, GPIO faults, and CAN robustness. DESAT self-test re-validated after initial power-on. These tests exercise the firmware under realistic conditions.
8	Integration & Coordination	I-01 to I-03 (CAN heartbeat) I-06 (Kickstand) I-12 to I-15 (Display, charger, ABS, BMS) I-16 (Multi-node fault) I-17 (CAN wiring) I-18 (Wake/ sleep) S-12 (Key cycle stress)	Medium	Full system integration tests with all external nodes. Coordination between subsystems validated. Key-cycle stress test (1000 cycles) as endurance screen. All previous tests passed before this group.
9	Performance Validation	S-06 (Full load thermal camera) S-07 (Thermal cycling) S-08 (Max regen) S-09 (Field weakening) S-13 (Power dip ride-through) S-14 (Reverse) S-15 (Overspeed) C-36 (Bearing current)	Medium-High	First full-power tests. Thermal camera survey identifies hotspots before extended run. Max regen validates overvoltage protection at worst case. Field weakening validates control stability at speed extremes. Power dip validates DC link capacitor sizing. Bearing current validates motor protection.

Order	Group	Tests	Risk Level	Justification
10	Fault Response Validation	S-01 to S-05 (System fault scenarios) C-45 (Thermal runaway) C-23 to C-25 (Power rail shorts) C-31 to C-34 (Phase shorts) I-05 (DC link OV) I-07 (Multi-CAN loss)	High	Full-power fault injection. Unintended tractive effort, loss of tractive effort, regen faults at realistic power levels. Phase-to-phase and phase-to-DC shorts at reduced voltage. Power rail shorts validate supply fault behavior. Requires all previous groups passed.
11	Environmental Stress	E-01 to E-03 (Vibration, shock) E-04 to E-07 (Thermal) E-08, E-09 (EMI) E-10, E-11 (ESD) E-12 (Water ingress)	Variable	Environmental tests are type tests (one unit, not every unit). Run on a unit that has passed all functional tests. Vibration and thermal can reveal mechanical weaknesses. EMI and ESD validate electromagnetic robustness. Water ingress validates sealing. Some tests (phase shorts, EMI) can damage the unit — these are final.

10.11.2 Stopping Criteria Between Groups

Advancement from one test group to the next requires **all** tests in the current group to pass. The following stopping criteria apply:

Table 22 — Inter-Group Stopping Criteria

Scenario	Required Action	Before Advancing
Any test in Group 1–3 fails	Stop. Do not apply HV. Debug hardware or software. Fix root cause. Re-run failed test + all preceding tests.	All tests in current group pass.
Any test in Group 4–5 fails	Stop. HV may be applied but do not connect motor or load. Debug sensor, isolation, or supply issue. Fix and re-run group.	All tests in current and all prior groups pass.
Any test in Group 6–8 fails	Stop. Motor is connected and spinning. Safe state may be untrusted. Debug safety mechanism. Fix and re-run from Group 5.	All tests in current and all prior groups pass.
Any test in Group 9 fails	Stop. Full power has been applied. Thermal or electrical damage may have occurred. Inspect hardware. Fix and re-run from Group 5.	All tests pass. Visual inspection confirms no damage.
Any test in Group 10 fails	Stop. Destructive fault testing may have damaged unit. Inspect thoroughly. If unit damaged, switch to fresh unit and re-run from Group 8.	All tests pass. Full functional verification (Groups 5–8) re-run and pass on same unit.

Scenario	Required Action	Before Advancing
Any environmental test fails	Document failure. Environmental failures are type-test issues, not unit-test blockers. Fix design, re-test on new unit.	All prior functional groups pass on test unit. Environmental failure documented.

10.11.3 Hardware Damage Risk Classification

Table 23 — Per-Test Hardware Damage Risk

Risk	Tests	Potential Damage	Mitigation
None	C-16, C-20, C-21, C-41, C-46, C-47, C-48, C-49, I-08, I-11	No electrical or mechanical stress.	Standard bench equipment.
Very Low	C-22, C-17, C-37, C-38, C-43, C-50, I-09, I-10	Possible MCU reset. No power stage stress.	Current-limited supplies. No motor load.
Low	C-26, C-27, C-35, C-42, C-44, I-04, I-12, I-17	Gate driver supply stress. Possible fuse blow.	Current-limited supplies. Fuses rated for test.
Medium	C-01–C-15, C-18–C-19, C-28–C-30, C-36, C-39–C-40, S-01–S-05, S-10–S-14, I-01–I-03, I-05–I-07, I-13–I-16, I-18, E-10–E-11	Motor movement. Low-level power applied. Thermal stress possible.	Dyno with mechanical guard. Low initial load. Thermal monitoring.
Medium-High	S-06–S-09, S-15, C-09–C-10, C-12, C-45	Full power thermal stress. Possible overtemperature if protection fails.	Thermal camera monitoring. Manual abort button. Pre-set temperature abort threshold.
High	C-23–C-25, C-31–C-34, C-07–C-08, S-13, I-05, E-01–E-09, E-12	Power stage damage from shorts. Thermal damage from full load. Mechanical damage from vibration.	Reduced DC link for shorts (50 V). Fused test fixtures. Remote operation. Vibration fixture rated for DUT mass. Fire suppression for full-power tests.
Very High	C-31–C-34 (full voltage)	IGBT explosion. DC link capacitor rupture. Arc flash. Fire.	**Never run at full DC link voltage.** Maximum 50 V for phase short tests. Blast shield. Remote operation only. Fire extinguisher present. No personnel in room during test.

> **Safety Warning for Phase Short Tests (C-31 through C-34)**
>
> Phase-to-phase and phase-to-DC short circuit tests are **inherently destructive if protection fails**. Even at reduced voltage, fault currents of 1000+ A are possible. The test procedure must include: (1) blast shield or enclosure around DUT, (2) remote-controlled contactor for short application (no manual connection), (3) current-limited DC supply or fuse rated below IGBT withstand, (4) fire suppression equipment, (5) no personnel in the test area during fault application, (6) video recording for post-test analysis, (7) pre-test verification that DESAT self-test (C-16) passed on the same day. If DESAT self-test has not been run within 24 hours, do not proceed with short-circuit tests.

11. Implementation Roadmap

This section provides a practical phased roadmap for implementing the safety requirements and closing the gaps identified in Section 9. The roadmap is organized by priority and effort level.

11.1 Phase 1: Safety-Critical Foundation (Immediate)

Table 24 — Phase 1 — Safety-Critical Foundation

Gap	Action	Effort	Validation
GAP-HW-01	**CLOSED** — Dual-MCU integrated monitoring detects overcurrent within 100 ms. LM397 comparators retained as non-safety redundant layer. No separate HW OCP required for safety case.	N/A	C-07, C-08, S-09, S-15: verify overcurrent detection and SSO response under dual-MCU monitoring
GAP-SW-01	Implement boot CRC-32 using STM32 CRC peripheral. Check safety-critical code + calibration data before PWM enable.	Low	C-20: verify corrupted flash prevents boot
GAP-SW-02	Implement reverse interlock: speed >100 rpm forward → negative tractive effort clamped to zero. Reverse only when stationary AND selected.	Low	S-02: verify reverse rejected at speed
GAP-SW-03	Define sensorless fallback policy: immediate SSO on encoder loss. Document decision. If sensorless implemented: bound to ≤2 s, ≤30% tractive effort.	Low	C-11: verify encoder loss → SSO <100 ms

11.2 Phase 2: Graceful Degradation (Short Term)

Table 25 — Phase 2 — Graceful Degradation

Gap	Action	Effort	Validation
GAP-HW-02	Implement tractive effort ramp-down at ≤200 Nm/s before SSO. Active torque decay on fault detection. Exception: <50 ms safety timing permits direct SSO.	Medium	S-03, S-04: verify ramp rate on dyno
FSR-06	Implement uncommanded regenerative braking monitor. Compare commanded vs. actual regen. >10 Nm for >200 ms → safe state.	Medium	S-05: verify detection and response
FSR-02	Implement tractive effort command plausibility: commanded current reference vs. measured phase currents. >20% for >100 ms → safe state.	Medium	C-07, C-08: verify plausibility catches sensor faults
FSR-17	Implement CAN heartbeat timeout handling with safe-state defaults. IO board loss → brake=pressed, kickstand=down. BMS loss → tractive effort zero.	Medium	I-01, I-02: verify timeout and defaults
FSR-08	Implement 2-out-of-3 IGBT temperature voter. Progressive derate at warning. SSO at critical. Thresholds in ECC memory.	Medium	C-09, C-10: verify voter behavior

11.3 Phase 3: Test Execution and Validation

Table 26 — Phase 3 — Test Execution and Validation

Activity	Effort	Description
Execute component tests (C-01 to C-20)	High	Full component-level test campaign on bench setup. Document all results. Fix any failures before proceeding.
Execute system tests (S-01 to S-05)	High	Dyno-based system tests. Measure ramp rates, detection times, and response characteristics. Compare against requirements.
Execute integration tests (I-01 to I-07)	Medium	CAN-based integration tests with simulators. Validate timeout behavior, safe defaults, and CAN-loss response.
Track testing (LIMIT-01)	High	Controlled track testing with experienced rider. Graduated lean angle tests with known fault injection. Rider controllability assessment.
Test report and safety case	Medium	Compile test results into safety case document. Justify achievable ASIL claims based on evidence. Document all known limitations.

11.4 Coprocessor Integration Validation

The STM32G474RCTx safety coprocessor is **part of the current hardware design** (ControlBoard.kicad_sch, Rev A). The following table maps coprocessor capabilities to safety goals:

Table 27 — Coprocessor Capability to Safety Goal Mapping

Feature	Description	Impact
Independent throttle monitoring	Coprocessor reads throttle pots (THROTTLE_A, THROTTLE_B) via independent ADC channels. Discrepancy >5% → GATE_DRIVE_PWR2_ENABLE low + GATE_DRIVE_RESET → SSO. Eliminates common-cause with main MCU.	SG-01 → ASIL D
Independent current monitoring	Coprocessor samples all 4 current sense signals (PH_U/V/W + DC link) + REF via independent ADC. Cross-checks against main MCU torque command. Overcurrent → SSO within 100 ms.	SG-06, SG-07 → ASIL target
Challenge-response watchdog	Coprocessor issues challenges via inter-MCU UART; main STM32 must respond within window. Failure → coprocessor asserts NRST → system reset → SSO. Not subject to on-chip common-cause.	SG-13 → ASIL D
PWM integrity monitoring	Coprocessor monitors all 6 PWM output pairs (PH_U/V/W_HIGH/LOW) for deadtime violations, stuck-on, stuck-off, and frequency anomalies. Independent of main STM32.	SG-12, SG-14, SG-15 → ASIL C
1oo2 gate drive power kill	GATE_DRIVE_PWR2_ENABLE (coprocessor) in logical-OR with GATE_DRIVE_PWR1_ENABLE (main). Either MCU deasserting its enable kills all six gate drive supplies. Independent feedback: GATE_DRIVE_PWR1_FB + GATE_DRIVE_PWR2_FB.	SG-13 → ASIL D
Independent CAN snooping	FDCAN2 + FDCAN3 monitor both CAN buses. Coprocessor cross-checks torque commands, heartbeats, and fault flags against main MCU actions. CAN anomaly → SSO.	SG-01, SG-04, SG-05 → ASIL target
Independent temperature monitoring	All 3 heatsink temps + motor temp monitored via independent ADC. 2oo3 voting cross-check with 100 °C hard cap. Thermal runaway → SSO.	SG-07 → ASIL target
FRAM logging	CY15B102Q-SXET 256 KB FRAM (attached to the main MCU via SPI) stores fault logs, configuration, hour meter, odometer. Hardware WP pin protects critical data.	SG-03 → ASIL C

The dual-MCU architecture makes **ASIL D achievable for SG-01 and SG-13 via ASIL B(D) + ASIL B(D) decomposition**. All other safety goals achieve their target ASIL. No hazards remain below target. The coprocessor firmware must be validated alongside the main MCU firmware — see test cases C-14 (coprocessor SSO), S-16 through S-19 (modulation transitions), and I-16 (multi-node fault with coprocessor participation).

References

Table 28 — Referenced Standards and Documents

Reference	Title / Description
ISO 26262-1:2018	*Road vehicles — Functional safety — Part 1: Vocabulary*. International Organization for Standardization.
ISO 26262-3:2018	*Road vehicles — Functional safety — Part 3: Concept phase*. Defines HARA methodology, ASIL assignment, and safety goal derivation.
ISO 26262-5:2018	*Road vehicles — Functional safety — Part 5: Product development at the hardware level*. Hardware architectural metrics and safety mechanisms.
ISO 26262-9:2018	*Road vehicles — Functional safety — Part 9: Automotive Safety Integrity Level (ASIL)-oriented and safety-oriented analyses*. ASIL decomposition and dependent failure analysis.
EN 50155:2021	*Railway applications — Electronic equipment used on rolling stock*. Qualification standard for onboard power supplies and electronic equipment in railway applications.
IEC 61508	*Functional safety of electrical/electronic/programmable electronic safety-related systems*. General functional safety standard; SIL terminology reference.
AEC-Q100	*Failure Mechanism Based Stress Test Qualification for Integrated Circuits*. Automotive Electronics Council. Qualification standard for the NCV57100 gate driver.
STM32H723ZG	*STM32H723/733 — Arm Cortex-M7 MCU Reference Manual (RM0433)*. STMicroelectronics. Main processor: 550 MHz, ECC RAM, dual FDCAN, HRTIM.
STM32G474RCTx	*STM32G474/484 — Arm Cortex-M4 MCU Reference Manual (RM0440)*. STMicroelectronics. Safety coprocessor: 170 MHz, FPU, 3x FDCAN, advanced motor-control timers, CORDIC.
CY15B102Q-SXET	*2-Mbit (256K x 8) Serial (SPI) F-RAM*. Infineon/Cypress. Non-volatile storage with unlimited write endurance, hardware WP pin.
NCV57100	*Isolated IGBT Gate Driver with Desaturation Protection* — ON Semiconductor / onsemi Datasheet. DESAT, UVLO, complementary inputs, active Miller clamp, soft turn-off specifications.
EC7BW-110S12	*EC7BW-110 Series — 20 W Isolated DC-DC Converter*. Cincon / Railway Grade. EN 50155 qualified onboard power supply.
Cossalter	*Motorcycle Dynamics (2nd ed.)*, Vittore Cossalter. Lulu Press, 2006. Motorcycle handling dynamics, tire friction envelopes, and lean angle mechanics reference.
NHTSA	*National Highway Traffic Safety Administration — Motorcycle Safety*. https://www.nhtsa.gov/motorcycles. Statistics and research on motorcycle accident causation.

Document History

Table 29 — Revision History

Version	Date	Changes
1.0	2026-06-13	Initial unified HARA document. Combined previous separate documents. Added railway terminology, mitigation strategy table, expanded cornering test rationale, fault injection test plan, implementation roadmap.
2.0	2026-06-13	Major expansion. Added 30 new component tests (brownout, power rail shorts, phase disconnect/short, bearing current/Class Y caps, DC link cap temperature, clock failure, PLL unlock, flash bit rot, GPIO stuck-at, ADC drift, SPI faults, CAN bus off, watchdog starvation, thermal runaway, pre-charge, Miller clamp, propagation delay, deadtime, isolation). Added 14 new system tests (10 original + 4 modulation transition) (full load thermal camera, thermal cycling, max regen, field weakening, startup/shutdown, key cycle stress, power dip ride-through, reverse, overspeed). Added 11 new integration tests (CAN fuzzing, bus load, bus off recovery, invalid frames, display, charger, ABS, BMS faults, multi-node, wiring faults, wake/sleep). Added 12 environmental tests (vibration random/sine/shock, thermal soak/high/low/shock, humidity, EMI radiated/conducted, ESD contact/air, water ingress). Added test execution order with progressive validation and hardware damage risk classification. Total: 75 tests across 4 categories.
3.0	2026-06-14	Added LIMIT-07 (STL Class B only), LIMIT-08 (no DFA), LIMIT-09 (no EMI/EMC pre-compliance). Added CORDIC subsection. Added two ASIL evaluations (with/without coprocessor). Clarified encoder and CAN nodes out of scope. Added gate driver FLT verification on PMIC EN power cut. Added complete state machine transition table. Added FOC tested on bench note. Added 17% statistic note.
3.1	2026-06-14	Added 4 system-level tests (S-16 to S-19) for multi-modulation scheme transitions. Total: 75 tests.
4.0	2026-07-08	Major architecture update: Dual-MCU is now the primary architecture. STM32G474RCTx coprocessor implemented. Six SSO pathways. GAP-HW-01 closed — dual-MCU integrated monitoring sufficient, HW OCP not required. 1oo2 gate drive power kill with independent feedback. ASIL D via ASIL B(D) decomposition.
4.1	2026-07-13	Editorial consistency pass: fault-injection test counts corrected to 99 (50 component / 19 system / 18 integration / 12 environmental); NCV57100 gate-driver part update; DRIVER_RESET turn-off mechanism corrected; bleeder assumption removed; TPS389006-Q1 rail supervisor integrated; stale ASIL and test-reference errors fixed; shared 3.3 V rail and gap-analysis staleness corrected; temperature voting updated from 1oo3 to 2oo3.

USER MANUAL

HW-C2-DOC-MANUAL-A · design document — rev A

USER MANUAL

Traction Inverter

Version	2.4
Prepared by	Thomas Liao
Date	July 13, 2026
Part	HW-C2-DOC-MANUAL-A · Rev A

University of California, Santa Cruz — Corzine Lab

Contents

Document Control 4
1. Overview 5
1.1 Product Variants 5
1.2 System Architecture 6
1.3 External Connections 6
1.4 Internal Debug Interfaces 6
1.5 Hardware Overview 7
2. Safety Information 9
2.1 High Voltage Warnings 9
2.2 General Safety 9
2.3 This Connector is NOT Sevcon-Compatible 9
3. Idle Power Draw and Storage 10
4. High Voltage Interlock (HVIL) 10
5. Thermal Management 11
6. Specifications 12
7. Ampseal 35-Pin Connector 13
7.1 CAN Bus 1 (BMS) 13
7.2 CAN Bus 2 (Dash / Charger) 14
7.3 Throttle Inputs 14
7.4 Motor Position / Temperature 14
7.5 Power and Keyswitch 15
7.6 HVIL 15
7.7 Precharge / HV Control 16
7.8 CAN Termination 16
7.9 Reserved Pins 16
8. Power Supply Architecture 16
8.1 Logic Power (VIN_Z) 17
8.2 Switched +12V Output 17
8.3 Isolated Sensor and Communication Rails 17
9. Gate Drive 17
10. Sensing and Acquisition 18
11. External Precharge 19
12. Dual Isolated CAN Bus 19
13. CAN Protocol 20
14. BMS Interface 21
15. USB-B Debug Port 21
16. Maintenance and Troubleshooting 22
16.1 Common Faults 22
16.2 Debug Interface Warning 23
17. Motor Control and Modulation Strategies 23

17.1 Available Modulation Schemes 23
17.2 Live Switching and Automatic Selection 24
17.3 Configuration via CAN Bus 25
18. Expansion Module 25

Document Control

- **MCUs:** STM32H723ZG + STM32G474RCTx
- **Operating Temp:** −40°C to +85°C
- **Reviewed by:** (not yet reviewed)

Revision History

Date	Version	Author	Description
July 13, 2026	2.4	Project Team	Editorial consistency pass: NCV57100 part update; USB debug description corrected; cross-references updated; precharge current limit corrected to 2 A; placeholder pinout made consistent; DC link cooling and expansion module sections added; hardware overview figures added; voting, companion-doc versions, and isolation ratings corrected.
July 8, 2026	2.3	Project Team	Dual-MCU architecture and CAN assignment updates. (entry added retroactively)
June 14, 2026	2.2	Project Team	Tone pass: removed defensive marketing language, blanked pinout (placeholder until harness finalized), reduced Sevcon warnings to one clean callout, removed all "Zero" brand references, toned down formal "KNOWN LIMITATION" language, renamed document to User Manual. Added Section 17: modulation strategies (planned feature, 10 schemes, live switching, automatic selection, hysteresis, transition torque blip mitigation, dyno validation plan).
June 14, 2026	2.1	Project Team	Removed "Sevcon replacement" marketing. Fixed encoder types (sin/cos or Hall only, not quadrature/resolver). Fixed IGBT spec (1200V standard, 600V optional). Clarified main contactor is driven by BMS/external system, not inverter. Fixed precharge description (high-side, isolated +12V through relay). Fixed switched +12V outputs. Removed all warranty language. Added parasitic drain note. Added no-guarantee disclaimers.
June 14, 2026	2.0	Project Team	Complete rewrite. Professional format, two-variant clarification, corrected HVIL description, expanded pinout tables, added safety callouts, troubleshooting and maintenance sections.
[Earlier]	1.x	Project Team	Initial draft versions.

Companion Documents

This manual covers the electrical interface, integration, and operation of the traction inverter hardware. The following companion documents provide additional detail:

- **HARA (v4.1):** Hazard Analysis and Risk Assessment per ISO 26262. 99 fault injection tests, 18 hazards, 15 safety goals, 21 FSRs. Dual-MCU architecture with STM32G474RCTx coprocessor. Six SSO pathways. ASIL D via ASIL B(D) decomposition.
- **TARA (v1.2):** Threat Analysis and Risk Assessment per ISO/SAE 21434. 9 cybersecurity tests, 7 CSRs, open-source trust model.
- **SWAD (v1.5):** Software Architecture Document. 4-layer architecture, FOC + multi-modulation, 6-state state machine, CAN protocol reference.

The **RTE (Real Time Examiner)** open-source tool connects via CAN bus for live monitoring, parameter configuration, and firmware updates.

1. Overview

The traction inverter is an open-source motor controller designed for electric drivetrain applications. It converts DC traction battery power to variable-voltage variable-frequency (VVVF) three-phase AC output to drive permanent magnet synchronous motors (PMSM). The unit integrates battery management communication, thermal monitoring, comprehensive safety interlocks, and multi-modulation field-oriented control (FOC).

This manual covers the 140V / 600A variant with the Ampseal 35-pin vehicle connector. This variant includes an onboard railway-grade isolated DC-DC converter that generates +12V logic power from the traction battery. A higher-voltage variant (up to 800V / 800A) is available; see Section 1.1 for details.

> **DANGER — High Voltage**
>
> This device contains lethal voltages up to 160 VDC (43-160 V range). All high-voltage connections are exposed on busbars. Only qualified personnel should install, service, or operate this equipment. Always verify zero energy state with a calibrated voltmeter before touching any terminal.

1.1 Product Variants

Two hardware variants are available. The variant described in this manual is the **140V / 600A unit with Ampseal 35-pin connector**.

Product Variant Comparison

Parameter	This Manual: 140V / 600A	Higher-Voltage Variant
DC Input Voltage	43 - 160 VDC	Capacitor-only swap to 450 V; PCB + capacitor change to 900 V (covers 800 V target)
Continuous Current	600 A RMS	Up to 800 A RMS
Peak Current	1200 A (time-limited)	Configurable
Vehicle Connector	Ampseal 35-pin (TE 776231-1)	User-provided
Logic Power	Onboard DC-DC from VIN_Z (included)	External +12V supply required
Gate Drive	+15 V / −9 V; >5 kV_{rms} (UL1577), 1424 V_{PK} / 1000 V_{rms} working (VDE 0884-11)	Same
Control Processor	STM32H723ZG	Same
IGBT Module	1200 V (standard)	Same 1200 V module

> **Variant Note**
>
> This manual describes only the 140V / 600A variant with Ampseal connector. The higher-voltage variant requires the user to provide a separate isolated +12V logic supply and does not include the Ampseal vehicle connector. Contact the project for documentation on other variants.

1.2 System Architecture

The traction converter consists of five PCB assemblies:

- **Main Board:** Interface components that repin the board-to-board connectors to the waterproof Ampseal 35 vehicle connector. Contains the 12V logic power module (railway-grade DC-DC), precharge and contactor relays, and signal conditioning for the external vehicle harness.
- **Control Board:** Dual-MCU control architecture. Main processor (STM32H723ZG, 550 MHz Cortex-M7) runs FOC motor control, sensor acquisition, CAN communication, and safe state management. Safety coprocessor (STM32G474RCTx, 170 MHz Cortex-M4+FPU) provides independent monitoring of all safety-critical signals, 1oo2 gate drive power kill, challenge/response watchdog, independent CAN bus snooping, and CY15B102Q-SXET 256 KB FRAM for fault logs and configuration.
- **Gate Driver Board:** Isolated gate drive power supplies, six-channel PWM gate drive with desaturation protection, active Miller clamp, and hardware PWM disable. Interfaces directly to the IGBT power module.
- **DC Bus Capacitor Board:** DC link capacitor bank (60× 330 µF / 200 V aluminium electrolytic capacitors, 19.8 mF total) on an aluminium heat-spreader plate for thermal coupling to the heatsink (Section 5). The 450 V capacitor-only upgrade swaps to 60× Nichicon UCS2W680MHD 68 µF / 450 V parts (4.08 mF total) and reduces the standoff length by 5 mm (e.g., 55 mm → 50 mm).
- **DC Bus Filter Board:** Snubber capacitor board mounted at the IGBT module phase terminals (U/V/W) to filter switching transients.

The Main Board aggregates all external vehicle connections through the Ampseal 35 connector and routes signals to the Control Board and Gate Driver Board via internal board-to-board cabling. All high-voltage circuitry is galvanically isolated from the low-voltage control domain.

1.3 External Connections

External Electrical Connections

Connection	Type	Description
Ampseal 35-pin	Sealed connector (TE 776231-1)	All control, communication, sensing, and auxiliary power signals
DC Link Positive	Busbar lug	HV DC input from battery pack (after precharge)
DC Link Negative	Busbar lug	HV DC return to battery negative
Phase U, V, W	Busbar lugs	Three-phase AC output to traction motor stator windings
USB-B	Exterior port	Debug log output, firmware update (Section 15)

1.4 Internal Debug Interfaces

Several internal connectors are available for debugging and firmware flashing. Using these requires partial disassembly of the inverter. When reassembling, ensure all waterproofing seals are correctly seated.

- **USB-B Internal:** Raw USB D+/D− routed directly to the coprocessor's (STM32G474RCTx) native USB 2.0 full-speed (12 Mbit/s) interface for debug logs and firmware updates (Section 15). The port is not bus-powered — the inverter must be externally powered for the USB interface to operate.
- **J-Link Connector:** Standard ARM debug connector for firmware flashing, debugging, and reading memory. RDP (Readout Protection) is never set on the STM32H723ZG — debugging is always possible.

WARNING — Do Not Set RDP

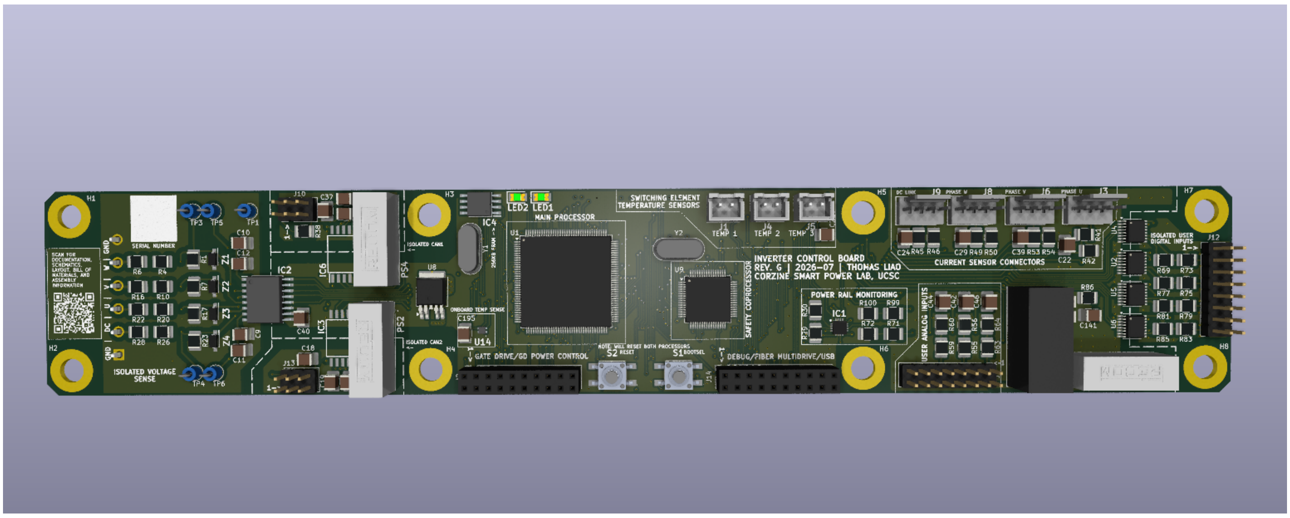
Setting RDP Level 1 or 2 on the STM32H723ZG will prevent firmware updates and debugging. The factory default is RDP Level 0 (no protection). If you use custom firmware, ensure your flashing tool does not set RDP. A bricked MCU can be recovered via the J-Link connector with RDP Level 0. With RDP Level 2, recovery is impossible.

CAUTION — Reassembly

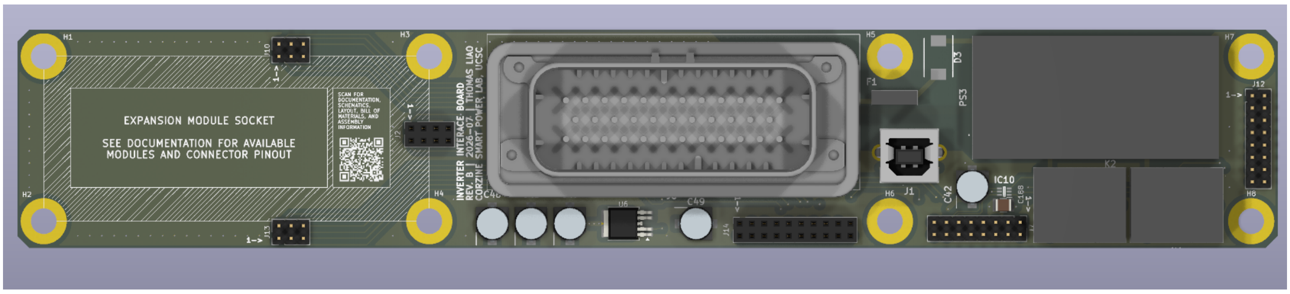
Damage caused by improper disassembly, connection, or reassembly is the user's responsibility. No warranty is provided — this is an open-source hardware project. Always verify waterproofing seals are intact after servicing. Modified firmware may compromise safety-critical functions; the user assumes all risk.

1.5 Hardware Overview

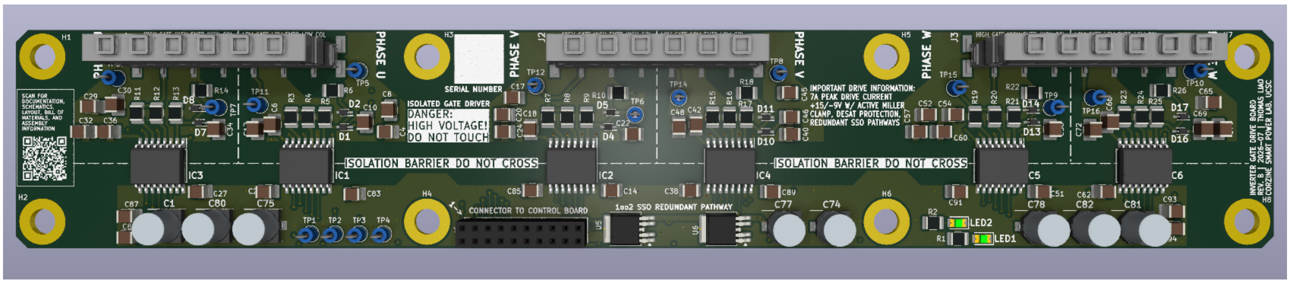
The renders below show the major PCB assemblies of the inverter.



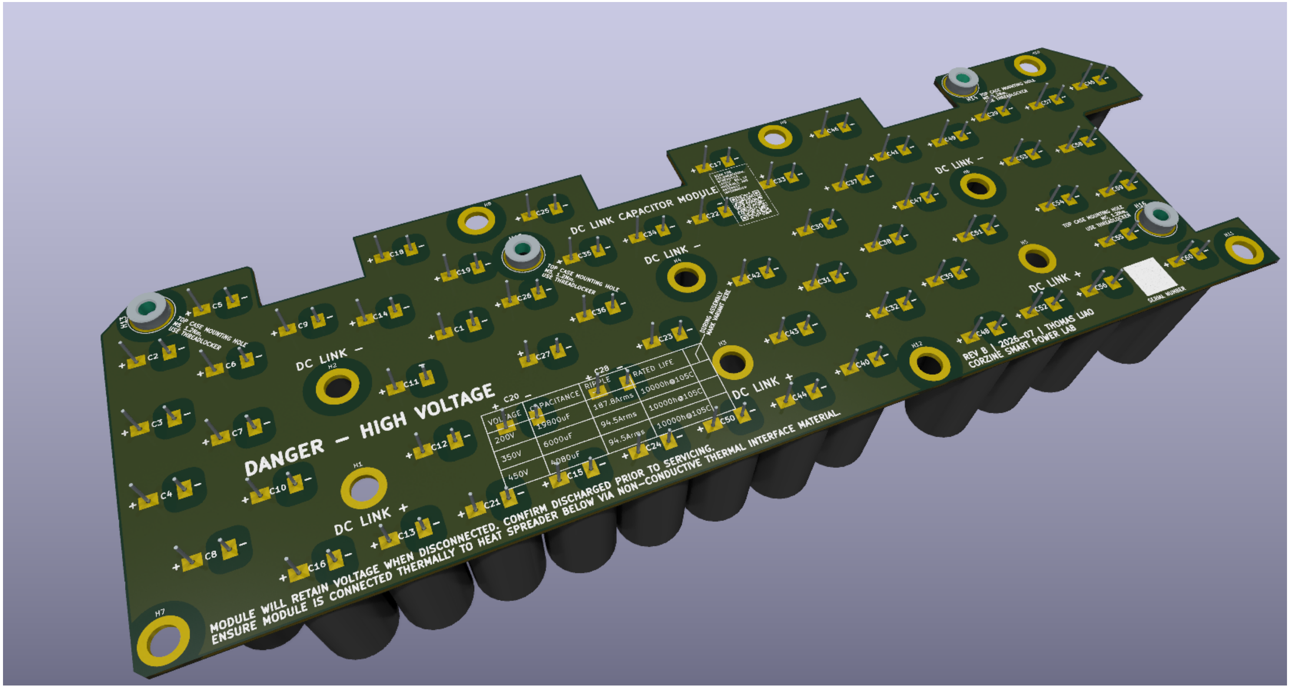
Control Board *— Dual-MCU control: STM32H723ZG main processor (FOC, sensor acquisition, CAN) and STM32G474RCTx safety coprocessor (independent monitoring, 1oo2 gate drive power kill).*



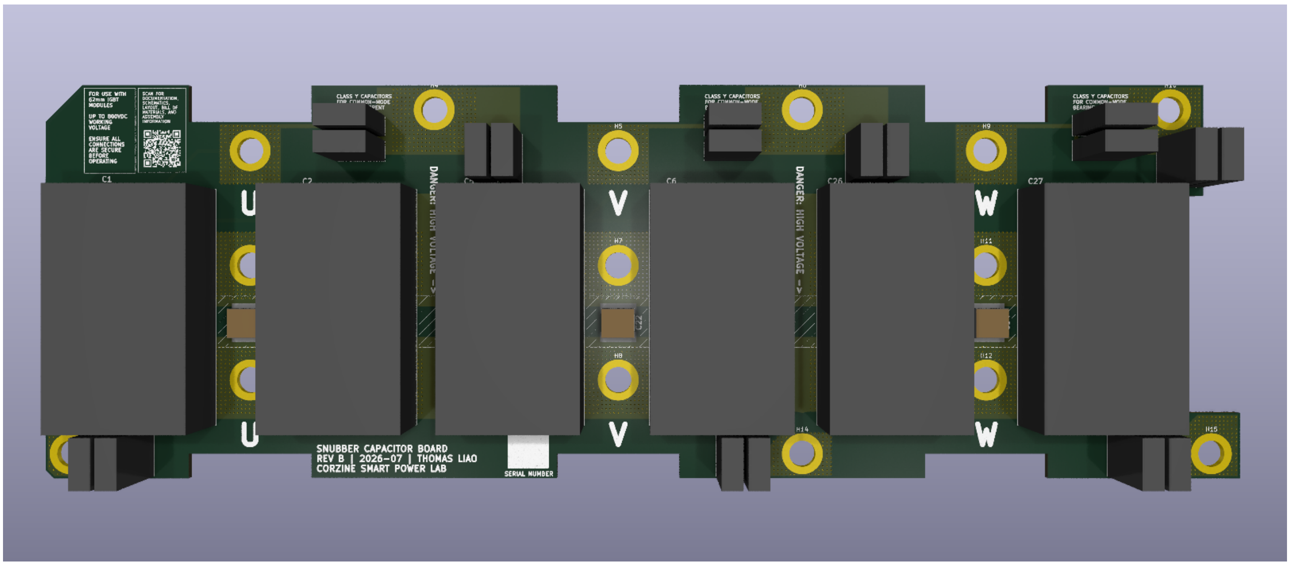
IO Board *(the "Main Board" of Section 1.2) — Vehicle interface: Ampseal 35-pin connector, railway-grade 43-160 V to +12 V DC-DC converter, precharge and auxiliary relay drives, J2 expansion connector (Section 18).*



Gate Driver Board *— Six isolated NCV57100 gate drive channels with +15 V / −9 V Murata supplies, DESAT protection, and active Miller clamp.*



DC Bus Capacitor Board *— DC link capacitor bank: 60× 330 µF / 200 V aluminium electrolytic capacitors (19.8 mF total). 450 V upgrade: 60× Nichicon UCS2W680MHD 68 µF / 450 V (4.08 mF total) with 5 mm shorter standoffs. No onboard bleeder — see Section 2.1.*



DC Bus Filter Board *— Snubber capacitor board mounted at the IGBT module phase terminals (U/V/W).*

2. Safety Information

2.1 High Voltage Warnings

> **DANGER — Lethal Voltage Present**
>
> The DC link busbars carry the full traction battery voltage (43-160 VDC). This voltage is lethal. Always treat the DC link as energized until verified otherwise with a calibrated voltmeter. Never work on the inverter with the battery connected. Remove the B+ fuse or open the main line contactor before servicing.
>
> **DANGER — Capacitor Stored Energy**
>
> The DC link capacitor bank (19.8 mF) has **no onboard bleeder** and remains charged at full bus voltage for hours after the battery is disconnected. Do not rely on any fixed wait time. Before touching any HV terminal, verify zero voltage with a calibrated voltmeter across DC Link Positive and Negative, and discharge the bank through a power resistor if any voltage remains.

2.2 General Safety

- Only qualified personnel should install, configure, or service this equipment.
- Always wear appropriate personal protective equipment (PPE) when working with high-voltage systems.
- Ensure the installation location provides adequate cooling. The inverter requires a user-provided heatsink (Section 5).
- The inverter draws quiescent power from the battery even when not actively inverting. For long-term storage, disconnect VIN_Z (Section 3).
- Do not connect or disconnect the Ampseal 35-pin connector while the system is powered.
- Do not operate the inverter without the enclosure lid properly sealed — the HVIL will detect this, but the interlock is a signal, not a physical barrier.

2.3 This Connector is NOT Sevcon-Compatible

> **DANGER — Pinout Incompatibility**
>
> The Ampseal 35-pin connector on this inverter is **NOT** pin-compatible with Sevcon Gen4 harnesses. Connecting a Sevcon harness to this inverter will cause immediate equipment damage, fire, and potential personal injury. Always verify pin-to-pin compatibility with the pinout tables in Section 7 before making any connection.

This inverter replaces legacy Sevcon Gen4 controllers in aftermarket conversion applications, but the vehicle harness must be rewired to match the pinout defined in this manual. A wiring adapter or new harness fabrication is required. Do not attempt to use an unmodified Sevcon harness.

3. Idle Power Draw and Storage

> **WARNING — Standby Power Consumption**
>
> The traction inverter draws quiescent power from the battery via VIN_Z even when not actively inverting. If left unattended, this standby consumption will discharge the battery pack and may cause permanent cell damage.

For rolling stock or vehicles requiring long-term unattended storage:

- **Complete isolation:** Disconnect or remove power to VIN_Z to eliminate all standby power draw during storage.
- **Storage mode:** The system can be configured to wake periodically (daily or weekly) to check battery state-of-charge (SOC) via the BMS CAN bus. If pack voltage drops below a configured threshold, the system initiates a maintenance charge cycle. In storage mode, the control processor wakes from low-power state, samples the DC link voltage, and commands the precharge/contactor logic if supplemental charging is required.

> **Note — No Hardware Sleep Pin**
>
> The inverter does not include a dedicated hardware sleep-enable pin or an ultra-low quiescent mode. The only way to eliminate standby power draw is to physically disconnect VIN_Z. A future hardware revision may add a low-power sleep state or a sleep-control input. For long-term storage, a battery disconnect switch or timer-controlled relay on the VIN_Z line is recommended.

For maximum safety during extended storage, always physically isolate the high-voltage input by removing the B+ fuse or opening the main line contactor.

4. High Voltage Interlock (HVIL)

The High Voltage Interlock Loop (HVIL) is a safety-critical signal path between the inverter and the Battery Management System (BMS). It is important to understand that **the inverter does not directly open the contactor** — the HVIL is a presence signal, and the BMS is responsible for contactor control.

How HVIL Works

1. The inverter provides a +5V source (O_+5V_HVIL_SOURCE_A) referenced to CAN1_GND.
2. This +5V is routed in series through all high-voltage connectors and the enclosure lid switch in the vehicle.
3. The return (I_+5V_HVIL_RETURN_A) comes back to the inverter, completing the loop.
4. When the loop is intact, the inverter asserts the SYSTEM_ON signal to the BMS, indicating it is safe to operate.
5. If any HV connector is unmated or the enclosure is opened, the loop breaks, SYSTEM_ON is deasserted, and the **BMS opens the main contactor**.
6. Upon SYSTEM_ON deassertion, the inverter immediately disables all PWM outputs and enters a safe state.

> **HVIL is a Signal, Not a Contactor Drive**

> The inverter does not control the contactor directly. The HVIL is a +5V interlock loop that the BMS monitors for integrity. When the loop opens, the BMS (not the inverter) opens the contactor. The inverter's role is to (1) provide the +5V source, (2) detect the return, (3) assert/deassert SYSTEM_ON, and (4) disable PWM when SYSTEM_ON goes away. Ensure your BMS is configured to open the contactor on SYSTEM_ON deassertion.
>
> **WARNING — Isolated Grounds**
>
> The BMS signal ground (CAN1_GND) is isolated from traction negative. Do not tie CAN1_GND or signal ground (GND_A) to the HV negative bus. Doing so compromises the safety isolation and may cause equipment damage or personal injury.

5. Thermal Management

The traction inverter does not include an integrated heatsink or cooling system. The device is provided with a flat mechanical backplate. The user must provide external thermal management:

- Apply thermal interface material (thermal grease or phase-change pad) between the device backplate and the heatsink or chassis mounting surface.
- Provide adequate heat removal (heatsink, liquid cooling plate, or chassis thermal path) sized for the continuous power output of the application.
- Ensure the mounting surface is flat and clean to minimize thermal contact resistance.

The IGBT junction temperature is monitored internally via three NTC thermistors (one per IGBT module). The firmware implements progressive thermal derating:

Thermal Derating Thresholds

Temperature	Action
< 80°C	Full torque available
80°C	Linear derate: 100% to 50% torque (80-90°C range)
90°C	Torque limited to 25%
95°C	Warning logged (non-critical)
100°C	Critical fault: ramp torque to zero, safe state

> **Thermal Design Margin**
>
> The 100°C hard cap provides a 75°C safety margin to the IGBT rated maximum junction temperature of 175°C. The real-time loss estimator predicts junction temperature from a thermal model and can warn the operator of impending thermal limits before the hard cap is reached. All thermal parameters are configurable via the RTE tool.

DC Link Capacitor Cooling

The DC link capacitor bank is thermally coupled to a 3.18 mm (1/8 in) aluminium heat-spreader plate, which mounts to the heatsink via six 55 mm aluminium standoffs (13 mm OD). The thermal path is sized for a 40 W ripple-current heat load at rated operation.

With thermal paste at the interfaces, the plate temperature rise is approximately 40°C above the heatsink base — about 80°C plate temperature at a 40°C heatsink base. Use **aluminium standoffs only** — steel standoffs are thermally unacceptable. The full analysis is documented in Docs/DC_LINK_THERMAL_ANALYSIS.md.

6. Specifications

Electrical Specifications — 140V / 600A Variant

Parameter	Value	Notes
DC Input Voltage	140 VDC nominal (43-160 VDC range)	43-160 VDC configurable range
Continuous Output Current	600 A RMS	Thermal limit depends on user cooling
Peak Output Current	1200 A	Configurable, time-limited
Output Type	3-phase AC, IGBT-based	VVVF drive for PMSM motors
Switching Frequency	300 Hz - 16 kHz	User-configurable via RTE; default 2 kHz SVPWM
Modulation Schemes	SPWM, SVPWM, ARSVPWM, SHEPWM, RCFM, RSPWM, N-Pulse / N-Pulse Wide / N-Pulse Custom, RSVM	SPWM and SVPWM implemented; remainder architected / in development — see Section 17
Power Device	Mitsubishi 1200V IGBT module	6-switch full bridge
Gate Drive Voltage	+15V / -9V	Positive turn-on, negative turn-off
Gate Drive Isolation	>5 kVrms (UL1577)	Galvanic, reinforced
Current Sensing	STM32H723 16-bit ADC	4 channels (I_U, I_V, I_W, I_DC)
Voltage Sensing	MAX22530 isolated ADC	4 channels (V_DC, V_U, V_V, V_W)
Control Interface	Dual isolated CAN 2.0B	Up to 1 Mbps each
Digital Inputs	8x ISO1212 isolated (2 populated in current build)	24-60V field inputs, galvanically isolated from logic domain
Relay Outputs	4x switched +12V	Precharge and auxiliary relay drive (not the main HV contactor)
Logic Power	VIN_Z via DC-DC	43-160V in, +12V out, 20W, railway grade
HVIL	5V inline interlock loop	Signal to BMS; BMS controls contactor
Operating Temperature	-40°C to +85°C	Ambient at backplate
Isolation (Logic to Traction)	Reinforced, railway grade	Full galvanic isolation
Control Processor	STM32H723ZG	Cortex-M7, 550 MHz, 1 MB flash
Non-Volatile Storage	SPI FRAM	Unlimited write endurance, WP pin

Parameter	Value	Notes
Debug Interface	USB-B (native USB 2.0 FS)	External port, isolated
Firmware Update	USB-B or CAN bus	Both use same protocol

7. Ampseal 35-Pin Connector

All vehicle-facing signals route through the Ampseal 35-pin sealed connector (TE 776231-1). The connector is keyed and cannot be rotated 180 degrees — the key prevents incorrect mating. Always verify the key orientation before mating.

> **DANGER — Pinout Not Yet Finalized**
>
> The pin assignments below are **placeholder only**. Pin numbers are shown as "—" until the harness design is finalized. Do not wire your vehicle harness based on this manual. A finalized pinout will be released in a future revision. Contact the project for the current draft pinout if you are building a test harness.

Connector Summary

Function Group	Pin Count
CAN Bus 1 (BMS)	4
CAN Bus 2 (Dash / Charger)	4
Throttle Inputs	2
Motor Position / Temperature	10
Power and Keyswitch	3
HVIL	2
Precharge / HV Control	4
CAN Termination	2
Reserved / Unused	4

7.1 CAN Bus 1 (BMS)

CAN Bus 1 — BMS and Peripheral Interface

Pin	Signal	Description
—	CAN1_L	CAN Bus 1 Low
—	CAN1_H	CAN Bus 1 High
—	CAN1_GND	CAN Bus 1 ground (isolated from traction negative)
—	GND_A	Signal ground (isolated from traction negative)

7.2 CAN Bus 2 (Dash / Charger)

CAN Bus 2 — Dashboard, Charger, and Diagnostic Interface

Pin	Signal	Description
—	CAN2_L	CAN Bus 2 Low
—	CAN2_H	CAN Bus 2 High
—	CAN2_GND	CAN Bus 2 ground (isolated from traction negative)
—	GND_A	Signal ground (shared with CAN1)

> **Dual Isolated CAN Buses**
>
> CAN Bus 1 is designated for BMS and powertrain-critical nodes. CAN Bus 2 is for dashboard, charger, and diagnostic tools. Both buses are galvanically isolated from traction negative and from each other. The CAN protocol is user-configurable via the RTE tool — frame IDs, scaling factors, and periods can be modified for vehicle-specific integration.

7.3 Throttle Inputs

Analog Throttle / Torque Demand Input

Pin	Signal	Description
—	THROTTLE	Analog throttle / torque demand input (redundant potentiometer, 0-5 V scaled to 0-3.3 V at the ADC)
—	THROTTLE_GND	Throttle signal return (single-ended, referenced to AGND)

> **WARNING — Throttle Wiring**
>
> The throttle signal is referenced to AGND, not to the traction battery negative. Do not connect throttle ground to HV negative. Use a 3-wire shielded cable for the throttle connection. The shield should be grounded at the inverter end only to avoid ground loops.

7.4 Motor Position / Temperature

Motor Position Sensor Interface (Sin/Cos or Hall)

Pin	Signal	Description
—	MOT_T1+	Motor temperature sensor 1 (+), 0.01" wire
—	MOT_T1-	Motor temperature sensor 1 (-), 0.01" wire
—	MOT_T2+	Motor temperature sensor 2 (+)
—	MOT_T2-	Motor temperature sensor 2 (-)
—	SIN+	Sin/Cos encoder sine channel (+) / Hall U

Pin	Signal	Description
—	SIN-	Sin/Cos encoder sine channel (-) / Hall V
—	COS+	Sin/Cos encoder cosine channel (+) / Hall W
—	COS-	Sin/Cos encoder cosine channel (-) / Hall common
—	REF+	Sin/Cos reference mark (+) / Hall VCC (+5V)
—	REF-	Sin/Cos reference mark (-) / Hall GND

> **Encoder Type Configuration**
>
> The firmware supports **sin/cos (synchro-style) encoders** and **digital Hall effect sensors**. Sin/cos encoders provide absolute position via analog sine/cosine differential signals. Hall sensors provide 60-degree electrical sector commutation. **Quadrature encoders and resolvers are NOT supported.** The pin names reflect the primary (sin/cos) function with the Hall mapping in parentheses. Configure the sensor type via the RTE tool before first run. Verify sensor type and pinout with the motor manufacturer.

7.5 Power and Keyswitch

Power Input and Keyswitch Interface

Pin	Signal	Description
—	VIN_Z	+43-160VDC logic supply input (via onboard DC-DC). This input powers all internal logic, gate drive, and auxiliary circuits.
—	KEYSWITCH	Keyswitch input (connect to battery positive via keyswitch)
—	KEYSWITCH_RET	Keyswitch return

> **WARNING — VIN_Z is Not Optional**
>
> VIN_Z must be connected for the inverter to operate. This input powers the onboard DC-DC converter that generates +12V for all internal logic and gate drive circuits. Without VIN_Z, the inverter is completely non-functional. For storage, VIN_Z can be disconnected to prevent battery drain (Section 3).

7.6 HVIL

High Voltage Interlock Loop

Pin	Signal	Description
—	O_+5V_HVIL_SOURCE_A	HVIL +5V source output (referenced to CAN1_GND)
—	I_+5V_HVIL_RETURN_A	HVIL +5V return input (from interlock loop)

The HVIL is a +5V series loop that runs through all high-voltage connectors and the enclosure lid switch. When the loop is intact, the inverter asserts SYSTEM_ON to the BMS. When the loop opens (connector unmated or lid opened), SYSTEM_ON is deasserted and the BMS opens the main contactor. See Section 4 for full HVIL description.

7.7 Precharge / HV Control

Precharge Control and HV Taps

Pin	Signal	Description
—	O_12V_PRECHARGE_ENABLE	Precharge relay drive output (+12V, high-side isolated)
—	O_12V_RELAY_AUX	Auxiliary relay drive output (+12V) — NOT the main HV contactor
—	I_VIN_HIGH_A	High-side HV tap for precharge voltage monitoring
—	I_VIN_LOW_A	Low-side HV tap for precharge voltage monitoring

7.8 CAN Termination

CAN Bus Termination Resistors

Pin	Signal	Description
—	CAN1_TERM	CAN Bus 1 120-ohm termination (jumper-selectable)
—	CAN2_TERM	CAN Bus 2 120-ohm termination (jumper-selectable)

> **CAN Termination**
>
> The termination resistors are enabled or disabled by on-board jumpers (CAN1/CAN2 termination jumpers). In a typical vehicle network, the BMS or another node provides termination at the opposite end of the bus. Enable termination on the inverter only if it is at the physical end of the bus. The jumpers allow termination to be fitted without external resistors.

7.9 Reserved Pins

Reserved / Unused Pins

Pin	Signal	Description
—	NC	Reserved for future use
—	NC	Reserved for future use
—	NC	Reserved for future use
—	NC	Reserved for future use

8. Power Supply Architecture

The power supply system consists of three subsystems that provide regulated and isolated power for all internal circuits.

8.1 Logic Power (VIN_Z)

VIN_Z is the primary input for all internal power. An onboard railway-grade isolated DC-DC converter (1500 Vdc I/O isolation) converts the 43-160 VDC traction battery voltage to regulated +12V logic power. This converter provides up to 20W, which is sufficient for the MCU, gate drive, CAN transceivers, sensors, and all auxiliary circuits.

The DC-DC converter is enabled by the keyswitch input. When the keyswitch is off, the converter is disabled and the inverter draws no power (except for microamp leakage current through the keyswitch detection circuit).

8.2 Switched +12V Output

The +12V from the DC-DC converter is distributed to four switched outputs for driving external relays:

- **O_12V_PRECHARGE_ENABLE:** Drives the precharge relay. The precharge relay controls a resistor-limited path that charges the DC link capacitor before the main contactor closes. The precharge relay is on the **high side** (battery positive path), not the low side. The +12V drive is isolated and routed through the relay coil.
- **O_12V_RELAY_AUX:** Auxiliary relay drive for vehicle-specific functions (e.g., cooling pump, fan, indicator light). **This is NOT the main HV contactor.** The main HV contactor is controlled by the BMS or other external safety system, not by this inverter.
- **CAN Bus 1 and 2 power:** Isolated +5V CAN transceiver power derived from the +12V rail.

8.3 Isolated Sensor and Communication Rails

From the +12V rail, multiple isolated DC-DC converters generate the regulated voltages required by specific subsystems:

- **Gate drive power (+15V / -9V):** Six isolated Murata converters provide gate drive voltage for the six IGBT switches. Positive rail (+15V) for fast turn-on; negative rail (-9V) for fast turn-off and Miller clamp.
- **CAN transceiver power (+5V isolated):** Isolated +5V for both CAN bus transceivers, maintaining galvanic isolation from traction negative.
- **Sensor power (+5V isolated):** Isolated +5V for current sensors, temperature sensors, and encoder interface.

> **Gate Drive Isolation**
>
> Each of the six gate drive channels has its own isolated DC-DC converter. This provides per-channel isolation — a fault on one channel cannot propagate to others through the gate drive power supply. Each Murata MGJ2D121509MPC-R7 gate-drive DC/DC provides 5.2 kVDC input-to-output isolation. The gate-driver isolation barrier (NCV57100) is rated >5 kV_{rms} per UL1577 and 1424 V_{PK} / 1000 V_{rms} working voltage per VDE 0884-11.

9. Gate Drive

Each IGBT is driven by an isolated gate driver channel with the following features:

- **+15V / -9V gate drive:** Positive voltage for rapid turn-on; negative voltage for rapid turn-off and active Miller clamp.

- **Active Miller clamp:** Prevents false turn-on during high dv/dt switching by clamping the gate to the negative rail during the off state.
- **Desaturation detection (DESAT):** Monitors the IGBT Vce during the on-state. If Vce exceeds the desaturation threshold, indicating a short-circuit condition, the driver immediately turns off the IGBT and asserts the FLT output.
- **Fault output (FLT):** Six individual FLT signals are OR'd together and fed to TIM1_BKIN. When any driver detects a fault, all PWM outputs are immediately disabled by hardware (SSO) within <100 ns.
- **Power supply monitoring (UVLO):** Each driver monitors its own isolated gate drive supply. If the supply drops below the UVLO threshold, the driver turns off the IGBT and asserts FLT.

> **Gate Drive Supply Kill — 1oo2 Architecture**
>
> Two independent MCUs can each disable all six gate drive power supplies: GATE_DRIVE_PWR1_ENABLE (main STM32H723) and GATE_DRIVE_PWR2_ENABLE (coprocessor STM32G474). Either enable going low shuts down the Murata supplies, the NCV57100 drivers detect UVLO, and force all IGBT gates to 0V via internal active pull-downs. This is a 1oo2 architecture — either MCU can achieve SSO independently. Independent feedback is provided: GATE_DRIVE_PWR1_FB (main MCU) and GATE_DRIVE_PWR2_FB (coprocessor). The FLT signal additionally confirms UVLO detection. Six total SSO pathways exist; see the HARA (v4.1) Section 2.2 for full details.

10. Sensing and Acquisition

Current Sensing

Phase currents (I_U, I_V, I_W) and DC link current (I_DC) are measured using Tamura LA37S Hall-effect current transducers with 1.042 mV/A sensitivity and +/-1200 A range. The outputs are conditioned to 0-3.3V for the STM32H723's 16-bit ADC.

Each phase current is sampled on the primary ADC at 16-bit resolution, with redundant 12-bit channels on secondary ADCs for fault detection. The ADC oversamples at an integer multiple (n x) of the PWM switching frequency, up to 48 kSPS, with sinc3 decimation for noise reduction. See SWAD (v1.5) Section 4.2 for full ADC architecture details.

Voltage Sensing

DC link voltage (V_DC) and all three phase voltages (V_U, V_V, V_W) are measured using the MAX22530 4-channel isolated 12-bit ADC. The MAX22530 provides reinforced isolation between the high-voltage measurement domain and the low-voltage logic domain via its internal reinforced barrier, with SPI interface to the STM32H723.

Each channel uses a 1001:1 high-voltage divider (4× 250 kΩ + 1 kΩ), giving a measurable range to ~1.8 kV full-scale.

Temperature Sensing

Three NTC thermistors are mounted to the IGBT module baseplate — one per module — for junction temperature estimation. The firmware uses 2-out-of-3 (2oo3) voting: the outlier sensor is excluded and the higher of the two agreeing readings is used, with progressive derating starting at 80°C and a hard cap at 100°C. Motor temperature sensors (if fitted) are connected via the MOT_T1/MOT_T2 pins on the Ampseal connector (Section 7.4; pin numbers are placeholder pending harness finalization).

11. External Precharge

> **WARNING — Main HV Contactor is NOT Driven by This Inverter**
>
> The main high-voltage contactor that connects the battery to the inverter DC link is **controlled by the BMS or other external safety system**, not by this inverter. The inverter only drives the **precharge relay** (a smaller relay that limits inrush current through a resistor). Do not connect the main contactor coil to any inverter output — this is a safety-critical function that must be managed by the BMS.

The precharge function uses a **high-side relay** in the battery positive path. A precharge resistor limits the current that charges the DC link capacitor when the system is first powered. The +12V relay drive is isolated from the traction high voltage. The relay closes the precharge path; once the DC link voltage reaches ~95% of battery voltage, the BMS closes the main contactor and the precharge relay opens.

> **CAUTION — Precharge Current Limit**
>
> The precharge resistor is external and user-supplied. Size it so that the peak precharge current stays below 2 A — the input fuse limit on the precharge path. Exceeding this limit will blow the fuse.

The inverter provides four signals to support the precharge function:

Precharge Interface Signals

Signal	Pin	Type	Description
O_12V_PRECHARGE_ENABLE	—	Output, +12V isolated	Drives the precharge relay (high-side, resistor-limited path)
O_12V_RELAY_AUX	—	Output, +12V	Auxiliary relay drive (NOT main contactor)
I_VIN_HIGH_A	—	Input, HV	High-side tap for precharge voltage monitoring
I_VIN_LOW_A	—	Input, HV	Low-side tap for precharge voltage monitoring

The firmware monitors the voltage difference between I_VIN_HIGH_A and I_VIN_LOW_A during the precharge sequence. When the DC link capacitor voltage reaches approximately 95% of the battery voltage, the inverter signals precharge complete to the BMS via CAN. The **BMS then closes the main contactor**. The inverter does not close the main contactor itself.

12. Dual Isolated CAN Bus

The inverter includes two independent CAN 2.0B buses, each with full galvanic isolation from traction negative and from each other. Each ISO1042BDWVR transceiver pair is rated for VIORM 2121 V_{PK}.

CAN Bus Assignment

Parameter	CAN Bus 1	CAN Bus 2
Pins	—	—
Nodes	BMS, IO board	Dashboard, charger, diagnostic tools

Parameter	CAN Bus 1	CAN Bus 2
Bitrate	500 kbps (default)	500 kbps (default)
Termination	Jumper-selectable	Jumper-selectable
Isolation	2.5 kVrms	2.5 kVrms

> **CAN Protocol is User-Configurable**
>
> The CAN frame definitions in this manual are a reference implementation only. All CAN IDs, scaling factors, periods, and frame layouts are stored in FRAM and can be modified via the RTE (Real Time Examiner) configuration tool. The VCU does not enforce a rigid protocol — the user defines their own communication scheme for vehicle-specific integration. See the SWAD (v1.5) Section 10 for the reference frame definitions.
>
> **CAUTION — Encoder and BMS are Out of Scope**
>
> This inverter does not include or supply the traction motor, rotor position sensor (sin/cos encoder or hall sensor), battery management system, IO board, charger, or vehicle display. These are external products that interface with the inverter via the CAN bus and analog/digital I/O. This manual documents the inverter-side interface only. Refer to the motor manufacturer, BMS manufacturer, and vehicle OEM for their respective documentation.

13. CAN Protocol

The CAN protocol is defined by the user via the RTE configuration tool. The following tables describe the default frame assignments. All IDs, periods, and scaling factors are user-modifiable.

VCU Transmitted Frames (TX)

CAN ID	Name	Period	Key Signals
0x18F00100	VCU_Status	100 ms	State, fault flags
0x18F00200	VCU_Motor	50 ms	Speed (rpm), torque (Nm), I_q, I_d
0x18F00300	VCU_Power	100 ms	V_DC, I_DC, power (kW)
0x18F00400	VCU_Thermal	500 ms	T_IGBT[0-2], T_junction_est
0x18F00500	VCU_LossEst	500 ms	P_total_loss, time_to_OT_sec
0x18F00600	VCU_Heartbeat	1000 ms	Seq counter, uptime
0x18F00700	VCU_FaultLog	Event	Event code + detail

VCU Received Frames (RX)

CAN ID	Source	Key Signals
0x18F10100	BMS	SOC, SOH, pack voltage, pack current
0x18F20100	IO Board	Throttle primary, throttle check

CAN ID	Source	Key Signals
0x18F20200	IO Board	Key on, brake, kickstand
0x18F20300	IO Board	Heartbeat (timeout: 1000 ms)
0x18F30100	Charger	Charger state, max current
0x18F40100	Diagnostic	UDS-style commands
0x18F50100	RTE Host	Parameter read/write

14. BMS Interface

The inverter communicates with the Battery Management System (BMS) via CAN Bus 1. The default CAN frame IDs are listed in Section 13, but these are user-configurable via RTE.

Heartbeat Supervision

The inverter monitors heartbeats from the BMS and IO board. If a heartbeat is not received within the timeout period, the inverter enters a fault state and ramps torque to zero:

Heartbeat Timeout Configuration (Defaults)

Node	Frame	Period	Timeout	Action
IO Board	IO_Heartbeat	100 ms	1000 ms	FAULT_CAN_TIMEOUT, ramp to zero
BMS	BMS_Status	100 ms	5000 ms	FAULT_CAN_TIMEOUT, ramp to zero
Charger	Chgr_Status	1000 ms	5000 ms	Stop charging session

SYSTEM_ON Signal

The inverter asserts SYSTEM_ON when:

- HVIL loop is intact (all connectors mated, lid closed)
- Keyswitch is on
- POST passed successfully
- No critical faults are active

SYSTEM_ON is deasserted when any of these conditions fails. The BMS must be configured to open the main contactor on SYSTEM_ON deassertion.

15. USB-B Debug Port

An external USB-B connector provides access to the debug and firmware update interface. The USB-B port routes raw USB D+/D− directly to the coprocessor's (STM32G474RCTx) native USB 2.0 full-speed (12 Mbit/s) interface. The port is **not bus-powered** — the inverter must be powered from its external supply for the USB interface to operate; plugging in a laptop alone will not power the board.

Functions

- **Debug log output:** Real-time status messages, fault logs, and diagnostic data
- **Firmware update:** Same protocol as CAN bus update (HMAC-SHA256 signed, chunk-based)
- **RTE tunnel:** RTE commands can be sent via USB-B when CAN is unavailable

Isolation

The USB-B port is galvanically isolated from the traction high-voltage domain. It is safe to connect a laptop to the USB-B port while the inverter is powered from the traction battery. However, for maximum safety, use an additional USB isolator between the laptop and the inverter when working on a live system.

> **WARNING — USB Isolator Recommended**
>
> While the USB-B port is internally isolated, an additional external USB isolator (e.g., Adafruit USB Isolator, 1 kV isolation) is strongly recommended when connecting a laptop to a live traction system. This provides defense in depth against ground fault paths through the laptop chassis.

16. Maintenance and Troubleshooting

16.1 Common Faults

Troubleshooting Guide

Symptom	Possible Cause	Action
No response to keyswitch	VIN_Z not connected or open fuse	Check VIN_Z voltage at the connector with a voltmeter. Verify fuse.
POST fails on boot	Gate driver fault pin stuck low; FRAM WP fault	Check FLT line at TIM1_BKIN. Run POST diagnostics via USB-B.
No torque output	Throttle plausibility fault; position sensor not detected	Check throttle wiring at the connector. Verify sin/cos or Hall signal on USB-B debug log.
BMS contactor won't close	HVIL loop open; SYSTEM_ON not asserted; BMS not closing contactor	Check HVIL continuity at the connector. Verify 5V source is present. Check BMS is configured to close contactor on SYSTEM_ON.
CAN communication fault	Wrong bitrate; missing termination	Verify CAN bitrate matches BMS (default 500 kbps). Check termination jumpers.
Thermal derating	Inadequate cooling; stuck-high temp sensor	Check heatsink mounting and TIM. Verify all three NTC readings via RTE.
Overcurrent fault	Shorted motor phase; incorrect current sensor zero	Disconnect motor and check phase-to-phase resistance. Run zero-cal via RTE.
PWM not outputting	Keyswitch off; brake applied; throttle deadband	Check keyswitch state, brake switch, throttle above deadband. Check state machine log.

16.2 Debug Interface Warning

> **WARNING — Authorized Use Only**
>
> The debug interface and firmware update capability are provided for development, diagnostic, and authorized repair purposes only. Access to these interfaces allows modification of safety-critical parameters including current limits, thermal thresholds, and fault response behavior. Modification of these parameters without proper understanding of the system can result in equipment damage, fire, or personal injury. Always maintain a known-good firmware image and parameter set before making changes.
>
> **Open Source**
>
> This project is open source. Firmware source code, schematics, and all documentation are available on the project GitHub repository. The STM32H723ZG is shipped with RDP Level 0 (no readout protection) — debugging and firmware modification are always possible. The user is the root of trust. The project explicitly rejects anti-user security mechanisms (OTP fuses, vendor-locked keys, encrypted bootloaders with unreplaceable keys). Physical tampering is an acceptable risk per the open-source trust model documented in the TARA (v1.2).
>
> **CAUTION — Third-Party Firmware**
>
> Only firmware obtained from the official project repository or built from audited source code is supported. Third-party firmware may disable safety features, override thermal limits, or introduce software faults. Use of unverified firmware is at the user's own risk. The HARA (v4.1) fault injection test results apply only to the official firmware. No fitness for purpose is guaranteed for any firmware configuration.

17. Motor Control and Modulation Strategies

> **Note — Planned Feature**
>
> The multi-modulation architecture described in this section is planned for a future firmware release. The current firmware implements SPWM and SVPWM. Expanded modulation support, live switching, and automatic selection are under development. Planned validation on a dynamometer at a university lab.

The firmware implements an adaptive pulse-width modulation framework. The user can select from multiple modulation schemes, switch between them during operation, and configure automatic selection rules based on operating conditions. All modulation parameters are accessible via the RTE tool over CAN bus.

17.1 Available Modulation Schemes

Modulation Schemes (Planned)

Scheme	Description	Typical Application
SPWM	Sinusoidal Pulse Width Modulation. Three-phase sine references compared with a triangular carrier. Simple, low harmonic content at high modulation indices.	General purpose; smooth operation at moderate speeds

Scheme	Description	Typical Application
SVPWM	Space Vector Pulse Width Modulation. Uses eight voltage vectors (six active, two zero) to synthesize the reference. Higher DC bus utilization than SPWM (~15% more).	High torque, efficient operation
ARSVPWM	Alternating Reverse Sequence Vector PWM. A variant of SVPWM that alternates the switching sequence pattern every other carrier period to spread harmonic energy and reduce peak current ripple.	Low-noise applications; spread-spectrum-like benefits
SHEPWM	Selective Harmonic Elimination PWM. Pre-computed switching angles eliminate specific low-order harmonics. Requires solve-at-runtime or pre-calculated angle tables.	High-speed operation where specific harmonics must be suppressed
RCFM	Random Carrier Frequency Modulation. The PWM carrier frequency is dithered randomly within a configured band to spread electromagnetic interference across a wide spectrum.	EMI-sensitive installations; reduces tonal acoustic noise
RSPWM	Random Sinusoidal Pulse Width Modulation. Combines random carrier frequency with random pulse position to maximize EMI spreading.	Maximum EMI reduction; aerospace and sensitive environments
N-Pulse	Low pulse-count operation for very high speeds. The inverter switches only a few times per electrical cycle. Requires careful current reconstruction.	Field weakening region; very high motor speeds
N-Pulse Wide	Extended N-Pulse with wider conduction angles. Trade-off between switching loss and current ripple at extreme speeds.	Extended field weakening beyond base N-Pulse range
N-Pulse Custom	User-defined pulse count and angle pattern. For experimental or application-specific modulation.	Research and custom motor applications
RSVM	Random Space Vector Modulation. Randomly selects between different valid vector sequences on each carrier period to spread harmonics.	Mid-speed range; alternative to RCFM with better DC utilization

17.2 Live Switching and Automatic Selection

The user can switch between modulation schemes at runtime via CAN bus commands. Two modes of operation are supported:

- **Manual mode:** The user explicitly selects the active modulation scheme via RTE. The switch occurs at the next zero-current or zero-voltage crossing to minimize disturbance.
- **Automatic mode:** The user defines a modulation map that selects the scheme based on real-time operating conditions. Parameters include motor speed, torque demand, DC link voltage, and thermal state.

Example automatic selection map:

Example Modulation Map (User-Configurable)

Speed Range	Torque	Selected Scheme	Rationale
0 - 20% base	Any	ARSVPWM	Low speed, low ripple, quiet operation
20% - 80% base	High	SVPWM	Maximum DC bus utilization
20% - 80% base	Low	RCFM	Spread EMI at light load
80% - 100% base	Any	SHEPWM	Eliminate low-order harmonics at high speed

Speed Range	Torque	Selected Scheme	Rationale
> 100% base (field weak)	Any	N-Pulse	Minimize switching losses

> **WARNING — Transition Torque Blips**
>
> Switching modulation schemes during active torque production can cause brief torque disturbances. The bumpless transfer logic crossfades the voltage reference over several PWM periods, but a small transient is unavoidable when the harmonic structure changes abruptly. The firmware gates scheme transitions during critical moments (high di/dt, regen/acceleration crossover) to prevent perceptible torque blips. Do not request a scheme switch during hard acceleration or braking unless you have validated the transition behavior on your specific motor.
>
> **Hysteresis on Scheme Boundaries**
>
> When operating near a speed or torque boundary between two modulation schemes, the automatic selector applies hysteresis to prevent rapid back-and-forth switching. The hysteresis band is user-configurable (default: 5% of the switching parameter). For example, if the SVPWM-to-SHEPWM boundary is at 80% base speed, the firmware will switch to SHEPWM at 80% but will not switch back to SVPWM until speed drops below 75%. This eliminates audible jitter and unnecessary torque transients at boundary conditions.

17.3 Configuration via CAN Bus

All modulation parameters are exposed through the RTE configuration interface over CAN bus:

- **Active scheme:** Read/write the currently selected modulation (manual mode) or the active map slot (automatic mode).
- **Modulation map:** Define up to 8 speed/torque regions with per-region scheme selection. Upload via RTE parameter write.
- **Transition timing:** Configure the crossfade duration (in PWM periods) for bumpless transfers.
- **Hysteresis bands:** Set the hysteresis width for each map boundary.
- **SHE angle table:** Upload pre-computed switching angles for SHEPWM. Supports live recalculation for variable DC link voltage.
- **RCFM band:** Set the minimum and maximum carrier frequency for random dithering.

> **WARNING — Validate on Your Hardware**
>
> Modulation scheme behavior varies with motor inductance, winding configuration, and mechanical resonance. Always validate modulation selection and transition behavior on your specific hardware before deploying in a vehicle. Torque ripple, acoustic noise, and EMI characteristics are motor-dependent. The automatic selection map should be tuned for your application.

18. Expansion Module

> **Note — Planned Feature**

> All expansion modules described in this section are planned. The J2 expansion connector exists on the IO board, but no plug-in modules are available yet.

The IO board provides a J2 expansion connector carrying MD_FIBER_TX, MD_FIBER_RX, MD_FIBER_SYNC, +5 V, +12 V, and GND. This connector supports the following planned plug-in modules:

- **Fiber IO multidrive module (planned):** Distributed drive configurations — motors with five or more phases, and parallel power stages with current sharing and load balancing, with hardware PWM synchronization between stages over fiber.
- **Fiber-optic gate-drive module (planned):** Isolated PWM delivered directly to the gate drivers over fiber optics.
- **WiFi/Bluetooth telemetry module (planned):** Wireless telemetry for monitoring and diagnostics.
- **Resolver interface module (planned):** Adds resolver position feedback. Without this module, native position inputs remain sin/cos encoder or Hall effect only (Section 7.4).

SOFTWARE ARCHITECTURE DOCUMENT

HW-C2-DOC-SWAD-A · design document — rev A

SOFTWARE ARCHITECTURE DOCUMENT

Traction Inverter

MCUs	STM32H723ZG + STM32G474RCTx
Operating Temp	−40°C to +85°C
Version	1.5
Prepared by	Thomas Liao
Reviewed by	(not yet reviewed)
Date	July 13, 2026
Part	HW-C2-DOC-SWAD-A · Rev A

University of California, Santa Cruz — Corzine Lab

Contents

Document History . . . 4
1. Introduction and Scope . . . 4
1.1 Purpose . . . 4
1.2 Scope . . . 5
1.3 Design Philosophy: User-Configurable, Safe, Reliable . . . 6
1.4 Target Hardware . . . 6
2. References and Standards . . . 7
3. Software Architecture Overview . . . 7
3.1 Design Principles . . . 7
3.2 Layer Diagram . . . 8
4. Hardware Abstraction Layer (HAL/BSP) . . . 8
4.1 TIM1 PWM Driver . . . 8
4.2 ADC Driver — Multi-Rate Oversampled . . . 9
4.3 Current Sensor Validation . . . 10
4.4 FDCAN Driver . . . 11
4.5 SPI FRAM — Non-Volatile Parameter Storage . . . 11
4.6 Power Rail Switching . . . 12
5. Safety Layer . . . 13
5.1 Fault Manager (FaultMgr) . . . 13
5.2 Safe State Manager (SSM) . . . 14
5.3 Watchdog Manager . . . 14
5.4 Power Supply Monitor . . . 14
6. Control Layer . . . 15
6.1 Field-Oriented Control (FOC) . . . 15
6.2 Modulation Scheme Library . . . 16
6.3 SVPWM (Default at 2 kHz) . . . 18
6.4 SHEPWM . . . 18
6.5 N-Pulse Family . . . 18
6.6 RSVM / RCFM . . . 19
6.7 PI Controllers . . . 19
6.8 State Machine . . . 19
6.9 Torque Commander . . . 21
6.10 Real-Time Loss Estimator and Thermal Model . . . 21
7. Application Layer . . . 23
7.1 Throttle Processing . . . 23
7.2 CAN Communication . . . 23
7.3 DC-Link Monitor . . . 23
7.4 Temperature Monitor (2oo3) . . . 23
7.5 POST (Power-On Self Test) . . . 24
7.6 RTE Configuration Interface . . . 24
7.7 Input Validation Framework . . . 25

8. Data Flow and Interfaces . . . 26
9. Memory Layout . . . 27
9.1 Flash Memory Map . . . 27
9.2 RAM Memory Map . . . 27
10. CAN Protocol . . . 28
11. Traceability Matrix . . . 28
12. Bootloader Architecture . . . 29
13. Firmware Update Protocol . . . 30
13.1 Update Protocol . . . 30
13.2 Security Requirements . . . 30
13.3 FRAM Protection During Update . . . 30
14. Scheduling and Timing . . . 31
14.1 Interrupt Priority Map . . . 31
14.2 1 ms System Tick (TIM6) . . . 31
14.3 WCET Summary . . . 31
15. Assumptions and Limitations . . . 32
15.1 Assumptions . . . 32
15.2 Known Limitations . . . 32
15.3 Future Work (V2.0+) . . . 33

Document History

Date	Version	Author	Description of Changes
June 13, 2026	1.0	Project Team	Initial release.
June 14, 2026	1.1	Project Team	TIM1 not HRTIM, multi-modulation, 50kHz oversampling, corrected ADC, 1oo3 temp, RTE interface.
June 14, 2026	1.2	Project Team	Major fixes: default 2kHz SVPWM (was 6kHz), min 300Hz (was 1kHz), ADC 0-3.3V scaling (not 1.67V), removed all GPIO pin assignments (refer to CubeMX), added SPI FRAM storage with write protection, input validation framework, real-time loss estimator + thermal model, power rail switching (CAN/sensor/gate driver kills), firmware update via UART/USB or CAN.
July 8, 2026	1.4	Project Team	Cover updated for dual-MCU (STM32H723ZG + STM32G474RCTx); body update pending. (entry added retroactively)
July 13, 2026	1.5	Project Team	Editorial consistency pass: FRAM part corrected to CY15B102Q-SXET 256 KB; temperature voting updated to 2oo3; current-sense bandwidth figures reconciled; cross-references updated to HARA v4.1 / TARA v1.2; gate-driver part corrected to NCV57100; rail supervisor corrected to TPS389006-Q1; watchdog part corrected

1. Introduction and Scope

1.1 Purpose

This Software Architecture Document (SWAD) defines the software architecture for the traction inverter / Vehicle Control Unit (VCU) — a dual-MCU design based on the STM32H723ZG main MCU and an STM32G474RCTx safety coprocessor (see the scope note below) — used in open-source aftermarket electric motorcycle conversion platforms. It implements the functional safety requirements specified in the companion Hazard Analysis and Risk Assessment (HARA, Version 4.1) and the cybersecurity requirements specified in the companion Threat Analysis and Risk Assessment (TARA, Version 1.2).

The architecture targets ISO 26262 ASIL-C safety integrity for motor control functions while remaining fully open-source, auditable, and free of vendor lock-in or OTP security fuses. All design decisions are traceable to specific Functional Safety Requirements (FSRs) or Cybersecurity Requirements (CSRs).

> **Scope Note: Safety Coprocessor (STM32G474RCTx)**
>
> The safety coprocessor (STM32G474RCTx) is present hardware on ControlBoard Rev A and is documented in HARA v4.1. This SWAD revision covers the main-MCU (STM32H723ZG) software architecture only; a full dual-MCU body update, including the coprocessor software architecture and inter-MCU interfaces, is planned for a future revision.

1.2 Scope

This document covers the complete software architecture:

- Four-layer software architecture (HAL/BSP, Safety, Control, Application)
- Six-state system state machine with full transition logic
- Field-Oriented Control (FOC) with multiple modulation schemes (SPWM, SVPWM, SHEPWM, N-Pulse, RSVM, RCFM)
- Multi-rate oversampled current sensing (~24 kHz effective bandwidth, 16-bit primary + 12-bit redundant)
- Real-time power loss estimator with thermal model for die temperature prediction
- Power rail switching — controlled shutdown of CAN, sensor, and gate driver supplies
- SPI FRAM for high-endurance non-volatile parameter storage with write protection
- Input validation framework for all user-configurable parameters
- All safety-critical modules with FSR traceability
- CAN protocol reference (user-tailorable)
- Flash and RAM memory layouts
- Bootloader with firmware update via UART/USB or CAN
- HMAC-SHA256 firmware signing (open-source trust model)
- RTE (Real Time Examiner) open-source configuration tool interface

This document does *not* cover:

- Hardware circuit design or PCB layout (see HW design files)
- Mechanical packaging or thermal design
- Detailed threat analysis (see TARA) or hazard analysis (see HARA)
- GPIO pin assignments (generated by STM32CubeMX, see project .ioc file)
- Implementation schedule (managed externally)
- **Traction motor design or construction** — the motor is an external product supplied by the vehicle manufacturer or aftermarket vendor
- **Encoder or resolver design** — the rotor position sensor is part of the external motor; this document only covers the interface to it
- **BMS, IO board, charger, or display design** — these are external CAN nodes; this document covers the CAN interface only

> **Two Evaluations: With and Without Safety Coprocessor**
>
> This document presents two safety integrity evaluations for the same hardware: (1) **Main-MCU-only configuration (no coprocessor):** ASIL-C achievable for most safety goals, ASIL-A for H-06/H-16/H-17. This is the configuration covered by this document revision. (2) **Dual-MCU configuration (with coprocessor):** ASIL-D achievable for SG-01 and SG-13 via ASIL decomposition. The coprocessor is present on ControlBoard Rev A hardware and is documented in HARA v4.1; its software architecture is not yet covered by this document (see scope note, Section 1.1). Where ASIL ratings differ between these two configurations, both are stated explicitly.

1.3 Design Philosophy: User-Configurable, Safe, Reliable

This system is designed from the ground up to be **user-configurable, safe, and reliable**. Every aspect of the control strategy that does not compromise safety is exposed to the user through the RTE (Real Time Examiner) open-source tool:

- **Modulation scheme:** SPWM, SVPWM, SHEPWM, N-Pulse (with variants), RSVM, RCFM — selectable at runtime
- **Switching frequency:** 300 Hz minimum, user-adjustable up to 16 kHz (default 2 kHz SVPWM)
- **FOC status:** Preliminary testing completed on Zero 75-10 motor (bench, open-loop V/Hz and closed-loop current control). Full torque control and dyno validation pending.
- **Torque maps:** Four user-defined 16-point curves, plus Sport and Eco presets
- **Safety thresholds:** DC-link OV/UV, temperature limits, throttle plausibility delta — all adjustable within hard bounds
- **PI gains:** Id and Iq controller gains, adjustable for different motors
- **CAN protocol:** Frame IDs, scaling factors, periods — all user-tailorable
- **Custom parameters:** 32 float + 32 uint32_t user-defined slots for application-specific tuning

However, **safety-critical parameters have hard-coded absolute bounds** that cannot be overridden. For example, the IGBT temperature hard cap of 100C cannot be raised above 100C even if the user requests it. The DC-link overvoltage threshold cannot be set above the hardware rating. These bounds are compiled into the firmware and are not modifiable via RTE.

Parameter storage uses an external SPI FRAM chip (not internal flash) for unlimited write endurance. A hardware write-protection pin prevents accidental parameter corruption during firmware updates or fault conditions.

1.4 Target Hardware

Table 1: Target Hardware Summary

Parameter	Value	Relevance
MCU	STM32H723ZG, Cortex-M7, 550 MHz	Single-core, FPU, cache, dual-bank flash
Flash	1 MB dual-bank (2 x 512 KB)	Bootloader + application + swap for OTA
RAM	564 KB total (320 KB AXISRAM, DTCM/ITCM, SRAM1-4)	FOC buffers, stack, ECC on AXISRAM
PWM + FOC	TIM1 Advanced Timer (6 comp. outputs)	PWM carrier 300Hz-16kHz; FOC loop runs at same frequency; hw dead-time, break input
Gate Driver	NCV57100 x 6 (AEC-Q100)	DESAT, UVLO, active Miller clamp
Current Sensors	Tamura LA37S x 4 (3-phase + DC link)	+/-1200 A, 1.042 mV/A, 0-3.3V to ADC
Temp Sensors	NTC thermistor x 1 per IGBT module (3 modules, 3 sensors total)	2-out-of-3 voting, outlier sensor excluded
CAN	Dual FDCAN, up to 8 Mbps	BMS, IO board, charger, diagnostic
FRAM	SPI-connected FRAM (Infineon/Cypress CY15B102Q-SXET, 2-Mbit / 256 KB)	Unlimited endurance param storage, WP pin
Encoder	Sin/Cos or resolver (configurable)	Rotor position for FOC

Parameter	Value	Relevance
DC-Link	102 V - 180 V (nominal range, 200 V-class hardware; 450 V with capacitor swap; 900 V with PCB + capacitor change)	Overvoltage/undervoltage monitoring
Motor Power	Up to 100 kW peak	Peak current ~280 A (at 350V)

2. References and Standards

Table 2: Normative References

Reference	Title	Usage
ISO 26262-1:2018	Functional safety -- Vocabulary	Definitions and terminology
ISO 26262-3:2018	Part 3: Concept phase (HARA)	Hazard analysis, ASIL assignment
ISO 26262-6:2018	Part 6: Product development -- software	Software architectural design
ISO/SAE 21434:2021	Cybersecurity engineering	Cybersecurity requirements (TARA)
STM32H723xx RM	Reference Manual (RM0468)	Register-level programming
NCV57100 DS	NCV57100 Datasheet	Gate driver timing and protection
Tamura LA37S	LA37S Current Transducer Datasheet	Current sensing specification
CY15B102Q-SXET	Infineon/Cypress 2-Mbit (256 KB) SPI FRAM	Non-volatile parameter storage

3. Software Architecture Overview

3.1 Design Principles

1. **User-configurable, safety-bounded:** Every non-safety-critical parameter is adjustable via RTE. Safety-critical parameters have hard-coded min/max bounds that cannot be overridden.
2. **Deterministic timing:** No dynamic memory allocation in safety paths. No RTOS. Superloop plus interrupts only.
3. **Hardware fault containment:** Gate driver faults trigger TIM1 break input for sub-microsecond hardware SSO, independent of software.
4. **Multi-path safe state:** Three complementary paths to SSO: (1) TIM1 break input (hardware, <100 ns, independent of MCU CPU), (2) MCU GPIO → PMIC EN gate driver supply kill (active, ~10 us, MCU-dependent, basic FLT feedback via NCV57100 VDD_UVLO), (3) loss of shared 3.3V rail → NCV57100 internal pull-down (passive, automatic, no software).
5. **Fail-safe defaults:** SSO is the hardware default. Any unrecognized condition results in SSO.
6. **Open-source trust model:** No OTP fuses, no vendor-locked keys. HMAC-SHA256 with user-managed keys. Physical access is acceptable risk.
7. **Multi-modulation:** SPWM, SVPWM, SHEPWM, N-Pulse family, RSVM, RCFM — selectable at runtime via RTE.
8. **Current loop at switching frequency:** FOC runs at the PWM carrier frequency (300 Hz - 16 kHz), not at a fixed rate. The current loop rate IS the switching frequency. ADC oversamples at an integer multiple (n×) of the PWM frequency, phase-locked to avoid beat noise.

9. **High-bandwidth sensing:** ADC oversampling up to 48 kSPS provides ~24 kHz effective current sense bandwidth, sufficient for low-inductance motors at low switching frequencies.
10. **Layered abstraction:** Safety mechanisms isolated in a dedicated layer, independent of control algorithms.
11. **Defensive programming:** All inputs range-checked. All state transitions validated. Full input validation framework (Section 7.7).

3.2 Layer Diagram

Table 3: Software Layer Summary

Layer	Modules	Safety Integrity	Timing
Application	Throttle, CAN Comms, DCLink Mon, Temp Mon (2oo3), POST, RTE, Input Validator	ASIL-B	1 ms - 100 ms
Control	FOC, Modulation Library, PI Controllers, State Machine, Torque Commander, Loss Estimator	ASIL-C	PWM period (3.3 ms @ 300 Hz - 62.5 us @ 16 kHz)
Safety	Fault Manager, Safe State Manager, Watchdog Manager, PSU Monitor, Rail Switch	ASIL-C	1 ms - 100 ms
HAL/BSP	TIM1, ADC, FDCAN, SPI (FRAM), GPIO, EXTI, Flash, CRC	ASIL-B	Hardware-driven

4. Hardware Abstraction Layer (HAL/BSP)

The HAL/BSP layer provides register-level drivers for all STM32H723ZG peripherals. This is the only code that directly accesses hardware registers. All upper layers call HAL functions.

> **GPIO Pinout — Refer to STM32CubeMX**
>
> All GPIO pin assignments, alternate function mappings, and peripheral pinouts are generated from the STM32CubeMX project configuration file (`.ioc`). The generated source files (`main.c` , `gpio.c` , `tim.c` , `adc.c` , `fdcan.c` , `spi.c`) are checked into version control and must not be hand-edited. Re-generate via CubeMX when changes are needed. This document does not list specific GPIO pins.

4.1 TIM1 PWM Driver

TIM1 (Advanced Control Timer) serves **dual roles**: (1) it generates six complementary PWM outputs at the user-selected carrier frequency with hardware dead-time and break input; (2) its update interrupt triggers the FOC current loop at the same frequency. The ADC is also triggered by TIM1 at an integer multiple (n×) of the PWM frequency for oversampled current sensing.

Table 4: TIM1 Timing Parameters

Parameter	Default	Range	Notes
TIM1 clock	275 MHz	Fixed	APB2 timer clock
Default PWM frequency	2 kHz	300 Hz - 16 kHz	User-configurable via RTE; SVPWM default

Parameter	Default	Range	Notes
Minimum PWM frequency	300 Hz	Hard limit	Below this: audible noise, excessive current ripple
Maximum PWM frequency	16 kHz	Hard limit	Above this: excessive switching losses
Dead time	2.0 us	500 ns - 4 us	User-configurable; prevents shoot-through
Break input	TIM1_BKIN	Active low	OR'd gate driver fault aggregation
Break response	SSO (MOE cleared)	Hardware	<100 ns, zero software latency
Re-arm	Software only	Required	Prevents auto-restart after fault
ADC trigger	TIM1_TRGO	n× PWM freq (up to 48 kSPS)	Phase-locked oversampling, sinc3 decimation
FOC trigger	TIM1 update (UIF)	PWM switching frequency	Current loop rate = carrier frequency

4.1.1 TIM1 Break Input — Hardware SSO

When any NCV57100 gate driver detects DESAT or UVLO, its FAULT output (active low) asserts. Six FAULT outputs are OR'd together and fed to TIM1_BKIN. TIM1 hardware immediately clears MOE (main output enable), forcing all six PWM outputs to the inactive (SSO) state within <100 ns. A break interrupt then fires to log the fault.

4.2 ADC Driver — Multi-Rate Oversampled

The ADC subsystem achieves ~24 kHz effective current sense bandwidth through multi-rate oversampling. An integer multiple (n×) of samples is taken per PWM period — n=24 at the 2 kHz default (see Table 12) — decimated via sinc3 filtering. This is essential for low-inductance motors running at low switching frequencies (300 Hz - 2 kHz) where current ripple would otherwise alias into the control loop.

Table 5: ADC Channel Assignment

Signal	ADC1 (16-bit)	ADC2 (12-bit)	ADC3 (12-bit)
I_U (phase U current)	INx (16-bit)	—	—
I_V (phase V current)	INx (16-bit)	INx (12-bit)	—
I_W (phase W current)	INx (16-bit)	INx (12-bit)	INx (12-bit)
REF_U (I_U reference)	INx (16-bit)	—	—
REF_V (I_V reference)	INx (16-bit)	INx (12-bit)	—
REF_W (I_W reference)	INx (16-bit)	INx (12-bit)	INx (12-bit)
V_DC (DC link voltage)	INx (16-bit)	—	—
Throttle main	INx (16-bit)	—	—
Throttle check	INx (16-bit)	—	—
T_IGBT[0-2]	INx (12-bit)	—	—
VREFINT (3.3V rail)	Internal	—	—

ADC1 at 16-bit is the primary measurement path for all control-critical signals. ADC2 and ADC3 at 12-bit provide lower-resolution redundant channels for cross-check. The exact ADC input channel numbers (INx) are assigned via STM32CubeMX and vary with board revision.

> **ADC Resolution: 16-bit Primary + 12-bit Redundant**
>
> The 12-bit redundant channels (ADC2, ADC3) have approximately 5.3x coarser resolution than the 16-bit primary (ADC1). For the purpose of fault *detection* — the safety-relevant function — this difference is not operationally significant. A 12-bit channel can reliably detect catastrophic sensor faults: broken wire (reading at rail), stuck output (constant value regardless of current), and large gain errors (>20% deviation). These are the failure modes that matter for safety. Subtle gain errors or small calibration drift are not safety-critical because the 16-bit primary channel remains the control input. The 12-bit channels serve as a "sanity check" on the primary, not as a precision backup. A second 16-bit ADC would provide higher-fidelity redundancy but is not required for ASIL-C fault detection.

4.2.1 Oversampling Configuration

At 2 kHz PWM (default), 24 ADC samples per period = 48 kSPS per channel. The sinc3 decimator produces one filtered output every 24 raw samples, giving an effective 2 kHz control update rate with ~24x noise reduction. The raw 48 kSPS stream is also analyzed for peak-to-peak ripple amplitude — excessive ripple flags a sensor or connection fault.

4.3 Current Sensor Validation

Each Tamura LA37S outputs a measurement signal and an independent reference signal. Both are conditioned to 0-3.3V for the 16-bit ADC.

Table 6: Tamura LA37S Electrical Characteristics

Parameter	Value	Note
Measurement range	-1200 A to +1200 A	Bidirectional
Sensitivity	1.042 mV/A	At sensor output pins
Center voltage (0 A)	2.5 V	Internal reference output
Input to ADC	0 - 3.3 V	After signal conditioning
Resolution (16-bit, 3.3V ref)	~36.6 mA/LSB	2400 A span / 65536

4.3.1 VREF Validation

The reference channel for each sensor is checked on every conversion. After signal conditioning (scaling to 3.3V range), the reference must be within:

Table 7: VREF Acceptance Window (After Conditioning to 3.3V)

	Min	Typical	Max
VREF (after conditioning)	1.63 V (2.48V scaled)	1.65 V (2.50V scaled)	1.67 V (2.52V scaled)

Deviation outside this window marks the sensor as degraded. FAULT_SENSOR_VREF is raised. The control loop continues using remaining healthy channels.

4.3.2 Zero-Point Check

During a controlled zero-current window, all phase sensors are read. Each must be within +/-20% of zero (i.e., +/-240 A equivalent). Gross deviations indicate a broken wire or stuck output. Results update the running zero-offset calibration.

4.3.3 DC-Link Back-Calculation

Independent sanity check of DC-link measurements:

$$V_{dc\_calc} = (3 \times V_{ph} \times I_{ph} \times \cos(\varphi)) / (I_{dc} \times \eta)$$

Where V_{ph} is AC phase voltage (RMS), I_{ph} is AC phase current (RMS), cos(φ) is motor power factor, η is inverter+motor efficiency (0.93-0.97, user-configurable), and I_{dc} is measured DC bus current. Runs at 100 Hz. Logs a warning if calculated V_{dc} deviates from measured V_{dc} by more than 15%.

4.4 FDCAN Driver

Dual FDCAN modules provide CAN communication. The protocol is intentionally user-tailorable — the frame definitions in Section 10 are a starting point. Users modify IDs, scaling, and layouts via RTE.

4.5 SPI FRAM — Non-Volatile Parameter Storage

An external SPI FRAM chip (Infineon/Cypress CY15B102Q-SXET, 2-Mbit / 256 KB) stores all user-configurable parameters, runtime counters, and fault logs. FRAM provides unlimited write endurance (10^14 write cycles) compared to flash (~10,000 cycles), making it suitable for frequently updated data.

Table 8: FRAM Storage Map

Address Range	Content	Update Frequency
0x0000 - 0x00FF	Parameter header (magic, version, CRC)	On parameter commit only
0x0100 - 0x04FF	Torque maps (4 x 16-point curves + metadata)	On user commit via RTE
0x0500 - 0x06FF	Safety thresholds (temp, DC-link, throttle)	On user commit via RTE
0x0700 - 0x07FF	CAN protocol config (IDs, scaling, periods)	On user commit via RTE
0x0800 - 0x08FF	Motor parameters (Ld, Lq, lambda_m, poles)	On user commit via RTE
0x0900 - 0x09FF	Modulation config (scheme, freq, dead time)	On user commit via RTE
0x0A00 - 0x0AFF	Custom user parameters (32 float + 32 uint32_t)	On user commit via RTE
0x0B00 - 0x0B3F	Odometer (total km, 64-bit)	Every 1 km
0x0B40 - 0x0B4F	Hour counter (motor run hours, 32-bit)	Every 1 hour
0x0B50 - 0x0B53	Boot counter (total resets, 32-bit)	Every boot
0x0B60 - 0x1F5F	Fault log ring buffer (256 entries x 20 bytes)	On every fault event
0x1F60 - 0x1FFF	Reserved	—

4.5.1 Write Protection

The FRAM chip has a hardware write-protect (WP) pin connected to a GPIO. WP is asserted (high) during normal operation, preventing accidental writes. WP is de-asserted only during:

1. Explicit parameter commit via RTE (user-initiated save)
2. Fault log write (atomic, WP toggled for single write cycle)
3. Odometer/hour counter update (periodic, WP toggled briefly)

> **FRAM WP Pin Failure Detection**
>
> If the WP pin GPIO fails (stuck low), FRAM is still protected because the SPI transaction must carry the correct write-enable (WREN) opcode. A stuck-low WP pin is detected during POST (test 13).

If the WP pin GPIO fails (stuck low), the FRAM is still protected from corruption because the SPI transaction must also carry the correct write-enable (WREN) opcode sequence. A failed WP pin is detected during POST.

4.6 Power Rail Switching

Three power rails can be switched off under software control. These complement the hardware SSO paths (TIM1_BKIN, shared 3.3V passive) but are not independent of the MCU:

Table 9: Power Rail Control

Rail	Supply	Control	Use Case
Gate driver power	Murata MGJ2D121509MPC-R7 x 6	PMIC EN (shared enable)	Primary SSO path: kills all 6 gate drive supplies simultaneously, forcing IGBT gates to 0V via NCV57100 active pull-down
Current sensor power	+5V sensor supply	GPIO-controlled load switch	Disables current sensors for safe state or diagnostics
CAN transceiver power	12V to 5V DC-DC for ISO1042	GPIO-controlled load switch	Disables CAN bus for EMI reduction or fault isolation

4.6.1 Gate Driver Supply Kill — Primary SSO Path

The gate driver supply kill is the most aggressive safe-state action. When the PMIC EN line is pulled low, all six Murata isolated DC-DC converters shut down simultaneously. Within microseconds, the NCV57100 gate drivers detect the loss of their isolated supply and activate internal pull-downs, forcing all IGBT gates to 0V.

FLT Verification on Supply Kill: When the NCV57100 detects VDD_UVLO (loss of gate drive supply), it asserts its FLT output (active low). Since all six FLT outputs are OR'd together and fed to TIM1_BKIN, the MCU can verify that a PMIC EN kill actually worked by reading the FLT line after a short delay (~100 us). If FLT is asserted (low), the gate driver supply has been successfully shut off. If FLT remains de-asserted (high), the supply kill failed and the system must use TIM1_BKIN SSO as the fallback. This provides a basic feedback mechanism — not as robust as a dedicated voltage sense on each supply, but better than no feedback at all. Independent supply voltage monitoring (via ADC or isolated status signal) is a recommended future enhancement for higher diagnostic coverage.

```
/* RailSwitch.c — software-controlled power rail management */
void RailSwitch_KillGateDrivers(void) {
```

```
    GPIO_WritePin(PMIC_EN_GPIO_Port, PMIC_EN_Pin, GPIO_PIN_RESET);
    /* All 6 Murata supplies off within ~10 us */
    /* NCV57100 internal pull-downs force gates to 0V within ~1 us */
    rail_status.gate_driver_power = false;
    FaultMgr_SetFault(FAULT_RAIL_GATE_DRIVER_KILL, 0);
}

void RailSwitch_KillCANPower(void) {
    GPIO_WritePin(CAN_PWR_EN_GPIO_Port, CAN_PWR_EN_Pin, GPIO_PIN_RESET);
    /* ISO1042 transceivers powered off */
    rail_status.can_power = false;
}

void RailSwitch_KillSensorPower(void) {
    GPIO_WritePin(SENSOR_PWR_EN_GPIO_Port, SENSOR_PWR_EN_Pin, GPIO_PIN_RESET);
    /* Current sensor +5V supply off */
    rail_status.sensor_power = false;
}
```

5. Safety Layer

5.1 Fault Manager (FaultMgr)

Global fault registry — single source of truth for system health. The State Machine queries this on every tick.

Table 10: Fault Classification

Fault Name	Class	Source	FSR	Response
FAULT_GATE_DRIVER	Critical	TIM1_BRK ISR	FSR-12,13	Immediate SSO + SAFE
FAULT_OVERCURRENT_HW	Critical	ADC analog watchdog	FSR-07	Immediate SSO + SAFE
FAULT_OVERCURRENT_SW	Critical	FOC ISR	FSR-07	Immediate SSO + SAFE
FAULT_ADC_FAIL	Critical	ADC cross-check	FSR-16	Ramp down + SAFE
FAULT_SENSOR_VREF	Non-critical	Current sensor VREF	FSR-16	Log, use healthy ch
FAULT_SENSOR_ZERO	Non-critical	Zero-point check	FSR-16	Log, flag recal
FAULT_THROTTLE_PLAUS	Critical	Throttle	FSR-01	Zero torque + FAULT
FAULT_DC_LINK_OV	Critical	DCLink Mon	FSR-11	Immediate SSO + SAFE
FAULT_DC_LINK_UV	Critical	DCLink Mon	FSR-21	Zero torque + FAULT
FAULT_IGBT_OT	Critical	TempMonitor	FSR-08	Ramp down + FAULT
FAULT_IGBT_OT_WARN	Non-crit	TempMonitor	FSR-08	Log, limit 50%
FAULT_ENCODER	Critical	ENC_Monitor	FSR-09	Ramp down + FAULT
FAULT_CAN_TIMEOUT	Critical	CAN heartbeat	FSR-17	Ramp down + FAULT
FAULT_WATCHDOG	Critical	IWDG/WWDG	FSR-15	Reset then POST
FAULT_ECC_RAM	Critical	NMI handler	FSR-20	Log + SAFE

Fault Name	Class	Source	FSR	Response
FAULT_POST_FAIL	Critical	POST	FSR-16,19	INIT then SAFE
FAULT_RAIL_GATE_DRIVER_KILL	Critical	RailSwitch	FSR-14	SSO via supply kill

5.2 Safe State Manager (SSM)

Three complementary paths to SSO:

1. **TIM1 break input (hardware, active):** Gate driver FAULT triggers TIM1_BKIN, MOE cleared, <100 ns. Independent of MCU CPU state.
2. **Gate driver supply kill (software + hardware, active):** MCU GPIO → PMIC EN low → Murata supplies off → NCV57100 UVLO pull-down, ~10 us. MCU-dependent. **Basic feedback:** NCV57100 asserts FLT on VDD_UVLO, which the MCU can read via TIM1_BKIN. Independent supply voltage monitoring is a recommended future enhancement.
3. **Shared 3.3V rail loss (hardware, passive):** Loss of 3.3V (shared by MCU and NCV57100 logic) → NCV57100 VDD lost → internal pull-down → SSO. No software intervention. Only effective for total power collapse.

Path 1 is the primary diagnosed path (TIM1_BKIN hardware). Path 2 complements it but depends on MCU GPIO functioning. Path 3 is a passive fallback for total power loss. Path 2 and Path 3 are related (both involve power) but operate under different conditions. A coprocessor with an independent switch would provide a fully independent third active path.

5.3 Watchdog Manager

Table 11: Watchdog Configuration

Watchdog	Type	Timeout	Service	Action
IWDG	Independent (32 kHz RC)	100 ms	Every 50 ms	System reset
WWDG	Window (sys clock)	83.3 ms	Every 50 ms	System reset
External	TPS389006-Q1 (6-ch window supervisor, I2C)	Per-rail window limits	Every 100 ms (I2C poll)	Gate-driver DRIVER_RESET, system reset

The external supervisor (TI TPS389006-Q1) monitors the 3.3 V, 5 V, and 12 V rails, the sensor 5 V rail, and both gate-drive power feedbacks. On any window violation it asserts the gate-driver DRIVER_RESET line directly, independent of MCU software.

5.4 Power Supply Monitor

Internal VREFINT channel monitors 3.3V regulator indirectly. Window: 3.0 V - 3.6 V. Outside this range: safe state entry.

6. Control Layer

6.1 Field-Oriented Control (FOC)

FOC runs in the **TIM1 update interrupt at the PWM switching frequency** (priority 0, cannot be preempted). The current loop rate IS the switching frequency: 300 Hz minimum, 16 kHz maximum, user-selectable via RTE. TIM1 generates the PWM outputs and triggers the FOC computation on every update event. Duty cycles are written to TIM1's preload (shadow) registers and applied at the next PWM period boundary — this is standard double-buffering.

6.1.1 Oversampled ADC Triggering

The ADC is triggered by TIM1 at an **integer multiple** (n×) of the PWM frequency, up to the hardware sampling rate limit. At each PWM period, n ADC samples are taken and decimated (boxcar or sinc3 filter) to produce one filtered current reading for the FOC loop:

Table 12: ADC Oversample Ratio vs. Switching Frequency

PWM Frequency	Oversample Ratio (n)	ADC Sample Rate	Effective BW	Filter
16 kHz	3	48 kSPS	~24 kHz	Sinc3
8 kHz	6	48 kSPS	~24 kHz	Sinc3
4 kHz	12	48 kSPS	~24 kHz	Sinc3
2 kHz (default)	24	48 kSPS	~24 kHz	Sinc3
1 kHz	48	48 kSPS	~24 kHz	Sinc3
500 Hz	96	48 kSPS	~24 kHz	Sinc3
300 Hz	160	48 kSPS	~24 kHz	Sinc3

The integer multiple ensures ADC sampling is phase-locked to the PWM carrier, avoiding beat-frequency noise. The ADC sample rate is capped at ~48 kSPS per channel (hardware limit of 16-bit mode with conversion time). The sinc3 decimator provides ~60 dB of alias rejection. Raw oversampled data is also analyzed for peak-to-peak ripple amplitude — excessive ripple flags a sensor or connection fault.

6.1.2 FOC ISR Implementation

```
/* TIM1_UP_IRQHandler — FOC runs at PWM switching frequency */
void TIM1_UP_IRQHandler(void) {
    TIM1->SR &= ~TIM_SR_UIF;

    /* Read decimated currents (sinc3 output, one sample per PWM period) */
    float iu = adc_decimated[0];
    float iv = adc_decimated[1];
    float iw = adc_decimated[2];

    /* Read rotor angle */
    float theta_e = Encoder_GetElectricalAngle();

    /* Clarke transform */
    float i_alpha = iu;
```

```
    float i_beta  = (iu + 2.0f * iv) * ONE_OVER_SQRT3;

    /* Park transform */
    float cos_t = cosf(theta_e), sin_t = sinf(theta_e);
    float id =  i_alpha * cos_t + i_beta * sin_t;
    float iq = -i_alpha * sin_t + i_beta * cos_t;

    /* PI controllers */
    float iq_ref = TorqueCmd_GetIqRef();
    float vd = PI_Update(&pi_id, 0.0f - id);
    float vq = PI_Update(&pi_iq, iq_ref - iq);

    /* Voltage limit */
    float v_max = DCLink_GetVdc() * ONE_OVER_SQRT3 * MAX_MODULATION;
    float v_mag_sq = vd*vd + vq*vq;
    if (v_mag_sq > v_max*v_max) {
        float scale = v_max / sqrtf(v_mag_sq);
        vd *= scale; vq *= scale;
    }

    /* Inverse Park */
    float v_alpha = vd * cos_t - vq * sin_t;
    float v_beta  = vd * sin_t + vq * cos_t;

    /* Modulation dispatch */
    float du, dv, dw;
    mod_scheme_table[active_scheme](v_alpha, v_beta, DCLink_GetVdc(), &du, &dv, &dw);

    /* Write to TIM1 preload registers (applied next PWM period) */
    TIM1->CCR1 = (uint16_t)(du * TIM1->ARR);
    TIM1->CCR2 = (uint16_t)(dv * TIM1->ARR);
    TIM1->CCR3 = (uint16_t)(dw * TIM1->ARR);
}
```

6.1.3 Trigonometric Functions: CORDIC vs. Software

The FOC ISR uses `cosf()`, `sinf()`, and `sqrtf()` from the CMSIS-DSP library. The STM32H723 includes a **CORDIC hardware accelerator** that computes sin, cos, and sqrt in ~10-20 cycles each with deterministic execution time:

- CORDIC: ~20-40 ns per operation (hardware, cache-independent)
- Software: ~150-200 ns per operation (library, cache-dependent)

CORDIC is recommended for production. Software trig is used during bench development for portability. Migration to CORDIC planned before vehicle testing. The 5 us WCET claim for SHEPWM worst-case must be validated with actual measurements on production hardware.

6.2 Modulation Scheme Library

Table 12a: Supported Modulation Schemes

Scheme	Type	Bus Util	Use Case
SPWM	Sinusoidal	78.5%	Baseline, simple
SVPWM	Space Vector	90.7%	Default (2 kHz), best utilization

Scheme	Type	Bus Util	Use Case
SHEPWM	Harmonic Elimination	Variable	Low noise, specific freq
N-Pulse	Synchronous	High	Min switching loss, low speed
N-Pulse Wide	Synchronous	High	Wide pulse, low speed high torque
Custom	Synchronous	Variable	User-defined via RTE
RSVM	Randomised	90.7%	EMI spreading
RCFM	Randomised	90.7%	Acoustic noise spreading

6.2.1 Modulation Scheme Selection and Transitions

The active modulation scheme is selected via the RTE configuration tool and stored as an enum in FRAM. The scheme can be changed at runtime through RTE, but only under safe transition conditions.

Transition Conditions

A modulation scheme transition is permitted only when:

- **Torque reference is zero** (no-load transition — safest, no current flowing), OR
- **Motor speed is below 10% of base speed** (low voltage demand, small disturbance from transition), AND
- **No fault flags are active** (healthy system state only)

If a transition is requested while conditions are not met, it is queued and executed automatically when conditions become valid. The user is notified via CAN.

Bumpless Transition Mechanism

1. **Snapshot:** Before transition, the current modulation index (|V_ref| / V_dc) and electrical angle are captured
2. **Crossfade:** Over 10 ms (10 PWM periods at 1 kHz, 160 periods at 16 kHz), duty cycles are linearly interpolated between the outgoing scheme's last output and the incoming scheme's first output
3. **Handover:** At 100% incoming scheme, the function pointer is atomically swapped
4. **Validation:** For 50 ms post-transition, the duty cycle delta between consecutive periods is monitored; excessive jump (>5% of ARR) logs a warning

Transition Matrix

Table 12b: Legal Modulation Scheme Transitions

From \ To	SPWM	SVPWM	SHEPWM	N-Pulse	RSVM	RCFM
SPWM	—	Anytime	Zero torque only	Below 10% speed	Anytime	Anytime
SVPWM	Anytime	—	Zero torque only	Below 10% speed	Anytime	Anytime
SHEPWM	Below 10% speed	Below 10% speed	—	Not allowed	Not allowed	Not allowed
N-Pulse	Below 10% speed	Below 10% speed	Not allowed	—	Not allowed	Not allowed
RSVM	Anytime	Anytime	Zero torque only	Below 10% speed	—	Anytime
RCFM	Anytime	Anytime	Zero torque only	Below 10% speed	Anytime	—

> **Asynchronous to Synchronous Transition**
>
> Transitioning from an asynchronous scheme (SPWM, SVPWM, RSVM, RCFM) to a synchronous scheme (SHEPWM, N-Pulse) requires the PWM frequency to be an integer multiple of the motor electrical frequency. The system automatically adjusts the PWM frequency to the nearest valid synchronous ratio before performing the transition. Transitioning back from synchronous to asynchronous is always allowed (below 10% speed).

6.3 SVPWM (Default at 2 kHz)

Seven-segment symmetric algorithm. Third harmonic injection. Overmodulation Mode 1 and 2 for extended speed range. Default switching frequency: 2 kHz, range 300 Hz - 16 kHz via RTE.

6.4 SHEPWM

Pre-calculated switching angles stored as lookup tables in flash, indexed by modulation index with linear interpolation. V2.0: live on-demand angle calculation (Newton-Raphson) reserved for dynamic operating points.

6.5 N-Pulse Family

1-pulse to 15-pulse synchronous modulation. In synchronous mode, the PWM frequency is phase-locked to an integer multiple of the motor electrical frequency:

Table 12c: N-Pulse Mode vs. Electrical Frequency (at 2 kHz PWM)

Mode	Pulses/Half-Period	Sync Ratio	Electrical Freq Range	Use Case
15-pulse	15	30:1	67 - 133 Hz	Near base speed
9-pulse	9	18:1	111 - 222 Hz	Mid speed range
5-pulse	5	10:1	200 - 400 Hz	High speed
3-pulse	3	6:1	333 - 667 Hz	Very high speed
1-pulse	1	2:1	1000 - 2000 Hz	Overmodulation region

> **Automatic Mode Transitions**

Transitions between N-pulse modes are automatic based on motor electrical frequency with configurable hysteresis to prevent mode chattering:

1. **Upshift** (lower pulse count): Triggered when f_elec rises above the upper threshold of the current mode. Hysteresis: +10% of threshold value.
2. **Downshift** (higher pulse count): Triggered when f_elec falls below the lower threshold. Hysteresis: -10% of threshold value.
3. **Boundary to asynchronous:** When f_elec falls below the minimum for 15-pulse mode, the system transitions to SVPWM (asynchronous). When f_elec rises above the maximum for 1-pulse mode, the system transitions to six-step (full square wave).

Sync/Async Boundary

The boundary between synchronous (N-pulse) and asynchronous (SVPWM/SPWM) operation is user-configurable via RTE. Default: synchronous above 67 Hz elec, asynchronous below. The transition bumplessly adjusts the PWM frequency to the nearest integer multiple of the electrical frequency.

6.6 RSVM / RCFM

LFSR-based pseudo-random sequence seeded from STM32 unique ID. Range configurable via RTE. RSVM randomizes the space vector sequence within each PWM period; RCFM dithers the carrier frequency around a center value. Both modes can be combined (RSVM + RCFM) for maximum EMI spreading.

6.7 PI Controllers

Two PI controllers (d-axis flux, q-axis torque) with anti-windup via back-calculation. Gains stored in FRAM, adjustable via RTE.

6.8 State Machine

Six states with fully defined transitions, guards, and entry/exit actions. Illegal transitions are rejected and logged as faults.

6.8.1 State Transition Table

Table 12d: State Transitions — Guards and Actions

From	To	Guard Condition	Entry Action	Exit Action
INIT	STANDBY	POST complete AND POST passed	—	—
INIT	SAFE	POST complete AND POST failed	Log failure code; SSO	—
STANDBY	READY	Key=ON AND Brake=on AND HV present AND FaultMgr healthy	Close precharge contactor; start precharge timer	—
STANDBY	SAFE	Critical fault detected during STANDBY	SSO; open contactor	—
READY	DRIVING	Key=ON AND Brake=off AND Throttle > deadband AND precharge complete	Enable torque command; start encoder validity timer	—
READY	STANDBY	Key=OFF	Open contactor; zero torque	—
DRIVING	READY	Key=ON AND Brake=on AND Throttle < deadband	Zero torque ref; idle PWM	Disable torque command
DRIVING	FAULT	Non-critical fault active (FAULT_MASK_NONCRITICAL)	Ramp torque to zero (500 ms)	—
DRIVING	SAFE	Critical fault active (FAULT_MASK_CRITICAL) OR Key=OFF	SSO (TIM1_BKIN or PMIC EN kill); open contactor; zero torque	Disable torque command
FAULT	SAFE	Ramp-down complete AND (critical fault active OR timeout 5s)	SSO; open contactor	—

From	To	Guard Condition	Entry Action	Exit Action
FAULT	READY	All faults cleared AND Key cycled OFF-then-ON AND Brake=on	Run partial POST (sensors, ADC, comms only)	—
SAFE	INIT	Key=OFF-then-ON AND Brake=on AND FaultMgr healthy AND all faults cleared	Run full POST	—

6.8.2 Illegal Transitions

The following transitions are explicitly forbidden and trigger FAULT_STATE_MACHINE if attempted:

- INIT -> DRIVING (POST not yet complete)
- STANDBY -> DRIVING (precharge not yet done)
- READY -> STANDBY while driving (must go through DRIVING -> FAULT/SAFE first)
- DRIVING -> STANDBY (must go through DRIVING -> READY first)
- SAFE -> any state except INIT (requires full POST cycle)
- Any state -> same state with transition attempt (logged as no-op)

6.8.3 State Machine Implementation

```
void SM_Run(void) {
    SystemState_t next_state = sm.current_state;
    uint32_t fault_flags = FaultMgr_GetActiveFlags();

    switch (sm.current_state) {
        case STATE_INIT:
            if (POST_IsComplete() && POST_Passed()) next_state = STATE_STANDBY;
            else if (POST_IsComplete() && !POST_Passed()) next_state = STATE_SAFE;
            break;
        case STATE_STANDBY:
            if (Inputs_KeyOn() && Inputs_BrakeOn() && DCLink_HVPresent()
                && FaultMgr_IsHealthy()) {
                next_state = STATE_READY; Contactor_ClosePrecharge();
            }
            if (fault_flags & FAULT_MASK_CRITICAL) next_state = STATE_SAFE;
            break;
        case STATE_READY:
            if (!Inputs_KeyOn()) { next_state = STATE_STANDBY; Contactor_Open(); }
            else if (Inputs_KeyOn() && !Inputs_BrakeOn()
                     && Throttle_GetPercent() > THROTTLE_DEADBAND
                     && Precharge_Complete())
                next_state = STATE_DRIVING;
            break;
        case STATE_DRIVING:
            if (fault_flags & FAULT_MASK_CRITICAL) {
                SSM_EnterImmediate(); next_state = STATE_SAFE;
            } else if (fault_flags & FAULT_MASK_NONCRITICAL) {
                SSM_RequestRampDown(); next_state = STATE_FAULT;
            } else if (Inputs_KeyOn() && Inputs_BrakeOn()
                       && Throttle_GetPercent() < THROTTLE_DEADBAND) {
                next_state = STATE_READY; TorqueCmd_SetZero();
            } else if (!Inputs_KeyOn()) { SSM_EnterImmediate(); next_state = STATE_SAFE; }
            break;
        case STATE_FAULT:
            if (fault_flags & FAULT_MASK_CRITICAL) { SSM_EnterImmediate(); next_state =
```

```
STATE_SAFE; }
            else if (SSM_RampDownComplete() && FaultMgr_IsHealthy()
                     && sm.fault_time + 5000 < HAL_GetTick())
                next_state = STATE_READY;
            break;
        case STATE_SAFE:
            if (!Inputs_KeyOn()) sm.reset_request = true;
            if (sm.reset_request && Inputs_KeyOn()
                && Inputs_BrakeOn() && FaultMgr_IsHealthy()) {
                next_state = STATE_INIT; POST_Start();
            }
            break;
    }

    /* Validate transition legality */
    if (next_state != sm.current_state) {
        if (!SM_IsTransitionLegal(sm.current_state, next_state)) {
            FaultMgr_SetFault(FAULT_STATE_MACHINE, (sm.current_state << 8) | next_state);
            next_state = STATE_SAFE; /* forced safe on illegal transition */
        }
        SM_ExitState(sm.current_state);
        sm.current_state = next_state;
        SM_EnterState(sm.current_state);
    }
}
```

6.9 Torque Commander

Converts throttle to I_q reference. Rate limits: ramp-up 500 A/s, ramp-down 2000 A/s, regen 1000 A/s. Four torque maps + Sport/Eco. All stored in FRAM via RTE.

6.10 Real-Time Loss Estimator and Thermal Model

The system continuously estimates the power dissipated in the IGBTs using a pre-programmed loss model. This model combines conduction losses, switching losses, and a thermal RC network to estimate junction temperature and predict thermal transients.

6.10.1 Loss Model

Table 13: Power Loss Components

Loss Type	Formula	Parameters (User-Configurable)
Conduction	P_cond = V_ce(sat) x I_c x D, where V_ce(sat) = V_ce0 + R_ce x I_c	V_ce0 (threshold, ~0.8V), R_ce (slope resistance, ~3 mOhm), from IGBT datasheet
Switching (turn-on)	P_sw_on = E_on x f_sw, where E_on from datasheet energy curve at operating V_dc and I_c	E_on vs (V_dc, I_c) lookup table from datasheet; gate resistance R_g
Switching (turn-off)	P_sw_off = E_off x f_sw	E_off vs (V_dc, I_c) lookup table; R_g
Diode (FWD)	P_diode = V_f x I_f x (1-D) + E_rec x f_sw	V_f, R_diode from datasheet; E_rec vs I_c

Loss Type	Formula	Parameters (User-Configurable)
Total per IGBT	P_total = P_cond + P_sw_on + P_sw_off + P_diode	All stored in FRAM, editable via RTE per module type

6.10.2 Thermal Model

A lumped-parameter thermal RC network models heat flow from junction to ambient:

T_junction = T_sensor + P_total x (R_th_jc + R_th_cs + R_th_sb) x (1 - e^(-t/tau))

Where T_sensor is the measured NTC temperature, R_th_jc is junction-to-case thermal resistance, R_th_cs is case-to-sink (TIM), R_th_sb is sink-to-baseplate, and tau is the thermal time constant of the sensor mounting location. All thermal parameters are user-configurable via RTE for different IGBT modules and cooling systems.

6.10.3 Estimated Junction Temperature

The model computes an estimated junction temperature every 1 ms. This estimate is compared against the measured sensor temperature:

- If T_junction_est > T_sensor_measured + 20C for > 5 seconds: flag FAULT_THERMAL_MISMATCH — the sensor may be detached or the thermal path degraded
- If T_junction_est > 125C: early warning — aggressive derating before the 100C sensor hard cap is reached

6.10.4 Time-to-Temperature Prediction

Using the thermal model, the system predicts how long until the sensor temperature reaches each threshold:

```
 /* LossEstimator.c — 1 ms tick */
void LossEstimator_Update(void) {
    /* 1. Calculate instantaneous power per IGBT */
    float i_phase = fabsf(adc_filtered[phase_index]); /* RMS approx */
    float duty = fabsf(duty_cycle[phase_index]);
    float v_sat = igbt_params.V_ce0 + igbt_params.R_ce * i_phase;
    float p_cond = v_sat * i_phase * duty;
    float p_sw = (igbt_params.E_on + igbt_params.E_off) * pwm_frequency;
    float p_total = p_cond + p_sw;

    /* 2. Update thermal model (discrete RC) */
    float t_ambient = temp_ambient; /* from sensor or estimated */
    float tau = thermal_params.tau_sensor; /* thermal time constant */
    float r_th = thermal_params.R_th_total;
    float dt = 0.001f; /* 1 ms */
    t_junction_est += (dt / tau) * ((t_ambient + p_total * r_th) - t_junction_est);

    /* 3. Predict time to thresholds */
    if (p_total > 0.0f) {
        float delta_to_warn = 95.0f - t_sensor_measured;
        float delta_to_fault = 100.0f - t_sensor_measured;
        float heating_rate = p_total * r_th / tau; /* C/s approximation */
        if (heating_rate > 0.1f) {
            time_to_warn_sec = delta_to_warn / heating_rate;
            time_to_fault_sec = delta_to_fault / heating_rate;
        }
    }
```

```
    /* 4. Publish to CAN/RTE */
    telemetry.p_junction_est = t_junction_est;
    telemetry.p_total_loss = p_total * 6.0f; /* all 6 IGBTs */
    telemetry.time_to_ot_sec = time_to_fault_sec;
}
```

The time-to-overtemperature prediction is published on CAN and visible in RTE. This gives the rider advance warning: "at current power, IGBT will reach thermal limit in 45 seconds." The system also uses this internally to pre-emptively reduce torque before the hard cap is reached (predictive derating).

7. Application Layer

Non-safety-critical modules providing vehicle-level functionality. Target: ASIL-B. All persistent data stored in SPI FRAM with write protection.

7.1 Throttle Processing

Dual potentiometer processing: simultaneous ADC read, range check (0.5V - 4.5V), plausibility check (<5% delta), map to 0-100%, deadband (0-5%, user-configurable), IIR filter. Failure: FAULT_THROTTLE_PLAUS, zero torque, FAULT state.

7.2 CAN Communication

User-tailorable protocol. Reference frames in Section 10 are defaults only. RX dispatch via ID-based callback table. RTE commands handled on dedicated CAN ID.

7.3 DC-Link Monitor

16-bit ADC, 10 ms IIR filter. Thresholds user-configurable via RTE within hard bounds (OV max = hardware rating, UV min = 60V). Cross-checked against back-calculation formula.

7.4 Temperature Monitor (2oo3)

2-out-of-3 voting: a single outlier sensor is excluded. No averaging, no median. If one sensor reports 95C and the others report 70C, the outlier is discarded and the system uses the higher of the two remaining (non-outlier) sensor readings.

> **Two-Sensor Fault Vulnerability**
>
> With 2-out-of-3 voting, a single noisy or stuck-high sensor is excluded as an outlier and can no longer trigger progressive derating (from 80C) or fault entry (at 100C) on its own. The sensor validity check (below) only catches readings outside the physical range (-40C to +175C). The residual risk is two sensors failing in the same direction (common-cause): two sensors stuck high within range would outvote the healthy sensor and cause unnecessary derating or fault entry. This is a known limitation (L-13): the design prioritizes thermal safety (never miss a real overheat) over availability (may derate on a false positive).

Table 14: IGBT Temperature Thresholds

Level	Temp	Response
Normal	< 80 C	Full torque
Derate 1	80 C	Linear derate: 100% to 50% torque (80-90C range)
Derate 2	90 C	Torque limited to 25%
Warning	95 C	FAULT_IGBT_OT_WARN logged
Fault	100 C	FAULT_IGBT_OT, ramp to zero, FAULT state

Each sensor checked against valid range (-40C to +175C). Two or more invalid sensors: FAULT_SENSOR_FAIL, FAULT state.

7.5 POST (Power-On Self Test)

POST runs once after every reset using X-CUBE-CLASSB (IEC 60730-1 Class B) for CPU register test, RAM March-C, and flash CRC. Additional application-specific tests:

Table 15: POST Test Checklist

Step	Test	Pass Criteria
1-4	X-CUBE-CLASSB: CPU registers, RAM walking-bit, RAM March-C, flash CRC	All Class B tests pass
5	ADC self-test (VREFINT, TSENSE)	Within expected range
6	Gate driver FAULT pins	All de-asserted (high)
7	Encoder signal presence	Valid sin/cos if rotating
8	CAN loopback	Frame received
9	Watchdog self-test (force timeout, verify reset cause)	Reset cause = IWDG
10	Current sensor VREF check	All REF within 2.48-2.52V window
11	Current sensor zero-point check	All within +/-20% of zero
12	FRAM read/write test	Pattern read back correctly
13	FRAM write-protection pin test	WP pin toggles correctly

7.6 RTE Configuration Interface

The **Real Time Examiner (RTE)** is an open-source host tool connecting via CAN bus. It provides live monitoring and runtime parameter configuration. Parameters stored in SPI FRAM (not flash) with unlimited write endurance.

Table 16: RTE-Configurable Parameters

Category	Parameters
Motor Control	Modulation scheme, PWM frequency (300 Hz - 16 kHz), PI gains, current limits, field weakening
Torque Maps	4 custom 16-point curves + Sport/Eco presets, ramp rates, deadband
Safety Thresholds	DC-link OV/UV, IGBT temp levels, throttle plausibility delta, CAN timeouts

Category	Parameters
Loss Model	V_ce0, R_ce, E_on/E_off curves, R_th values, tau — all per IGBT module type
Vehicle Data	Odometer (km), hour counter, boot counter, fault log (256 entries)
CAN Protocol	All frame IDs, scaling factors, periods — fully user-defined
Custom	32 float + 32 uint32_t user parameters

Changes validated before application (Section 7.7). Committed to FRAM only on explicit "save" command. FRAM WP pin asserted during normal operation, de-asserted only during commit.

7.7 Input Validation Framework

Every user-configurable parameter passed via RTE is validated before being applied. The validation framework has three layers:

7.7.1 Layer 1: Hard Bounds (Compile-Time)

Safety-critical parameters have absolute min/max values compiled into the firmware. These **cannot be overridden** by RTE or any other interface. They represent the physical limits of the hardware.

Table 17: Hard Bounds Examples

Parameter	Hard Min	Hard Max	Rationale
IGBT temp fault threshold	60 C	100 C	100 C hard cap (75 C margin to Tj_max 175 C)
DC-link OV threshold	60 V	200 V	200 V = current hardware voltage rating limit (450 V with cap swap; 900 V with PCB + cap change)
DC-link UV threshold	60 V	150 V	Below 60 V: ADC quantization dominates
PWM frequency	300 Hz	16 kHz	Below 300 Hz: excessive ripple/noise; above 16 kHz: thermal
Dead time	500 ns	4 us	Below 500 ns: shoot-through risk; above 4 us: excessive loss
Max torque (I_q_ref)	0 A	350 A	350 A = hardware current limit with margin
Current sensor VREF min	2.40 V	—	Below 2.40 V: sensor fault or supply collapse

7.7.2 Layer 2: Cross-Parameter Consistency

Parameters are validated against each other for logical consistency:

- DC-link UV threshold must be < DC-link OV threshold (minimum 10 V separation)
- Temperature derate start must be < temperature fault threshold (minimum 10 C separation)
- Throttle deadband must be < 20%
- PI gains must be positive (negative gain = positive feedback = runaway)
- Modulation scheme must be valid enum value (0-7)
- Switching frequency must be compatible with selected modulation (SHEPWM has minimum frequency)

7.7.3 Layer 3: Rate-of-Change Limiting

To prevent abrupt changes that could destabilize the control loop:

- PI gain changes: maximum 10% per RTE commit
- Switching frequency changes: only at zero torque
- Modulation scheme changes: only at zero torque or below 10% base speed
- Safety threshold changes: only in STANDBY or READY state (not while driving)

7.7.4 Validation Framework Implementation

```
/* InputValidator.c — three-layer validation */
ValidationResult_t InputValidator_Check(const ParamDef_t* def, float value) {
    /* Layer 1: Hard bounds */
    if (value < def->hard_min || value > def->hard_max) {
        return VALIDATION_HARD_BOUND_VIOLATION;
    }
    /* Layer 2: Cross-parameter consistency */
    if (def->consistency_check != NULL) {
        if (!def->consistency_check(value)) {
            return VALIDATION_INCONSISTENT;
        }
    }
    /* Layer 3: Rate of change */
    float delta = fabsf(value - def->current_value);
    if (delta > def->max_delta) {
        return VALIDATION_RATE_LIMITED;
    }
    return VALIDATION_OK;
}

/* Example: IGBT temp threshold validation */
bool Validator_TempFaultConsistent(float new_threshold) {
    /* Fault threshold must be > derate start + 10C */
    return (new_threshold > params.igbt_derate_start + 10.0f);
}
```

8. Data Flow and Interfaces

Table 18: Module-to-Module Data Interfaces

Source	Data	Consumer(s)	Freq	Method
ADC oversampler	I_U, I_V, I_W (decimated)	FOC, LossEstimator	PWM freq	DMA double buffer
ADC oversampler	I_U, I_V, I_W (raw samples)	Ripple monitor	n x PWM freq	DMA circular
Encoder	theta_e	FOC	PWM freq	Function call
Throttle	Throttle %	TorqueCmd	1 kHz	Function call
TorqueCmd	I_q ref	FOC	100 Hz	Atomic float
DCLink Mon	V_DC	FOC, CAN	100 Hz	Function call
Temp Mon	T_IGBT (highest)	FaultMgr, LossEstimator	10 Hz	Function call

Source	Data	Consumer(s)	Freq	Method
LossEstimator	T_junction_est, P_loss, time_to_OT	CAN, RTE, FaultMgr	1 kHz	Global struct
FaultMgr	Fault flags	SM, CAN, RTE	On change	Volatile uint32
State Machine	Current state	FOC, TorqueCmd, CAN	100 Hz	Global enum
CAN RX	BMS/IO/charger data	Handlers	Event	Callback dispatch
RTE CAN	Param read/write	RTE handler	Event	FRAM access

> **Cache Coherency**
>
> DMA buffers for ADC results placed in SRAM4 (non-cacheable via MPU). All DMA buffers use cache-line-aligned addresses. The FOC ISR reads ADC DMA results — cache invalidation is not required because the DMA buffer resides in non-cacheable SRAM4.

9. Memory Layout

9.1 Flash Memory Map

Table 19: Flash Memory Map

Region	Address	Size	Content
Sector 0	0x0800_0000	8 KB	Bootloader
Sector 1	0x0800_2000	8 KB	Bootloader config (HMAC key)
Sector 2	0x0800_4000	8 KB	Reserved
Sectors 3-7	0x0800_6000	40 KB	Reserved
Bank 1 App	0x0801_0000	448 KB	Application firmware
Bank 2 Swap	0x0808_0000	512 KB	OTA update swap

No RDP (open-source). No OTP. WRP on bootloader sectors only.

9.2 RAM Memory Map

Table 20: RAM Memory Map

Region	Size	Usage	ECC
AXISRAM	320 KB	Stack, heap, FOC buffers, fault log runtime	Yes (SECDED)
SRAM1	16 KB	DMA buffers, CAN FIFOs (non-cacheable)	No
SRAM2	16 KB	ADC oversample buffers, DMA descriptors	No
SRAM4	16 KB	Warm-boot retention (fault, cycle count)	No

> **ECC Handling (FSR-20)**
>
> AXISRAM has SECDED ECC. Single-bit errors corrected by hardware. Double-bit errors trigger NMI: log address, set FAULT_ECC_RAM, trigger safe state, avoid re-reading corrupted address.

10. CAN Protocol

Reference implementation — user-tailorable via RTE. All IDs, scaling factors, and periods are defaults stored in FRAM and modifiable.

Table 21: VCU TX Frame Reference (Defaults)

CAN ID	Name	Period	Key Signals
0x18F00100	VCU_Status	100 ms	State, fault flags
0x18F00200	VCU_Motor	50 ms	Speed (rpm), torque (Nm), I_q, I_d
0x18F00300	VCU_Power	100 ms	V_DC, I_DC, power (kW)
0x18F00400	VCU_Thermal	500 ms	T_IGBT[0-2], T_junction_est
0x18F00500	VCU_LossEst	500 ms	P_total_loss, time_to_OT_sec
0x18F00600	VCU_Heartbeat	1000 ms	Seq counter, uptime
0x18F00700	VCU_FaultLog	Event	Event code + detail

11. Traceability Matrix

Table 22: FSR-to-Module Traceability

FSR	Description	ASIL	Module(s)
FSR-01	Read dual redundant tractive effort control pots on both MCUs. Discrepancy >5% or either OOR detected by either MCU → safe state.	D	Throttle module
FSR-02	Tractive effort command plausibility: commanded current reference vs. measured phase currents. Deviation >20% for >100 ms → safe state.	C	FOC ISR, CurrentSensor_Validate
FSR-03	Tractive effort control input rate-limited. Max tractive effort rate: 500 Nm/s (calibration parameter).	C	SSM, FaultMgr
FSR-04	Reject reverse tractive effort when speed >0. Reverse only when stationary AND explicitly selected.	B	State Machine
FSR-05	On fault, ramp down tractive effort at ≤200 Nm/s before SSO. Abrupt removal prohibited unless <50 ms safety timing requires faster response.	C	SSM, FaultMgr
FSR-06	Monitor regenerative braking tractive effort continuously. Uncommanded >10 Nm for >200 ms → safe state.	C	ADC multi-rate, CurrentSensor_Validate
FSR-07	Max tractive effort limit: software LUT on both MCUs with cross-check. Dual-MCU independent current monitoring detects overcurrent within 100 ms. DESAT handles hard short-circuit (<2 us).	C	FOC ISR, CurrentSensor_Validate

FSR	Description	ASIL	Module(s)
FSR-08	One NTC temp sensor per IGBT module (3 total), 2-out-of-3 voting with 100 °C hard cap. Critical threshold in ECC memory.	B	TempMonitor (2oo3)
FSR-09	Encoder loss → safe state within 100 ms. Sensorless fallback if implemented: ≤2 s, ≤30% tractive effort.	C	Encoder, ENC_Monitor
FSR-10	HVIL monitored continuously. Interruption → immediate PWM disable + BMS contactor open request within 50 ms.	B	Isolated DI
FSR-11	DC link bus voltage monitored with isolated ADC. OV threshold → immediate regen disable. Critical OV → SSO within 50 ms.	B	DCLink_Monitor
FSR-12	NCV57100 complementary inputs (IN+/IN−) prevent HS+LS simultaneous conduction. Power-on self-test confirms.	A	TIM1 break ISR, POST
FSR-13	NCV57100 DESAT detection active on all six IGBTs. DESAT event → local PWM disable within <2 us, independent of MCU.	A	TIM1 break ISR
FSR-14	STM32 breakpoint input → HW PWM disable within <10 us on any critical fault. Independent of main CPU execution.	C	TIM1 break ISR
FSR-15	Independent windowed watchdog. Failure to service → automatic PWM disable + system reset. Timeout ≤50 ms.	C	Watchdog Manager
FSR-16	Power-on self-test (POST): ADC references, current sensor offsets, gate driver communications, encoder signal, HVIL continuity, watchdog, NCV57100 DESAT self-test. All must pass before PWM enable.	C	POST
FSR-17	IO board CAN heartbeat 1 s timeout. Loss → safe-state defaults (brake pressed, kickstand down) and tractive effort restricted to zero.	C	CAN RX ISR, State Machine
FSR-18	Tractive effort control limit switch monitored independently of analog pots. Activation overrides any analog value and commands zero tractive effort.	C	Isolated DI, Throttle module
FSR-19	Boot-time CRC-32 over safety-critical code and calibration data. Mismatch → PWM enable prevented, fault logged.	C	Bootloader
FSR-20	ECC RAM for all safety-critical variables. Single-bit errors corrected (SECDED). Double-bit errors → safe state.	C	NMI handler, FaultMgr
FSR-21	Monitor DC link bus undervoltage. UV → tractive effort derate, critical UV → safe state. Prevents overcurrent due to insufficient DC link voltage.	B	DCLink_Monitor

12. Bootloader Architecture

Minimal second-stage bootloader in flash Sector 0. Runs on every reset before the application.

```
void Bootloader_Main(void) {
    SystemClock_Config_Basic();
    ResetReason_t reason = Boot_GetResetReason();

    /* Boot CRC check (FSR-19) */
    uint32_t app_crc = Boot_CalculateCRC32(APP_START_ADDR, APP_SIZE);
    uint32_t stored_crc = *(uint32_t*)APP_CRC_ADDR;
    if (app_crc != stored_crc) { Boot_EnterUpdateMode(); return; }

    /* Check for update request (500 ms window) */
```

```
    if (reason == RESET_POWER_ON || reason == RESET_PIN) {
        FDCAN_Init_Basic();
        if (Boot_CheckForUpdateRequest(500)) { Boot_EnterUpdateMode(); return; }
        UART_Init_Basic();
        if (Boot_CheckForUSBUpdateRequest(500)) { Boot_EnterUpdateMode(); return; }
    }
    Boot_JumpToApplication(APP_START_ADDR);
}
```

13. Firmware Update Protocol

Firmware updates accepted over **UART/USB** or **CAN** — both use the same protocol. The update interface is selected automatically during the 500 ms bootloader window (CAN frame or USB enumeration detected).

13.1 Update Protocol

Table 23: Firmware Update State Machine

State	Description	Transitions
IDLE	Waiting for update request	RX request -> AUTH
AUTH	HMAC verify request, check rollback counter	OK -> RX; Fail -> IDLE
RX	Receiving chunks, write to Bank 2	All chunks -> VERIFY; Timeout -> IDLE
VERIFY	SHA-256 full image + HMAC signature	OK -> COMMIT; Fail -> IDLE
COMMIT	Atomic bank swap, update rollback counter	OK -> REBOOT
REBOOT	System reset	-> Bootloader

13.2 Security Requirements

- Update only when stationary (CSR-04: speed=0, brake=on, key-on)
- Chunks: sequence number + CRC-32; missing chunks re-requested
- Complete image SHA-256 before flash commit
- HMAC-SHA256 signature with user-managed key from bootloader config
- Anti-rollback counter (CSR-03); user-resettable via JTAG
- Dual-bank atomic swap

13.3 FRAM Protection During Update

During firmware update, the FRAM write-protection (WP) pin is **asserted** (high) to prevent any accidental parameter corruption. The bootloader does not access FRAM. After a successful update, the application verifies FRAM integrity on first boot and restores from a shadow copy in flash if corruption is detected.

> **Open-Source Trust Model**

> No OTP fuses. No vendor-locked keys. HMAC key stored in plaintext bootloader config. Physical access allows key modification via JTAG. User is root of trust.

14. Scheduling and Timing

14.1 Interrupt Priority Map

Table 24: NVIC Priority Assignment

Prio	IRQ	Source	Max Latency
0	TIM1_UP	FOC loop + PWM update	0 us (no preemption)
1	ADC_DMA	ADC oversample buffer full	5 us
2	TIM1_BRK	Gate driver fault (SSO)	<10 us (hw handles)
3	TIM6	1 ms system tick	100 us
4	FDCAN1/2_RX0	CAN receive	200 us
5	TIM7	Watchdog service	10 us
6	FDCAN_TX	CAN TX complete	N/A
7	USART	Debug / USB	N/A

14.2 1 ms System Tick (TIM6)

```
void TIM6_DAC_IRQHandler(void) {
    HAL_TIM_IRQHandler(&htim6);
    SafetyMonitor_ADCCheck();
    Throttle_Process();
    DCLink_Monitor();
    ENC_Monitor();
    LossEstimator_Update();          /* junction temp + time-to-OT */
    if ((tick_count % 50) == 0) { WWDG_Refresh(); IWDG_Refresh(); ExternalWDT_Toggle(); }
    if ((tick_count % 10) == 0) { SM_Run(); TorqueCmd_Update(); }
    if ((tick_count % 100) == 0) { CAN_TX_Heartbeat(); CAN_CheckHeartbeats(); }
    if ((tick_count % 100) == 0) { Temp_Monitor_2oo3(); CurrentSensor_Validate(); }
    tick_count++;
}
```

14.3 WCET Summary

Table 25: Worst-Case Execution Time

Task	Period	WCET	Util
FOC ISR (TIM1_UP)	62.5 us - 3.3 ms	5.0 us (SHE worst case)	8.0% @ 16 kHz / 0.15% @ 300 Hz
ADC DMA callback	20.8 us	0.5 us	2.4%

Task	Period	WCET	Util
TIM1 break ISR	Event	0.3 us	Negligible
TIM6 tick	1 ms	60 us (incl. LossEstimator)	6.0%
CAN RX ISR	Event	10 us	<1%
Total @ 16 kHz PWM (worst case)			**<18%**
Total @ 2 kHz PWM (nominal)			**<9%**

15. Assumptions and Limitations

15.1 Assumptions

ID	Assumption
A-01	Gate driver FAULT outputs OR'd to TIM1_BKIN
A-02	16-bit ADC1 primary + 12-bit ADC2/ADC3 redundant
A-03	Motor parameters (Ld, Lq, lambda_m, poles) known; stored in FRAM via RTE
A-04	3.3V supply stable within +/- 10%
A-05	Physical tampering out of scope (open-source trust model)
A-06	Single-core, no RTOS
A-07	Single CAN bus (no redundancy)
A-08	User has STM32CubeMX for GPIO and peripheral configuration
A-09	Loss model parameters (V_ce0, R_ce, E_on, E_off, R_th) entered via RTE for user's specific IGBT module
A-10	FRAM chip present and functional (verified by POST test 12)

15.2 Known Limitations

ID	Limitation	Mitigation
L-01	Single-core main MCU, no lockstep	External watchdog; POST; 3-path SSO; safety coprocessor present on ControlBoard Rev A (see scope note, Section 1.1)
L-02	SHEPWM: pre-computed tables only (no live calc in V1.0)	Tables cover typical range; V2.0 reserved
L-03	Single CAN bus	Safe state on timeout
L-04	No field weakening in V1.0	Planned V2.0
L-05	Gate drivers AEC-Q100, not ASIL-rated	DESAT+UVLO+dead time; ASIL C max
L-06	HMAC key in plaintext flash	Intentional; user is root
L-07	No regen current limit in V1.0	BMS backstop; V1.1

ID	Limitation	Mitigation
L-08	STL is Class B (X-CUBE-CLASSB), not Class D	Documented in HARA LIMIT-07; 99 fault injection tests
L-09	Loss model accuracy depends on user-entered IGBT parameters	Defaults from datasheet; user tunes via RTE; thermal model cross-checks against sensor
L-10	No Dependent Failure Analysis (DFA) per ISO 26262-9	Acknowledged in HARA. Common-cause between SSO paths (e.g., PCB delamination affecting both TIM1_BKIN and PMIC EN routing) not formally analyzed. Required for ASIL C claim.
L-11	No EMI/EMC pre-compliance assessment	Environmental EMI tests defined (E-08, E-09) but no pre-compliance testing or design margin analysis completed. Risk in unknown vehicle installations.
L-12	12-bit ADC redundancy has coarser resolution than 16-bit primary	The 5.3x resolution difference is not a safety issue for fault detection (broken wire, stuck output are reliably caught). Subtle gain drift is not safety-critical. A second 16-bit ADC would provide higher-fidelity redundancy but is not required for ASIL-C.
L-13	2oo3 temp voting: common-cause failure of two sensors in the same direction can outvote the healthy sensor	A single outlier sensor is excluded; one stuck-high sensor no longer causes derating. Residual risk requires two sensors stuck high (within valid range), which triggers derating even if the third reads normal. Trade-off: thermal safety over availability.

15.3 Future Work (V2.0+)

- Live on-demand SHE angle calculation (Newton-Raphson/homotopy)
- Field weakening for extended speed range
- Sensorless FOC as encoder backup
- CAN FD at 2 Mbps
- Auto-tuning of PI gains
- Bluetooth LE for wireless diagnostics
- SD card data logging

THREAT ANALYSIS AND RISK ASSESSMENT

HW-C2-DOC-TARA-A · design document — rev A

THREAT ANALYSIS AND RISK ASSESSMENT

Traction Inverter

Version	1.2
Prepared by	Thomas Liao
Date	July 13, 2026
Part	HW-C2-DOC-TARA-A · Rev A

University of California, Santa Cruz — Corzine Lab

Contents

1. Scope and Philosophy . . . 3
1.1 The Trust-the-User Model . . . 3
1.2 What We Actually Protect Against . . . 4
1.3 What We Explicitly Do Not Claim . . . 4
2. Asset Identification . . . 4
2.1 Attack Surface Map . . . 5
3. Threat Scenarios . . . 5
4. Risk Assessment . . . 6
5. Cybersecurity Requirements . . . 7
6. Test Plan . . . 8
7. Implementation Guidance . . . 11
7.1 HMAC Key Management . . . 11
7.2 CAN Update Protocol . . . 11
7.3 Enabling RDP for Security-Conscious Users . . . 11
8. Gap Analysis . . . 12
8.1 HARA Cross-Reference . . . 12
References . . . 13
Document History . . . 13

> **MCUs:** STM32H723ZG + STM32G474RCTx
> **Operating Temp:** −40°C to +85°C
> **Reviewed by:** (not yet reviewed)

1. Scope and Philosophy

This Threat Analysis and Risk Assessment (TARA) covers the cybersecurity aspects of an open-source aftermarket electric motorcycle traction inverter and vehicle control unit (VCU). It follows the methodology of **ISO/SAE 21434** (Road vehicles — Cybersecurity engineering) but adapts it for the realities of open-source hardware.

1.1 The Trust-the-User Model

This project is **open source**. The end user has full access to source code, schematics, firmware, and build tools. This fundamentally changes the cybersecurity posture compared to a proprietary automotive product:

Table 1: Trust Model: Open Source vs. Proprietary

Aspect	Proprietary (OEM)	This Project (Open Source)
Source code access	Restricted to manufacturer	Fully public; user can inspect, modify, rebuild
Firmware signing keys	Manufacturer-controlled, burned into OTP	User-generated, user-managed; no vendor keys
Physical tamper resistance	Security screws, potting, encrypted flash	None by design; user has physical access to everything
Attack model	Remote attackers, supply chain, dishonest dealers	CAN bus attackers, malicious accessories, rider error
Remediation authority	Manufacturer issues OTA patches	Community issues patches; user decides when to update
Liability	Manufacturer assumes product liability	User assumes full responsibility (open-source disclaimer)

> **Design Principle: User Sovereignty Over Anti-User DRM**
>
> This project explicitly rejects all anti-user security mechanisms: OTP fuses, write-once flash regions, vendor-signed firmware with unreplaceable keys, encrypted bootloaders that prevent user modification, and any form of DRM that treats the hardware owner as an adversary. These mechanisms are incompatible with the open-source philosophy and provide no meaningful security benefit when the attacker already has physical access and the full source code. If a user wishes to add their own tamper protection (RDP Level 1/2, encrypted external flash, physical tamper switches) for their specific threat model, they are free to do so — the base design imposes no barriers and documents how. The security model is **trust the user, protect the bus**: the legitimate owner is never the threat; remote CAN bus attacks are.

1.2 What We Actually Protect Against

With physical access = game over (acceptable for open source), the meaningful attack surfaces are:

1. **CAN bus remote attacks:** A malicious or compromised node on the CAN bus (aftermarket accessory, diagnostic tool, charger) should not be able to cause unintended tractive effort, disable safety features, or flash malicious firmware without authorization.
2. **Firmware integrity during CAN updates:** Bit errors, EMI, or bus faults during a legitimate firmware update should not brick the device or install corrupted safety-critical code.
3. **Replay and downgrade:** An attacker capturing a valid firmware update should not be able to replay it to install older, potentially vulnerable firmware.
4. **Denial of service:** CAN bus flooding should not prevent the VCU from receiving critical safety messages (BMS heartbeat, brake input).

1.3 What We Explicitly Do Not Claim

Table 2: Out-of-Scope Security Claims

Claim	Why Not
Physical tamper resistance	Open hardware; user has PCB, schematic, JTAG access. Physical tamper resistance is the user's responsibility if desired.
Supply chain security	Open BOM; user sources their own components. Component authenticity verification is the user's responsibility.
Side-channel attack resistance	No power analysis countermeasures, no timing randomization. Research-level attacks are out of scope.
Firmware extraction protection	Source code is public; extracting firmware from the device provides no advantage. STM32 RDP Level 0 by default.
Anti-cloning	Open schematic and Gerbers; anyone can build a copy. This is a feature, not a bug.

2. Asset Identification

Table 3: Cybersecurity-Relevant Assets

ID	Asset	Location	Why It Matters	Compromise Impact
A-01	Application firmware	STM32 internal flash	Contains all safety logic, FOC control, fault handling	Arbitrary tractive effort control, safety bypass
A-02	Bootloader	STM32 internal flash (separate sector)	Verifies firmware before boot; root of trust	Bootloader bypass = any firmware runs unchecked
A-03	Firmware signing key	User possession of build key; symmetric key compiled into STM32 bootloader flash (readable under RDP Level 0)	Used to sign valid firmware updates	Key leak = anyone can sign valid firmware
A-04	Calibration data	STM32 flash (torque LUT, thresholds)	Defines safety limits and motor parameters	Altered limits = unsafe operation

ID	Asset	Location	Why It Matters	Compromise Impact
A-05	CAN bus communication	CAN1 (BMS), CAN2 (IO board, display, charger, ABS)	All external safety inputs and commands	Forged messages = false safety clearance
A-06	Runtime state (RAM)	STM32 ECC RAM	Active safety variables, fault flags, tractive effort command	Corruption = incorrect safety decisions

2.1 Attack Surface Map

Table 4: Attack Surfaces and Interfaces

Interface	Access	Risk	Mitigation Strategy
CAN1 (BMS)	Vehicle-internal, BMS node required	Medium: compromised BMS or malicious node on bus	Firmware signing; heartbeat timeout; safe defaults on loss
CAN2 (IO board, display, charger, ABS)	Vehicle-internal, multiple nodes	Higher: more nodes = larger attack surface	Firmware signing; message freshness; heartbeat timeout
SWD/JTAG	Physical access to debug header	Low (physical access required = user is owner)	None by design; user owns the hardware
USB (if present)	Physical access	Low (physical access required)	None needed; physical access = owner
UART console	Physical access or accessible header	Low	Disabled in release builds; debug-only

3. Threat Scenarios

Threat scenarios follow ISO/SAE 21434 threat analysis methodology: identify threat sources, attack vectors, and resulting damage scenarios.

Table 5: Threat Scenarios

ID	Threat	Source	Attack Vector	Affected Asset	Damage Scenario
T-01	Malicious firmware upload via CAN	Compromised accessory node, malicious diagnostic tool	Inject firmware update frames on CAN1 or CAN2; bootloader accepts unsigned firmware	A-01, A-02	Firmware disables all safety mechanisms; arbitrary tractive effort; no safe state
T-02	Firmware corruption during CAN update	EMI, loose connector, bus fault	Bit flip in CAN frame during legitimate update	A-01	Corrupted safety threshold or tractive effort limit; latent fault until triggered
T-03	Firmware downgrade (replay attack)	Attacker with CAN bus access and captured update log	Replay previously captured valid firmware update	A-01	Roll back to older firmware with known vulnerabilities; re-introduce patched bugs
T-04	CAN bus flooding (DoS)	Compromised node, malicious accessory	Flood CAN bus with high-priority frames	A-05	BMS heartbeat lost; safe state entry at speed; potential rear collision

ID	Threat	Source	Attack Vector	Affected Asset	Damage Scenario
T-05	Forged safety-critical CAN frames	Compromised display/ charger/IO node	Inject fake BMS "all clear" or fake brake-released message	A-05, A-06	VCU operates based on false safety data; tractive effort when it should be zero
T-06	Firmware update during vehicle motion	Buggy update tool, malicious update trigger	Initiate firmware update while speed > 0	A-01	Update aborts mid-flash; corrupted firmware; undefined behavior on next boot
T-07	Bootloader replacement via application	Malicious firmware already running	Application firmware erases and rewrites bootloader sector	A-02	New bootloader skips signature verification; permanent compromise

4. Risk Assessment

Risk rating uses ISO/SAE 21434's impact/likelihood framework adapted for the open-source trust model. **Impact** considers vehicle-level safety consequences. **Likelihood** considers attack feasibility given the open-source context (physical access already assumed = user).

Table 6: Risk Rating Matrix

Threat	Safety Impact	Attack Likelihood	Risk Level	Rationale
T-01: Malicious firmware upload	**Severe**	**Possible**	**HIGH**	CAN bus is accessible from multiple vehicle locations. Without signature verification, any node can flash malicious firmware. However, attacker must craft valid STM32 binary specifically for this hardware.
T-02: Firmware corruption	**Severe**	**Likely**	**HIGH**	CAN is inherently unreliable. EMI, vibration, and bus contention cause frame errors. Without per-chunk integrity checks, corruption goes undetected.
T-03: Firmware downgrade	**Major**	**Possible**	**MEDIUM**	Requires attacker to capture a previous valid update (via CAN sniffing) and replay it. Attacker must have had bus access during the original update.
T-04: CAN bus flooding	**Moderate**	**Possible**	**MEDIUM**	Safe state entry is the outcome, which is safe but disruptive. Risk is nuisance and potential rear collision from sudden deceleration at highway speed (same as H-03 in HARA).
T-05: Forged safety frames	**Severe**	**Unlikely**	**MEDIUM**	Requires compromising an existing CAN node or adding a malicious node to the bus. Physical access to the vehicle's CAN wiring required.
T-06: Update during motion	**Major**	**Unlikely**	**LOW**	Update tool should enforce preconditions; this is a defense-in-depth measure against buggy tools, not a primary attack vector.
T-07: Bootloader replacement	**Severe**	**Unlikely**	**LOW**	Requires already-compromised application firmware. If application is compromised, bootloader protection is irrelevant (attacker already controls the system).

5. Cybersecurity Requirements

These cybersecurity requirements (CSRs) are independent of the HARA's Functional Safety Requirements (FSRs). Where a CSR and FSR overlap (e.g., CAN heartbeat timeout), the CSR references the FSR rather than duplicating it.

Table 7: Cybersecurity Requirements

ID	Requirement	Threat	Risk	Implementation Notes
CSR-01	CAN firmware update requires cryptographic signature verification. Unsigned or incorrectly signed firmware → reject update, log event, keep current firmware active.	T-01	HIGH	Algorithm: HMAC-SHA256 (recommended) or Ed25519. Key is user-generated, stored in a JSON config file during build, compiled into the bootloader. **No OTP or write-once fuses.** User can rotate keys by re-flashing bootloader with new key.
CSR-02	CAN firmware update protocol: each chunk carries a 32-bit CRC. Overall firmware image carries a cryptographic hash (SHA-256) verified before flash commit. Mismatch → abort update, erase partial image, log event.	T-02	HIGH	Chunk structure: [seq_number:2B][data:up to 64B][crc32:4B]. Final frame: [sha256:32B]. Abort on any CRC or hash mismatch. Partial flash is erased, not left in indeterminate state. Application-layer chunks are segmented into 0–6 byte CAN frames per Table 8; CRC in CHUNK_ACK covers each CAN segment, and the 32-byte SHA-256/HMAC digest is sent as four 8-byte frames.
CSR-03	Anti-rollback counter stored in flash sector reserved for configuration data. Firmware with counter ≤ current counter → reject. Counter increments on every successful update. Counter is user-resettable via JTAG/SWD (physical access required).	T-03	MEDIUM	Stored as 64-bit value in designated flash page. Not write-once: user can reset via debugger if they brick the counter. Normal updates increment automatically. Physical access = can reset = acceptable for open source.
CSR-04	Firmware update preconditions: vehicle speed = 0 (encoder < 10 rpm for >2 s), brake applied (IO board), key-on, tractive effort = 0. Any precondition violated → abort update immediately.	T-06	LOW	All preconditions must be verified by bootloader (not application). Update aborts mid-transfer if preconditions change. This is defense-in-depth; the primary protection is the user's good judgment.
CSR-05	CAN bus heartbeat timeout (FSR-17) serves as DoS protection. IO board loss → safe state within 1 s. BMS loss → safe state within 5 s. These timeouts are independent of bus load.	T-04	MEDIUM	References FSR-17 from HARA. No additional CAN-specific DoS protection needed; safe-state timeout is the defense. Test: CT-09 (CAN DoS / heartbeat timeout), I-10 (CAN bus off recovery), I-09 (CAN bus load).
CSR-06	CAN message freshness: safety-critical frames (BMS heartbeat, IO board inputs) include a 32-bit rolling counter. Frames with stale counter (>5 s old) or unexpected counter gap (>10) → treat as lost (safe defaults).	T-05	MEDIUM	Lightweight alternative to full SecOC. No MAC (would require shared key on every node). Rolling counter detects replay and sequence gaps. Shared counter seed established at system startup. Not cryptographically strong but sufficient for the threat model.

ID	Requirement	Threat	Risk	Implementation Notes
CSR-07	Bootloader sector is not write-protected by hardware. This is intentional: the user must be able to update the bootloader. The bootloader's integrity is protected by CSR-01 (signed bootloader updates) and the user's physical custody of the hardware.	T-07	LOW	Explicit design decision for open source. STM32 RDP Level 0 (no read protection) by default. User can enable RDP Level 1 or 2 if they choose. Document how to do so, but do not mandate it.

> **Why HMAC-SHA256 Over RSA/ECDSA**
>
> HMAC-SHA256 is recommended over asymmetric algorithms (RSA, ECDSA, Ed25519) for three reasons: (1) **Smaller code size** in the bootloader — critical for size-constrained bootloaders; (2) **Faster verification** — SHA-256 is hardware-accelerated on STM32H7; (3) **No PKI complexity** — the user generates one key, stores it in their build environment, and signs their own firmware. The threat model does not require protection against key compromise (the user owns the key) so the non-repudiation benefits of asymmetric crypto are irrelevant here.

6. Test Plan

Nine test cases validate the cybersecurity requirements. CT-01 through CT-06 and CT-09 require a CAN interface and the VCU. CT-07 and CT-08 additionally require SWD/JTAG access to the bootloader flash.

CT-01: Unsigned Firmware Update Rejected

Objective: Verify CSR-01 — Unsigned firmware update via CAN is rejected.

Threat: T-01 (malicious firmware upload)

Setup: VCU with CAN update enabled. Bootloader built with known HMAC key.

Procedure:

1. Enter update mode: key-on, brake applied, speed=0.
2. Send firmware chunks over CAN without HMAC signature.
3. Send commit command.
4. Verify bootloader rejects update. Current firmware remains active.

Pass: Rejected. DTC logged: "signature verification failed." No partial flash write.

CT-02: Signed Firmware Update Accepted

Objective: Verify CSR-01 positive path — Signed update accepted.

Threat: T-01 (legitimate update)

Procedure:

1. Sign firmware with correct HMAC key.
2. Send signed chunks with per-chunk CRC (CSR-02).
3. Send SHA-256 hash of complete image.

4. Verify accepted, flash committed, system boots.
5. Verify POST passes. New version active.

Pass: Full update chain succeeds.

CT-03: Corrupted Chunk Detected

Objective: Verify CSR-02 — Bit flip in CAN chunk aborts update.

Threat: T-02 (firmware corruption)

Procedure:

1. Begin signed firmware update.
2. Flip one bit in a chunk's data payload.
3. Verify chunk CRC detects error. Update aborts.
4. Repeat with corrupted CRC field and corrupted sequence number.

Pass: All corruption detected. Update aborted. Partial image erased.

CT-04: Anti-Rollback Counter

Objective: Verify CSR-03 — Downgrade attack rejected.

Threat: T-03 (replay / downgrade)

Procedure:

1. System running firmware with counter = 5.
2. Attempt flash of older firmware with counter = 4. Verify rejected.
3. Flash newer firmware with counter = 6. Verify accepted. Counter = 6.
4. Attempt reflash with counter = 6. Verify rejected (must be > current).
5. Via debugger, reset counter to 0. Verify older firmware now accepted. *(Physical access = user can reset)*

Pass: Counter logic correct. User can reset via debugger (documented behavior).

CT-05: Update Preconditions

Objective: Verify CSR-04 — Update only when stationary.

Threat: T-06 (update during motion)

Procedure:

1. All preconditions met. Verify update accepted.
2. Speed > 0. Verify rejected.
3. Brake = off. Verify rejected.
4. During update, release brake. Verify aborts immediately.

Pass: All precondition violations rejected or aborted.

CT-06: CAN Freshness Counter — Replay Detected

Objective: Verify CSR-06 — Stale CAN frames treated as lost.

Threat: T-05 (forged safety frames)

Procedure:

1. Normal operation with freshness counters active.
2. Replay captured BMS heartbeat from 10 s ago. Verify detected as stale.
3. Verify VCU applies safe defaults (treats as heartbeat lost).
4. Verify DTC logged: "stale CAN frame detected."

Pass: Replay detected. Safe defaults applied. No tractive effort from stale frames.

CT-07: Bootloader Mutable (Design Verification)

Objective: Verify CSR-07 — Bootloader can be updated by user.

Threat: T-07 (bootloader replacement — intentional user action)

Procedure:

1. Via SWD/JTAG, read current bootloader flash sector. Verify readable (RDP Level 0).
2. Via SWD/JTAG, write modified bootloader (e.g., change version string). Verify succeeds.
3. Boot system. Verify modified bootloader runs.
4. Restore original bootloader. Verify system returns to original state.

Pass: Bootloader is mutable via SWD. User has full control. Document this as intentional.

CT-08: HMAC Key Rotation

Objective: Verify CSR-01 — user can rotate firmware signing keys.

Threat: T-01 (key compromise recovery)

Procedure:

1. Build bootloader with HMAC key K1. Flash to device.
2. Sign firmware with K1. Verify accepted.
3. Sign firmware with K2 (different key). Verify rejected.
4. Build and flash new bootloader with HMAC key K2 (via SWD).
5. Sign firmware with K2. Verify accepted.
6. Sign firmware with K1. Verify rejected.

Pass: Key rotation works. Old key rejected after rotation. Documented in user manual.

CT-09: CAN DoS / Heartbeat Timeout

Objective: Verify CSR-05 — CAN bus flooding causes heartbeat loss and safe state entry.

Threat: T-04 (CAN bus flooding / DoS)

Procedure:

1. Establish normal operation with valid IO board and BMS heartbeat traffic.
2. Flood CAN2 with high-priority frames at >80 % bus load.
3. Verify IO board heartbeat timeout triggers safe state within 1 s.
4. Repeat on CAN1 with BMS heartbeat and verify safe state within 5 s.
5. Verify normal operation resumes automatically after the bus flood stops and heartbeats return.

Pass: Safe state is entered within the FSR-17 timeout under bus flood, and the system recovers cleanly when flooding stops.

7. Implementation Guidance

7.1 HMAC Key Management

The user manages their own signing key. The project will provide a build script that:

1. Generates a random 256-bit key if none exists (stored in `keys/fw_signing_key.bin`).
2. Embeds the key into the bootloader binary at build time.
3. Provides a signing tool (`sign_firmware.py`) that takes a compiled firmware binary and produces a signed update package.

The key is never transmitted over CAN. It never leaves the user's build machine. Key compromise means someone gained access to the user's build environment, at which point they can already build and flash arbitrary firmware (physical access or remote access to build machine).

7.2 CAN Update Protocol

Recommended protocol (kept simple for open-source implementability):

Table 8: CAN Update Frame Format

Frame Type	CAN ID	Payload	Description
INIT	0x7F0	[0x01][version_major][version_minor][size_kb:2B]	Start update session
CHUNK	0x7F1	[seq:2B][data:0-6B]	Firmware data chunk
CHUNK_ACK	0x7F2	[seq:2B][crc32:4B]	Chunk received, CRC attached
HASH	0x7F3	[sha256_first_8B]	Image hash (send 4 frames for full 256 bits)
SIGN	0x7F4	[hmac_first_8B]	HMAC signature (send 4 frames for full 256 bits)
COMMIT	0x7F5	[0x02]	Verify and commit
STATUS	0x7F6	[status_code][progress_pct]	Bootloader response
ABORT	0x7F7	[error_code]	Error / user abort

7.3 Enabling RDP for Security-Conscious Users

For users who want additional protection, document (but do not mandate) how to enable STM32 Read Protection:

> **Optional: STM32 RDP Level 1**
>
> RDP Level 1 prevents external debuggers from reading flash memory via SWD/JTAG. The bootloader and application remain fully functional. Flash can still be mass-erased (which also erases the key).
>
> To enable: `ST-Link Utility → Target → Option Bytes → RDP = Level 1`

> **Warning:** RDP Level 2 is irreversible and disables debug permanently. Do not recommend Level 2 in documentation; mention it exists but strongly warn against it for a development-friendly open-source project.

8. Gap Analysis

Table 9: Cybersecurity Gaps

Gap	Risk	Mitigation	Effort
No HMAC signature verification yet	**HIGH**	Add HMAC-SHA256 to bootloader; provide signing script	Medium
No CAN update integrity protocol yet	**HIGH**	Define chunk format + CRC + SHA-256 hash	Low
No anti-rollback counter yet	MEDIUM	Add 64-bit counter to config flash page	Low
No freshness counter on CAN frames yet	MEDIUM	Add 32-bit rolling counter to BMS/IO board heartbeat frames	Low

8.1 HARA Cross-Reference

This TARA is a companion document to the HARA. Safety-relevant cybersecurity threats (those that could cause unintended tractive effort or prevent safe state entry) are linked to HARA hazards:

Table 10: TARA-to-HARA Cross-Reference

TARA Threat	HARA Hazard	Shared Mitigation
T-01 (malicious firmware upload)	H-01 (unintended tractive effort), H-15 (software error)	CSR-01 (firmware signing) supplements FSR-19 (boot CRC) and FSR-16 (POST) for boot-time integrity; it does not supplement throttle-input FSRs.
T-02 (firmware corruption)	H-15 (software error), H-14 (latched tractive effort)	CSR-02 (integrity) supplements FSR-19 (boot CRC)
T-04 (CAN DoS)	H-03 (loss of tractive effort)	CSR-05 references FSR-17 (heartbeat timeout)
T-05 (forged CAN frames)	H-01 (unintended tractive effort)	CSR-06 (freshness) supplements FSR-17 (safe defaults)

> **Note on Separation of Concerns**
>
> The HARA addresses **random hardware failures** and **systematic design faults** that cause hazardous events. This TARA addresses **intentional malicious actions** that cause the same hazardous events. The requirements are separate (FSRs vs. CSRs) because they are verified against different fault models (random fault injection vs. adversarial attack simulation), but they share the same safe-state destination: SSO when anything goes wrong.

References

Table 11: Referenced Standards and Documents

Reference	Title / Description
ISO/SAE 21434:2021	*Road vehicles — Cybersecurity engineering*. Defines threat analysis methodology, cybersecurity requirements, and validation for road vehicle electrical/electronic systems.
SAE J3061	*Cybersecurity Guidebook for Cyber-Physical Vehicle Systems*. Informative guidelines for automotive cybersecurity engineering.
ISO 26262-3:2018	*Road vehicles — Functional safety — Concept phase*. HARA methodology companion; safety/cybersecurity overlap addressed in both documents.
FIPS 198-1 / NIST SP 800-107	*FIPS 198-1 / NIST SP 800-107* — HMAC-SHA256 implementation reference.
STM32H723 Reference Manual (RM0433)	*STM32H723/733 — Arm Cortex-M7 MCU*. Flash protection (RDP, WRP), CRC peripheral, and bootloader application note (AN2606).
HARA (this project)	*Hazard Analysis and Risk Assessment — Traction Inverter/VCU*. Companion document covering functional safety. Cross-referenced for safety-relevant cybersecurity threats.

Document History

Table 12: Revision History

Version	Date	Changes
1.0	June 13, 2026	Initial TARA release. 7 threat scenarios (T-01 through T-07), 7 cybersecurity requirements (CSR-01 through CSR-07), 8 test cases (CT-01 through CT-08). Open-source trust model: no OTP, no vendor lock-in, user-managed HMAC-SHA256 keys. ISO/SAE 21434 aligned. Cross-referenced to HARA companion document.
1.1	July 8, 2026	User-sovereignty security model; 7 threat scenarios (T-01 through T-07); 7 Cybersecurity Requirements; 8 cybersecurity test cases. (entry added retroactively)
1.2	July 13, 2026	Editorial consistency pass: restored missing v1.1 revision-history entry; cross-reference and formatting fixes; CT-08 objective aligned to CSR-01 (HMAC key rotation); CT-09 added for CSR-05 CAN DoS / heartbeat timeout.